

Quasipolynomial Bounds for Arithmetic Progressions

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Abstract

We prove Erdős’s conjecture that every set of positive integers with divergent reciprocal sum contains arithmetic progressions of every finite length. More quantitatively, for every fixed $k \geq 3$, we show

$$r_k(N) \leq C_k N \exp(-c_k (\log N)^{\varepsilon_k})$$

with $C_k, c_k, \varepsilon_k > 0$, where $r_k(N)$ is the largest size of a subset of $\{1, \dots, N\}$ with no nonconstant k -term arithmetic progression.

Contents

1	Introduction	4
1.1	Density thresholds and summability	5
1.2	Prior density bounds	5
1.3	Preserving structure through density increments	6
1.4	Coloring thresholds	8
1.5	How to read the proof	8
2	Polynomial cells and the increment theorem	8
2.1	Precision conventions	8
2.2	Cells and certificates	9
2.3	The triangular increment	9
2.4	Polynomial patches	10
2.5	The absolute starting rule	11
2.6	Scalar rank and scale conventions	12
2.6.1	Formal rank	12
2.7	Descending precision budgets	12
2.8	From a cell to the next certificate	12
3	Constrained paths and their detection estimates	14
3.1	Normalized lift charts	15
3.2	The coefficient law and the actual path ensemble	16
3.3	Comparing sampled and ambient tests	21
3.3.1	Niltests	21
3.3.2	Cube normalization	21
3.4	The geometry needed in lifting	23
3.5	The stopped tree of a triangular cell	23

4	Preparing a cell by rank cuts	24
4.1	The relation needed for a rank cut	25
4.2	A rank cut preserving the box certificate	25
4.3	Two recovery estimates	28
4.4	The sharpened major statement	32
4.5	Transferring the failed relation through earlier layers	33
4.6	Termination and the order of the budgets	35
5	Detection independent of the current width	36
5.1	A pseudorandom chart weight	36
5.2	Densification without paying the chart mass	37
6	Forward sampling of a prepared cell	40
6.1	Individual moduli and selected affine planes	41
6.2	Descent of the density certificate	42
7	Returning the increment: comparison and passive layers	44
7.1	Paired cutoffs and the inductive score	44
7.2	The comparison measure	46
7.3	Proof of the selected-plane estimates	48
7.4	Scalar transfer with labels and flags	50
7.5	The group representation and its fixed menus	52
7.6	Reducing degree while retaining a marked projection	54
7.6.1	Symbols and real or rational control	54
7.7	Warm matching at a passive layer	56
7.8	Returning an ordinary orbit	57
7.9	Scalar return with a small retained modulus	59
8	Active globalization with exact marking	60
8.1	The inductive statement	60
9	Parameter extraction and exact restoration	69
9.1	Removing flags and solving inactive equations	70
9.2	Real pivots and boundary losses	71
9.3	Eliminating the full active modulus	72
9.4	Quantitative freezing	73
9.5	Localizing the domain in normalized coordinates	74
9.6	Exact slots and fixed weight inflation	75
10	Triangular iteration	76
10.1	Returning to simultaneous residual boxes	76
10.2	Structural choices come first	77
10.3	The numerical schedule	78
10.4	The increment and the density bound	80

11	Weighted consequences	82
11.1	Dyadic summation	82
11.2	Weights larger than the reciprocal	83
11.3	Logarithmic weights and dense subsets of the primes	83
11.4	Uniform harmonic tails	84
A	Quantitative conventions and symbols	86
A.1	Filtered nilmanifolds	86
A.2	Symbols and separation	89
A.3	Gowers norms and the normalized square	90
B	Dropping the top frequency of a biased symbol	93
B.1	Recovering coordinate derivatives	94
B.2	Proof of the step-drop lemma	96
C	Shift comparison	100
C.1	The degree-one argument	101
C.2	The finite-list reduction	107
C.3	Modeling and a common anchor	108
C.4	Recovering tests from two fixed observations	109
D	Positive counting and absolute increments	114
D.1	From positive comparison to counting	115
D.2	Patches and the lifting statement	119
E	Unconstrained sampling and scalar transfer	123
E.1	Scalar comparison without constraints	123
E.1.1	Stability of the structured baseline	124
E.1.2	Residue contraction and low-degree projections	128
E.1.3	Proof of scalar transfer	131
F	Constrained affine sampling	136
F.1	The constrained sampler and its parameter order	136
F.2	The geometry of small lifts	138
F.3	The coefficient tilt	139
F.4	Fourier modes and Boolean cube evaluation	141
G	Cube comparison and pathwise detection	146
G.1	The cube functional to be compared	146
G.2	Evaluating the product blocks	147
G.3	Counting integer fibers	152
G.4	Kernel tuples and finite evaluation covers	153
G.5	Stability of nonsingular polynomial images	155
G.6	Proof of the cube comparison	158
G.7	Detection and finite-dimensional approximation	161
G.8	Cube comparison in normalized chart units	163

H	Positive score recovery from constrained samples	164
H.1	A positive comparison weight for a fixed path	165
H.2	Modular rank and the bounds for the comparison weight	166
H.2.1	A modular Fourier estimate	166
H.2.2	Producing the modular rank hypotheses	168
H.2.3	Bounds for the comparison weight	171
H.3	Transferring the score and counting the new slots	172
H.4	Completion of the constrained sampling theorem	175
I	Relative lifting with additive rank	178
I.1	The weighted major decomposition	179
I.1.1	The translation group and its potential	180
I.1.2	Proof of the weighted major decomposition	182
I.2	Globalization in a fixed group	183
I.3	Removing the lowest patch layer	189

1 Introduction

Erdős asked whether every set A of positive integers for which $\sum_{a \in A} 1/a$ diverges contains arithmetic progressions of every finite length [14, Problem 4.33.6]. We prove a quantitative density bound that implies this assertion. For integers $k \geq 3$ and $N \geq 1$, let $r_k(N)$ be the largest cardinality of a subset of $[N] = \{1, \dots, N\}$ containing no nonconstant k -term arithmetic progression $a, a+d, \dots, a+(k-1)d$, with $d > 0$. All logarithms are natural.

Theorem 1.1. *For each fixed integer $k \geq 3$ there are constants $C_k, c_k, \varepsilon_k > 0$ such that, for every $N \geq 2$,*

$$r_k(N) \leq C_k N \exp(-c_k (\log N)^{\varepsilon_k}). \quad (1.1)$$

Equivalently, there is $A_k \geq 1$ such that an α -dense subset of $[N]$ contains a nonconstant k -term progression whenever

$$\log N \geq A_k (2 + \log(1/\alpha))^{A_k}, \quad 0 < \alpha \leq 1. \quad (1.2)$$

The constants may depend on the fixed length k ; the exponent is not optimized. The reciprocal-sum consequence follows immediately from summing the density bound over dyadic intervals.

Corollary 1.2 (Divergent reciprocal sums). *Every set $A \subseteq \mathbb{N}$ with $\sum_{a \in A} a^{-1} = \infty$ contains nonconstant arithmetic progressions of every finite length.*

Proof. Fix $k \geq 3$, and suppose that A contains no nonconstant k -term progression. For $m \geq 0$, translation identifies $[2^m, 2^{m+1}) \cap \mathbb{Z}$ with an interval of length 2^m . Thus $|A \cap [2^m, 2^{m+1})| \leq r_k(2^m)$. Theorem 1.1 implies, for $m \geq 1$,

$$\sum_{a \in A \cap [2^m, 2^{m+1})} \frac{1}{a} \leq 2^{-m} r_k(2^m) \leq C_k \exp(-c_k (m \log 2)^{\varepsilon_k}).$$

For every $d, \varepsilon > 0$, the sequence e^{-dm^ε} is summable: since $m^\varepsilon / \log m \rightarrow \infty$, its terms are at most m^{-2} for all sufficiently large m . The $m = 0$ interval contributes at most one. The reciprocal sum of A is therefore finite, a contradiction. Divergence also implies that A is infinite, which supplies lengths one and two. $\square$

1.1 Density thresholds and summability

The threshold in (1.2) is quasipolynomial in $1/\alpha$: the logarithm of a sufficient interval length is a fixed power of $2 + \log(1/\alpha)$. The inversion between the two forms of Theorem 1.1, including the endpoints, is proved in Section 10.

The infinite-set consequence depends on more than a density tending to zero. Divergence of a reciprocal sum allows zero-density sets, and the qualitative estimate $r_k(N) = o(N)$ does not by itself settle Erdős's question. The proof of Corollary 1.2 uses the stronger condition

$$\sum_{m \geq 1} \frac{r_k(2^m)}{2^m} < \infty.$$

Our bound also gives a stronger divergence criterion: for every fixed $B \geq 0$, a set $A \subseteq \mathbb{N}$ satisfying

$$\sum_{a \in A} \frac{(\log(2+a))^B}{a} = \infty$$

contains progressions of every finite length. Section 11 proves this criterion, uniform harmonic bounds and tails, and estimates for larger weights whose parameters depend on k . It also recovers the theorem of Green and Tao that every subset of the primes of positive relative upper density contains infinitely many progressions of every finite length [21, Theorem 1.2].

1.2 Prior density bounds

Van der Waerden's theorem concerns a partition: every finite coloring of the positive integers has a monochromatic arithmetic progression of each finite length [47]. The density problem asks for a condition on a prescribed set. Erdős and Turán began the study of the corresponding extremal functions in 1936 and explicitly conjectured that $r_3(N) = o(N)$ [15]. The all-length positive-density conjecture appears explicitly in Erdős's later account [13, Section I.19]. Roth proved the three-term case by Fourier-analytic methods [39]; Szemerédi proved the four-term case in 1969 and the general theorem in 1975 [44, 45]. Furstenberg's subsequent proof recast the assertion as multiple recurrence in a measure-preserving system [16]. The reciprocal-sum conjecture extends this density problem to sparser sets, including the primes [14, Problem 4.33.6].

For three terms, successive quantitative advances sharpened Roth's density-increment method: a shortage of progressions forces a density increase on a smaller structured set. Roth obtained $r_3(N) \ll N/\log \log N$. Heath-Brown and Szemerédi replaced the double logarithm by a positive power of $\log N$ [26, 46], and Bourgain developed density increments on regular Bohr sets [9, 10]. Sanders reached $r_3(N) \ll N/(\log N)^{1-o(1)}$, and Bloom improved the remaining logarithmic factors [41, 4]. Bloom and Sisask gave an almost-periodicity approach to this near-logarithmic range [5]. Their later bound $r_3(N) \ll N/(\log N)^{1+c}$, for an absolute $c > 0$, crossed the dyadic summability threshold and established the three-term reciprocal-sum conjecture [6].

Kelley and Meka obtained the stronger bound

$$r_3(N) \ll N \exp(-c(\log N)^{1/12})$$

by combining positivity, dependent-random-choice sifting and almost-periodicity to control the loss in a density increment [30]. Bloom and Sisask simplified the argument and then improved the exponent to $1/9$ [8, 7]. Raghavan subsequently obtained $r_3(N) \leq N \exp(-(\log N)^{1/6-o(1)})$ [38, Theorem 1.4]. These refinements concern three terms. On the lower-bound side, Behrend's

construction already gives $r_3(N) \geq N \exp(-C\sqrt{\log N})$, so a fixed power saving in N is impossible even in this case [1].

Gowers's approach to longer progressions detects higher-order structure. He introduced the uniformity norms that control progression counts and extended density increments beyond linear Fourier analysis [17, 18]. His general theorem gives $r_k(N) \ll_k N/(\log \log N)^{c_k}$ for every fixed k , with $c_k > 0$. Green and Tao obtained $r_4(N) \ll N/(\log N)^c$ [24]. Leng, Sah and Sawhney first treated five terms and then proved, for every fixed $k \geq 5$,

$$r_k(N) \ll_k N \exp(-(\log \log N)^{c_k}), \quad 0 < c_k < 1$$

[33, 34]. Their general argument combines a quasipolynomial inverse theorem with density increments. The inverse theorem controls the structured correlation obtained from a large uniformity norm; the cost of localizing and iterating that correlation remains a separate issue. The resulting saving in $\log \log N$ does not give the dyadic summability above.

Theorem 1.1 gives a stretched-exponential saving in $\log N$ for every fixed length. Its contribution to Erdős's conjecture is this all-length, summable bound; no improvement of the three-term exponent is claimed. We next explain the quantitative constraint on the density-increment regions that allows the argument to be iterated at this scale.

1.3 Preserving structure through density increments

A fixed multiplicative density gain can be applied only $O_k(1 + \log(1/\alpha))$ times before the density exceeds one. To obtain Theorem 1.1, the total logarithmic loss of interval length over these increments must be polynomial in $2 + \log(1/\alpha)$. We therefore need to control both the number of coordinates describing each new region and the precision of the constraints inherited from previous regions.

The regions are triangular polynomial cells. A spatial integer variable u determines integer blocks $b_1, \dots, b_D$ successively through constraints of the form

$$\|b_h - C_h(u, b_1, \dots, b_{h-1}) - l_h\|_\infty \leq w_h, \quad 1 \leq h \leq D, \quad 0 < w_h \leq 1/32.$$

Here D depends only on k , each l_h is a fixed center, and C_h is a polynomial. The spatial variable has weight one, the block b_h has weight h , and C_h has weighted degree at most h . The narrow widths make each integer block unique when it exists. Keeping these integer blocks, and substituting them exactly into all later polynomials, preserves the equations defining the old region when a new constraint is added. No bound on the coefficients of the C_h is imposed.

Write $Q_h = \log(2/w_h)$ for the precision at weight h . The main increment theorem bounds the *additional* precision at weight h by a polynomial in the dimension data, the logarithmic density parameter, and the precisions at strictly higher weights. In particular, this bound does not use the current Q_h or the lower precisions. The dimensions obey an analogous descending recurrence. Since the number D of weights is fixed, descending induction bounds all dimensions and precisions by polynomials through the required number of increments. Section 2 states these recurrences precisely; Section 10 proves that the construction satisfies them.

Two different arguments supply these bounds. The first produces an *absolute increment* for a $[0, 1]$ -valued function on an integer box with sufficiently long sides. The function must have positive mean and contain no nonconstant k -term progression in its support. The increment adds only a number of determining coordinates polynomial in the logarithmic inverse mean, independently of the dimension of that box. The second returns such an increment from a sampled box to the original cell, preserving the old determining equations and the descending dependence of their width losses.

An absolute increment with uncontrolled dependence on the old widths would not suffice for this second task.

The absolute increment. The higher-order inverse theorem converts a large uniformity norm into correlation with a nilsequence, a Lipschitz function evaluated along a polynomial orbit on a compact nilmanifold. This description of structured correlations has both ergodic and additive-combinatorial origins. Host and Kra developed seminorms and nilsystem characteristic factors, Ziegler proved the universal-factor theorem, and Bergelson, Host and Kra isolated nilsequences in multiple correlations [27, 48, 2]. Green and Tao obtained locally quadratic structure for the U^3 norm [20]; Green, Tao and Ziegler then proved the all-degree interval inverse theorem [25]. Manners gave explicit quantitative inverse bounds on prime cyclic groups [36]. The stronger quasipolynomial bounds of Leng, Sah and Sawhney are the interval input used here [35, Theorem 1.2]; the appendices prove the box and product-cyclic reductions needed in the argument.

To pass from structured tests to positive progression counting, we prove a shift-comparison theorem: lowering the testing degree permits multiplication by a translated nonnegative function outside a small exceptional set of shifts. Its degree-one argument adapts Kelley–Meka unbalancing and sifting, as developed on Bohr sets by Bloom and Sisask [30, 8]. The almost-periodicity input is Schoen–Sisask’s Theorem 5.4 [43]. It combines the probabilistic method of Croot and Sisask with Sanders’s local control of large spectra [12, 40]; see also Sanders’s Bogolyubov–Ruzsa theorem [42]. The Schoen–Sisask radius bound retains the incoming radius as a multiplicative factor, which is essential for the shift comparison.

Higher-degree comparison requires control of polynomial orbits. Leibman established a qualitative distribution theorem [31]. Green and Tao made this theory quantitative and factored polynomial orbits into smooth, equidistributed and rational components [23]. Leng’s efficient estimates reduce the nilpotency step rather than repeatedly reducing the dimension [32]. We use this mechanism, together with the inverse theorem, to remove top-degree frequencies in the comparison argument. The appendices prove the required forms with explicit dependence on all growing dimensions and parameter counts.

The comparison and counting arguments first give the absolute rule in a fixed dimension. A relative lifting theorem extends it with only an additive number of new determining coordinates. Applied with the constant old region, this yields the dimension-independent fresh-coordinate bound in Lemma 2.2. This foundational lifting argument supplies the absolute rule; preservation of the old width budgets requires the separate return argument.

Returning to the original cell. A density increment found on a sampled parameter box must be expressed again in the original variables. The cell can have very small mass, and an error in one of its determining integers would enter every later polynomial that uses that integer. The return therefore keeps the old equations exact.

For tests of degree below the current layer weight, we use a cube version of Conlon–Fox–Zhao densification to replace sparse factors by bounded ones before applying inverse theory [11]. The comparison proved here has bounds independent of that layer’s chart mass. At layers where the testing degree is at least the layer weight, polynomial families retain the old equations through an exact projection while allowing the newly added region to have higher degree. Preparation, ascent and extraction then control the relative loss of each width by structural data and higher-weight precision budgets. The general route from inverse theory to density increments is classical [22]; retaining this dependence through the return is the quantitative issue addressed here.

1.4 Coloring thresholds

Let $W_r(k)$ be the least integer N such that every coloring of $[N]$ with r colors contains a monochromatic nonconstant k -term arithmetic progression. A largest color class has density at least $1/r$, so Theorem 1.1 gives

$$W_r(k) \leq \left\lceil \exp(A_k(2 + \log r)^{A_k}) \right\rceil \quad (r \geq 1), \quad (1.3)$$

after increasing A_k . Thus, for fixed k , the sufficient length is quasipolynomial in the number of colors. The companion paper [37, Theorem 7.2] gives the complementary lower bound $W_r(k) > \exp((\log r)^2/(64 \log 2))$ for all integers $r \geq 256$ and $k \geq 3$. Together these estimates show that for each fixed $k \geq 3$, growth in the number of colors is superpolynomial but at most quasipolynomial. The upper constant and exponent may depend on k .

1.5 How to read the proof

Section 2 defines the cells and precision budgets, states the triangular increment theorem, and explains the dimension-independent absolute rule. Figure 3 gives the main route, and Table 1 identifies the analytic statements and their full proofs. Section 3 constructs the constrained path law and states its sampling guarantees. Section 4 prepares the cell by rank cuts, and Section 5 proves the detection bound independent of the current width. Section 6 transfers the certificate to terminal boxes and obtains an absolute increment. Section 7 sets up its return, develops the comparison estimates, and handles the passive layers. Section 8 treats the active layers. Section 9 extracts the next cell, and Section 10 closes the recurrences and proves Theorem 1.1. Section 11 establishes the weighted consequences.

The appendices contain the rational and symbol calculus, shift comparison, progression counting, constrained sampling, cube comparison and relative lifting arguments. The two principal external analytic inputs to the increment proof are the interval inverse theorem of Leng, Sah and Sawhney and the radius-sensitive almost-periodicity theorem of Schoen and Sisask [35, 43]. Their precise hypotheses are stated at the points of application.

2 Polynomial cells and the increment theorem

Fix $k \geq 3$ and put $s = k - 2$. We seek density increments on regions that retain every polynomial constraint introduced by an earlier increment. The regions will be triangular polynomial cells. Their integer coordinates are determined successively, and their residual widths are indexed by weight. The central estimate will bound the relative width loss at each weight using only higher-weight widths.

Every degree and descent depth below is fixed in terms of s ; dimensions may grow. A degree bound may be replaced by any fixed larger one. We work on rectangular integer boxes, with uniform probability averages and max norms, allowing dimension factors in norm comparisons. A *slice* is an integer-affinely reindexed residue subbox. Different coordinate strides are permitted. Its logarithmic cost includes their logarithms and the logarithms of inverse relative side lengths. Sufficient lower bounds on the original sides absorb all endpoint rounding errors. A function is *progression-free* when its support contains no nonconstant integer-vector k -term progression.

2.1 Precision conventions

Write $\mathcal{P}(P) = (2 + P)^C$ and $\mathcal{B}(P) = \exp((2 + P)^C)$, with C depending only on the fixed degrees and allowed to increase. A quantity has polynomial *logarithmic* cost when its size and, when needed,

inverse size are at most $\mathcal{B}(P)$. Dimensions, numbers of structural variables, and polynomial degrees are not measured logarithmically. Only a bounded number of successive polynomial changes of budget is implicit; a polynomial number of multiplicative factors is permitted because their logarithmic costs add. The height of rational data bounds both numerators and positive denominators; a denominator bound alone leaves numerators unrestricted. Multiplying a density threshold by a fixed positive constant changes its logarithmic budget by only a fixed additive amount. We absorb such changes in the displayed polynomial bounds, including the fresh rank bound d_0 , whenever a discounted threshold is used.

2.2 Cells and certificates

Choose a fixed $D \geq s$, whose existence is verified by the weight argument in Section 9. Give the spatial variable u weight one and integer blocks $b_h \in \mathbb{Z}^{d_h}$ weight h , for $1 \leq h \leq D$. The weighted degree of a monomial is the sum of its variable weights, counted with multiplicity. A triangular cell is specified by polynomials

$$C_h(u, b_{<h}) \in \mathbb{R}^{d_h}, \quad \deg_{\text{wt}} C_h \leq h,$$

bounded centers l_h , and widths $0 < w_h \leq 1/32$. There is no bound on the coefficients of C_h . Integer translations keep the centers bounded without changing this convention. Zero-dimensional blocks are allowed. Put

$$Q_h = \log(2/w_h).$$

Thus Q_h measures the logarithmic precision of the residual at weight h .

For a small $\gamma > 0$, let B^- and B^+ be the indicators that there is an integer tuple satisfying respectively

$$\|b_h - C_h(u, b_{<h}) - l_h\|_\infty \leq w_h, \quad \|b_h - C_h(u, b_{<h}) - l_h\|_\infty \leq (1 + \gamma)w_h \quad (1 \leq h \leq D). \quad (2.1)$$

Triangularity and the narrow widths imply uniqueness of the tuple, also on slightly padded boxes. Boundary conventions may be chosen consistently and do not require any regularity of the point distribution. A *density certificate* is

$$\mathbb{E}fB^- > a\mathbb{E}B^+, \quad 0 \leq f \leq 1, \quad (2.2)$$

where f is progression-free. Its excess is not assumed to have a quantitative lower bound.

We may replace the two indicators temporarily by product smooth cutoffs: the small cutoff is below the strict indicator, and the large cutoff is one on the enlarged box and supported slightly outside it. Their plateaus and transitions are proportional to the width and gap. When translating small boxes, the ratio of enlarged to strict coverage volumes is $(1 + O(\gamma))^{O(\sum_h d_h)}$. We choose γ from dimension forecasts, independently of all widths, and pay this ratio by a small multiplicative discount of the target. Thus no hidden boundary regularity assertion is needed.

2.3 The triangular increment

The new cell will retain the old determining coordinates and may add two kinds of integer coordinates. Fresh *absolute slots* come from the analytic increment. Preparation may also copy a higher-weight block into a lower-weight block. The following theorem controls both additions and the loss of residual width.

Theorem 2.1 (Triangular increment). *There are fixed $D = D_k$, $\eta = \eta_k > 0$, and polynomials with degrees depending only on k having the following property. Suppose $p \geq 2$, $a \geq e^{-p}$, and a progression-free input on a positive-dimensional integer box has a certificate (2.2), with width logs $Q_h = \log(2/w_h)$. If every root side is sufficiently large with logarithmic cutoff polynomial in p , the dimension data, and all Q_h , then either the hypotheses are impossible or a root slice carries a new certificate at threshold $(1 + \eta)a$.*

The new cell retains the determining data through exact integer reparametrizations and adds at most $d_0 = (2 + p)^{C_k}$ fresh absolute slots together with downward preparation copies. Its dimensions d'_i and width logs Q'_i satisfy

$$d'_i - d_i \leq d_0 + P_i(p, d_{i+1}, \dots, d_D), \quad (2.3)$$

$$Q'_i - Q_i \leq P'_i(p, \text{all dimension forecasts}, Q_{i+1}, \dots, Q_D). \quad (2.4)$$

The root slice cost has polynomial logarithmic bound in all the same data, now including every Q_i . Polynomial exponents and the fixed gain are independent of the number of increments. Gap and discount choices can be made in advance from the dimension and cut-count forecasts.

The main point is the absence of $Q_i, Q_{<i}$ from the right-hand side of (2.4). For the interval application, reserve $T = O_k(p)$ increments, starting with no slots and fixed width data. The dimension recurrence first bounds the top block, then each lower block. Proposition 10.2 verifies this recurrence for the actual preparation copies and bounds all within-round structural counts. It supplies one polynomial forecast $d_*(p)$ for the entire run, before any numerical precision is chosen.

With that forecast fixed, choose nondecreasing polynomials R_i dominating the width losses in (2.4). Define

$$V_D = Q_D(0) + TR_D(p, d_*), \quad V_i = Q_i(0) + TR_i(p, d_*, V_{i+1}, \dots, V_D) \quad (i = D - 1, \dots, 1).$$

Descending induction gives $Q_i(t) \leq V_i$ for all $t \leq T$. Since D is fixed, every V_i is polynomial in p . The logarithmic root-side requirements and the sum of all logarithmic slice costs are therefore polynomial as well. Meanwhile the threshold rises by a fixed factor at every increment and, after $O_k(p)$ steps, exceeds one. A certificate at threshold at least one is impossible because $fB^- \leq B^+$ pointwise. Section 10 makes this deduction precise and proves the required dimension and width recurrences.

It remains to construct the increment with these particular dependencies. A bound polynomial in all current width logs would not suffice: repeated substitution could raise the polynomial degree at every round.

2.4 Polynomial patches

The analytic increment produces a smooth test rather than a cell indicator. These tests are polynomial patches; their integer slots will supply the new absolute coordinates. Give integer slots $b_1, \dots, b_d$ nondecreasing weights $1 \leq \omega_i \leq q$, and give each spatial parameter weight one. A degree- q patch is

$$B(u) = \sum_{b \in \mathbb{Z}^d} \Phi(S_u(b)), \quad S_u(b)_i = b_i - A_i(u, b_1, \dots, b_{i-1}), \quad (2.5)$$

where $\deg_{\text{wt}} A_i \leq \omega_i$ and $\Phi : \mathbb{R}^d \rightarrow [0, 1]$ is Lipschitz and supported in $[-1/4, 1/4]^d$. Its rank is the number d of integer slots. The number of parameters, d , and $\log(2 + \text{Lip } \Phi)$ enter its logarithmic complexity; the coefficients of the A_i do not. Rank zero includes the constant patch 1. Successively determining the integer slots shows that at most one summand is nonzero, so $0 \leq B \leq 1$.

Figure 1 illustrates the distinction between integer determining slots and small real residuals. For a nonnegative test B , its *score at target a* is $\mathbb{E}(f - a)B$. Positivity means that the mean of f under the probability weight $B/\mathbb{E}B$ exceeds a ; a quantitative lower bound on the score also controls the available mass.

2.5 The absolute starting rule

Positive progression counting first gives an increment on boxes of a fixed dimension. Relative lifting transfers that rule to arbitrary dimension while keeping the number of new slots separate from the other complexity costs. The following form is used at the terminal boxes of the cell argument.

Lemma 2.2 (Dimension-independent fresh rank). *There are $\xi > 0$ and $C = C_k$ with the following property. Let $p \geq 2$, and let f be a progression-free $[0, 1]$ -valued function of mean at least $a \geq e^{-p}$ on a box of positive dimension n . Suppose every side is sufficiently large, with a sufficient logarithmic cutoff polynomial in $2 + p + n$. Then, unless the hypotheses are impossible, f admits a degree- s polynomial patch on a slice with positive score at target $(1 + \xi)a$. The number of slots is at most $d_0 = (2 + p)^C$, independent of n . The logarithms of inverse score, Lipschitz norm, slice cost, and sufficient side cutoff are polynomial in $2 + p + n$.*

Proof. Fix the base dimension $n_0(s)$ and prime-side comparability constant in Proposition D.7. Proposition D.3, Lemma D.5, and Theorem C.1 give an absolute rule on these fixed-dimensional boxes, with gain $1 + \xi_0$ and rank $(2 + p)^{C_k}$. The side cutoff absorbs all three requirements; the prime sides can be odd.

Choose a small fixed $c > 0$. At any actual mean $\alpha \geq (1 - c)a$, the absolute rule has logarithmic budget $O(1 + p)$ and gives positive score at the universal smaller target $\Lambda = (1 + \xi_0)(1 - c)a$. If $\Lambda \geq 1$, its nonnegative test gives an impossibility rule instead. In the relative theorem use the constant old test 1, of rank zero, at threshold $(1 - c)a$. Its old score is at least ca . Enlarge the resource parameter to a polynomial in $p + n$ to cover the dimension, absolute cutoff, and $\log(1/(ca))$. Theorem I.15 proves Proposition D.7, including its separate impossibility clause. Its rank conclusion

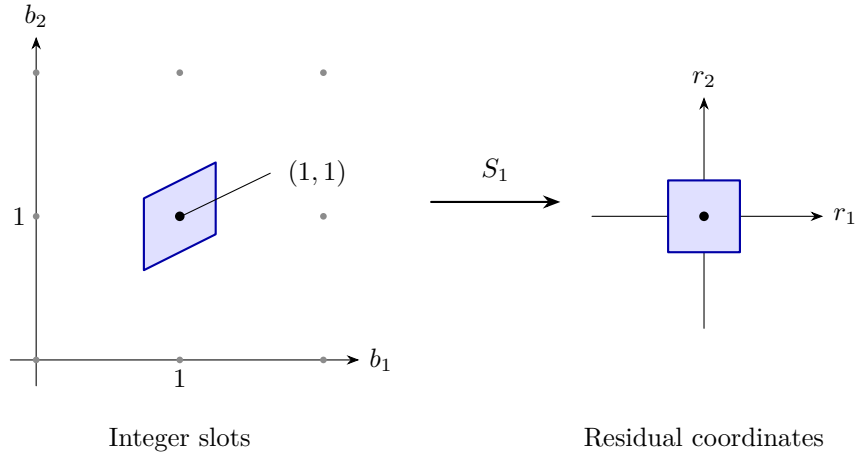


Figure 1: A degree-two patch of rank two, with slot weights 1, 2, at $u = 1$. Here $A_1(u) = u$, $A_2(u, b_1) = (u^2 + ub_1)/2$, and $S_1(b_1, b_2) = (b_1 - 1, b_2 - (1 + b_1)/2)$. The shaded parallelogram is the inverse image of $[-1/4, 1/4]^2$, the permitted support of the kernel. Only the integer pair $(1, 1)$ lies in it. For a general patch, at most one integer tuple satisfies the support constraints, by successive determination of the slots.

is $d + d_0$, so the enlarged resource parameter does not enlarge the fresh rank d_0 . Choose its fixed target discount ϵ_0 and then c so that $(1 - \epsilon_0)(1 + \xi_0)(1 - c) > 1 + \xi$ for a fixed $\xi > 0$. This gives the claimed rule and all its quantitative bounds. $\square$

2.6 Scalar rank and scale conventions

Preparation tests the homogeneous parts of the determining polynomials. For a box with side lengths $H = (H_1, \dots, H_n)$, write $H^\nu = \prod_i H_i^{\nu_i}$. A homogeneous scalar or vector polynomial is *slow* at budget p if its coefficient of u^ν has magnitude at most $\mathcal{B}(p)H^{-\nu}$. It is *rational* if its coefficients have a common denominator at most $\mathcal{B}(p)$, with no numerator bound. The homogeneous part at the assigned degree or weight is called its *symbol*. A polynomial is *fully slow* when the same coefficient bounds hold at every degree, including the constant coefficient, after the box has been translated to the origin. Thus real estimates use normalized variables, while denominator estimates use the raw integer variables. Group-valued symbols are introduced with the ascent representation in Section 7.

2.6.1 Formal rank

Suppose a degree- j polynomial C_j takes values in a rational space $W_j \subseteq \mathbb{R}^{d_j}$ on a box of sides H . It has rank at least R at this scale if every integer row $\lambda \in \mathbb{Z}^{d_j}$ with $\|\lambda\| \leq R$ and $\lambda|_{W_j} \neq 0$ has the following property: the degree- j homogeneous part of λC_j cannot be written as a polynomial whose coefficient of u^ν is bounded by R/H^ν plus a rational polynomial with common denominator at most R . For a system, this condition is imposed at every layer. This is the rank condition of Definition F.1; it is different from the slot count called the rank of a patch.

2.7 Descending precision budgets

Definition 2.3 (Descending budgets). At layer j , the *warm data* consist of p , forecasts for all dimensions and structural variable counts, gap parameters depending only on these data, and the widths and numerical budgets already chosen at layers strictly above j . A warm logarithmic bound is polynomial in these data and does not use any Q_i with $i \leq j$.

The *cold preliminary data* may additionally use the current width $\log Q_j$, including its forecast preparation losses, but not the eventual rank threshold, long path scale, height of the prepared value space, or parent-side cutoff at layer j . The *late data* are the perturbation, range, rank, and side choices made after all preliminary modeling and algebraic requirements at that layer. Their logarithms are polynomial in the cold preliminary budget.

The descending choice order is opposite to the order in which paths are sampled through the layers. Auxiliary models may have cold complexity. Their eventual return to the original variables must nevertheless have a warm *relative width loss*; the extraction argument preserves the defining polynomial maps exactly. Figure 2 displays the permitted dependence.

2.8 From a cell to the next certificate

Preparation removes low-rank relations while preserving the certificate (Proposition 4.1). Constrained paths then carry it to terminal boxes (Proposition 6.2), where Lemma 2.2 supplies a quantitative score. The passive and active return arguments carry that score back through the layers (Proposition 7.8 and Theorem 8.2). Extraction turns the returned score into a triangular cell (Proposition 9.1). Finally, Section 10 verifies the dimension and width bounds and iterates the increment.

During preparation the polynomial C_j takes values exactly in a rational subspace $W_j \subseteq E_j = \mathbb{R}^{d_j}$, initially the full space. The rank test defined above and in Definition F.1 requires that no integer

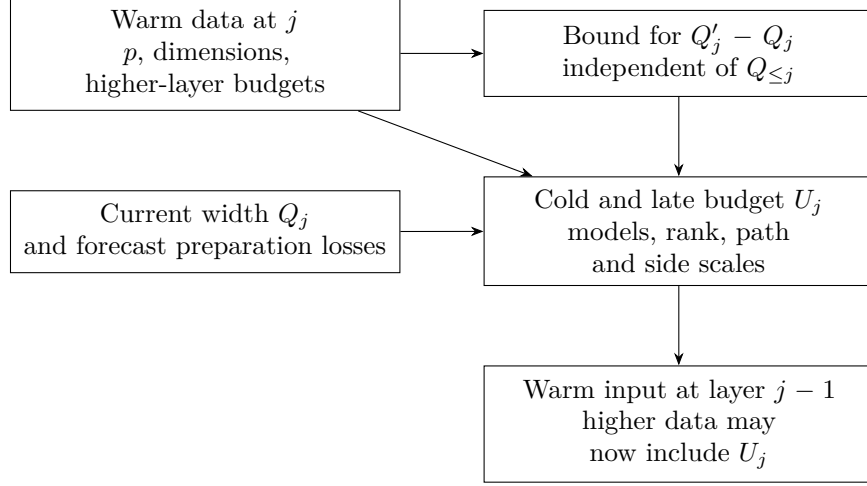


Figure 2: The allowed dependence is triangular. There is no arrow from the current width or its cold outputs back to its relative width loss. The full dependency schedule is proved in Section 10.

row of norm at most R_j , nonzero on W_j , has a degree- j symbol that is slow plus rational at that threshold. The test is made on the current path box at a history that has survived the earlier tests. The histories arriving at each layer carry the subprobability law of Section 3.5; stopping removes mass without renormalizing the survivors. Its failure probability may charge $Q_{\geq j}$ but not lower widths. There is no single root equidistribution test at the finest precision of all layers.

At a successful layer use a Haar center in $W_j/(W_j \cap \mathbb{Z}^{d_j})$, conditioned to the feasible range specified in Section 3.5. The lift has exact identity

$$C_j - c = y + \beta,$$

where $y \in \mathbb{R}^{d_j}$ is the small real residual and $\beta \in \mathbb{Z}^{d_j}$ is the integer lift. The raw determining slot is $b_j = \beta + b_{j,0}$ for a constant integer vector. This convention matters: every later old polynomial uses the actual integer slot, not an approximation to its value. Structural numbers of path columns are fixed polynomials of the input dimensions before any numerical precision is chosen. Consequently all ambient dimensions along a branch are polynomial in the current dimension forecasts. Sides are selected to support the next layer and finally Lemma 2.2.

The return processes layers from D down to 1. Layers $j > s$ are passive: the new patch does not acquire their integer arguments. At the final s active layers its weights may increase by fixed factors, while the old determining equations and their weights stay exact. This distinction is why a fixed D can be chosen before the iteration, even though the number of slots grows.

The input certificate and the returned score carry different information. Preparation needs only the strict inequality in the input certificate: its selections do not charge the inverse of that inequality's excess. Proposition 6.2 then gives terminal means near the current threshold on a quantitatively large set of paths. Lemma 2.2 supplies the quantitative score needed for ascent. Thus all subsequent approximation accuracies can be chosen from the recorded budgets, independently of the original excess.

The analytic proofs are grouped separately. Table 1, at the entrance to that part, locates each input statement and its proof.

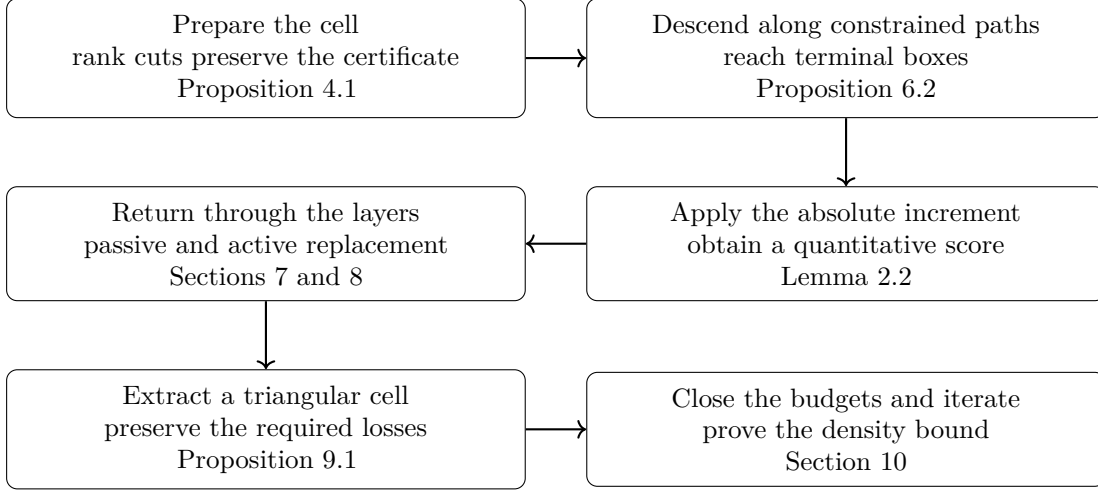


Figure 3: The main proof route. Preparation and extraction preserve the triangular dependence of the width losses; the absolute increment controls the number of fresh determining coordinates.

3 Constrained paths and their detection estimates

At one layer of a triangular cell, we seek many integer-affine maps $\psi(t) = x + Vt$ along which the current polynomial has a small polynomial residual and an exact integer-polynomial lift. The affine maps preserve arithmetic progressions; the polynomial identity will preserve the determining equations when a new patch is returned to the parent. We construct their probability law in this section.

There are two comparison estimates to prove for that law. With the center fixed, tests chosen separately on the sampled boxes must be detectable on the parent. After averaging the center, the one-site distribution must approximate the parent distribution. Below the current degree, the first comparison must be measured relative to the small chart: charging its reciprocal mass to the output complexity would charge the current width to its own relative loss.

The coefficient density will be a product law on an abstract coefficient torus. The actual ensemble is the integer spatial law weighted by that density. These are different probability spaces. The construction below defines both; high rank will supply the comparison between them. The general estimates for coefficient modes and polynomial images are proved in Appendices F and G.

Throughout this section, j is the current layer, $d = d_j$, $W \subseteq \mathbb{R}^d$ is rational, and

$$U = \prod_{a=1}^n \{0, \dots, H_a - 1\}, \quad C : \mathbb{R}^n \longrightarrow W, \quad \deg C \leq j.$$

Lower layers have already been substituted. Rank means the ordinary rank condition of Definition F.1, with the single tag j : no integer row of norm at most R_j , nonzero on W , takes the degree- j symbol of C to a slow-plus-rational polynomial at bound R_j . Warm, cold-pre, and late choices have the meanings fixed in Definition 2.3. All degrees and cube orders below belong to a fixed finite range. Structural numbers of columns are polynomial in dimensions and are fixed before numerical precisions.

3.1 Normalized lift charts

Put $\Lambda = W \cap \mathbb{Z}^d$ and $\Lambda' = \text{proj}_{W^\perp} \mathbb{Z}^d$. Choose the basis v_i , $1 \leq i \leq \dim W^\perp$, of the projected lattice Λ' supplied by Lemma F.4, and set

$$K_i = \max(1, \lceil |v_i|^{-1} \rceil), \quad k_i = \max(1, \lceil K_i v \rceil), \quad m_j = v^{\dim W} \frac{\prod_i k_i}{\text{covol}(\Lambda)}. \quad (3.1)$$

Here v is a sufficiently small warm fraction of w_j . All lift coordinates use this same physical width. More precisely, v/w_j has logarithmic cost polynomial in dimensions and $\log \gamma^{-1}$. Principal amplitudes in normalized lift units, and the proportionality constants in inactive side lengths, can have logarithmic costs depending only on dimensions. They are chosen small enough that all true and ideal lifts lie well inside a buffered chart of radius comparable to v , itself inside the required gap. These charts are narrow enough that every Boolean difference of an order used below has components of magnitude less than $1/2$.

For a center $c \in W/\Lambda$, choose a representative in W . On a uniqueness chart write

$$C(u) - c = y(u) + \beta(u), \quad \beta(u) \in \mathbb{Z}^d.$$

The sheet coordinates satisfy

$$\text{proj}_{W^\perp} y = \sum_i v_i z_i = -\text{proj}_{W^\perp} \beta.$$

Choose integer lifts of the basis vectors v_i , and let $l(z)$ be their integer linear combination. Then $\beta + l(z) \in \Lambda$. Choose an integer basis of Λ and let w be the coordinates of $\beta + l(z)$ in that basis. Primitivity of Λ shows that $\beta \mapsto (z, w)$ is an integer unimodular change of coordinates. Here w is a deck coordinate, not a width. A residue character of (z, w) is therefore a residue character of β at the same modulus; the real size of the basis coefficients does not enter this assertion.

Lemma 3.1 (Chart mass and normalized coordinates). *The quantities m_j and m_j^{-1} have cold-pre bounds independent of the height of W . On a buffered chart the coordinates y_W/v and z_i/k_i have dimension-controlled norms in terms of y/v . The row defining z_i on the ambient lift has norm at most $\exp(\text{poly}(d))K_i$, is integral on $\mathbb{Z}^d$, and annihilates W .*

A bounded warm enlargement of the chart has Haar measure at most a warm multiple of m_j . A ball of a warm fraction of radius v , centered at a feasible torus point and with the prescribed room, has Haar measure at least a warm inverse multiple of m_j . These relative bounds do not depend on any width when the radius ratios use only dimensions and γ .

Proof. Lemma F.4 gives

$$\text{covol}(\Lambda) \asymp_{\exp(\text{poly}(d))} \prod_i K_i$$

and dimension-controlled maps between the projected Euclidean coordinates and z_i/K_i . Since $k_i \asymp \max(1, K_i v)$, multiplying the latter coordinates by $K_i v/k_i \leq 1$ proves the asserted upper norm bound. The dual projected lattice is the integral annihilator of W , giving the integral rows and their stated norms. If $q = \dim W^\perp$, then

$$\exp(-\text{poly}(d))v^d \leq m_j \leq \exp(\text{poly}(d))v^{\dim W}.$$

In particular its logarithmic costs involve $\log v^{-1}$ but not subspace height.

Haar measure in these coordinates is Lebesgue measure on W times counting measure on the sheets, divided by $\text{covol}(\Lambda)$. Counting sheet coordinates in a dimension-controlled parallelepiped

and integrating on W gives the upper bound. Around a feasible sheet, a smaller such parallelepiped contains at least a dimension-controlled multiple of $\prod_i \max(1, \tau k_i)$ sheet points when its radius is τv ; this is at least $\tau^q \prod_i k_i$ up to dimension factors. Integration on W supplies the other $\tau^{\dim W}$ factor. This proves the lower bound, including arbitrary fixed real translations of the chart. Feasibility is necessary: no assertion is made that an arbitrary translated small box meets the torus. $\square$

3.2 The coefficient law and the actual path ensemble

First consider the model $W = \mathbb{R}^d$, where there are no discrete sheet coordinates. We want the spatial directions and the small-lift directions to be recoverable separately. The reference lift law therefore uses disjoint blocks of j parameters with a common side L , to be selected later. On a lift axis a , its principal part has the form

$$\frac{y_a^{\text{ideal}}(t)}{v} = \eta_a + \sum_B c_{aB} \prod_{\ell \in B} \frac{t_\ell}{L},$$

with the normalized c_{aB} bounded away from zero. The constant η_a supplies the local translation. The blocks for different axes are disjoint, and a separate kernel block controls the spatial image. Fixing all but one variable in each selected product at normalized values bounded away from zero gives independent directions in this ideal model. Repeating the products on further disjoint blocks also smooths their one-site and cube evaluation laws.

The reason to recover lift directions rather than raw integer coefficients comes from the lift identity we require. We will arrange that $C(\psi(t)) - c = y(t) + \beta(t)$ is a polynomial identity of degree at most j . Its top homogeneous parts then give

$$\beta_{[j]}(t) - C_{[j]}(\psi_{[1]}(t)) = -y_{[j]}(t).$$

Here brackets denote ordinary homogeneous degree. Thus the products control the deviation of the integer variable from its prescribed polynomial value without any bound on the coefficients of C . Small remaining lift coefficients and small nonkernel spatial displacements perturb this ideal recovery map; they do not give exact independence to the full map.

For a rational W , the projected lift coordinates are integer sheets. Their available ranges determine whether the preceding products can use long parameters or must instead use coefficient one and shorter parameters. We now prescribe those ranges and the additional columns needed for residue tests and exact sheet translations. All independence in this prescription belongs to the reference coefficient law and the separate spatial law. The actual path law couples them by the tilt below.

After the pre requirements are fixed, choose a perturbation parameter $0 < \sigma < 1$ and a range parameter $A \geq 2$, in that order. Rank thresholds and parent sides will be chosen later. Choose a scale gap with no $k_i^{1/j}$ between L and AL . For example, among more than $\dim W^\perp$ consecutive disjoint multiplicative intervals of ratio A , at least one contains none of these numbers. Hence this choice is possible with $A \leq L \leq A^{O(1+d)}$. With a fixed sufficiently large degree-dependent C_* , call a sheet axis

$$\begin{array}{ll} \text{inactive} & k_i \leq L^j, \\ \text{moderate} & (AL)^j \leq k_i \leq LC_*, \\ \text{enormous} & k_i > LC_*. \end{array}$$

There are only polynomially many numbers to avoid, so $\log L$ can be chosen with the permitted late bound without using their magnitudes. Rounding is performed with fixed buffers at the cutoffs. An active sheet axis means a moderate or enormous axis.

The parameter box is $\mathcal{T} = \prod_{\ell} \{0, \dots, T_{\ell} - 1\}$. We prescribe its columns, the independent reference law of the lift coefficients, and the supports of the separate spatial law as follows. The actual coefficients of $C(x + Vt)$ are determined by (x, V) ; the tilt defined below combines these two laws.

1. To keep the spatial image nondegenerate independently of the principal lift directions, spatial kernel columns have side L . In normalized spatial coordinates their matrices lie in buffered neighborhoods of fixed invertible matrices, at a common warm spatial scale. The inverse norms are warm-bounded throughout the supports.
2. Continuous and active lift axes have independent dedicated blocks of j columns, all of side L . Their principal monomials are squarefree products on disjoint blocks. In normalized units y_W/v and z_i/k_i , the principal amplitudes have buffered supports bounded away from zero at the chosen dimension scale. Equivalently, a principal raw coefficient has size comparable to v/L^j on a continuous axis and k_i/L^j on an active sheet axis, with the corresponding smooth continuous or integer law.
3. On an inactive axis, integral coefficients cannot be made arbitrarily small. We therefore keep the product coefficient exactly one and shorten the dedicated sides to quantities comparable to $k_i^{1/j}$ at the prescribed dimension factors. If these sides would be below a specified warm minimum, omit the axis: set *every* coefficient, including the constant, equal to zero, and supply no columns for it. Hence omitted k_i are warm-bounded. Otherwise supply one additional nonkernel column $t_{i,0}$, with side comparable to k_i . Its coefficient is one so that translating it translates the sheet output by exactly the same amount, as needed for the later affine sections. Set

$$z_i(t) = z_i(0) + t_{i,0} + \sum_{\text{dedicated blocks } B} \prod_{\ell \in B} t_{\ell}. \quad (3.2)$$

There are no other terms on this axis. The common proportionality constants are small enough for chart containment.

4. To test congruence relations without sharing the principal smoothing variables, add a structural polynomial number of free nonkernel columns, of side L , disjoint from the preceding blocks. Their independent leading coefficient data supply the residue-test vectors in the reference comparison law. On moderate axes, put independent small perturbations on all pure-free nonconstant monomials, with integer coefficient widths comparable to $k_i \sigma / T^{\nu}$. Other nondesignated moderate coefficients remain zero. The scale gap makes the selected integer ranges broad enough for later individual-modulus tests. On continuous and enormous axes, each remaining nonconstant monomial t^{ν} has an independent small coefficient whose width in the normalized output coordinates is comparable to σ / T^{ν} . The continuous coefficients have smooth laws. The enormous-axis coefficients are integers, with the smooth discrete laws specified below.

Here and below fixed structural factors may multiply these widths. Every constant lift coefficient has an independent law of fixed warm support in its normalized units. When its integer range is too short for that law it is fixed at zero; its corresponding k_i is then warm-bounded. Counts of kernel and product blocks are fixed polynomials in the structural dimensions, large enough for the fixed cube orders and smoothing orders used below, with spare blocks for smoothing. Numerical precisions never increase these counts. Every nonomitted side exceeds the specified warm minimum, which may include higher-layer parent and separation requirements. Every side is $O(L^j)$.

We specify the coefficient laws as well as their supports. Fix a nonnegative, nonzero smooth bump φ supported in $(-1, 1)$ and positive on a neighborhood of zero. An integer coefficient with

center a and scale S has law

$$\Pr(\alpha = k) = \frac{\varphi((k - a)/S)}{\sum_{r \in \mathbb{Z}} \varphi((r - a)/S)}.$$

The center and scale place this support strictly inside the prescribed interval and ensure that its positive region contains an integer, so the denominator is positive. Coefficients fixed at zero or one have their stated point-mass laws. Continuous coefficients are their centers plus their prescribed widths times independent variables with fixed smooth buffered densities. All these choices are independent across coefficient positions and output axes.

For a monomial t^ν , let f_ν be the resulting density of its coefficient in the orthogonal W coordinates, and let $p_{i,\nu}$ be the probability mass of its coefficient on sheet axis i . On the coefficient torus W/Λ , the local Haar measure is $du_W d\#z/\text{covol}(\Lambda)$. Consequently

$$\text{covol}(\Lambda) f_\nu(u_W) \prod_i p_{i,\nu}(z_i)$$

is the density of this coefficient law relative to Haar. The selected supports lie in a buffered coefficient chart. Their product over $|\nu| \leq j$ defines a probability density $\mathcal{D}_c$ on $\prod_{|\nu| \leq j} W/\Lambda$; the center c shifts only the constant coefficient. In particular $\int \mathcal{D}_c = 1$.

We also choose a law $\rho(x, V)$ on integer spatial coefficients. The root x is nearly uniform on a buffered interior box, with a smooth transition near its boundary. In each spatial kernel block the normalized matrix lies in a product neighborhood of a fixed invertible matrix at the selected warm spatial scale. Its integer entries are sampled independently by smooth buffered bumps in their normalized intervals, with the same normalization as the integer coefficient law above. Every nonkernel entry is independent and has a smooth profile supported in

$$|V_{a\ell}| \leq \frac{c_0 \sigma H_a}{m T_\ell},$$

where m is the number of columns and $c_0 > 0$ is a fixed small constant. These widths become long integer intervals when the parent sides are chosen. Their total normalized displacement is $O(\sigma)$. All supports have enough room that every path lies in U .

The actual law of the affine maps is

$$d\mathbb{P}_c(\psi) = \frac{\rho(x, V) \mathcal{D}_c(\text{coeff } C(x + Vt))}{Z_c}, \quad \psi(t) = x + Vt. \quad (3.3)$$

The normalizer in this formula is $Z_c = \mathbb{E}_\rho \mathcal{D}_c(\text{coeff } C(x + Vt))$. The identity $\int \mathcal{D}_c = 1$ concerns abstract Haar coefficients; the conclusion $Z_c = 1 + o(1)$ will use high rank of the actual polynomial coefficients.

There is a second distinction when residue information is needed. On the finite cover $W/M\Lambda$, Haar measure above a coefficient in W/Λ is uniform on its $M^{\dim W}$ deck choices. Thus the deck coefficients modulo M are independent uniform variables conditional on the small-lift coefficient data under this comparison law. The actual integer spatial law will only approximate the resulting joint residue law at each prescribed individual modulus. It need not have this exact conditional independence after real coefficient values are fixed.

By contrast, the lift identity on a path is exact. On the support of $\mathcal{D}_c$, subtracting the small real coefficient lifts from $C(x + Vt) - c$ leaves integer monomial coefficients. Their sum is a polynomial $\beta(t)$, and the small coefficients sum to $y(t)$. Chart containment identifies these polynomials with the site lifts throughout the path.

We will use high rank through the following Fourier estimate. Consider coefficient characters on a prescribed integer cover, represented by rational ambient rows λ_ν whose denominators divide its modulus. For fixed parameter sites t_ω , evaluation sends a W -valued polynomial coefficient array $(a_\nu)_\nu$ to $(\sum_\nu a_\nu t_\omega^\nu)_\omega$. A coefficient mode $e(\sum_\nu \lambda_\nu a_\nu)$ *factors through these sites* if its real linear functional is a linear combination of the evaluation functionals; equivalently, it vanishes on their common kernel. Here $e(z) = \exp(2\pi iz)$.

For one site, or sites $t_\omega = t_0 + \sum_{i=1}^q \omega_i t_i$ forming an affine Boolean cube of fixed order with independent differences, Lemma F.7 removes every mode which does not factor through the sites:

$$\left| \mathbb{E}_\rho e \left(\sum_\nu \lambda_\nu \text{coeff}_\nu(C(x + Vt)) \right) \prod_\omega F_\omega(x + Vt_\omega) \right| \leq \varepsilon \quad (|F_\omega| \leq 1).$$

The rank and parent-side thresholds depend polynomially in logarithmic scale on the prescribed Fourier frequencies and coefficient mass, cover modulus, spatial smoothness, inverse error, $\log L$, and $\log \sigma^{-1}$. They do not depend on the height of W . Spatial residue conditions may be included. With no site insertions, every nontrivial coefficient character is removed. The estimate applies to the Fourier expansion of $\mathcal{D}_c$ before dividing by Z_c .

Factoring through real evaluation need not give a character on the original evaluation torus. Clearing the denominators of a rational right inverse gives a finite cover on which it does. The cube comparison below retains this cover. For a single site the constant coefficient already gives an integral right inverse. These observations specify the estimate used in the following lemma; its proof remains in the appendix.

Lemma 3.2 (The normalized one-layer ensemble). *For the law just defined, sufficiently large permitted rank and parent side cutoffs give the following conclusions, uniformly in the center.*

1. $Z_c = 1 + o(1)$, and on every retained path

$$C(\psi(t)) - c = y(\psi(t)) + \beta(\psi(t)) \tag{3.4}$$

is a polynomial identity. The small lift is fully slow at width v ; $\beta \circ \psi$ has integral monomial coefficients.

2. *The non-site-mode removal of Lemma F.7, and the Haar and individual-cover conclusions of Corollaries F.9 and F.10, hold with cold-pre requests and late errors. In particular, on each prescribed individual modulus, the relevant coefficient residues have the product chart/deck law, independently of the unweighted spatial residues, to arbitrarily small prescribed total variation. On moderate axes the pure-free nonconstant coefficients are independently uniform to this accuracy once the range parameter is chosen sufficiently large.*
3. *Noninjective paths can be discarded at any prescribed cold-pre probability. At fixed center the averaged one-site law is supported on the buffered chart and has density at most B/m_j relative to uniform measure on U , apart from arbitrarily small excess mass, where B is warm-bounded.*
4. *Given a warm spatial error $\epsilon > 0$, the root law can be chosen so that Haar mixing of centers gives the one-site upper bound $(1 + \epsilon)$ times the uniform law, and the corresponding lower bound $(1 - \epsilon)$ away from a warm-thickness boundary strip, each with arbitrarily small cold-pre additive error against every nonnegative function bounded by one. The spatial scale and strip thickness do not depend on Q_j .*

Proof. The independent coefficient density has Haar integral one. On its support subtracting the specified small coefficient lifts leaves integral coefficients. Summing the resulting monomials proves

Equation (3.4); uniqueness of the chart identifies this polynomial lift with the site lift throughout the path.

We check the quantitative hypotheses for Fourier removal rather than deduce $Z_c = 1$ from the abstract normalization. In a coefficient chart the Haar density has the factor $\text{covol}(\Lambda)/(v^{\dim W} \prod_i k_i) = m_j^{-1}$. The remaining factors are normalized continuous densities and normalized discrete bumps. On axes with $k_i \leq L^{C_*}$, singleton interpolation costs only $O(\log L)$ in addition to cold-pre costs. This includes omitted axes. On enormous axes all coefficient ranges contain many grid points. Since $T^\nu \leq L^{j^2}$ for degrees at most j , their normalized smooth interpolation has logarithmic costs polynomial in the cold-pre budget, $\log \sigma^{-1}$, and $\log L$, when C_* is sufficiently large. Lemma 3.1 controls all normalized coordinate maps independently of orientation and height. Buffered ambient chart extensions can therefore be periodized and Fourier approximated with just these costs.

On a modulus cover multiply by the residue character of the chart index β . Rows defining (z, w) reduce modulo that modulus, so no real deck-basis norm is needed. The same ambient extension works on the cover with its individual logarithmic modulus cost. The displacement proof of Lemma F.7 only changes by replacing its parameter bound L with $O(L^j)$. Its translating boxes and the parent cutoff still have permitted late logarithmic costs. Thus it removes each non-site mode with arbitrary bounded site insertions. With no insertions it removes every nontrivial coefficient mode; the trivial mode is the Haar integral one. This proves $Z_c = 1 + o(1)$. Translating c changes Fourier phases but none of the bounds, proving uniformity.

For each prescribed residue vector apply this argument on its own cover and sum per-atom errors. The logarithm of the atom count is polynomial in dimensions and the individual modulus log. This is the total-variation argument of Corollary F.10, with the ambient extension just verified. A moderate pure-free coefficient of top degree has width at least a structural factor times $A^j \sigma$; coefficients of smaller degree have no smaller range. Choosing A after the individual-modulus requests makes every such smooth integer law uniform modulo that modulus to the required accuracy. Parent sides similarly make spatial free entries broad. The deck laws are uniform independently of the sheet data. This proves the full joint assertion, not merely a claim about designated top monomials.

The independent constant coefficient has Haar density at most B/m_j for a warm B . The surviving one-site weight under non-site removal is its convolution with the probability law of the other coefficient contributions, so has the same cap. The independent spatial root has a warm cap conditional on all slopes. Removal with arbitrary site indicators consequently bounds the excess mass above a fixed warm multiple of m_j^{-1} by the requested error. All actual supports are buffered.

When centers are mixed, a mode factoring through one site has its site row as its constant coefficient row. Center integration kills every nontrivial such row; removal kills the nonfactoring modes. Only the unweighted spatial law remains. Choose its root nearly uniform on an interior box and make all spatial displacements a smaller warm fraction of the sides. Its translate has the asserted upper density everywhere and lower density off the indicated strip. Only the Fourier removal error, not that strip, is requested at cold precision.

Finally, for a fixed nonzero difference $a = t - t'$, the equation $Va = 0$ determines at most one value of an entry in a column for which $a_\ell \neq 0$. Smooth broad spatial laws, a union bound over at most $O(L^{2jm})$ pairs, and the permitted late supremum bound of the tilt make collisions negligible by increasing parent sides. This completes all the claims in the permitted choice order. $\square$

3.3 Comparing sampled and ambient tests

3.3.1 Niltests

A rational filtered nilmanifold of degree q consists of a connected, simply connected nilpotent Lie group G , a cocompact lattice Γ , and connected rational subgroups

$$G = G_1 \supseteq \cdots \supseteq G_{q+1} = \{1\}, \quad [G_i, G_j] \subseteq G_{i+j}.$$

Rationality is measured in fixed exponential coordinates on the Lie algebra $\mathfrak{g} = \log G$, with rational bases for $\mathfrak{g}_i = \log G_i$. A polynomial map $g : \mathbb{Z}^n \rightarrow G$ is adapted when $\log g$ modulo $\mathfrak{g}_{i+1}$ has degree at most i , using weighted degree when variable weights are assigned. A *niltest* is the function $u \mapsto F(g(u)\Gamma)$, where F is bounded and Lipschitz on G/Γ . The metric is the quotient of the right-invariant Riemannian metric making the chosen coordinate basis orthonormal at the identity.

Complexity at most p means that $\dim G$, the logarithmic heights of the structural constants and filtered bases, and $\log(2 + \|F\|_\infty + \text{Lip } F)$ are at most p , with

$$l\mathbb{Z}^{\dim G} \subseteq \log \Gamma \subseteq l^{-1}\mathbb{Z}^{\dim G} \quad \text{for some integer } 1 \leq l \leq e^p.$$

A metric change includes the logarithm of its comparison constant in this budget. No coefficient bound is imposed on the orbit g . On $\mathbb{Z}/N\mathbb{Z}$, evaluation uses representatives $0, \dots, N-1$; degree-zero tests are constants, and dimension zero is allowed. Definition A.1 repeats these conventions where the rational calculus is developed. The equivalence with adapted Taylor/binomial polynomial sequences is proved there.

3.3.2 Cube normalization

For a finite integer box Q , let $\text{Cub}_j(Q)$ be the set of tuples $(x, h_1, \dots, h_j)$ for which every vertex $x + \omega \cdot h$, $\omega \in \{0, 1\}^j$, belongs to Q . Writing $\mathcal{C}z = \bar{z}$, we use

$$\|f\|_{U^j(Q)}^{2^j} = \frac{1}{|\text{Cub}_j(Q)|} \sum_{(x, h) \in \text{Cub}_j(Q)} \prod_{\omega \in \{0, 1\}^j} \mathcal{C}^{|\omega|} f(x + \omega \cdot h).$$

On a finite abelian group, all parameters instead have independent uniform averages. In particular, the constant function 1 has norm one under both conventions. Formula (A.2) and the identities following it give the exact comparison with zero extension to a product of cyclic groups and the cube-count weights required when taking derivatives.

A *normalized site twist* is a unit-bounded function localized to a buffered chart which is a product, or a bounded-complexity combination of products, of a smooth function of $(u/H, y/v)$ and a bounded-modulus function of (u, β) . Smoothness is measured in these normalized variables. Such a twist need not have warm Lipschitz cost as a function on the unlocalized ambient torus.

Lemma 3.3 (Cube comparison in chart units). *Fix slice costs, a fixed cube order q , and a requested accuracy. For a fixed permitted parameter slice B , the sampled functional*

$$(P_\omega)_\omega \longmapsto \mathbb{E}_\psi \mathbb{E}_{\mathbf{t} \in \text{Cub}_q(B)} \prod_{\omega \in \{0, 1\}^q} P_\omega(\psi(t_\omega))$$

is, uniformly for $|P_\omega| \leq 1$, approximated to the requested accuracy by

$$\sum_\alpha c_\alpha \mathbb{E}_{\mathbf{u} \in \text{Cub}_q(U)} \prod_{\omega \in \{0, 1\}^q} P_\omega(u_\omega) J_{\alpha, \omega}(u_\omega).$$

The site twists $J_{\alpha,\omega}$, coefficient mass $\sum_{\alpha} |c_{\alpha}|$, and number of terms have cold-pre bounds. If $q \leq j$ and the request is warm, its ambient output can instead be written

$$\sum_{\alpha} c_{\alpha} \mathbb{E}_{\text{ambient cubes}} \prod_{\omega \in \{0,1\}^q} P_{\omega}(u_{\omega}) \frac{J_{\alpha,\omega}(u_{\omega})}{m_j}, \quad (3.5)$$

with warm total coefficient mass, warm normalized site twists $J_{\alpha,\omega}$, and an arbitrarily small prescribed warm additive error, uniformly for arbitrary $|P_{\omega}| \leq 1$. Finite sums suffice. Only the error transfer and removal, not the output twists or coefficient mass, use cold or late costs.

The proof appears after the general cube comparison in Appendix G. Its factors depend only on individual ambient sites. This is the feature that lets a large sampled cube average yield an ambient niltest correlation.

Choose a nonnegative smooth cutoff χ supported in a buffered uniqueness chart and equal to one on all path and twist supports. For fixed center c , normalized chart measure on U is the probability measure

$$\mu_c(\{u\}) = \frac{\chi(C(u) - c)}{\sum_{v \in U} \chi(C(v) - c)}.$$

The denominator is positive for the sufficiently large rank and side choices below. This measure is used only for the warm detection assertion; the approximation seminorm is defined from the actual path law.

Fix a degree r , permitted slice costs, and local test complexities. For a site input $g : U \rightarrow \mathbb{C}$, put

$$X_r(g) = \mathbb{E}_{\psi \sim \mathbb{P}_c} \sup_{B, F} |\mathbb{E}_{t \in B} g(\psi(t)) F(t)|,$$

where B ranges over the permitted slices of $\mathcal{T}$ and F over unit-bounded local niltests of degree r and the prescribed complexity. Each path may choose its own slice and niltest. An ambient atom is a product $J(u)N(u)$, where J is a normalized site twist and N is an ordinary ambient degree- r niltest; both factors may be taken unit-bounded.

Theorem 3.4 (Pathwise detection and approximation). *Prescribe finitely many testing degrees in the fixed allowed range, local test complexities, slice costs, input caps, and accuracies. The later parameters of the one-layer construction can be chosen once so that the following assertions hold for this same ensemble, uniformly in the center and the site input.*

For cold-pre requests, if a cold-pre-bounded input satisfies $X_r(g) \geq a > 0$, with $\log(1/a)$ bounded by cold-pre data, then g correlates in uniform parent measure with an ambient atom of cold-pre complexity, and the logarithm of the reciprocal correlation has a cold-pre bound. Every $|g| \leq 1$ can also be approximated in X_r , to the prescribed cold-pre accuracy, by a linear combination of such atoms with cold-pre coefficient mass and number of terms.

If $r < j$ and the requests are warm, every warm-bounded input with $X_r(g) \geq a > 0$ and a warm bound for $\log(1/a)$ correlates in μ_c with an ambient atom of warm complexity, with a warm bound for the logarithm of the reciprocal correlation. The atoms, coefficient mass, and number of terms in the X_r approximation also have warm bounds. These bounds may use the omission threshold but do not use Q_j or the height of W .

For a specified larger input cap, include its logarithm in the request budget and rescale. Increasing the requested accuracy, test, or path budgets changes the corresponding warm or cold-pre output logarithms polynomially. All path probabilities and testing suprema refer to the law before productive paths are selected.

Proof of the cold clause. The ordinary argument of Lemma G.10 applies to the ensemble above by Lemma 3.3. Indeed, mesh endpoints and enumerate strides to reduce the separately chosen slices to finitely many possibilities. A short side has a bounded number of points and can be enumerated directly. A positive fraction of detecting paths therefore has a large U^{r+1} norm on one common slice. The niltests on these paths need not be the same.

Lemma 3.3 transfers their averaged cube power to a mixed ambient cube correlation. When all physical chart costs are included in the cold preliminary budget, its site factors have controlled caps. Gowers–Cauchy–Schwarz and Theorem A.7, as in the cited detection proof, give an ambient degree- r niltest times a permitted site twist correlating with the input. Degree zero uses the two independent sites of a one-cube and gives a twist alone.

To pass from detection to approximation, let m_U be the full one-site density relative to uniform measure on U . If every permitted slice has at least fraction δ_{sl} of the full parameter box, then

$$X_r(\phi) \leq \delta_{\text{sl}}^{-1} \mathbb{E}_U m_U |\phi|.$$

Lemma 3.2 bounds the excess mass above a cap of cold preliminary size. Doubling the cap bounds the mass of the resulting tail by twice that excess. Select this cap first, then request the tail accuracy needed by Lemma G.11. That lemma produces the finite approximation, with controlled atom count and coefficient mass. All requests precede the late rank and side choices. This proves the cold clause for the same ensemble as the warm clause. Section 5 proves its bounds independent of the current chart mass. $\square$

3.4 The geometry needed in lifting

There is a deliberate distinction from the packaged geometry in Property 1. Mixed-center spatial comparison here has a warm-thickness boundary strip; it is not claimed with a strip as thin as every cold detection accuracy. This distinction does not change the path deductions used below. In the proof of Proposition I.1, local correlations are generated at fixed centers by freezing pre-short axes and using a chart-supported phase input. The rest of that argument uses the exact fully slow/integral identities, fixed-center testing detection, translation groups, and the ambient rank estimate. Lemma 3.2 and the cold clause of Theorem 3.4 provide precisely these data; physical-coordinate or cover twists then have cold-pre logarithmic Lipschitz and modulus costs. Changing the upper parameter side bound to $O(L^j)$ changes only late cutoffs, and omitting axes removes variables from these arguments. The same inspection applies to the symbol and modeling steps in Proposition I.6. For passive partner selection below, it is the warm *approximation* clause of Theorem 3.4, rather than a whole-box detector of warm mass, that will be used.

3.5 The stopped tree of a triangular cell

Fix the current triangular cell, its value spaces W_j , and one collection of numerical requests for all uses of the one-layer ensemble. We now define the probability measure in which preparation will test rank failures. The layers are sampled in the order $1, \dots, D$, although their numerical budgets are chosen in the opposite order.

At entry to layer j , a history consists of the earlier affine path maps, their centers and their integral lift polynomials. Substituting these maps and lifts into C_j gives an ordinary polynomial on the current parent box, still taking values in W_j . The same exact substitutions have been made in every higher determining polynomial. We use C_j for this pulled-back polynomial.

First test the rank of C_j at the current parent scale. If the test fails, stop that history. Otherwise choose its center with Haar probability conditioned on

$$\mathcal{C}_j = \left\{ c \in W_j / (W_j \cap \mathbb{Z}^{d_j}) : \|c + l_j - b_{j,0}\|_\infty < (1 + \gamma/2)w_j \text{ for some } b_{j,0} \in \mathbb{Z}^{d_j} \right\}. \quad (3.6)$$

The condition is independent of the chosen representative of c : changing it by an element of $W_j \cap \mathbb{Z}^{d_j}$ changes $b_{j,0}$ by the same integer vector. Once a representative is fixed, $b_{j,0}$ is unique. Sample ψ from the fixed-center ensemble $\mathbb{P}_c$. Its polynomial lift identity is

$$C_j(\psi(t)) - c = y_j(t) + \beta_j(t), \quad b_j(t) = \beta_j(t) + b_{j,0}.$$

The second equation specifies the raw determining slot used in all higher polynomials. It gives

$$b_j(t) - C_j(\psi(t)) - l_j = b_{j,0} - c - l_j - y_j(t).$$

The lift width is chosen sufficiently small inside the gap that the enlarged weight- j constraint holds on the whole retained path. Substitute this affine map and this exact integer polynomial into the remaining determining blocks and proceed to layer $j + 1$.

For applications that discard additional sampled paths, fix the discard rules as part of the current tree being tested. At each history a rule specifies a subset of the finite support of $\mathbb{P}_c$; outcomes outside that subset are killed before proceeding. In the forward argument these subsets impose injectivity and the modular conditions defined there, and their lost probability must satisfy the bounds prescribed in the forward-mass proof. They delete probability mass; they do not renormalize the distribution on the remaining histories. A mask of the input function, such as a site-dependent flag, is likewise distinct from conditioning the path law.

Thus the histories arriving at each layer carry a subprobability measure. The rank-stop probability at j means the mass of histories that reach that test and fail it; histories already stopped contribute zero. All conditional probabilities used in backward transfer are ordinary conditional probabilities within this one tree. After a rank cut the cell and its polynomials change, so preparation builds this tree again and repeats the tests. It does not assume that an earlier successful rank test survives the cut.

Write e_j for the Haar measure of $\mathcal{C}_j$. It depends on W_j, l_j, w_j, γ and not on the sampled prefix. The strict certificate ensures that $\mathcal{C}_j$ is nonempty with positive measure. The quantitative lower bound for e_j , and the comparison of strict and enlarged masses down this tree, will be proved with the forward-mass argument.

4 Preparing a cell by rank cuts

The comparison estimates for the paths of Section 3 require sufficiently high rank of the current determining polynomial. We now modify the cell until failures of these rank tests have the required small probabilities. If a failure probability exceeds its prescribed bound, we obtain a relation at the root and use it to remove one dimension from a rational value space. The modification must retain the old integer tuple, since every higher determining polynomial is evaluated on that tuple and can have arbitrarily large coefficients.

The following proposition is the preparation statement needed for the forward passage through the cell. Its probability refers to the stopped tree defined in Section 3.5. After each modification we test the new tree; successful tests from an earlier tree are not assumed to persist.

Proposition 4.1 (Prepared system with triangular costs). *Given a triangular two-box certificate, forecast dimension bounds and descending numerical budgets in the sense of Definition 2.3. After polynomially many rank-cut transactions, there is a certificate with an arbitrarily small prescribed total proportional target discount such that every rank-stop probability in the actual current sampling procedure is at most its prescribed bound. The procedure may restart its tests after every transaction.*

The dimensions added at block i are polynomial in the dimensions strictly above i . The increase in its width logarithm is polynomial in dimension data, p , and higher-level numerical budgets, and is independent of its own initial width logarithm except for carrying that logarithm additively. Root slice costs and sufficient root-side cutoffs have polynomial logarithms in the full eventual budgets.

Throughout this section the degree bound is fixed. Polynomial dependence on dimensions is permitted. A dimension-controlled factor means a quantity whose size and inverse size are at most the exponential of a fixed polynomial in the total relevant dimensions. The polynomial exponents depend only on the fixed degree bounds. Warm and cold-pre budgets have the descending meaning of Definition 2.3; a warm bound at j does not use $Q_i = \log(2/w_i)$ for $i \leq j$. All finite families of requests are forecast before the late perturbation, range, rank, and side choices.

4.1 The relation needed for a rank cut

Return to a triangular system with blocks b_i of weight i and C_i taking values in rational spaces W_i . Put

$$A_i = [C_i]_{\text{wt}=i}.$$

Write v_i for the physical lift-chart width at layer i , the quantity denoted by v in Section 3. Let $K_i \supset W_i$ be the space obtained by setting the omitted sheet coordinates at level i to zero. These spaces and their coordinate classifications are fixed for the current candidate cell, independently of the sampled paths. Their log heights are cold-pre at their own levels: an omitted integral row has norm at most a warm factor times v_i^{-1} .

Lemma 4.2 (Backward rank transfer). *Suppose the actual stopped sampling procedure fails the rank test at block h with probability larger than the prescribed small bound at that block. Then there is an integral row λ , nonzero on W_h and of norm at most R_h , for which, on $\mathbb{R}^n \times \bigoplus_{i < h} K_i$,*

$$\lambda A_h = S \left(\frac{u}{\tilde{H}}, \left(\frac{b_i - A_i(u, b_{<i})}{w_i \varepsilon_i} \right)_{i < h} \right) + R(u, b_{<h}). \quad (4.1)$$

The log coefficient bound for S and the log common denominator of R in the raw integer variables are polynomial in the cold budget at h alone, including dimensions. Each $\log(\varepsilon_i^{-1})$ is warm at i , and the spatial shrinkages from H to $\tilde{H}$ have polynomial log cost. The slow and rational terms have ambient extensions with these respective bounds. The identity is asserted on the displayed subspaces, not outside them.

We first prove the geometric consequence of this relation. This identifies the two estimates that its proof must provide: the real term must be controlled in normalized residual coordinates, whereas the rational term needs a common denominator in integer coordinates. The proof of Lemma 4.2 then occupies the next three subsections.

4.2 A rank cut preserving the box certificate

We now turn the relation into a geometric transaction. At every site the strict and enlarged tests count integral tuples satisfying the triangular residual conditions. The radii are small enough for

uniqueness, including all the fixed buffers used below. We retain the actual tuple throughout; higher polynomials are never estimated by continuity in an integer slot.

Lemma 4.3 (Relative rank-cut transaction). *Suppose a triangular two-box certificate*

$$\mathbb{E}fB^- > a\mathbb{E}B^+$$

has a relation (4.1) with λ nonzero on W_h . After a spatial slice, lower residual refinements, and a prescribed small proportional discount of a , there is another certificate with the following properties.

1. *The new polynomial at h takes values in $W_h \cap \ker \lambda$. If $h > 1$, at most one full copy of $\mathbb{R}^{d_h}$ is added at weight $h - 1$; if $h = 1$, there is no incoming copy.*
2. *At weight h , the relative width loss uses only dimension and gap factors, independently of R_h , Q_h , and the size of the rational section used in the cut. At weights above h , no width loss is necessary. Lower relative losses are polynomial-log in their warm data and the cold data at h .*
3. *Every old higher polynomial is transformed by an exact weighted polynomial substitution on its actual integer arguments. All weights remain valid. Rational value spaces are preserved, apart from the stated intersection and full incoming copy.*
4. *Spatial slice costs and sufficient root-side cutoffs have polynomial logarithms in the full budgets. There is no loss proportional to the inverse of the actual mass of the old test.*

Proof. Choose a rational $e \in W_h$ with $\lambda e = 1$, and set

$$C_h^{\text{new}} = (1 - e\lambda)C_h.$$

All subsequent changes of lower variables are composed into this formula. Its values belong to $W_h \cap \ker \lambda$ exactly. We first construct, on suitable lower and spatial cells, the identity

$$C_h^{\text{old}} = C_h^{\text{new}} + C_{\text{in}} + c + I + E, \quad \|E\|_\infty \ll \gamma w_h. \quad (4.2)$$

Here I is integral-valued on the reindexed variables and has weight at most h ; c is constant; and, when $h > 1$, C_{in} has weight at most $h - 1$ and depends only on arguments strictly below $h - 1$.

Prescribe the spatial and lower integer variables modulo a common integer clearing the denominator of eR in (4.1). Its log size is polynomial in the cold budget at h and the section bound. On reindexing a residue class, every positive-degree coefficient of this rational polynomial becomes integral; its constant is absorbed into c . Arbitrarily large numerators do not affect this assertion.

Subdivide the lower residuals by warm relative factors. Each omitted sheet row annihilates C_i and is integral on b_i . Its norm is at most a warm factor times v_i^{-1} , so it varies by less than 1 on a sufficiently small enlarged lower cell. It is therefore constant there. Thus each nonempty enlarged lower cell puts b_i on a fixed affine translate of K_i .

First take the weight- h parts of both terms in (4.1). The left side is homogeneous of weight h , and each deviation $b_i - A_i$ is homogeneous of weight i , so this preserves the identity and does not enlarge either the coefficient bound or the common denominator. Thus we may take S and R homogeneous of weight h in their respective weighted coordinates.

Let u_* be the spatial cell center, and let r_i be the center of the chosen lower box in the residual coordinate $b_i - C_i$. Define the completed slow polynomial by

$$\widehat{S}(u, b_{<h}) = S \left(\frac{u - u_*}{\widetilde{H}}, \left(\frac{b_i - C_i(u, b_{<i}) - r_i}{w_i \varepsilon_i} \right)_{i < h} \right).$$

Since C_i has top weighted part A_i , these replacements and constant translations preserve the top weighted symbol. In the joint spatial and residual coordinates $\hat{S}$ has controlled coefficients. Refining those coordinates makes its oscillation, even after multiplication by e , much less than γw_h . The additional relative precision uses only dimension factors, γ^{-1} , w_h^{-1} , $\|e\|$, and the coefficient bound in (4.1). For every lower level these are permitted higher-level costs. No derivative bound for the original C_i is needed: the residual coordinates themselves are being localized.

The difference between λC_h and this completed slow term plus R has vanishing weight- h part on $\mathbb{R}^n \times \bigoplus_{i < h} K_i$. Off that subspace it need not have smaller weight. Compose the difference with rational affine projections onto the affine spaces of the chosen cell. This gives a formal polynomial of weight at most $h - 1$ agreeing with the difference on the whole cell. Multiplying by e produces a lower-weight remainder, while the slow term contributes a constant and the indicated small error and the rational term contributes I and a constant.

Implement a lower residue restriction by $b_i = Mb'_i + b_{i0}$ and use $C'_i = C_i/M$ after the corresponding substitutions; put the constant offset in the residual center. This preserves W_i . If $h > 1$, the lower-weight remainder has only a constant-coefficient linear dependence on the variables of weight $h - 1$. Reduce that coefficient matrix modulo integers, putting the removed integral term into I . Its remaining entries are bounded. Replace b'_{h-1} there by C'_{h-1} plus the chosen residual center. A further lower refinement to width $\ll \gamma w_h / (1 + d_{h-1})$ makes the error suitably small. What remains is C_{in} , depending only on lower arguments. It need not take values in the line spanned by e , which is why we allow a full d_h -dimensional incoming copy. When $h = 1$, the remainder is constant instead. This proves (4.2).

Choose a narrow residual box modulo 1 for a new integer block b_{in} around C_{in} , with radius much less than γw_h . Introduce b_h^{new} by the exact identity

$$b_h^{\text{old}} = b_h^{\text{new}} + b_{\text{in}} + I, \quad (4.3)$$

with a possible constant integer adjustment. Apply this exact substitution in every C_i for $i > h$. Since I has weight at most h and b_{in} has weight $h - 1$, their weighted degree bounds are preserved. No continuity estimate in an old integer argument occurs, and all higher widths can be kept.

We give the selection argument, including its uniform width control. At lower residual boxes use continuously shifted smaller boxes, allowing centers in a small neighborhood of the old strict box but well inside its enlargement. Every old strict point receives the full smaller strict coverage volume, and every new enlarged point lies in the old enlarged box. The same construction works after the uniform residue scaling.

For the incoming block choose its center l_{in} uniformly in a unit cube. For any old strict tuple, the total center measure of admissible incoming strict choices is exactly their box volume, including integer indexing across wrap. Let the new h -center range within radius $(1 + \gamma/2)w_h$ about $l_h + c - l_{\text{in}}$. Write $r_{\text{in}} = b_{\text{in}} - C_{\text{in}}$ and $r_h^{\text{new}} = b_h^{\text{new}} - C_h^{\text{new}}$. Equations (4.2) and (4.3) give

$$b_h^{\text{old}} - C_h^{\text{old}} = r_h^{\text{new}} + r_{\text{in}} - c - E.$$

Choose the two new radii and $\|E\|_\infty$ well below $\gamma w_h/8$. An old strict tuple then receives full strict coverage in the new h -center, and every transported enlarged tuple lies in the old enlarged h -box. The new h radius can be a fixed dimension/gap fraction of w_h , independently of e or R_h . Within weight $h - 1$ take a common decreased width for the existing and incoming coordinates. This may make the incoming box arbitrarily smaller than γw_h , but does not require the error in (4.2) to be small at that new incoming scale: the error is used only in the old h -gap.

Let c_- be the product of the full strict coverage volumes and c_+ the corresponding enlarged upper coverage constant, including the residue and spatial averaging. For every site, counting its

exact tuples gives

$$\int B_{\text{new}}^- \geq c_- B_{\text{old}}^-, \quad \int B_{\text{new}}^+ \leq c_+ B_{\text{old}}^+.$$

The first inequality persists after multiplication by $f \geq 0$. The ratio satisfies

$$c_+/c_- \leq (1 + O(\gamma))^{O(\sum_i d_i + d_h)}.$$

Unchanged coordinates have ratio 1 in this coverage calculation. Consequently, with $a' = ac_-/c_+$, integration of $\mathbb{E}fB_{\text{new}}^- - a'\mathbb{E}B_{\text{new}}^+$ is strictly positive. One cell gives the desired certificate. There is no division by the actual mass of either old test. Sum over residue classes and average spatial slices with their actual box masses; equal or nearly equal subdivisions have controlled relative sides once the parent cutoff is imposed. All constants, affine projections, and incoming polynomials may depend on the selected lower cell. They are measurable, for example by using fixed projections for each discrete sheet value.

Finally shift centers and integer indices into their bounded representative ranges. This only changes constant integer adjustments in (4.3), not a value space. Every asserted cost follows from the refinements above. Large section norms are charged only to lower/spatial precision; the own layer width uses only its gap and dimensions. This proves all parts of the lemma. $\square$

4.3 Two recovery estimates

Consider one invocation of the sampler, at level j , satisfying the parent rank and side hypotheses of Lemma 3.2 at all the requested accuracies, with ordinary polynomial $C : U \rightarrow W \subset E = \mathbb{R}^d$, and write $A = C_{[j]}$. The sides of U are H_i , and the common lift-chart width is v . Write (w, z) for the unimodular integer coordinates on the lift integer β , with z the projected sheet coordinates. The corresponding projected lattice basis vectors are denoted by v_i , and $k_i = \max(1, K_i v)$, with the harmless rounded convention of the sampler. No relation between this symbol w and a block width is intended.

Remove every omitted sheet direction and, if needed, some additional inactive directions. Removing a direction always means removing *all* its parameter columns, including its additional linear column. Let $K \supset W$ be the subspace on which the removed sheet coordinates vanish. The leading substitution after these columns have been set to zero is

$$\Phi(t) = (\psi_{[1]}(t), \beta_{[j]}(t)) \in \mathbb{R}^n \times K.$$

We use the full primitive lattice $K \cap \mathbb{Z}^d$, not an arbitrary rationally scaled basis of K .

The real recovery estimate must be independent of the unrestricted coefficients of A . We therefore compare a lift variable with its predicted value $A(u)$ before measuring coefficients. This is different from choosing an integer basis for K , whose real conditioning may be poor.

Lemma 4.4 (Normalized real recovery). *Let a_j be the common normalized spatial scale of the kernel blocks. On the space of weighted homogeneous polynomials of any prescribed bounded degree, substitution by Φ has a bounded inverse on its image when coefficients on $\mathbb{R}^n \times K$ are measured in*

$$\left(\frac{u_i}{a_j H_i} \right)_i, \quad \frac{b - A(u)}{v}, \quad (4.4)$$

and coefficients on the parameter box are measured in t_ℓ/T_ℓ . Both the forward operator norm and the inverse norm are bounded by dimension-controlled factors. The assertion holds for coefficient tensors in an additional real normed coefficient space.

If all normalized parameter variables are restricted to radius ζ , replace a_j by $a_j\zeta$ and v by $v\zeta^j$ in (4.4). The same conclusion holds. The constants do not depend on the widths or on the coefficients of C .

Proof. The homogeneous identity supplied by the lift is

$$b - A(u) = -y_{[j]}.$$

Consequently the unbounded coefficients of A disappear after the change of coordinates (4.4). In these coordinates the ideal map has spatial kernel blocks and lift principal blocks supported on disjoint parameter columns. Fix all but one variable in a suitable dedicated product for each retained lift direction, at points bounded away from zero in their normalized ranges. Together with an invertible spatial kernel block, the remaining variables then give a full-dimensional affine image whose sides are bounded below by dimension-controlled constants.

For an inactive axis, the product coefficient is 1, and the product of the sides is comparable to k_i within dimension-controlled factors. The extra linear column can be set to zero; this includes the case $j = 1$. Axes whose rounded scale could obstruct this comparison have already been omitted by the minimum-side convention. The split-basis conditioning for the remaining lift directions is dimension-controlled by the lattice chart construction. Thus a fixed box in the real coordinates (4.4) is covered with a dimension-controlled inverse parametrization.

At fixed degree, interpolation on this box bounds every coefficient by a dimension-controlled multiple of the supremum of the polynomial on the box. Conversely, the normalized principal substitution has dimension-controlled coefficients, so it bounds the coefficients of every pullback. There are only polynomially many coefficients at fixed degree. These two observations give the asserted norms on the finite-dimensional coefficient spaces. They also apply with a vector-valued coefficient, by the same interpolation formulas.

The perturbations are chosen small in these normalized units. The resulting operator on the coefficient space is therefore a small perturbation of an injective operator with the displayed inverse bound. Choosing the late perturbation sufficiently small preserves the bounds. Deleting entire inactive column families does not alter the retained principal blocks. Finally, the leading spatial map has degree 1 and the leading lift map has degree j , so rescaling all parameters by ζ gives exactly the two stated powers of ζ . $\square$

The arithmetic recovery estimate uses the extra pure-free columns. Their structural number is chosen before any accuracy budget. On a separate pure-free unit axis, the spatial degree-1 coefficients and the degree- j deck and active-sheet coefficients form one output vector. The coefficient families used for different such axes are disjoint. Lemma 3.2 makes these vectors jointly uniform modulo any one prescribed modulus, to an arbitrarily small preassigned error. In the abstract Haar-cover comparison law, deck residues are uniform even conditional on the sheet data. For the actual sampler we use the resulting joint finite-residue comparison, not conditional uniformity given exact real sheet coefficients. On a moderate axis the added top-degree integer coefficient range has size comparable to $k_i\sigma/L^j \geq A^j\sigma$; hence the late scale gap gives the same conclusion for these coefficients. Spatial free columns have broad integer supports as well. Only separate modulus tests, not a test at the product of all requested moduli, are used below.

Lemma 4.5 (Prime-power polynomial test). *For each fixed r there are constants $c_r, C_r > 0$ such that, for every prime p , integer $e \geq 1$, and polynomial P over $\mathbb{Z}/p^e\mathbb{Z}$ of total degree at most r with at least one unit coefficient,*

$$\Pr(P(X) = 0 \pmod{p^e}) \leq C_r(p^e)^{-c_r},$$

where X has independent uniform coordinates. The constants are independent of the number of variables.

Proof. First consider one variable, allowing a coefficient of valuation at most $(1 - \eta)e$. Put $B = \binom{r+1}{2}$ and, for $r \geq 1$, choose $\ell = \lceil \eta e / (2B) \rceil$. If the zero set occupied more than r classes modulo p^ℓ , choose $r + 1$ zeros in different such classes. Their Vandermonde determinant has valuation at most $(\ell - 1)B < \eta e / 2$. Multiplying the evaluation vector by the adjugate shows that every coefficient has valuation at least $e - (\ell - 1)B > (1 - \eta)e$, a contradiction. Thus the zero set has measure at most $rp^{-\ell}$. Bounded depths and degree zero are absorbed into constants.

The same calculation proves the following induction statement: for an m -variable polynomial with a specified coefficient of valuation $v < e$, its zero probability is at most

$$mr p^{-(e-v)/(2B 2^{m-1})}.$$

For the induction step, use the coefficient polynomial in the first $m - 1$ variables containing the specified coefficient and test it to depth $h = \lceil (e + v)/2 \rceil$. The induction hypothesis bounds its exceptional probability using $h - v \geq (e - v)/2$. Outside that set its value has valuation at most $h - 1$, and the univariate calculation has gap $e - h + 1 \geq (e - v)/2$. The constants add to at most mr . A constant coefficient polynomial has empty exceptional set and satisfies the same estimate.

For many variables, start with a monomial having a unit coefficient. If a strictly componentwise dominating monomial has coefficient valuation below the midpoint of the present valuation and e , move to that monomial. Each move strictly increases total degree, so there are at most r moves. At termination the chosen coefficient has valuation at most $(1 - 2^{-r})e$, and every strictly dominating coefficient has larger valuation, separated from it by a fixed degree-dependent fraction of the remaining depth.

Fix all variables not occurring in the chosen monomial. In the coefficient of that monomial, every new contribution comes from a strictly dominating monomial, and hence cannot cancel the chosen coefficient. Its valuation is unchanged. There remain at most r variables. If the chosen monomial is constant, all nonconstant contributions have larger valuation and the same observation already settles the claim. Otherwise use the displayed induction statement with $m \leq r$ and $e - v \geq 2^{-r}e$. For example, the deliberately weak choices $C_r = r^2$ and $c_r = 1/(2^{2r}B)$ suffice for $r \geq 1$. $\square$

Lemma 4.6 (Simultaneous denominator recovery). *Prescribe a cold-pre upper bound H for the least common denominator of the coefficients of a rational polynomial of bounded degree, and an exceptional probability $\delta > 0$. The late sampler parameters can be chosen so that, outside an event of probability at most δ , there is an integer D_0 with*

$$\log D_0 \leq \text{poly}(d + n + \log(2/\delta))$$

having the following property simultaneously for every allowed subspace K . If F is weighted homogeneous on $\mathbb{R}^n \times K$, its coefficients in the primitive lattice coordinates have least common denominator at most H , and d' is a positive integer for which $d'F \circ \Phi$ has integral coefficients, then $d'D_0F$ has integral coefficients in those coordinates. There is no upper-bound requirement on d' . The same assertion holds for polynomially many components and formal coefficient tensors with a common initial denominator. The bound on D_0 is independent of H .

Proof. First set the inactive parameter columns to zero. Let N be the number of scalar monomials in the remaining output coordinates at the prescribed degree bound; N is polynomial in their dimension. For $q = p^e$ above a degree-dependent constant, choose sufficiently many disjoint pure-free axes that Lemma 4.5 and a union bound over at most q^N coefficient vectors give probability at most

q^{-10} that *some* primitive polynomial vanishes at all their output vectors. Indeed, if there are M axes, the bound is

$$q^N (C_r q^{-c_r})^M,$$

and M can be a fixed polynomial in the structural dimensions. An identically zero substituted polynomial vanishes at each unit axis, so the same bound applies to bad polynomial substitution. The union over all polynomials is what permits F to be chosen after the sampled path.

For each prime let b_p be the greatest tested exceptional depth, also including a fixed baseline for prime powers below the preceding constant. Under the uniform residue law, CRT makes bad events at distinct primes independent. Thus simultaneous bad powers $q_1, \dots, q_t$ at distinct primes have probability at most $(q_1 \cdots q_t)^{-10}$, apart from the fixed baseline. There is a finite-witness bound uniform in the tested range. If $\prod_p p^{b_p} > R$, either one factor exceeds R , or a subproduct lies in $(R, R^2]$. Summing respectively over prime powers and over integers gives an $O(R^{-9})$ upper bound, with degree-dependent constants. Taking R polynomial in $2/\delta$ bounds this exceptional product outside probability at most $\delta/2$.

For the actual sampler, use its joint finite-residue law separately at every required prime power and every product witness in $(R, R^2]$. These form a finite family. The logarithms of their number and largest modulus are bounded by the pre-bound on H and $\log(2/\delta)$. Choose the individual approximation errors so that their sum is below $\delta/2$. Thus the input precision may depend on H , but the bound on the exceptional product does not. This argument uses no approximation at the product of all tested primes.

On a good path, the following integral consequence holds on the pure-free outputs. If substitution makes an integral polynomial divisible by p^a , then all its coefficients are divisible by p^{a-b_p} . Otherwise divide by its least coefficient valuation; the resulting primitive polynomial would give an exceptional depth larger than b_p .

Restore inactive outputs by increasing total degree in those coordinates. After setting unused nonfree columns and the extra linear columns to zero, each retained inactive coordinate is a sum of disjoint dedicated degree- j products, each with coefficient 1. Suppose the least inactive degree not yet removed is a_{in} , and write its terms as $\sum_{|\alpha|=a_{\text{in}}} z^\alpha P_\alpha$ in the other outputs. Choose one dedicated product for each relevant coordinate and extract the product with exponent α_i in its parameters. Its degree in these parameters is ja_{in} , and its coefficient is exactly P_α evaluated on the pure-free outputs. Higher inactive degrees cannot contribute. Dependence of other outputs on these dedicated parameters adds positive parameter degree and also cannot contribute at degree ja_{in} . No factorial division is involved.

Consequently, if the whole substituted integral polynomial is divisible by p^a , this extraction recovers the coefficients at the least inactive degree with loss at most b_p . Subtract the corresponding integral coefficient terms. Their substitutions retain that divisibility, so the residual is divisible to the reduced depth. Continue through the bounded number of possible inactive degrees. The total loss is at most a degree-dependent multiple of b_p . The same argument works for any subset of removed inactive directions with the same pure-free goodness; it does not require a new probability estimate for every subspace.

Finally let d be the least common denominator of F and put $P = dF$. For a prime dividing d , set $e = \max(0, v_p(d) - v_p(d'))$. The hypothesis implies that $P \circ \Phi$ is divisible by p^e , and $e \leq \log_p H$. Since P has a coefficient prime to p , the preceding divisibility conclusion forces $e \leq Cb_p$ for a fixed-degree constant C . Thus d divides $d'(\prod_p p^{b_p})^C$. Absorb this fixed power into D_0 . Apply the same argument to scalar tensor entries; their simultaneous polynomial test and common denominator give the claimed joint conclusion. $\square$

4.4 The sharpened major statement

The next lemma permits preliminary restrictions to entire short inactive directions. Their lengths, and the height of the subspace they determine, are part of the cold-pre input. The important output is sharper than that input budget.

Lemma 4.7 (Relative major refinement). *Let $D(u, b)$ be a fixed weighted homogeneous scalar polynomial of bounded degree on $\mathbb{R}^n \times K$, with no coefficient bound. Suppose that on at least a prescribed cold-pre fraction of paths its full leading substitution has a slow-plus-rational decomposition: the slow coefficient norm on the normalized parameter box of radius ζ is at most B , and the rational term has a common denominator at most ρ . These input bounds, the inverse fraction, and ζ^{-1} have cold-pre logarithms. Assume the paths have Lemma 4.6's goodness, with their respective path-dependent multipliers bounded by D_0 , through all denominator bounds forecast for the argument.*

Then on $\mathbb{R}^n \times K$,

$$D(u, b) = S \left(\left(\frac{u_i}{H_i a_j \zeta} \right)_i, \frac{b - A(u)}{v \zeta^j} \right) + R(u, b), \quad (4.5)$$

where the coefficient norm of S is at most $B \exp(\text{poly}(d + n))$, and the logarithm of a common denominator of R in primitive lattice coordinates is bounded by

$$\text{poly}(d + n + \log(2 + D_0 + \rho)).$$

The assertion is simultaneous for polynomially many coefficient tensors. Their real norm may contain a specified pre-scaling, and their integral coordinates may have specified pre-height. These heights do not enter the sharpened real bound.

The two terms can be extended separately to ambient polynomials: orthogonally in deviation coordinates for S , and by a primitive lattice extension for R . Their identity is asserted only on the specified subspace.

Proof. We first recover the two terms on a subspace K' for which the remaining parameter sides are long enough to separate small real coefficients from rational ones. We then restore the removed directions by extracting their coefficient-one products. The restoration is an induction on degree, so its depth is independent of the number of directions.

Apply Proposition I.1 at cold-pre cost to a fixed ambient extension of D , after freezing the already excluded short coordinates. The proposition allows unrestricted coefficients. Its proof applies to the present sampler: the local correlation is obtained from a buffered chart-supported phase, fully slow and integral lift identities from Lemma 3.2, and the cold detection conclusion of Theorem 3.4. It does not use a fine mixed-center spatial boundary estimate. We obtain a decomposition with cold-pre bounds on a rational space $K_0 \supset W$ of cold-pre height.

Only after this output, increase the pure-long threshold, deleting every further inactive direction with a dedicated side below it. Delete its linear column too; its length is bounded by a fixed power of a dedicated side, so all these deleted columns are still pre-bounded. Denote the resulting subspace of K by K' . Then $K' \subset K_0$. Indeed an integral annihilating row of K_0 vanishes on W . Its value on a retained projected lattice vector v_i is an integer, since v_i is the projection of an integer vector. Its absolute value is less than 1 once K_i exceeds the prescribed height bound, because $|v_i| \ll 1/K_i$. It is therefore zero. This proves the inclusion.

Restrict the candidate decomposition to the remaining pure-long parameters and compare it with the given decomposition. In every positive-degree coefficient their rational difference is the negative of a slow difference. The denominator is cold-pre bounded, and all the remaining sides

can exceed the corresponding separation scale. Lemma A.5, with zero target subspace, makes both differences zero. Hence the candidate slow and rational terms have, separately, the original bounds B and ρ on these evaluations. Apply Lemma 4.4 to the first and Lemma 4.6 to the second. This gives the sharpened conclusions on K' . In degree zero, simply take a split of the fixed constant from one productive path; no separation of constants is asserted.

Extend the two sharpened terms separately to K . Orthogonal projection of a deviation onto K' does not enlarge its real norm. For the rational term, $K' \cap \mathbb{Z}^d$ is primitive, so a basis of this lattice extends to an integral basis of the ambient lattice. Extending the polynomial independently of the complementary coordinates preserves its denominator. The resulting numerator sizes may be arbitrarily large and are not bounded or used in a slow estimate. The slow extension retains its real coefficient bound on the full retained path substitutions, by the forward bound in Lemma 4.4. The rational extension needs only its denominator bound there: integral substitution preserves it regardless of numerator size. No real bound on that evaluation is claimed or needed. Subtract the two terms. The residual vanishes on K' .

We finish by induction on the polynomial degree. Expand the residual in the extra sheet coordinates for K/K' and take its least remaining total degree $a \geq 1$:

$$\sum_{|\alpha|=a} z^\alpha D_\alpha(u, b'), \quad b' \in K'.$$

At this least degree the coefficient tensors are intrinsic; changing the linear section changes only higher extra-sheet degrees. Set the new short linear columns to zero and extract separate dedicated products of total parameter degree ja , as in the denominator proof. Each normalized product has size comparable to $k_i \zeta^j$. Thus the evaluated D_α have slow norm at most the current norm times

$$\exp(\text{poly}(d+n)) \prod_i (k_i \zeta^j)^{-\alpha_i}, \quad (4.6)$$

and rational denominator at the current sharpened bound. The coefficient-1 extraction introduces no rational division. Other outputs' dependence on these parameters contributes only above degree ja .

Each D_α has smaller weighted degree, so apply the present lemma inductively, simultaneously to its tensor entries, starting with K' . The newly frozen sides and subspace description are pre-bounded before this invocation. Extend the two answers separately, multiply them by z^α , and subtract. Since $z_i(A(u)) = 0$, multiplication by the normalized powers of z_i cancels precisely the losses in (4.6). Since the z_i are integral lattice coordinates, the rational denominator changes only by the common-denominator products from boundedly many stages. Continue to the next extra-sheet degree. The recursion depth is bounded by the fixed polynomial degree, not by the dimension. The number of simultaneous components is polynomial in dimension. This proves the stated bounds and the tensor assertion. $\square$

4.5 Transferring the failed relation through earlier layers

Proof of Lemma 4.2. There are at most $(1 + 2R_h)^{d_h}$ possible failed rows. Pigeonhole one of them. Its failure fraction still has inverse logarithm polynomial in the cold budget at h . Prune to prefixes whose conditional successful continuation fraction is at least a fixed power of this fraction; ordinary conditional averaging gives such prefixes at every level. All earlier rank tests on them have passed. For this calculation discard modular-bad lower paths, with an exceptional product bound D_0 whose logarithm is polynomial in the same budget at h . The exceptional probabilities are small enough not to remove most of the pruned branch. The tested denominator ranges may use the larger cold-pre

budgets of those lower layers. Lemma 4.6 is simultaneous on these ranges, so its output D_0 does not acquire those larger costs. At each node, fix one center for which the required conditional fraction remains. The center does not affect any leading map.

Lift the failed relation backwards, one layer $j < h$ at a time. The later variables b_i , $j < i < h$, are now formal variables on K_i . Lower histories are fixed at this node. The induction hypothesis supplies local slow/rational splittings on the productive children, with slow coordinates

$$\frac{t}{T} \quad \text{and} \quad \frac{b_i - A_i(\psi_{[1]}, \beta_{[j]}, b_{>j, <i})}{w_i \varepsilon_i}, \quad j < i < h,$$

allowing a common warm shrink ζ of the normalized t box. The initial such splitting is precisely the failed rank certificate.

At each stage we keep two expressions for the same residual: its raw polynomial in the later integer blocks, and its slow-plus-rational splitting on productive children. A degree tuple is removed from both expressions before the next tuple is considered. The slow coefficient will be multiplied by deviation monomials; the rational coefficient will be multiplied by raw integer monomials. Before subtraction, the newly recovered slow coefficient must agree exactly with the old local slow coefficient, and likewise for the rational coefficient. A change between those monomials can involve unrestricted coefficients of the A_i .

Order the tuples of degrees in the later blocks lexicographically, with the highest layer first, in decreasing order. At a current tuple $\alpha = (a_i)_{i>j}$, extract the coefficient tensor of the raw polynomial residual. Higher tuples have already been removed from the raw polynomial and from the local splitting. The local slow tensor has norm bounded by its prescribed coefficient bound times $\prod_{i>j} (w_i \varepsilon_i)^{-a_i}$; the local rational tensor has the accumulated common denominator in primitive coordinates. Apply Lemma 4.7 simultaneously to these coefficients. Extend its slow tensor orthogonally and its rational tensor primitively. Multiply the slow tensor by the corresponding deviation monomials $(b_i - A_i)^\alpha$, and the rational tensor by the raw monomials b_i^α , and subtract.

We verify the exactness needed for this subtraction. In positive remaining degree in t , the old local tensor split and the newly constructed split sum to the same evaluated raw tensor. Their slow bounds, expressed on the full normalized t box, use only dimensions, $Q_{>j}$, and already fixed upper-layer data. This is because the forward operator in Lemma 4.4 has a dimension-controlled bound: evaluating a deviation divided by v_j does not introduce an inverse power of v_j . Their rational denominators have the same allowed dependence. The primitive bases for later K_i are also part of the higher-level data. The minimum-side convention at j can therefore ensure separation using only warm information at j . Applying Lemma A.5 coefficientwise shows that the two slow tensors agree exactly and the two rational tensors agree exactly. No assertion of approximate equality is used before multiplication by a possibly very large coefficient of a later A_i .

In remaining degree zero there is no u, b_j argument. Pigeonhole the exact pair of constant tensors instead. To see that only pre-many choices occur, compare two such pairs: their rational difference has bounded denominator, and its size is bounded by the slow norms, in the fixed later lattice coordinates. A bounded region of a bounded-denominator lattice is finite with controlled log cardinality. The common center of this region can be arbitrarily large, which does not affect that count. The extra fraction loss is a pre cost for the local application, not a change to the already imposed modular-goodness multiplier.

These arguments justify subtraction from the local splitting as well as from the raw polynomial. They prevent unbounded lower coefficients arising from the changes between raw and deviation monomials. Primitive extensions of homogeneous tensors remain homogeneous. Orthogonal extension is compatible with every later A_i , since those values lie in $W_i \subset K_i$. Continue through the degree

tuples. There are only boundedly many tuples, because the number of layers and all degrees are fixed; entries of a tensor are processed simultaneously rather than one monomial at a time.

Each lift inserts the new deviation scale $v_j \zeta^j$ and spatial scale $a_j \zeta$. Thus its new ε_j has warm inverse logarithm at j . The sharpened coefficient and denominator bounds are changed only by polynomial dimension factors and the bounds already charged at h . Fixed-depth composition gives (4.1). When $h = 1$, there is no backwards step and the rank certificate is already this relation.

For completeness, successive local selections do not create a probability circularity. At the beginning use the pruned branch with the selected row and modular goodness. At any of its nodes one may subsequently retain a smaller pre fraction to choose a constant split or perform a major calculation. The conclusion at that node is existence of a split with the uniform sharpened bounds. For the preceding ancestor it is this existence statement that is used; the smaller internal fraction need not be propagated as an output coefficient or denominator cost. All the finitely nested denominator ranges were forecast before the late sampler choices. The argument only requires positive conditional fractions at the nodes where the corresponding major theorem is applied. $\square$

4.6 Termination and the order of the budgets

Proof of Proposition 4.1. If a stop probability is too large, apply Lemma 4.2 at that level and then Lemma 4.3. Retest the resulting actual system. Earlier tests are not assumed to remain valid after a restriction.

To count transactions, let c_h be the number of cuts at level h . Each such cut removes one dimension of W_h . New dimensions there arise only from incoming copies at higher levels. Thus c_h is at most the initial dimension at h plus its total incoming dimension. The total dimension added at $h - 1$ is at most c_h times the largest ambient d_h encountered. Starting at the highest level and descending gives polynomial bounds at each level, because the depth is fixed. These bounds are structural and are fixed before choosing numerical accuracies. In particular, dimensions added at a given level use only higher-level dimensions.

The rational heights do not undergo one polynomial exponentiation per cut. At a fixed level the starting and incoming value spaces are full coordinate spaces. Stack all previously imposed integer rows, extending old rows by zeros as copies arrive. The current W_h is their joint kernel. To choose the next section, solve this stacked system together with $\lambda e = 1$. A nonzero minor and Cramer's rule bound both the section and the rational kernel height polynomially in the total dimension, cut count, and $\log R_h$. This is the simultaneous calculation in Lemma I.12, with the simpler full incoming spaces here. Lower polynomial substitutions and uniform scalar residue rescalings do not change this value-space description.

We spell out the absence of a width circularity. At a level j , first fix all requirements inherited from levels above it. Using only those data and dimensions, choose a_j , the ratio v_j/w_j , and the minimum-side threshold for omission. The latter must cover higher-level parent cutoffs and the separation scales in Lemma 4.2. In that lemma the evaluated bounds use only higher data, not the height or width of K_j . Auxiliary cold threshold increases inside Lemma 4.7 merely remove further pre-short directions temporarily; they do not change this warm minimum-side choice.

Forecast a per-cut relative log-width loss A_j using the same data, without the initial Q_j . A cut at j uses only its own dimension and gap factors. A cut above j may charge the upper failure's coefficient, denominator, section, and width bounds, the already fixed deviation ratios and spatial shrinkages, and the omission cutoff. These are warm at j . If a common lower width must lie below an upper width, multiply its current width by the required small upper-scale factor; since all widths are less than 1, this also supplies the required absolute bound. A cut below j causes no shrinkage at j . With N_c the structural cut bound, the preparation width is therefore at most $Q_j + N_c A_j$ in

logarithmic units.

Only now choose cold-pre and late budgets at j , permitting this additive width bound. They cover the bounded-depth major calculations, denominator witness tests, perturbations, ranges, rank thresholds, and parent cutoffs. They also record the requirements for lifting a failure at j through its ancestors. All calls at an ancestor can forecast the required pre ranges before that ancestor's late choices. Because there are only fixedly many layer levels and bounded-depth calculations within each call, this is a bounded downward composition of polynomial log bounds, not a recursion through numerical precision at the same level.

Choose gaps and individual target discounts using N_c and the dimension forecasts so that their total is as small as prescribed. The positive selections are geometric coverage arguments and therefore do not require a numerical lower bound on the initial strict surplus. Each spatial transaction has a polynomial log cost in the full budgets. Summing over at most N_c such costs and imposing the corresponding initial root-side cutoff sustains all reindexings and subdivisions. The finite cut bound then forces termination with the stated probability tests and costs. $\square$

5 Detection independent of the current width

We prove the warm clause of Theorem 3.4. The layer is j and the testing degree is $r < j$. The desired detection and approximation bounds and accuracies are warm; auxiliary Fourier comparisons and transfer errors may use the stated cold and late costs. The coefficient law and its late parameters are the same as in Section 3; no new conditioning on productive paths is made.

Lemma 3.3 writes the sampled cube average as a bounded combination of ambient cube averages whose site factors are bounded by a warm multiple of m_j^{-1} . Applying an inverse theorem directly to those factors would charge the reciprocal chart mass to the output. We instead use high rank to control products of the chart weight, replace all but one of the sparse factors by bounded functions, and apply the inverse theorem to bounded conditional averages. The resulting detector and approximation have costs independent of Q_j .

5.1 A pseudorandom chart weight

Let $\mathcal{H} = \prod_a \mathbb{Z}/(4H_a\mathbb{Z})$ and use representative coordinates to evaluate C outside U . An additive cube all of whose vertices lie in U has no wrap in its two-face identities; the conversion between box and group normalizations costs only dimension factors. Use the cutoff χ chosen in Section 3, equal to one on a neighborhood of all twists and actual path supports. Define

$$a_\chi = \int_{W/\Lambda} \chi, \quad \nu(u) = \frac{\chi(C(u) - c)}{a_\chi}. \quad (5.1)$$

Here the cutoff is read through the buffered ambient chart. By Lemma 3.1, $a_\chi \asymp_{\text{warm}} m_j$. Every sparse factor in Equation (3.5) has absolute value at most $B\nu$ for warm B .

Lemma 5.1 (Bounded-grid moments). *For $1 \leq q \leq j$ and any bounded Cartesian grid of distinct choice vectors S , the late rank and side choices can ensure*

$$\mathbb{E} \prod_{\alpha \in S} \nu \left(\sum_{a=1}^q x_{a,\alpha_a} \right) = 1 + o(1), \quad (5.2)$$

where all $x_{a,b}$ are independent uniform variables of $\mathcal{H}$. The error can be made smaller than any specified pre bound, simultaneously for the bounded grid enlargements used below. In particular

$\mathbb{E}\nu = 1 + o(1)$. *Marginal spatial boundary and smooth chart estimates obey the same permitted cold costs.*

Proof. Fourier approximate the normalized cutoffs at cold cost in their ambient charts. A nontrivial resulting character has phase

$$\sum_{\alpha \in S} \lambda_{\alpha} C \left(\sum_a x_{a, \alpha_a} \right),$$

where at least one row is nonzero on W ; rows annihilating W may be deleted. Its degree- j tensors are independent in the required sense. Indeed fix α , substitute $x_{a, \alpha_a} = t_a z$ and all other $x_{a, b} = 0$, and extract the coefficient of $t_1^{j-q+1} t_2 \cdots t_q$. All other choice vectors disappear, and the result is

$$\frac{j!}{(j-q+1)!} \lambda_{\alpha} C_{[j]}(z).$$

Thus a slow-plus-rational certificate for the joint symbol gives one for a nonzero row of $C_{[j]}$, at cold cost.

To apply this to the cyclic representatives, partition wrap cells into sufficiently fine boxes and discard their boundary strips. All caps, Fourier masses, and inverse cell sizes are charged at cold scale. On each cell the wrap changes are constant translations and leave the preceding homogeneous calculation unchanged. If a nontrivial character had nonnegligible bias, Lemma B.1 on a last-layer torus, followed by the coefficient extraction just given and restoration of the box scales, would contradict rank R_j . The trivial character gives the product of Haar means, namely one. Choosing the errors before the late cutoffs proves the estimate. Inserting a prescribed smooth spatial weight, or approximating a boundary strip, gives the marginal assertions by the same argument. Actual coincidences of randomly sampled sites are included in these averages; distinctness here concerns the formal choice vectors. $\square$

5.2 Densification without paying the chart mass

We use a cube version of the densification method of Conlon, Fox and Zhao [11, Sections 6.2–6.3]. The bounded-grid estimates above supply the required estimates for products of the chart weight. We retain the details because the conclusion concerns normalized chart measure and warm niltest complexity.

Lemma 5.2 (Relative cube detection). *Suppose a mixed q -cube average as in Equation (3.5) has absolute value at least δ , and write its factors as $F_{\omega} = P_{\omega} J_{\omega} / m_j$, where $|P_{\omega}| \leq 1$ and J_{ω} are warm normalized site twists supported where $\chi = 1$. Suppose $|F_{\omega}| \leq B\nu$, where $1 \leq q \leq j$ is fixed and B, δ^{-1} are warm-bounded. Fix a detector vertex s . Its input P_s correlates, in normalized chart measure, with a warm normalized twist times an ordinary degree- $(q-1)$ niltest, by a warm inverse amount. For $q = 1$ the niltest can be constant.*

Proof. Extend all factors by zero from U to $\mathcal{H}$. The group cube average is the box cube average multiplied by

$$\kappa = \frac{|\text{Cub}_q(U)|}{|\mathcal{H}|^{q+1}}.$$

Both κ and its reciprocal have warm dimension bounds. Absorb this factor into the correlation threshold and work with group cubes throughout the proof.

Conjugations can be absorbed into the factors. First replace all vertices other than the designated detector by unit-bounded functions, one at a time. At a vertex i , write the average as $\mathbb{E} F_i D_i$, where

D_i is the conditional average of the other factors given that site. Every current factor has cap $B'(1 + \nu)$, with warm B' . Cauchy–Schwarz gives

$$\mathbb{E}(1 + \nu)|D_i|^2 \geq \frac{\delta^2}{(B')^2 \mathbb{E}(1 + \nu)}.$$

We claim

$$\mathbb{E}(\nu - 1)|D_i|^2 = o(1), \quad \mathbb{E}|D_i|^4 = O_{\text{warm}}(1). \quad (5.3)$$

For the fourth moment expand four cubes with a common root. Represent them by sums of q inputs, sharing the root input on each axis and using a separate alternate input for each cube. Every nonroot vertex has a copy-specific alternate coordinate, so these vertices have distinct choice vectors. Their product majorant is bounded by Equation (5.2).

For the first estimate use two such rooted cubes. Every nonroot factor misses at least one of the q root inputs. Apply weighted Cauchy–Schwarz successively to these root inputs. At a step, put each factor independent of the duplicated input outside the inner average under its majorant $B'(1 + \nu)$. A previously introduced positive majorant is carried once at this step, not squared. A dependent factor is copied to distinct choices of the duplicated input. Distinctness persists: descendants of different original vertices remain distinct after forgetting the clone labels, and descendants of one vertex differ in a cloned coordinate on which that vertex depends. Every nonroot descendant retains an alternate coordinate, so never coincides with a root descendant. After q steps the copies of $\nu - 1$ occupy distinct root-only choices and all other factors have been majorized on distinct nonroot choices. The outer masses are bounded by Equation (5.2). Expanding the remaining $(\nu - 1)$ and $(1 + \nu)$ factors, the main terms cancel, while their errors are arbitrarily small. Fixed-depth Cauchy–Schwarz proves the first estimate in Equation (5.3).

Consequently $\mathbb{E}|D_i|^2$ is warm-large. Interpolation yields

$$\mathbb{E}|D_i| \geq \frac{(\mathbb{E}|D_i|^2)^{3/2}}{(\mathbb{E}|D_i|^4)^{1/2}}.$$

Choosing a unit phase in place of F_i preserves a warm-large cube average. At this stage no pointwise cap for a conditional dual containing several sparse factors has been asserted.

Now only the detector vertex s is sparse. For another vertex $i \neq s$, conditional on its site u , the sparse site is

$$u + \sum_{a=1}^q (s_a - i_a) h_a.$$

One coefficient is 1 or -1 , so this site is exactly uniform on $\mathcal{H}$. Dropping the other bounded factors proves the pointwise estimate

$$|D_i(u)| \leq B \mathbb{E}_{\mathcal{H}} \nu = O_{\text{warm}}(1). \quad (5.4)$$

This uses pairwise uniformity of group cube sites, not a pointwise conditional pseudorandomness assertion for ν . Zero extensions, localization, and representative wraps are included in the bounded factors and do not change the estimate.

If the current correlation is at least δ' , then $\mathbb{E}|D_i|^2 \geq (\delta')^2$. The sparse factor satisfies

$$\|F_s\|_{U^q}^{2q} \leq B^{2q} \mathbb{E}_{\text{cube}} \prod_{\omega} \nu(u_{\omega}) = O_{\text{warm}}(1)$$

by Lemma 5.1. Inserting the appropriate conjugate of D_i at vertex i and applying Gowers–Cauchy–Schwarz gives

$$(\delta')^2 \leq \|D_i\|_{U^q} \|F_s\|_{U^q} \prod_{\omega \neq i, s} \|F_{\omega}\|_{U^q}.$$

Thus D_i has a warm-large Gowers norm. For $q > 1$, apply the form of Theorem A.7 for products of cyclic groups on $\mathcal{H}$ to the bounded function in Equation (5.4), after warm rescaling. It gives a correlating ordinary degree- $(q - 1)$ niltest. Replacing the factor at i by that niltest preserves a warm correlation; repeat at each nondetector vertex. The inverse theorem has never been applied to a function with cap m_j^{-1} .

The representative-coordinate box version of this step is obtained by partitioning coordinates into short representative intervals, using the Gowers triangle inequality to choose a large piece, and smoothing its boundaries, as in Theorem A.7. The dual inputs are bounded. After all replacements, fix the increments to obtain a correlation of the sparse detector with a product of translated niltests. Wrap and localization boundaries in this last selection can be smoothed using the spatial and chart marginal bounds of Lemma 5.1. Products and linear localizing masks have the permitted degree and warm complexity. When $q = 1$, the two group sites are independent, so a large cube average already gives a large detector mean and the same conclusion with the constant niltest.

Finally the sparse twist is supported where the cutoff in Equation (5.1) is one. Dividing that factor by ν there changes it only by a warm normalization. The detector is supported in U . Restricting the group average to U and normalizing by $\mathbb{E}_U \nu$ therefore gives the measure μ_c defined in Section 3; these normalizations have warm cost. The resulting twist is unit-bounded after warm rescaling, as claimed. $\square$

Lemma 5.3 (Weighted detection implies approximation). *Lemma G.11 remains valid with any positive probability measure μ on a finite set in place of its uniform measure. Its coefficient-mass and atom-count bounds depend only on the same quantitative cap, tail, and detection data, not on the smallest atom of μ .*

Proof. Remove zero-mass points. Suppose $X(g) \leq C\mathbb{E}_\mu m|g|$, $\mathbb{E}_\mu m \leq C$, and the cap and detector hypotheses hold. Write $B = CC_0/\eta$ and $\delta_0 = \delta(B, \eta/2)$, and request

$$\mathbb{E}_\mu m 1_{m > C_0} \leq \frac{\eta}{C} \min(1/8, \delta_0/4).$$

For $M > \max(1, 2/\delta_0)$, if $(9/8)f$ is outside $M\overline{\text{aconv}}(\mathcal{J})$ plus the X -ball of radius η , finite-dimensional separation in the μ pairing gives χ with

$$\Re \langle (9/8)f, \chi \rangle_\mu = 1, \quad X^*(\chi) \leq \eta^{-1}, \quad \sup_{J \in \mathcal{J}} |\langle J, \chi \rangle_\mu| \leq M^{-1}.$$

Testing at a single point gives $|\chi| \leq Cm/\eta$; the positive atom mass cancels from this inequality. Radially clip χ at B to obtain v . The tail bound gives $\mathbb{E}_\mu |\chi - v| \leq \min(1/8, \delta_0/4)$ and $\mathbb{E}_\mu |v| \geq 55/72$. Hence

$$X(v) \geq \eta \Re \langle v, \chi \rangle_\mu \geq \eta \mathbb{E}_\mu |v|^2 > \eta/2.$$

Detection contradicts the last bound on the pairings with χ . This proves the convex-hull approximation with X -error at most $8\eta/9$. Approximate the closed hull finitely, then sample its unit-bounded atoms with their phases. The expected squared $L^2((1+m)\mu)$ sampling error is at most $(1+C)M^2/N$, and $X(g) \leq C^{3/2} \|g\|_{L^2((1+m)\mu)}$. Choosing N from C, M, η uses the remaining error margin. This proves the assertion without dependence on atom masses. $\square$

Proof of the warm clause of Theorem 3.4. Mesh and enumerate the allowed pathwise slices as in the cold proof. Lemma A.9 and Lemma 3.3 give ambient cube detection at order $q = r + 1 \leq j$. Lemma 5.2 therefore gives warm detection in normalized chart measure.

Here is the precise change of measure for approximation. Let m_U be the full one-site density relative to uniform measure on U , and set

$$d\mu = d\mu_c = \frac{\nu}{\mathbb{E}_U \nu} dU, \quad m_{\text{rel}} = (\mathbb{E}_U \nu) \frac{m_U}{\nu}.$$

Sites of zero ν mass are ignored; no path or test support uses them. On actual supports $\nu \asymp_{\text{warm}} m_j^{-1}$, and $\mathbb{E}_U \nu = 1 + o(1)$ by rank. The cap and excess-mass statement in Lemma 3.2 therefore gives a warm cap and arbitrarily small excess mass for m_{rel} under μ . Enlarge that cap from C_0 to $2C_0$; the inequality $m_{\text{rel}} 1_{m_{\text{rel}} > 2C_0} \leq 2(m_{\text{rel}} - C_0)_+$ gives the required arbitrarily small tail. If every permitted slice has at least fraction δ_{sl} of full parameter mass, then

$$X_r(g) \leq \delta_{\text{sl}}^{-1} \mathbb{E}_U m_U |g| = \delta_{\text{sl}}^{-1} \mathbb{E}_\mu m_{\text{rel}} |g|.$$

This bound holds before taking the supremum; it exchanges no supremum with conditional expectation. The detector of Lemma 5.2 is a μ pairing with an allowed twist times niltest. Apply Lemma 5.3, choosing its cap and detection request first and the one necessary tail accuracy afterwards. All are warm requests, whereas the late choices make the required tail as small as needed. This proves the full arbitrary-input approximation assertion and its stated uniformities. $\square$

6 Forward sampling of a prepared cell

The prepared certificate must first produce terminal inputs to which the absolute increment applies. If Y denotes their mean, with stopped paths counted as zero, our immediate goal is $\mathbb{E}Y \geq (1 - \eta)a$. This estimate uses one-site sampling and proportional control of the points removed at each layer. It does not use the scalar comparison for scores.

The points removed at active layers are specified by a finite-ring construction. We give its full definition and the estimates needed for descent here. Its proof appears in Section 7, where the same construction is used to return the increment: its affine planes solve the congruence constraints without subdividing at the full modulus.

We keep the one-layer notation of Section 3. In particular, the layer is j , the parent box is U , the chart width is v , and

$$C(\psi(t)) - c = y(t) + \beta(t), \quad m_j = \frac{v^{\dim W} \prod_i k_i}{\text{covol}(W \cap \mathbb{Z}^{d_j})}.$$

Write I for the inactive sheet axes, including the omitted axes; the latter satisfy $z_i(t) = 0$ identically. The parameters belonging to these axes form t_I ; all other parameters form $t_{\parallel}$. Throughout this section a true site means a point $u \in U$ in the buffered uniqueness chart, with its unique integer lift $\beta(u)$, and $X = (u, \beta(u))$.

The congruence outputs will be denoted by

$$b(t) = (\psi(t), w(t), z_{\text{act}}(t)), \quad b(X) = (u, w(X), z_{\text{act}}(X)). \quad (6.1)$$

Here w is the deck coordinate, not a width. Their total dimension is m_b , and their respective degree bounds, or tags, are $1, j, j$. Equal tags are combined when using modular estimates. The congruence output b in Equation (6.1) is distinct from the determining block b_j .

6.1 Individual moduli and selected affine planes

The eventual bookkeeping modulus is

$$\mathfrak{M} = \text{lcm}(1, \dots, Z) = \prod_{\varpi \leq Z} \varpi^{a_\varpi}, \quad a_\varpi = \left\lfloor \frac{\log Z}{\log \varpi} \right\rfloor. \quad (6.2)$$

The logarithm of Z has a pre bound: warm pre at a passive layer and cold pre at an active layer. The much larger number $\log \mathfrak{M}$ is not charged to that bound. No equidistribution assertion below uses the whole modulus $\mathfrak{M}$, except when it happens to have an allowed size. We first forecast the detection, modeling, and symbol bounds, and only then choose Z to contain their required prime powers.

We shall repeatedly use the following consequence of Lemmas H.3 and H.2. Increase the structural number of pure-free columns before choosing any numerical accuracies. Apart from an arbitrarily small prescribed warm probability of paths, the leading homogeneous parts of Equation (6.1), in $t_{\parallel}$, satisfy the modular rank hypotheses through Z , with exceptional-power product

$$\prod_{\varpi \leq Z} \varpi^{b_\varpi} \leq D'_0, \quad \log D'_0 \text{ warm bounded.} \quad (6.3)$$

Indeed the pure-free coefficients contain the independent disjoint monomials required by Lemma H.3. Lemma 3.2 transfers their product residue laws at each individual witness modulus. The exceptional-product proof uses only such witnesses, so its bound is independent of the number of tested primes. We call paths satisfying Equation (6.3) modular-good.

At an active layer we mark exceptional parameter values using the number of suitable affine sections of the congruence map. The sections satisfy an exact identity: their s coordinates prescribe the change in the output, while their h coordinates change the inactive linear variables. These same sections will be used in the return argument.

Lemma 6.1 (Selected planes and flags). *Let x be a vector of v_f pure-free columns and let r_0 be the vector of additional coefficient-one linear inactive columns. Fix all other parameter entries. Put $\ell = m_b + \dim r_0$, and let the i th component of b have degree bound $d_i \in \{1, j\}$. Set*

$$J = \sum_{i=1}^{m_b} \left\{ \binom{\ell + d_i}{d_i} - 1 \right\}, \quad n_{\text{der}} = \sum_{i=1}^{m_b} \binom{\ell + d_i - 1}{d_i - 1}.$$

For every sufficiently small fixed $c_\star > 0$, a polynomial structural choice of v_f has the following properties. At a prime ϖ , let $\epsilon_\varpi = c_\star \varpi^{-2}$. Count full-column-rank matrices $V \in \text{Mat}_{v_f \times \ell}(\mathbb{F}_\varpi)$ satisfying the formal identity

$$b(x + V(s, h), r_0 + h) = b(x, r_0) + s, \quad (s, h) \in \mathbb{F}_\varpi^{m_b} \times \mathbb{F}_\varpi^{\dim r_0}. \quad (6.4)$$

There are J nonconstant scalar coefficient equations in this identity. Select a solution V only if both of the following conditions hold at every basepoint on its plane: the full number of solutions is at most $(1 + \epsilon_\varpi) \varpi^{v_f \ell - J}$, and the coefficient-equation Jacobian in V has rank J at the transported solution.

At depth $a \geq 1$, let $w_{\varpi, a}(t)$ be the number of solutions of Equation (6.4) modulo ϖ^a with selected reduction, divided by $\varpi^{a(v_f \ell - J)}$. Then:

1. *Selection is invariant under changing basepoint along the plane, and $w_{\varpi, a}(t) = w_{\varpi, 1}(t \bmod \varpi)$ exactly.*

2. Always $0 \leq w_{\varpi,a} \leq 1 + \epsilon_{\varpi}$. Under the uniform nonconstant pure-free coefficient law, at every fixed basepoint,

$$\Pr(|w_{\varpi,1} - 1| > C\epsilon_{\varpi}) \leq c_{\star}\varpi^{-10}.$$

Here C is fixed, and the exponent 10 can be increased by enlarging the structural column count.

3. Define a bad flag at ϖ by the failure of the lower bound $w_{\varpi,1} \geq 1 - C\epsilon_{\varpi}$, and put

$$W_{\star}(t) = \prod_{\varpi \leq Z} w_{\varpi,a_{\varpi}}(t). \quad (6.5)$$

Then $W_{\star} \leq 1 + O(c_{\star})$ everywhere and $W_{\star} \geq 1 - O(c_{\star})$ when there is no bad flag. For a squarefree product d of primes, the uniform-model probability of joint bad flags is at most d^{-10} ; the sum of the single-prime probabilities is $O(c_{\star})$.

4. In the actual ensemble these estimates transfer, with negligible errors, at the individual moduli used to test them. After averaging the center over unrestricted Haar measure, the total one-site loss from flags, with any nonnegative bounded site insertion, is at most $O(c_{\star})$ times its parent mean, plus an arbitrarily small cold-pre error.

All flags depend only on nonconstant coefficient data and on $t \bmod \varpi$, never on the constant coefficient of b .

The proof of Lemma 6.1 is given in Section 7, after the comparison measure is defined. For the descent we use its one-site loss bound; the full count and translation invariance enter the return.

6.2 Descent of the density certificate

We pass the density certificate down the stopped tree of Section 3.5, using the feasible-center set (3.6) and its Haar measure e_j . Once the absolute increment supplies a quantitative score, the return will use Proposition 7.4 in the reverse order of layers.

The density certificate supplies a strict root tuple. Its layer- j torus point lies in the feasible set with room at least $\gamma w_j/2$ in each coordinate. Lemma 3.1 applied to a smaller ball about that point therefore gives a cold-pre bound on $\log(1/e_j)$ at layer j . With v sufficiently small inside the gap, every such path satisfies the enlarged weight- j constraint. Conversely, any evaluation satisfying its strict constraint must have a center in Equation (3.6), with the raw determining slot specified in the stopped tree.

Proposition 6.2 (Forward mass). *For any sufficiently small fixed $\eta > 0$, a prepared certificate at threshold a gives a surviving set of terminal injective paths whose expected terminal input mean, with stopped paths counted as zero, is at least $(1 - \eta)a$. The input is pulled back successively and masked by no bad flag at every active layer. Its support is progression-free. At least a fraction ηa of terminal paths have mean at least $(1 - 2\eta)a$, after harmless adjustment of the loss constants.*

The rank-stop probability at layer h need only have inverse logarithmic cost polynomial in the data at layers $\geq h$. Discards immediately after that layer can be requested warm-small after charging the feasible-center proportions of layers $> h$. Thus this proposition does not require a lower-layer width in a higher-layer rank budget.

Proof. Stop a prefix when it fails its current rank test. Also discard noninjective paths and failures of Equation (6.3). Preparation tests the actual conditioned procedure, including earlier stops. Its requested rank-stop probabilities are arbitrarily small at cold cost in the current and higher layers. Modular-goodness discards can be made arbitrarily warm-small uniformly over feasible centers, with

D'_0 still warm. Injectivity errors can be requested at the same accuracy by the later side choice. Choose these errors small enough to pay for the indicated products of later e_i .

Fix a nonstopped, rank-sufficient parent at layer j . Average first over unrestricted Haar centers. The one-site calculation in Lemma 3.2 says that, with any bounded nonnegative site insertion, this average is at least $(1 - o(1))$ times the parent mean off a warm-thickness spatial boundary strip, up to an arbitrarily small cold-pre error. It also has the corresponding uniform upper density bound. Apply this first with the input times all remaining strict constraints. Lemma 6.1 bounds the bad-flag loss by a small proportional part of the parent mean and a negligible additive error. In estimating the boundary loss it suffices to keep the current strict constraint, since all other factors are at most one. Choose a smooth torus majorant G_j , equal to one on that strict constraint and supported inside the feasible set (3.6). Then $0 \leq G_j \leq 1$ and $\int G_j \leq e_j$. If a spatial boundary cutoff has thickness θ and mean $O(n\theta)$ on this parent, the marginal rank estimate proved with Lemma 5.1 gives

$$\mathbb{E}_U 1_{\text{boundary}} 1_{\text{current strict}} \leq O(n\theta)e_j + \delta,$$

with an arbitrarily small cold-pre error δ . Smooth spatial majorants may be used for the boundary indicator. Choose θ warm-small relative to $a \prod_{i>j} e_i$, and then choose δ cold-small relative to $a \prod_{i \geq j} e_i$. This makes the boundary contribution sufficiently small without charging e_j^{-1} to the warm thickness; only later feasible-center proportions enter after cancelling e_j .

Every strict contribution uses a feasible center. Conditioning on those centers, and then dropping the now processed strict constraint, shows that the expected new strict-and-input mass is at least

$$e_j^{-1}(1 - \epsilon) S - O\left(\epsilon a \prod_{i>j} e_i\right), \quad (6.6)$$

where S is its parent mass and $\epsilon > 0$ is an arbitrarily small fixed step loss. For enlarged constraints the comparison has the opposite direction. On every feasible-center path the processed enlarged constraint holds. The unrestricted mixed one-site upper bound therefore gives parent enlarged mass at least

$$(1 - \epsilon)e_j T_{\text{child}} - O\left(\epsilon \prod_{i \geq j} e_i\right). \quad (6.7)$$

This upper comparison of nonnegative child masses remains valid after discards. All absolute errors in these statements are requested before the late rank and side choices, much smaller than the displayed scales.

Here is the tree bookkeeping explicitly. Let S_j and T_j be the strict-and-masked-input mass and the enlarged-only mass, respectively, averaged over prefixes arriving at the rank test at layer j , with the remaining constraints evaluated on their parent. The probability space retains stopped prefixes as zero mass. Write

$$s_j = \frac{S_j}{\prod_{i \geq j} e_i}, \quad t_j^* = \frac{T_j}{\prod_{i \geq j} e_i},$$

with empty products at $D + 1$. A rank stop at h loses at most its probability divided by $\prod_{i \geq h} e_i$ from normalized strict mass; discarding the output of layer h loses at most its probability divided by $\prod_{i > h} e_i$. Request the former errors much smaller than $\epsilon a \prod_{i \geq h} e_i$ and the latter much smaller than $\epsilon a \prod_{i > h} e_i$. These are exactly the cold and warm dependencies in the statement. Integration of Equations (6.6) and (6.7) now gives

$$s_{j+1} \geq (1 - O(\epsilon))s_j - O(\epsilon a), \quad t_j^* \geq (1 - O(\epsilon))t_{j+1}^* - O(\epsilon). \quad (6.8)$$

In the second inequality restrict first to good parents; any later stopping can only decrease its right-hand nonnegative mass.

At the terminal level t_{D+1}^* is the full survival probability, which is $1 - O(D\epsilon)$ by the requested errors. Thus Equation (6.8) implies

$$\mathbb{E}B^+ = T_1 \geq (1 - O(D\epsilon)) \prod_{i=1}^D e_i.$$

The certificate $S_1 > aT_1$ gives $s_1 > at_1^*$, and the other recurrence yields

$$s_{D+1} = \mathbb{E}[\text{terminal input mean; survival}] \geq (1 - O(D\epsilon))a.$$

Choose ϵ and $c_\star$ so that the total is at most η . If $Y \in [0, 1]$ denotes this terminal mean, with $Y = 0$ on stopped paths, then

$$\Pr(Y \geq (1 - 2\eta)a) \geq \mathbb{E}Y - (1 - 2\eta)a \geq \eta a.$$

Finally, an injective integer-affine pullback of a function with progression-free support again has progression-free support; each flag mask only removes points. The terminal sides were chosen to satisfy the dimension-dependent absolute-rule cutoff. That rule therefore applies on the productive paths at threshold $(1 - 2\eta)a$, yielding its uniform multiplicative gain and quantitative surplus, or its impossibility alternative. All losses here may be fixed much smaller than that gain. $\square$

7 Returning the increment: comparison and passive layers

We now return from a productive terminal path to its ancestors, in the order $D, D - 1, \dots, 1$. The absolute increment has supplied a target $\lambda \leq 1$ separated from the desired final target by a fixed proportional margin. At boundedly many places in the ascent we decrease λ by a factor $1 - \epsilon$. The total decrease is chosen smaller than this margin. If the absolute rule instead gave impossibility, there is nothing to return.

Two features of the return will be maintained exactly. First, the determining slots continue to use the original prepared polynomials; they are not replaced by approximations in their integer values. Second, we transport the positive and negative payoff together. A favorable choice for one of them alone need not be favorable for their difference.

7.1 Paired cutoffs and the inductive score

For $c \in \mathbb{R}^d$ and $r > 0$, choose product cutoffs $U_{r,c}^-, U_{r,c}^+ : \mathbb{R}^d \rightarrow [0, 1]$. The small cutoff is one when $\|z - c\|_\infty \leq (1 - \gamma)r$ and vanishes when $\|z - c\|_\infty \geq r$. The large cutoff is one when $\|z - c\|_\infty \leq (1 + \gamma)r$ and vanishes when $\|z - c\|_\infty \geq (1 + 2\gamma)r$. Piecewise linear transitions, or smooth transitions with the same bounds, suffice. Their full coverage volumes

$$V_\pm = \int_{\mathbb{R}^d} U_{r,c}^\pm(z) dc$$

are independent of z , and

$$1 \leq \frac{V_+}{V_-} \leq (1 + C\gamma)^d, \quad \log(2 + \text{Lip } U_{r,c}^\pm) \ll \log(2/r) + \log(2/\gamma) + \log(2 + d). \quad (7.1)$$

Small cutoffs lie below the corresponding strict box indicator; large cutoffs dominate the corresponding enlarged box indicator.

Lemma 7.1 (Pairing the absolute patch). *The absolute patch on a productive terminal path may be replaced by a pair of product cutoffs as above, after a small proportional decrease of its target. The resulting score, counted with full parameter-box mass, is positive with inverse logarithm polynomial in p and the dimension bounds. The absolute radii have the same type of inverse logarithmic bound, without dependence on any determining width. The number and weights of the absolute slots are unchanged.*

Proof. Write the absolute patch as

$$B(t) = \sum_{a \in \mathbb{Z}^{d'_0}} \Phi(S_t(a)), \quad d'_0 \leq d_0,$$

as in Definition D.4, with Φ extended by zero. Suppose its full-mass score at target λ_0 is at least σ . Include the explicit mass of its parameter slice in σ . For either sign, convolution with the cutoff gives

$$\left| V_{\pm}^{-1} \int \Phi(c) U_{r,c}^{\pm}(z) dc - \Phi(z) \right| \leq C \text{Lip}(\Phi) \text{poly}(d'_0) r.$$

The union of the supports in this comparison is still a uniqueness box if r is sufficiently small. Consequently the sum over integer tuples has at most one nonzero term, even on this enlarged union. Uniform convolution error therefore bounds the error in each tested average, without an uncontrolled sum over tuples.

Choose a slightly smaller target λ , and then choose γ so that $\lambda V_+/V_- \leq \lambda_0$. Finally choose r so that the displayed uniform error is smaller than a sufficiently small multiple of σ . Averaging the paired score over c with the nonnegative weight $\Phi(c)$ gives at least $V_- \sigma/2$. The center region has bounded, or dimension-exponential, volume. Hence one center has a positive paired score with the asserted quantitative bound. The logarithms of V_-^{-1} , the inverse score, and r^{-1} have polynomial bounds in the absolute patch data. These data depend on p and dimensions, not on old widths. The chosen bounded center can either be retained in the cutoffs or absorbed into the target polynomials. Extending a polynomial from a reindexed affine slice by inverse affine substitution causes no coefficient restriction, since polynomial coefficients are unrestricted. $\square$

Fix a productive prefix at layer j . Its parent is an integer box with variable u , and all previous path maps and pulled-back input functions are now fixed. In particular C_j is an ordinary polynomial on this parent. Earlier substitutions have also been made in the later C_i ; we keep the notation C_i for those exact polynomials. Fix a feasible center at layer j and write

$$C_j(\psi(t)) - c = y(\psi(t)) + \beta(\psi(t)), \quad b_j = \beta + b_{j,0}.$$

The path ensemble and this identity are those of Lemma 3.2.

Definition 7.2 (Ascent format). At entry to layer j , the following data and properties constitute the ascent format.

- (i) A warm-cost slice of the parameter box carries a full-mass score at least δ , with $\log(1/\delta)$ warm. Its positive payoff contains the fixed pulled-back input $f_0(\psi(t))$, where $0 \leq f_0 \leq 1$. When $j \leq s$, it also contains the current factor $1_{\text{no bad}}(t)$. Its negative payoff has multiplier $-\lambda$.
- (ii) Both payoffs use, respectively, the small and large cutoffs of the absolute slots and of all determining slots b_i with $i > j$. For a determining block these are cutoffs of the *actual* residual

$$b_i - C_i(u, b_{\leq i-1})$$

at a bounded constant center l'_i and a radius whose logarithmic loss relative to w_i is warm at layer i . Their enlarged supports lie inside fixed padded versions of the old boxes. The block may also be restricted to a residue class of warm cost at layer i . Payoffs are summed over their allowed integral tuples. Each unsigned sum, with or without f_0 , is bounded by one.

- (iii) Determining polynomials never use absolute slots. Absolute polynomials may use determining slots b_i with $j < i \leq s$, but not passive slots $i > s$. The combined system has a fixed joint weighted-degree bound, with determining weights exactly i . Absolute weights are unchanged at passive steps and may be inflated at active steps. Once active determining slots are present, absolute weights exceed their weights.
- (iv) Polynomial extensions and determining identities are formal identities. Equality of integer slot arguments is never inferred merely from closeness of the corresponding real polynomials.

The terminal absolute return, after Lemma 7.1, supplies the initial instance. The surviving paths of Proposition 6.2 are modular-good at every layer. All productive subsets used below are subsets of these surviving paths; the active argument therefore starts with modular-good children even when its later selections have cold cost. The output of a layer will have this format at layer $j - 1$, with the current raw block of d_j determining slots included. Its full-mass score may contain a factor m_j . Thus its inverse logarithm may charge Q_j , which is warm at the next ancestor. The relative loss in the new width at layer j , by contrast, must be warm at layer j itself.

We only need the return at parents having a warm-inverse fraction of productive children. This follows by conditional averaging: if the total productive probability is at least κ , a set of parents of probability at least $\kappa/2$ has conditional productive probability at least $\kappa/2$. At each such parent, average once more over its feasible centers to fix a center with a comparable conditional probability. Repeat at the bounded number of layers. Inside one fixed parent argument a subsequent choice may lose a cold-pre fraction of its paths; this does not multiply the already obtained measure of parents where an output exists.

These choices require no simultaneous measurable choice of a polynomial witness over a continuum of centers. For fixed input data, earlier integer affine maps, integral lift polynomials and flags have finite possibilities. They index the discrete states on which the rank tests and descendant witnesses depend; center integration itself is measurable. After fixing a center, the path law again has finite discrete support. Witnesses can consequently be selected at these states after the probability estimates.

7.2 The comparison measure

The score in Definition 7.2 is an average on a child parameter box and still contains polynomial data chosen separately on each child. At a passive layer, Proposition 7.8 replaces these data by one ambient polynomial; scalar comparison then transfers site payoffs from one finite list. At active layers, Section 8 uses this comparison at the degree-zero end of its induction. We first define the weighted parent average needed for these steps. The inactive parameters retain their actual integer values. Each parallel parameter has a real copy and a residue copy; these are averaged independently before the site and incidence constraints are imposed. The common site then relates the two copies.

For a modular-good path and each fixed actual t_I , let $d(b; t_I)$ be the density of $b(t) \bmod \mathfrak{M}$, relative to uniform output measure, when $t_{\parallel} \bmod \mathfrak{M}$ is uniform. Lemma H.2 gives a warm cap for d , and an absolutely convergent Fourier expansion whose tail above individual character order T is bounded by a warm constant times an arbitrarily large fixed inverse power of T . A pre-cost stride restriction on $t_{\parallel}$ has the same conclusions with a pre-cost prefactor. Different strides can be

combined into one divisible stride and disjoint residue classes. These conclusions also hold for any complementary subset of the primes. In particular none of these caps grows with Z .

Fix a modular-good path. Define the nonnegative functional $\mathcal{K}$ on bounded site-and-label functions by the following integration rule.

1. Use the full-parent average $\mathbb{E}_{u \in U}$ with the indicator of the buffered uniqueness chart, not the average conditioned on that chart. Independently average over one actual integer vector t_I in its original ranges, and over an independent uniform $t_{\parallel}^{\text{res}} \bmod \mathfrak{M}$. Impose

$$z_I(X) = z_I(t_I), \quad b(X) = b(t_{\parallel}^{\text{res}}, t_I) \pmod{\mathfrak{M}}, \quad (7.2)$$

and multiply by

$$m_j^{-1} \left(\prod_{i \in I} k_i \right) \mathfrak{M}^{m_b}. \quad (7.3)$$

2. Use normalized real parallel parameters in $[0, 1]$. Keep the spatial kernel output and the continuous and active principal lift outputs, with the path's actual constant shifts. Their required output is

$$(u/H, y_W(u)/v, z_{\text{act}}(u)/k). \quad (7.4)$$

Choose one spatial kernel block and one variable in a dedicated product block for each lift output direction as pivots. Integrate the remaining real parameters, solve linearly for the pivots giving (7.4), and retain only solutions in their parameter ranges. Weight each solution by the inverse absolute determinant; a zero determinant contributes nothing. Evaluate the real labels at this solved parameter vector. This defines the integration on each output fiber even for an arbitrary bounded function of the labels.

Use the orthonormal coordinates and measure convention defining m_j . Nonkernel spatial displacement and nonprincipal lift perturbations are omitted only in this real density, not in the integer congruences.

3. A slice imposes its real interval and residue conditions on the two parallel copies, and its actual integer membership condition on t_I . Unless explicitly normalized, its mass is retained. Extra normalized labels use the real parallel copy and t_I/T_I ; extra integer labels use the residue parallel copy and the same actual t_I .

Lemma 7.3 (Comparison measure and forecasting weights). *For site functions alone these are the normalized-chart forecasting weights of Equation (H.1). The unrestricted site weight is at most Bm_j^{-1} for a warm B . Normalized pre-cost slices have cap $B_{\text{pre}}m_j^{-1}$ and uniform approximations, on buffered charts, by pre-cost combinations of order-zero site twists times m_j^{-1} . Both approximation error and coefficient bounds are measured in these normalized units and are independent of Z . Smoothness is measured in $(u/H, y/v)$.*

Proof. For site functions, the pivot integral is the output density by the change-of-variables formula. Spare independent smoothing blocks ensure that this formula gives the same continuous version of the density at every output, also with rectangular interval restrictions: a possibly singular block law is convolved with a bounded Lipschitz density. Thus no value of an arbitrary almost-everywhere density representative is being evaluated at a true site. Lemma G.4 supplies the required smoothing, while independent nonsingular spatial kernel blocks give integrable Fourier decay with a first moment. This continuity assertion concerns the site weight. Arbitrary label insertions are evaluated by the explicit pivot integral above.

The same convolution observation controls deletion of pivot choices with very small determinant, or pivot coordinates in narrow boundary strips. Condition on the exceptional pivot group before convolving with spare blocks. Its contribution to every output density is bounded by a controlled constant times its parameter mass. The determinant is a product of bounded-degree factors and fixed nondegenerate principal amplitudes; keeping its factors outside appropriately small strips therefore has polynomial logarithmic cost. These facts concern density in normalized output coordinates, not a point mass at a critical image.

We now check the cap and approximation. Conditional on t_I , the congruence density has the cap following Equation (6.3). The inactive product-block law has a bounded density relative to its grid volume. Its extra linear column is handled by conditioning and translating; omitted axes have warm-bounded k_i and are enumerated. Multiplying these caps by Equation (7.3), and by the bounded real-output density, gives Bm_j^{-1} . The corresponding slice proof charges its explicit normalization and pre-cost restrictions.

For the approximation, truncate the absolutely convergent congruence Fourier series by individual character order. A term of order q has coefficient depending only on $t_I \bmod q$, because b has integer coefficients. Split t_I into these residue classes. The order-zero product-block expansion on each class gives the remaining smooth and residue twists; an axis too short after the restrictions has pre-bounded k_i and can instead be enumerated. Characters of sheet and deck coordinates are characters of β at the same modulus. The real kernel and dedicated blocks have the stated normalized smoothness. Summing the finitely many retained orders and dividing the error among them preserves a pre bound. This is the forecasting-weight proof preceding Proposition H.1, now in units v, k_i . Its absolute Fourier tail and all caps are uniform in Z . $\square$

7.3 Proof of the selected-plane estimates

The comparison measure retains the congruence data of the whole path. The identity in Equation (6.4) provides a family of sections of that data: the variable s changes the output by a prescribed amount, while h changes the inactive linear variables. The selection tests ensure both controlled counting weight and the translation invariance needed when these sections are used in extraction.

Proof of Lemma 6.1. We first prove the finite-field assertions, including the characteristic dependence. Temporarily write $P = \varpi$, $N = v_f$, and fix the other coefficients. Every nonconstant pure-free coefficient of every output is independently uniform. An affine change of free variables acts invertibly on polynomial coefficients modulo constants, in any characteristic. For a full-rank V , restriction to $x + V(s, h)$ maps these coefficients surjectively onto all polynomials in (s, h) modulo constants of the indicated degree. This follows, for example, by choosing a linear left inverse of V . Consequently Equation (6.4) has probability exactly P^{-J} . The fixed constant coefficients do not affect this assertion, since the identity subtracts $b(x, r_0)$.

Let N_0 be the unselected solution count at a given basepoint and $H = P^{N\ell-J}$. Restrictions to two direction spaces with direct sum have independent nonconstant coefficients. The proportion of rank-deficient matrices is $O(\ell P^{\ell-N})$, and the proportion of pairs whose column spaces do not have direct sum is $O(\ell P^{2\ell-N})$. Using the trivial joint probability bound on these bad pairs gives

$$\mathbb{E}(N_0/H) = 1 + O(\ell P^{\ell-N}), \quad (7.5)$$

$$\mathbb{E}[(N_0/H)^2] = 1 + O(\ell P^{2\ell-N+2J}). \quad (7.6)$$

The harmless error in the first line is also absorbed into the second.

Condition next on Equation (6.4) for a specified full-rank first matrix. At a specified translated point of its plane, two further direction spaces retain independent uniform nonconstant restrictions

whenever all three free direction spaces have direct sum. Indeed use basis coordinates for their direct sum, centered at that translated point; translation along the first plane changes its fixed restriction but leaves the other nonconstant coefficient arrays free. The same first- and second-moment argument has error at most

$$O(\ell P^{3\ell-N+2J}). \quad (7.7)$$

This bound allows the crude estimate on all nondirect triples.

Here is the Jacobian check. In free coordinates adapted to the first plane, write the transverse variables as $y_1, \dots, y_{N-\ell}$. Conditional on its restriction, the transverse derivatives in these directions are independent uniform vectors in

$$E_{\text{der}} = \bigoplus_{i=1}^{m_b} \text{Pol}_{\leq d_i-1}(\mathbb{F}_P^\ell), \quad \dim E_{\text{der}} = n_{\text{der}}.$$

The probability that $N - \ell$ such vectors fail to span is at most $O(P^{n_{\text{der}}-(N-\ell)})$, by a union bound over nonzero dual vectors. If they span, varying the q th column of V realizes that space multiplied by the q th plane coordinate. Every nonconstant monomial of degree at most d_i has such a factor, so these variations span all J coefficient equations. This proof differentiates only terms $y_a(s, h)^\alpha$, whose transverse exponent is one. It uses neither factorial division nor a derivative of a high power and is valid also when $P \leq j$.

There are P^ℓ translated basepoints on a plane. Apply Chebyshev's inequality at tolerance $\epsilon_P/2$ using Equation (7.7), and union bound over these points together with the derivative failures. Since J, ℓ , and n_{der} do not involve N , taking N a sufficiently large polynomial in the structural dimensions makes the conditional probability that a solution is unselected at most, for example, $c_\star \epsilon_P^2 P^{-20}$. Summing over solutions bounds the expected deleted count divided by H by the same quantity. Markov's inequality for the deleted count and Equations (7.5)–(7.6) for the full count prove the asserted concentration. The exact upper test in the selection rule also gives $w_{\varpi,1} \leq 1 + \epsilon_\varpi$: if any matrix is selected, that test holds at the original basepoint.

The formal plane identity persists at every point of its plane. Both tests defining selection inspect that whole plane, so the selected incidence relation is translation invariant. At a selected solution modulo ϖ , the Jacobian is surjective. A lift from depth a to depth $a + 1$ requires solving J independent linear equations in the $N\ell$ next digits of V . There are exactly $\varpi^{N\ell-J}$ solutions. Induction proves the claimed depth independence at every compatible lift of the basepoint, regardless of its higher digits or of those of the fixed polynomial coefficients.

The product bounds follow from $\sum_\varpi \epsilon_\varpi = O(c_\star)$. Chinese remaindering makes the nonconstant coefficient arrays at different primes independent, even after conditioning on the remaining nonconstant data. This gives the joint flag bounds. Notice that all equations and their Jacobians ignore the constant coefficient of b .

For the actual path law, Lemma 3.2 gives the needed finite-residue comparisons. Uniformity is required in all nonconstant pure-free degrees, not only the leading degree. On moderate axes these are precisely the added pure-free perturbations; the other axes use the original continuous/enormous laws and uniform deck choices. The late range choices make every required coefficient interval broad at each individual modulus.

For completeness, the proportional estimate with an arbitrary site insertion needs the analytic-removal assertion as well as the residue law. Fix a parameter value t and average the unnormalized tilted one-site integral over all centers. Its normalizer is $1 + o(1)$ uniformly. Center averaging removes the constant-coefficient density. Enumerate the spatial nonconstant residues of the flag event and Fourier approximate its nonconstant lift-coefficient condition on the individual prime cover. A nontrivial coefficient mode factoring through the site has its site row as its constant row. Since the

flag condition has constant row zero, such a surviving factoring mode is trivial. Lemma F.7, in the form verified in Lemma 3.2, removes all other modes even with the arbitrary site insertion. What remains is the Haar-chart flag probability conditional on the specified spatial nonconstant coefficient residues modulo ϖ . Denote this residue array by A_{sp} . Analytic removal is applied on each residue class, not after fixing the exact values of all spatial slopes. In the remaining unweighted spatial average, the independent root law has an upper density arbitrarily close to uniform for every fixed slope tuple. Hence, if $q_{\varpi}(A_{\text{sp}})$ is the conditional Haar-chart flag probability, the bound on each residue class, for a nonnegative insertion F , is

$$q_{\varpi}(A_{\text{sp}})(1 + o(1))\mathbb{E}_U F.$$

Only after averaging the broad spatial slope residues A_{sp} do we use $\mathbb{E}q_{\varpi}(A_{\text{sp}}) \leq c_{\star}\varpi^{-10}$; the conditional probability itself need not be small. Summation over individual primes proves the result. The approximation error per prime can be chosen small enough that its sum through Z is still arbitrarily cold-small. No simultaneous law modulo $\mathfrak{M}$ was used. $\square$

7.4 Scalar transfer with labels and flags

A menu below is a finite list of bounded functions of the true site, with no regularity assumed in that variable. At a fixed parent and center, the same menu is used for every path. Its entries may depend on ambient maps already selected in the return argument; a path may then select an entry and a permitted slice. The logarithmic cardinality, cap and required accuracy are forecast before the late sampler choices. This forecasts bounds on the eventual menu, rather than requiring the ambient maps themselves to be chosen before the ensemble is constructed.

Proposition 7.4 (Scalar comparison). *Suppose a set of productive modular-good paths has probability at least τ , with $\log(1/\tau)$ pre bounded, and each has a specified pre-cost slice. Consider bounded scores depending on site and auxiliary labels. Assume that, after partitioning into pre-count many pre-cost slices, each score is uniformly approximable at the required pre accuracy, both on actual path arguments and on the joint support of $\mathcal{K}$, by the same menu of bounded site functions. A bounded number of successive increases of these pre bounds is allowed.*

Then, outside a set of paths smaller than the productive probability, the actual scalar averages and their $\mathcal{K}$ integrals differ by an arbitrarily prescribed pre-small error. This holds simultaneously for a pre-count menu of scores and for path-dependent slice choices. At active layers it also holds after multiplying a score by the no-bad-flag indicator. It is valid either for normalized slice averages or with full-box mass retained, paying the explicit slice fractions. The budget is warm for passive uses and cold pre otherwise.

Proof. First omit flags. Enlarge the degree-zero local testing seminorm by the supremum of pairings with all normalized sliced forecasting weights of Lemma 7.3. Their cap and uniform order-zero twist approximation give detection for the enlarged seminorm. Apply the separation argument of Lemma G.11, using Theorem 3.4. In the warm case its inner product is the normalized chart measure, and the one-site cap and tail are taken relative to that measure. Every fixed bounded masked site input therefore has an order-zero twist model whose residual is small both in the pathwise slice seminorm and against every forecasting weight. Fix the number of atoms, their coefficient mass, and their complexity before Z is chosen. Markov's inequality and a union bound over the menu make all the actual residual errors small outside a set of paths of probability less than any prescribed fraction of τ . The supremum in the seminorm permits slices to be chosen after the path.

Choose Z to include every prime power needed by these atom moduli, slice strides, and label periods. For a model twist

$$J(u) = F(u/H, y(u)/v)R(u, \beta(u) \bmod q),$$

the actual and comparison integrals agree at the required accuracy. Here are the normalization and modulus details. Truncate the rational Fourier density at individual order q' . Each retained term and the atom require only the cover of modulus $\text{lcm}(q, q')$. Apply single-site Haar-cover equidistribution on this cover. Exact inactive equalities can be extended by buffered cutoffs at the actual grid resolution; since inactive $k_i \leq L^j$, this has permitted late cost. In orthonormal lift coordinates, Haar sheet measure is Lebesgue measure on W times counting measure on the sheet coordinates, divided by $\text{covol}(W \cap \mathbb{Z}^{d_j})$. Multiplication by $m_j^{-1} \prod_{i \in I} k_i$ leaves normalized Lebesgue measure in y_W/v , exact inactive counting measure, and active grid volumes $\prod_{i \notin I} k_i^{-1}$. The spatial and deck residues are independent uniform variables. On each active axis the normalized spacing in the needed residue class is at most $\text{lcm}(q, q')/k_i$. Since $k_i \geq (AL)^j$, the later choice of A makes the grid sum converge uniformly to the real density.

Restore the rational density by its uniform absolute tail. Its integral against R depends only on the marginal modulo q , not on the whole $\mathfrak{M}$. Actual Riemann summation on each parallel parameter in its prescribed residue class gives that same marginal jointly with the real argument. The omitted nonkernel spatial terms and normalized lift perturbations have size $O(\sigma)$ and therefore change the smooth input negligibly. Every residue input throughout uses the true integer polynomial. No rounding of a principal real output is used to infer its residue. These are precisely the individual-cover steps of Equation (H.10), with the present normalization. Combining them with the two model residual estimates proves the unflagged assertion.

The common-menu hypothesis is stable under the label operations we shall use. Independent bounded-length integer axes can be enumerated; integer labels can be split at pre-cost moduli dividing $\mathfrak{M}$; and pre-Lipschitz dependence on long normalized labels can be meshed inside the original slice while the integer labels are held fixed. Use the same mesh value on actual and comparison supports. Axes too short for a mesh or stride are enumerated exactly. Parallel lengths were chosen after these requirements, so endpoint rounding there is negligible. These operations cost a union bound over menu entries, not a simultaneous pigeonhole of all their pathwise choices.

We next prove the flagged assertion and record its choice order. At a given approximation step choose a large pre threshold T , before choosing Z , and replace the no-bad indicator by

$$\sum_{\substack{d \leq T \\ \text{squarefree} \\ \varpi | d \Rightarrow \varpi \leq Z}} \mu(d) 1_{\{\text{bad at every } \varpi | d\}}. \quad (7.8)$$

This is exact if the product of the bad primes is at most T . Otherwise its absolute error is at most $1 + T$. The exceptional event has a witness consisting either of one bad prime $\varpi > T$, or of a squarefree product $d \in (T, T^2]$ of bad primes: in the latter case multiply primes at most T until the product first crosses T . Lemma 6.1 and summation give expected exceptional mass $O(T^{-9})$ in the uniform model and hence error $O(T^{-8})$. The same conclusion holds in actual unrestricted path/parameter averages at fixed center, up to negligible errors, by the individual finite-residue laws for nonconstant coefficients. For a normalized slice of probability ρ , divide the full-mass error by ρ ; this charges only the permitted pre factor ρ^{-1} .

We need the same statement for $\mathcal{K}$, where incidence weights cannot simply be ignored. On modular-good paths, retain the matching at the witnessing primes and bound matching at all complementary primes by their modular Fourier cap, conditional on t_I . Flags see only first digits,

but retain matching to the full tested depth at each witness prime. For a squarefree witness $d \leq T^2$, this full modulus is

$$Q_d = \prod_{\varpi|d} \varpi^{a_\varpi}, \quad \log Q_d \leq \frac{2 \log T \log Z}{\log 2}. \quad (7.9)$$

For a single prime witness its full modulus has logarithm at most $\log Z$. Thus each comparison has allowed late cost after Z , even though $\mathfrak{M}$ itself need not be affordable. The prime set of a witness does not depend on the site.

On the Q_d cover integrate the real output density and the exact inactive constraints with the prescribed residue data, by the individual-cover computation above, increasing A also for these moduli. Integration removes the retained incidence-density factor and leaves uniform parameter residues on the witness primes. The actual t_I can be held fixed throughout. Complementary parallel residues were bounded by their separate cap, and the Chinese remainder theorem separates them from the witness residues. This cover calculation is uniform in the fixed path data: the real principal amplitudes have bounded normalized ranges, exact inactive cutoffs have grid resolution k_i^{-1} with $k_i \leq L^j$, and all integer coefficients in the witness condition are reduced modulo Q_d . Their Fourier costs are therefore among the forecast cold and late costs, independently of unreduced integer coefficient sizes. In particular, writing E_d for the witness flag event, the full-mass inequality on a modular-good path is

$$\mathcal{K}(1_{E_d}) \leq C_{\text{pre}} \mathbb{E}_{t_I} \mathbb{E}_{t_{\parallel}} \bmod Q_d 1_{E_d} + o(1). \quad (7.10)$$

There is no factor $Q_d^{m_b}$ left in this inequality: the retained matching factor cancels against uniform output mass on the cover, and the remaining full real density and normalized inactive-grid mass integrate to at most one. A normalized slice instead divides this bound by its slice probability. The errors can be chosen uniformly small enough to sum over all witnesses, since their number is at most $Z + T^2$ and $\log Z, \log T$ have the prescribed bounds. Average over paths and drop their goodness restriction for this nonnegative upper bound. The finite-residue comparison at the witness then gives its uniform flag probability. Summing witnesses gives $O(T^{-8})$ error, now with a fixed pre prefactor from D'_0 and the normalized real densities. This is the required forecast-side bound.

Markov's inequality makes both truncation errors small on enough productive paths, taking T large relative to the current score, probability, and cap. Each retained intersection in Equation (7.8) uses a modulus $d \leq T$ and is constant on a further pre-cost residue slice, separately for each path. The unflagged comparison therefore applies to it. Subsequent models and menus may charge $\log T$: the truncation just proved used only the unexpanded score and the current cap and accuracy, not those later menu sizes. A bounded number of required truncations and models can consequently be forecast successively before Z . Then Z is chosen to contain their periods, and only then are the individual witness comparisons and late sampler parameters fixed. This order avoids any dependence on the unaffordable size of $\mathfrak{M}$. The proof also shows that actual-path flag truncation alone may be used in positive-degree detection and pivot selection. $\square$

7.5 The group representation and its fixed menus

Let $X = (u, \beta)$, with u of weight one and β of weight j . As actual site data in the buffered chart, β is the uniquely determined lift integer. As an argument of a polynomial identity, it is a formal variable.

Exact marked projections. A rational homomorphism $\pi : G \rightarrow Y$ between filtered groups is *strong* if $\pi(G_i) = Y_i$ at every layer. A *marked* polynomial map is supplied with a fixed polynomial P_Y and an exact identity $\pi(g(X)) = P_Y(X)$ in its formal variables X . Retaining that identity

preserves the old determining equations when a new test is constructed. The rational refiltration statement that preserves this structure is Lemma 7.6.

Rational kernels, simultaneous intersections, sections, controlled lattices and fixed-step Baker–Campbell–Hausdorff calculations have polynomial logarithmic cost by Lemma A.2. All uses of this calculus allow replacement of the degree by any fixed larger degree; none bounds arbitrary orbit coefficients.

Lemma 7.5 (Representation with exact determining projection). *At a fixed parent and feasible center, the ascent data admit the following representation, after pigeonholing warm structural data.*

At a passive layer $j > s$, there is one triangular shear group G^0 for the absolute slots, of degree at most s , and path-dependent ordinary filtered polynomial maps $g_\psi(t)$ into it.

At an active layer $j \leq s$, there is one joint triangular group G^0 for the active determining slots $j < i \leq s$ and the absolute slots, together with a strong rational projection $\pi^0 : G^0 \rightarrow Y$ to the determining shear group. There is a fixed weighted filtered polynomial $P_Y(X)$ such that

$$\pi^0 g_\psi(t) = P_Y(X_\psi(t)), \quad X_\psi(t) = (\psi(t), \beta(\psi(t))). \quad (7.11)$$

In both cases the two payoffs, excluding the slice, current flag and target multiplier, are evaluations of fields

$$F_{\psi,u}^-, F_{\psi,u}^+ : G^0/\Gamma^0 \longrightarrow [0, 1].$$

They have a common warm logarithmic Lipschitz bound. At every specified accuracy they have meshes of polynomial logarithmic size in two norms. A point-value mesh approximates the nilmanifold functions $F_{\psi,u}^\pm$ in supremum norm on G^0/Γ^0 . A field mesh approximates $u \mapsto F_{\psi,u}^\pm$ simultaneously over all parent sites, with the supremum also over G^0/Γ^0 . None of these bounds depends on orbit coefficient sizes. The original cutoff functions may be retained exactly in the output; the meshes are only devices for selecting partners and comparing scores.

Proof. We first separate the passive determining conditions from the group variables. Outside the required current chart choose fixed common values, which the local and forecasting supports never test. At each site in the chart use fixed padded boxes, for example radii $2w_i$ around the prepared offsets, to compute the later old slots successively. Triangularity gives at most one tuple. If a required padded lift does not exist, set the resulting passive multiplier equal to zero. Evaluate the passive cutoffs and congruences on this canonical tuple, obtaining site multipliers $L_\psi^\pm$.

This procedure can use active determining slots, but it does not alter a nonzero payoff. On the actual orbit, every nonzero active cutoff forces exactly its canonical padded lift. The same is true on any returned orbit having the exact determining projection (7.11). Thus the separated expression agrees with the original joint expression at every point at which it will be tested, including when either expression is zero. On arbitrary points of the full group quotient we simply keep $L_\psi^\pm$ as site multipliers independent of that group point; no identity with unconstrained slot evaluation is claimed there.

The canonical tuple is fixed as a function of the site. Only the bounded centers, cutoff radii, moduli and classes vary with the path. Meshing those real parameters and enumerating the discrete ones gives a field mesh in site supremum norm. Its cost is warm at j , since only widths Q_i with $i > j$ enter. The fixed input f_0 can be included without increasing this mesh size.

For active determining classes write $b_i = M_i k_i + \alpha_i$ and scale their residuals by $1/M_i$. Pigeonhole the classes and moduli, as well as the absolute rank and weight assignments, before fixing the group. This has warm cost. Order determining slots before absolute slots. At the first active step there are no previously processed active determining slots; at later steps the larger absolute weights permit this order.

Apply the shear construction in the proof of Lemma D.5. In the residual triangular map, normalize all pure full-destination-weight slot coefficients by triangular integral changes of the summation variables, in increasing slot order. These changes are independent of t , preserve the integer tuples bijectively, and are substituted exactly into every later formula. Large coefficients created by an earlier change are reduced when their own slot is reached. Write the resulting residual map as

$$S^0 \circ g_\psi(t),$$

where S^0 consists of its full-destination-weight slot terms. Those coefficients are bounded; its determining part is fixed at the parent, since full-weight slot coefficients cannot depend on the positive-weight variables X . The remaining transformations strictly lower slot weight and form the filtered group G^0 , with filtration given by weight deficit.

Forgetting absolute slots is a group homomorphism to Y : a determining transformation has no absolute argument. A determining shear of any deficit lifts by acting on determining slots and leaving absolute slots fixed. The projection is therefore strong at every filtration layer, with controlled rational sections. Taking the determining part of $(S^0)^{-1}$ composed with the exact residual map defines $P_Y(X)$ and proves (7.11) as a polynomial identity. Cutoff offsets are placed in the tested functions, not in these target polynomials. All integral normalizations can be undone when raw determining coordinates are restored.

The two functions on the quotient evaluate the absolute and active determining cutoffs, multiplied respectively by $f_0(u)L_\psi^-(u)$ and $L_\psi^+(u)$. They exclude the current flag, slice and λ . The integer-lattice action permutes tuples, so these are well-defined quotient functions. Triangular uniqueness gives their cap one. The proof of Lemma D.5 bounds their Lipschitz constants: on bounded group representatives, the inverse of the bounded full-weight map S^0 and its derivatives have dimension-controlled logarithmic bounds; contributing integer tuples and their derivative estimates have the same bounds. Class scalings and inherited cutoff radii have warm cost here.

Finally, S^0 and all cutoff parameters lie in bounded, polynomial-dimensional parameter sets with controlled moduli of continuity. Their grids, together with the passive field mesh already constructed, give both claimed meshes. No mesh is taken over the polynomial coefficients of g_ψ or over arbitrary site functions. Keeping the original functions after a mesh-based selection costs only the prescribed approximation error, so their exact widths and buffers remain available. $\square$

7.6 Reducing degree while retaining a marked projection

7.6.1 Symbols and real or rational control

For a filtered group G of degree q , write $\mathfrak{g}_i = \log G_i$. Its associated graded Lie algebra is $\bigoplus_{i=1}^q \mathfrak{g}_i / \mathfrak{g}_{i+1}$. The *symbol* of g is the map into its simply connected group whose logarithmic component in grade i is the degree- i homogeneous part of $\log g$ projected to that grade; with assigned variable weights, homogeneous degree means weighted homogeneous degree. In the scalar case, the symbol is just the homogeneous polynomial at the assigned weight. On a box with side lengths $H = (H_1, \dots, H_n)$, write $H^\nu = \prod_i H_i^{\nu_i}$. A symbol is *slow* at budget p if its logarithmic coefficient of u^ν has magnitude at most $\mathcal{B}(p)H^{-\nu}$ in the declared coordinates. It is *rational* if its logarithmic coefficients have a common denominator at most $\mathcal{B}(p)$, without a numerator bound. A polynomial map is *fully slow* if the corresponding bounds hold for every coefficient of its logarithm, including the constant coefficient, after translating the box to the origin. Thus slow control concerns normalized real variables, whereas rational control concerns denominators in the raw integer variables. The compatible splittings and their lifts to actual maps are proved in Section A.

The group representation separates two requirements. The functions must retain a positive signed score, and the orbit must retain the fixed determining projection. The algebraic reason these requirements can coexist is that the projection can record the top coordinate removed in a degree reduction. In the passive case take Y to be the trivial group and π^0 the trivial projection. Write r for the degree of the represented group G^0 .

Suppose a strong projection $\pi : G \rightarrow F$ is injective on its top layer G_r . Then an element $g \in G$ is determined by gG_r and $\pi(g)$: two elements with the same first coordinate differ by an element of G_r , and equality of their projections makes that difference trivial. A prescribed polynomial projection therefore leaves only a degree- $(r-1)$ orbit to construct. The original group need not satisfy this injectivity. The next lemma allows us to obtain it on the rational subgroup selected by the symbol argument.

Lemma 7.6 (Refiltration under a strong projection). *Let $\pi : G \rightarrow F$ be a rational homomorphism of connected simply connected filtered nilpotent Lie groups such that $\pi(G_i) = F_i$ for every i . Let $\mathfrak{v}$ be a graded rational subalgebra of $\text{gr } G$, and let $\mathfrak{v}_F = \text{gr}(\pi)(\mathfrak{v})$. Define $H_i \subset G_i$ and $H_{F,i} \subset F_i$ by taking the preimages of $\mathfrak{v}_i$ and $(\mathfrak{v}_F)_i$ in each graded quotient. Then $H = H_1$ and $H_F = H_{F,1}$ are filtered rational groups, and $\pi(H_i) = H_{F,i}$ for every i .*

If the original degree is r and the map $\mathfrak{v}_r \rightarrow (\mathfrak{v}_F)_r$ is injective, then

$$H \longrightarrow (H/H_r) \times_{H_F/H_{F,r}} H_F, \quad h \longmapsto (hH_r, \pi(h)) \quad (7.12)$$

is a rational isomorphism, onto at each filtered layer.

Proof. Every H_i contains G_{i+1} , so the H_i are nested. The bracket inclusion follows by reducing a bracket modulo G_{i+j+1} and using $[\mathfrak{v}_i, \mathfrak{v}_j] \subset \mathfrak{v}_{i+j}$. Rationality follows from the rationality of the quotient subspaces. For $y \in H_{F,i}$ choose $x \in G_i$ whose graded image lies in $\mathfrak{v}_i$ and projects to that of y . The discrepancy $\pi(x)^{-1}y$ lies in F_{i+1} , and hence has a lift in $G_{i+1} \subset H_i$. Correcting x proves strongness.

At the top, $G_{r+1} = F_{r+1} = 1$, so $H_r \rightarrow H_{F,r}$ is both injective and surjective. The kernel of (7.12) is therefore trivial. Given a compatible pair (xH_r, y) , compatibility says $\pi(x)^{-1}y \in H_{F,r}$. There is a unique $z \in H_r$ with $\pi(z) = \pi(x)^{-1}y$, and xz maps to the pair. The same correction lies in every required earlier layer, proving filtered surjectivity. In logarithmic coordinates the map and its inverse are rational linear maps between rational Lie algebras, followed by exponential coordinates. This proves rationality and the claimed isomorphism. $\square$

Here is how the local correlations supply the hypothesis of the lemma. Truncate the top Fourier expansion of each payoff and retain those modes whose tested averages are significant relative to the score. The total of the discarded averages is charged to the error. Each retained frequency can be paired, by path detection, with a top frequency of an ambient partner $Q_i(u)$ in a group G_{Q_i} . Step drop puts the joint symbol in a rational graded algebra on which the sum of those two frequencies vanishes, up to slow and rational outer factors. Choose a basis among the payoff frequencies restricted to $G_r^0 \cap \ker \pi^0$, and let L be their connected joint annihilator there. After averaging over L , form

$$G = (G^0/L) \times \prod_i G_{Q_i}, \quad F = Y \times \prod_i G_{Q_i},$$

with projection induced by π^0 and the identities on the partners. Lemma A.6 intersects the pair constraints in one graded algebra $\mathfrak{v}$.

A vector in $\mathfrak{v}_r$ whose projection to F is zero has zero partner coordinates. Each pair constraint then says that the corresponding payoff frequency vanishes on its G^0 coordinate. That coordinate lies in L , so it is zero in G^0/L . Thus $\mathfrak{v}_r \rightarrow (\mathfrak{v}_F)_r$ is injective, exactly as required in Lemma 7.6. Only a basis of restricted frequencies is needed, so the number of partner groups is bounded by the dimension of the top kernel.

The next task is to factor the joint local orbit in G as

$$g = EhR, \quad \pi E = E_F, \quad \pi h = P_H, \quad \pi R = R_F,$$

where the projected factors are fixed ambient polynomials and h is adapted to the refiltration H . Matching symbols alone does not give these identities; the proofs below construct exact kernel corrections. Once they hold, (7.12) reconstructs h from hH_r and the fixed value P_H . On suitable lattice covers, a tested function of EhR becomes a function of this compatible pair. Specializing its second coordinate at P_H leaves a field of tests on the lower-degree quotient H/H_r .

Proposition I.6 carries out this construction when all auxiliary slice and score losses have the same early budget. We need two refinements. At passive layers the losses in the scoring slices must all be warm; the next lemma recovers that bound in the symbol-matching step. At active layers the auxiliary construction may be cold, so Section 8 retains its short variables and outer factors as polynomial labels and preserves the original score by identities of expected payoffs.

7.7 Warm matching at a passive layer

At $j > s$, all degrees that occur in ordinary partner selection are strictly below j . The warm approximation conclusion of Theorem 3.4 therefore applies. Its ambient partners are ordinary niltests times twists localized to the chart. On a path, their smooth arguments $\psi(t)/H$ and $y(\psi(t))/v$ have bounded normalized polynomial coefficients. Warm subdivision and bounded-modulus restriction freeze these twists to any warm accuracy. In particular, freezing them does not cost Q_j .

The remaining issue is the major calculation in marked-symbol matching. The following observation is the reason it has warm output at passive layers despite using a cold auxiliary theorem.

Lemma 7.7 (Warm ordinary matching). *Fix $j > s$ and $1 \leq d \leq s$. Consider polynomially many fixed weighted homogeneous scalar forms of degree d whose leading path evaluations have slow-plus-rational decompositions with common warm bounds on a warm-inverse fraction of paths. The forms themselves have slow-plus-rational decompositions on u , with coefficients bounded in u/H and with a common denominator of warm cost. The auxiliary cold-pre requests can be forecast before the late path scales and rank thresholds. They cause no actual cold-cost restriction of a productive scoring slice.*

Proof. A weighted form of degree $d < j$ has no β argument. Apply Proposition I.1, or its quantitative refinement in Lemma 4.7, at cold-pre cost. It yields, on the ordinary u variables,

$$A(u) = S(u/H) + R(u),$$

with cold-pre coefficient and denominator bounds. Any restrictions on lift-value spaces in that conclusion are irrelevant to an expression with no lift argument.

For this auxiliary argument retain only the parallel long parameter variables, whose sides are L . Choose L after the just-obtained cold-pre output bounds. The displayed decomposition and the original warm decomposition have equal evaluations there. Their difference is simultaneously slow and rational, with cold-pre bounds. Separation, Lemma A.5, makes the positive-degree difference

zero once L exceeds those bounds. Thus the evaluation of S has the original warm normalized coefficient bound, and that of R has the original warm denominator bound. There is no constant term to separate at this positive homogeneous grade.

The spatial kernel contains an invertible block at the common warm normalized scale. Restricting to this block recovers the coefficients of S on u/H with a warm inverse norm, giving the claimed real bound. For the rational bound, multiply R by the warm common denominator of its evaluation and apply Lemma 4.6 to the pure-free spatial outputs. Its loss depends on dimensions and the inverse retained path probability, not on the cold a priori denominator ceiling. Use the unconditional uniform-goodness estimate, choosing its exceptional probability smaller than the warm-inverse fraction under consideration. This does not assert equidistribution after conditioning on the productive subset. At least one retained path is good, and its denominator conclusion is a warm bound for the fixed R . Apply the same argument to the polynomially many coordinates jointly, with common denominators throughout.

The enlarged upper ceilings and comparison accuracies are only cold-pre requests. They are fixed before the late choices of L , perturbation sizes, rank and parent sides. At each successive matching grade the modular loss is chosen from the current warm probability and accuracy, rather than from an output accuracy not yet known. There are only boundedly many grades. This order avoids any circular requirement. Finally, the restriction to parallel variables above proves an identity of the fixed scalar forms; it is not a restriction of the scoring slice. Its long thresholds therefore do not become cold subslice costs in globalization. $\square$

7.8 Returning an ordinary orbit

Proposition 7.8 (Passive micro replacement). *Suppose $j > s$ and the ascent format holds on a warm-inverse fraction of paths at a fixed parent and feasible center. Use the group and the two fields of Lemma 7.5. There is a single ambient ordinary filtered polynomial*

$$g^U(u) \in G^0$$

of degree at most s , with no β argument, for which a warm-inverse fraction of the productive paths have warm-cost subslices on which replacing g_ψ by $g^U \circ \psi$ preserves a positive signed full-mass score of warm size, after the allowed small target discounts. Both inherited cutoff functions and all their exact determining conditions are retained.

Proof. We adapt the proof of Proposition I.6 and record the points needed for the two payoffs and the warm budget. This is a bounded-degree induction inside the original triangular group. No new absolute slots are introduced by it.

Top modes and a fixed menu of inputs. Truncate the top vertical Fourier expansions of both fields with total uniform error much smaller than the current score. On each productive path call a restricted top frequency significant if the absolute value of its tested average in either payoff exceeds the score divided by a sufficiently large multiple of the total number of retained modes. The sum of all discarded averages is negligible. Take the joint connected annihilator L in the kernel top torus and choose a basis among all significant restricted frequencies from the two lists. Averaging both fields over L preserves their range $[0, 1]$ and changes the signed score only by the discarded error. The number of pivots is at most the top dimension. As in the cited proof, pigeonholing bases, rather than arbitrary subsets of frequencies, costs only a warm fraction.

For each pivot use the payoff in which it is significant. Path-dependent external fields require care here. First choose a simultaneous field mesh at the required site-supremum accuracy. For every member of that fixed menu form point-value bins using a common point-value mesh. The

masked site inputs belonging to these bins now form a fixed controlled menu over all paths. Apply Theorem 3.4 to each bounded masked input, with error small enough to pay for the menu size, pivot threshold and retained path probability. A union bound and Markov's inequality make the models simultaneously valid on a retained warm-inverse fraction of paths, for their separately chosen slices and tests. The local testing degree is at most $s < j$, so every output bound in this step is warm.

For clarity, if there are J inputs, retained probability κ and required local tolerance τ , choose testing-seminorm errors at most $\kappa\tau/(4J)$. If $e_h(\psi)$ is the pathwise supremum error for input h , then

$$\Pr\{\max_h e_h(\psi) > \tau\} \leq \sum_h \frac{\mathbb{E}e_h(\psi)}{\tau} \leq \kappa/4.$$

This controls path-dependent tests without attempting to pigeonhole arbitrary site functions or orbit coefficients.

Splitting and matching. A model term supplies an ambient partner for each pivot. Truncate its top modes, retain a biased pairing, and freeze its chart twist by the warm subdivision described above. Freeze axes shorter than the warm step-drop threshold. Step drop gives pure-long symbol splittings for the local map and its partners. Refer the homogeneous identities back through their affine reindexings; translation does not affect the leading symbols and residue dilations cost warm factors. The separate slices for different pivots need not meet, since these are homogeneous identities. Separation joins their fast algebras in the product of the quotient group and the polynomially many selected partner groups.

The fixed marked polynomial in that product consists initially only of the ambient partner maps, all functions of u alone. At a matching grade $d \leq s$, its residual modulo the projected fast algebra has a warm slow-plus-rational path evaluation. Lemma 7.7 gives the same bounds for its ordinary ambient scalar coordinates. Rational sections lift them to the required left slow and right rational factors. Updating these factors changes no lower grade. There are at most s grades, so the induction uses boundedly many polynomial increases of warm budgets. The argument has no lift argument to project away: one may take the full lift-value space throughout. All full factors and marked maps are ordinary polynomials in u .

Exact reconstruction and field meshes. Refilter and reconstruct the lower-degree problem as in Proposition I.6. Its filtered rational fiber isomorphism preserves the marked projection exactly. Slow kernel factors can be frozen on warm subboxes, and rational kernel factors on warm residue classes. Common denominators, not coefficient numerators, control the latter restrictions. The source lattice and cover may therefore be chosen at warm cost independently of the unrestricted orbit coefficients.

We verify that the function-menu hypothesis persists under these operations. After partners, common marked maps and frozen shifts have been selected, their site-dependent specializations are fixed functions of the site. On the rational joint image used in the reconstruction proof, a test has the form

$$f_{\phi,e,r}(i(h)) = \phi(ehr\Gamma).$$

Here e is bounded and r ranges over finitely many bounded right-lattice residues. These functions have one warm Lipschitz bound. For a fixed residue r , their supremum distance is at most

$$\|\phi - \phi'\|_\infty + \text{Lip}(\phi')d(e, e').$$

Use the same clipped McShane extension from the same joint image for all these tests. Its infimum formula is nonexpansive in the supremum norm. Specialization at the compact marked coordinate has the common Lipschitz bound proved in the cited reconstruction argument. Consequently the original parameter meshes still control whole fields once the site maps have been fixed. Point values

have meshes obtained also by gridding that compact marked coordinate. No coefficient bound on its site polynomial is needed. Equivalently, path-variable cutoff parameters may first be meshed in field supremum norm and pigeonholed, with the prescribed error restored after the comparison.

The lower-degree induction now applies to both reconstructed payoffs at once. Lifting through the rational fiber isomorphism and restoring the fixed marked projection give one ordinary ambient map in the quotient of G^0 . A filtered logarithmic section lifts it to G^0 . Its degree remains at most s .

Deaveraging the combined score. At every top-torus averaging step, deaverage only after the lower degree return, and use the *signed* score. If its average is at least $\delta_1 > 0$, the score is bounded above by a warm constant B and the set of translations giving score at least $\delta_1/2$ has measure at least $\delta_1/(2B)$. Fubini supplies a single translation working on a warm-inverse fraction of paths. Translating in the kernel torus preserves the marked projection. Repeat at the bounded number of induction steps. This produces the asserted single g^U and retains the original two fields, with only the allotted approximation errors and target discounts. Subslices and their full-mass fractions have warm cost throughout. $\square$

7.9 Scalar return with a small retained modulus

Corollary 7.9 (Passive scalar return). *After Proposition 7.8, one can select a modular-good path and its retained warm-cost subslice for which the forecasting measure $\mathcal{K}$ of Lemma 7.3 has a signed surplus of warm size. After one further small proportional target discount, its parallel residue matching can be imposed modulo an integer M_* with warm logarithmic cost. The site payoffs still evaluate the original cutoffs on $g^U(u)$, with the exact old determining conditions.*

Proof. Once g^U is fixed, evaluation of the two warm field menus on it gives menus of bounded site functions with the same supremum accuracy. These site functions need not have any regularity. The scalar comparison, Proposition 7.4, therefore transfers their full-mass signed score to $\mathcal{K}$ on one modular-good path, including its retained subslice mask. There are no active flags in this passive step. The fixed cutoff functions themselves are recovered after their temporary mesh approximations, within the prescribed comparison error.

Write $\mathfrak{M} = \text{lcm}(1, \dots, Z)$ for the forecasting modulus. In this passive use choose Z at warm-pre cost, after the warm model and slice requirements; auxiliary cold major comparisons have their own individual-modulus requests and do not enlarge this Z . The slice strides divide $\mathfrak{M}$. Condition on the actual inactive parameters and on the stride classes. At a prime ℓ which is neither exceptional for modular decay nor a divisor of the stride, Lemma H.2 gives the uniform estimate

$$\|d_\ell - 1\|_\infty \leq 2\ell^{-2}.$$

Choose a sufficiently large fixed cutoff P_0 so that the product of these densities over $\ell > P_0$ lies in $1 \pm \eta$, with η smaller than the allotted proportional discount. Chinese remaindering separates these primes from the remaining ones; mixtures over stride classes cause no change in the uniform comparison.

Here the comparison is multiplicative separately on the two nonnegative masses. If $A_\pm$ denote the masses with all matching factors and $B_\pm$ those after removing the indicated primes, then

$$(1 - \eta)B_\pm \leq A_\pm \leq (1 + \eta)B_\pm.$$

Consequently

$$B_- - \lambda \frac{1 - \eta}{1 + \eta} B_+ \geq \frac{A_- - \lambda A_+}{1 + \eta}.$$

A fixed proportional discount thus retains a comparable positive surplus. In particular P_0 need not grow with the inverse of that possibly tiny surplus.

Let M_* retain the full tested prime powers at primes dividing the exceptional product D'_0 , at primes dividing a joint slice stride q , and at primes at most P_0 . Every such prime power has logarithm at most $\log Z$. Hence

$$\log M_* \leq (C_{P_0} + \log_2 D'_0 + \log_2 q) \log Z,$$

which is warm. This bounds the full retained depths, not merely their squarefree product. The discarded-prime comparison is pointwise in the site and inactive data, so the exact inactive equalities and all cutoff conditions are unchanged. These are the scalar data needed for the parent-box extraction. $\square$

8 Active globalization with exact marking

We now return a score through an active layer $j \leq s$. The local orbit includes the higher active determining slots, so its replacement must retain the formal identity $\pi^0 g = P_Y$ from Lemma 7.5. This identity will restore the old integer arguments exactly in Section 9.

The passive proof could restrict the scoring slice whenever it froze a slow factor or a rational factor. Here the auxiliary bounds may depend on Q_j . Such a restriction could reduce the score by a cold factor, which the eventual relative width loss cannot afford. We may select a cold-small fraction of paths on which one construction works, but on each selected path we must keep the original slice and its warm score.

We therefore retain short integer parameters in polynomial families instead of fixing their values in the score. We also retain the slow and rational factors as polynomials in two sets of labels: normalized real labels and integer labels. On an actual path both describe the same parameters. In the forecasting functional $\mathcal{K}$, the two parallel labels are separate integration variables, with independent base measures before the incidence and output weights are imposed; each inactive label and its normalization still come from one actual integer. The argument below preserves this distinction through the degree induction.

8.1 The inductive statement

Write $t = (v, z)$ for the current parameter variables. The entries of z are distinguished short variables: they are actual integer variables among the inactive parameters t_I , and their lengths have pre-cost. The list can initially be empty. The other variables v are currently long. Further inactive variables may become short during the argument. A parallel variable never becomes short, because its length L is chosen after all the pre-thresholds below. We retain the labels

$$\theta = (t_i/T_i)_i, \quad \tilde{t} = t$$

on an actual path. In the forecasting measure $\mathcal{K}$, these mean separate normalized real and residue labels for the parallel variables, and the actual integer inactive variables with their normalizations. All references to residue representatives below concern only the parallel integer labels.

The induction changes the functions tested on the group as well as the orbit. We need two different finite approximations for these functions. At one site, a point-value mesh approximates the nilmanifold function itself. Across sites, a field mesh approximates one path-indexed field uniformly over the parent box. The latter lets us choose a controlled list of fields first; binning the values of each chosen field then gives fixed site masks to which the sampler applies. Neither assertion requires

regularity in the site variable, or a mesh of orbit coefficients. We state the precise conditions using the quotient metrics and complexity conventions of Definition A.1.

Definition 8.1 (Controlled external fields). Let G^0/Γ^0 have controlled rational filtered data. A pair of fields

$$F_{\psi,u}^{\pm}(\theta, \tilde{t}) : G^0/\Gamma^0 \longrightarrow [0, 1]$$

is controlled if it has the following properties at every requested pre-accuracy.

1. Its Lipschitz constants on the quotient, and in the normalized real labels with all other arguments fixed, have pre-cost. In the long integer labels it is periodic with a common pre-bounded modulus for each path. The moduli belong to a pre-bounded list and divide the eventual modulus $\mathfrak{M}$.
2. The point-value family over sites, paths, and admissible labels has a supremum-norm mesh of pre-logarithmic size.
3. For fixed short labels and fixed mesh and residue arguments in the remaining labels, the family indexed by ψ has a mesh in supremum norm over the sites, again of pre-logarithmic size.

The bounds may charge the accuracy, the auxiliary dimensions, and the short lengths. The assertion concerns only the chart and label ranges in use. Integer labels can be represented by arbitrary integers. No mesh of polynomial orbit coefficients is required.

The initial fields of Lemma 7.5 satisfy this condition. They include the passive site multipliers and, on the positive side, the input f_0 , but exclude the current flag, slice, and target.

At a positive degree r , the degree reduction is governed by Lemma 7.6. For a strong projection $H \rightarrow H_F$ that is injective on its top layer, an element h is uniquely determined by its lower-degree image hH_r and its marked image in H_F . We will arrange this situation for the middle factor of the local orbit. The fixed marked polynomial then supplies the second coordinate, leaving a problem of degree $r - 1$.

The original projection need not have this injectivity. Our first step uses the top frequencies visible to the payoffs to construct a rational graded algebra whose top projects injectively. Our second step converts that symbol information into exact factorizations over one common marked input. The last step passes to the lower quotient and restores the original two payoffs by averaging. Throughout this construction, restrictions used to discover identities change the selected paths, not the slice in the objective score.

Theorem 8.2 (Active replacement). *Fix the prepared parent and a feasible center at a layer $j \leq s$. Assume the modified sampler, scalar comparison, and major-decomposition contracts established above. Let G^0 be a controlled filtered group of degree r , with a strong rational projection $\pi^0 : G^0 \rightarrow Y$. Suppose a pre-inverse fraction of the available paths are modular-good in the sense of Equation (6.3), and each such path has a polynomial $g_\psi(v, z)$ whose logarithmic coordinates have bounded total degree, adapted in v at each fixed z , and satisfy the formal identity*

$$\pi^0 g_\psi(v, z) = P_Y(X_\psi(v, z); z), \quad X_\psi = (\psi, \beta \circ \psi). \quad (8.1)$$

Here $P_Y(X; z)$ is fixed on those paths, weighted filtered in X , and polynomial in z , with bounded total degree. There is no bound on the coefficients of either polynomial.

Let the two external fields satisfy Definition 8.1. Suppose their signed score, with the original slice mask 1_S and full parameter mass, is at least $\delta > 0$:

$$\mathbb{E}_t 1_S(t) \left[1_{\text{no bad}}(t) F_{\psi, \psi(t)}^-(\theta, \tilde{t})(g_\psi(t)\Gamma^0) - \lambda F_{\psi, \psi(t)}^+(\theta, \tilde{t})(g_\psi(t)\Gamma^0) \right] \geq \delta, \quad (8.2)$$

where $0 < \lambda \leq 1$. Include the original slice cost, the short-label range bounds, and $\log(1/\delta)$ in the prescribed preliminary data. All path probabilities refer to the original ensemble, with discarded paths retained as zero mass; subsequent selections are subsets of the modular-good productive paths just specified. For any prescribed $0 < \eta < \delta/2$, include $\log(1/\eta)$ in that request. Then one retained modular-good path admits a stochastic polynomial

$$g^\sharp(X, \theta, \tilde{t}) \in G^0, \quad \pi^0 g^\sharp(X, \theta, \tilde{t}) = P_Y(X; z) \quad (8.3)$$

whose $\mathcal{K}$ -score, with the same slice and flag masks and the expectation of the two payoffs, is at least $\delta - \eta$.

A stochastic polynomial means a polynomial formula with random constant parameters, sampled independently of its arguments. Its total logarithmic-coordinate degree is bounded in terms of the input degree bounds only; its coefficients need not be bounded. A filtered-degree bound in the original parameters is not asserted. On the joint chart and exact-inactive locus

$$z_I(X) = z_I(t_I), \quad (8.4)$$

each expected payoff separately is independent of the representatives chosen for the parallel residue labels, with all other data fixed. This is a pointwise statement for every admissible label choice.

The retained path fraction and all auxiliary logarithmic costs may have cold pre-cost. The number of successive polynomial budget increases and all degree increases depend only on the fixed input degrees. Thus, when δ , the original slice, and η have warm bounds, the resulting score and slice still have those bounds.

Proof. We induct on r . Every loss of score below is a prescribed additive part of η . There are boundedly many such losses at each of the boundedly many degree drops. Productive path fractions, unlike the score in (8.2), may decrease by pre factors. We always budget their inverse logarithms before the next modeling or separation request. Every retained subset consists of modular-good paths, so the eventual scalar comparison receives its stated goodness hypothesis even when the retained fraction has cold preliminary cost.

If $r = 0$, the group is trivial. At fixed short labels, subdivide normalized long-label ranges and integer residue classes according to the controlled field meshes. On every piece the same bounded site function approximates the payoff both on an actual path and on the corresponding forecasting support. Axes shorter than the mesh and residue cutoffs are enumerated exactly; all other pieces have pre-relative sides. Parallel sides exceed these cutoffs, so endpoint rounding has negligible mass. Apply Proposition 7.4, including its flag version, to the fixed menu of site functions. A union bound over menu entries and testing slices gives the claimed comparison on retained paths. We do not pigeonhole an arbitrary list of mesh choices across all sites. The trivial polynomial is the output, and the representative assertion follows from periodicity of the fields.

Assume henceforth $r > 0$.

1. Constructing an injective top projection

Keeping short parameters in a polynomial family. Use compatible rational splittings of the filtrations of G^0 and Y . Form the Lie group $\widehat{G}^0$ of logarithmic polynomial functions in dummy short variables z' : in a split original layer i , impose a degree cap Ci in z' . The filtration is by pointwise membership in the original layers. Choose the fixed integer C large enough for the maps below. A bracket of layer- i and layer- h coordinates has dummy degree at most $C(i + h)$, so the caps are closed under BCH. Thus $\widehat{G}^0$ is a rational nilpotent Lie group of polynomial dimension. Compatible

sections give a strong projection to the analogous group $\hat{Y}$. Evaluation at any specified z' is strong before refiltration.

Choose one controlled deep lattice $\hat{\Gamma}^0$ whose evaluation at every integral z' lies in Γ^0 . Indeed a deep rational integral grid of monomial coefficients has this property, and the bounded-step rational group operations preserve it after one further fixed divisibility. This is a simultaneous lattice choice, not a separate choice at each short label.

The families $z' \mapsto g_\psi(v, z')$ and $z' \mapsto P_Y(X; z')$ fit the caps. The common-site correction used below also fits: it uses only the given maps, whose site components have degrees at most 1 and j , a pointwise filtered logarithmic section, and bounded-step BCH. Consequently C can be fixed from the input degrees before partner selection.

There are now two site inputs to distinguish. At an actual short label z , the local family has projected value

$$z' \mapsto P_Y(X_\psi(v, z'); z'),$$

whereas reconstruction at the actual site $X_\psi(v, z)$ will require

$$z' \mapsto P_Y(X_\psi(v, z); z').$$

These families agree at $z' = z$. Their pure-long leading symbols also agree, because the leading site input is independent of the short labels. Their full polynomial maps need not agree. We may therefore use the uncorrected family to find the top algebra now; after matching the fixed marked polynomial, we will correct the full family at the actual site.

Averaging over the top kernel. Apply Lemma A.4 in the pointwise top torus G_r^0 to both external functions. Use a fixed convolution and Fourier expansion with uniform error smaller than the assigned tolerance. Taking a retained vertical component preserves controlled binning, with pre amplification of approximation errors. At each actual short label z , call a truncated mode significant when the absolute value of its conditional tested average over the remaining variables, including that term's masks, exceeds the tolerance divided by a sufficiently large constant times the number of modes. The sum of discarded averages, integrated over all short labels with their original masses, is then negligible.

Let L_0 be the connected joint annihilator, in $\hat{G}_r^0 \cap \ker \hat{\pi}^0$, of the significant pointwise frequency rows composed with evaluation at their respective z . These rows are integral on the family lattice. Pigeonhole a basis of them, together with its frequencies, short labels, and payoff choices. There are polynomially many basis rows and a pre number of possible basis lists. In particular, we do not enumerate all subsets of the truncated modes. The rational central subgroup L_0 is now fixed on a pre-inverse fraction of paths.

Average both payoffs by left translation over $L_0/(L_0 \cap \hat{\Gamma}^0)$, evaluating a family translate at the actual short label. A character is unchanged if its row vanishes on L_0 and is killed otherwise. Hence every significant mode survives, and the full score on the original slice changes by only the discarded-average and Fourier-tail errors. Both averaged functions remain in $[0, 1]$.

Partners and the joint fast algebra. For each pivot row we obtain a step drop paired with a fixed ambient partner. We give the selection details because these auxiliary restrictions must not be imposed on the objective score.

If the pivot comes from the positive flagged term, first truncate the inclusion-exclusion expansion of the no-bad flag as in the proof of Proposition 7.4, using Lemma 6.1. The error may charge the short label mass, mode count, and current productive probability. Split each retained term into its pre residue subpieces, so its coefficient is constant on each piece for the chosen path.

At the pivot's short label, subdivide the remaining normalized labels and integer classes for controlled binning. First approximate the site field by its path-field mesh. Then bin its point values using the point-value mesh. The triangle inequality supplies a biased pairing of a fixed vertical function with an arbitrary bounded masked site input from a fixed pre menu. Pieces too small to support the bias can be discarded for this selection. All slices used for detection therefore have pre-relative sides, including the specialization of the short variables. Their orbit is still filtered of degree r in the remaining parameters.

Use the sampler's pathwise testing-seminorm approximation at degree r on all these masked inputs. Markov's inequality and a union bound pay for the menu and the current productive probability. One model term gives correlation with a fixed ambient degree- r partner $Q_i(u)$, its niltest function, and a site twist. Truncate the partner's top Fourier expansion, freeze the twist on a finer slice, and select a biased pair of vertical functions. Freezing is permitted because normalized spatial and lift coordinates have controlled path coefficients and the lift residues can be fixed. Freeze further axes that are too short for Lemma B.1. That Lemma gives a splitting of the pair's pure-long symbol into a rational fast algebra annihilated by the joint top frequency. Translation, restriction, and dilation refer this homogeneous identity back to the original parameters with pre-cost.

Perform these selections simultaneously for the polynomially many pivots, exactly as in the partner-selection part of Proposition I.6. Pigeonhole partners, top rows, and the controlled-height rational fast algebras. The input test complexities in the modeling request were fixed before partners were selected. Symbol slices for distinct pivots need not intersect: we use only their common remaining pure-long variables. At each original layer the total degree in the previous long variables is bounded by that layer, so specializing newly short variables or changing their origins cannot change these pure-long top terms. Jointly raise any interpolation and separation thresholds after the selections. No new slice is imposed on (8.2). If there are no pivots, use no partners and the full fast algebra.

The pure-long leading site input $X_{\text{top}} = (\psi_{[1]}, \beta_{[j]})$ is independent of all short labels. Therefore evaluation and coordinate restriction of the joint family symbol give exactly the pair symbols just obtained. Pull each pair algebra back under those strong maps, lift its outer factors, and join the splittings using Lemma A.6. After quotienting by L_0 , set

$$\begin{aligned} G &= (\widehat{G}^0/L_0) \times \prod_i G_{Q_i}, & F &= \widehat{Y} \times \prod_i G_{Q_i}, \\ \pi : G &\longrightarrow F, & P_F(X) &= ((z' \mapsto P_Y(X; z')), (Q_i(u))_i). \end{aligned} \tag{8.5}$$

Let $\mathfrak{v} \subset \text{gr } G$ be the joined fast algebra and $\mathfrak{v}_F = \text{gr } \pi(\mathfrak{v})$. The constraints from the partners now give the required injectivity:

$$\mathfrak{v}_r \longrightarrow (\mathfrak{v}_F)_r \quad \text{is injective.}$$

Indeed a top vector in its kernel has zero partner coordinates; its family component in the marking kernel is annihilated by every pivot and thus lies in L_0 before taking the quotient. The marked projection of the joint symbol is the weighted symbol of P_F evaluated on X_{top} . Thus the top-layer ambiguity has been removed. The remaining work before degree reduction is to factor the full maps with common outer projections. The marked identity established so far concerns only symbols: the two full family projections displayed above still have to be reconciled.

2. Exact factors over one common site

Matching the fixed marked symbol. Apply the marked-symbol argument of Proposition I.6 in (8.5). It gives slow and rational outer symbols $\mathcal{E}, \mathcal{R}$ for P_F on a controlled-height tag space

$K \supset W$ in the β variable, with residual in $\exp \mathfrak{v}_F$. For completeness, at each grade raise the long threshold so that the leading β -evaluation lies in the currently imposed space. Its projection off that space has bounded-denominator rational coefficients, because β is integral, and is fully slow, because $C - c \in W$. Separation makes the pure-long coefficients vanish. Compare the projected local splitting with the already matched grades, reset their outer factors through $\mathfrak{v}_F$, and pass to the quotient at the next grade. The remaining fixed scalar coordinates evaluate to slow plus rational polynomials. Proposition I.1, with the cold detection and chart identities verified above, supplies their ambient decompositions. Intersect the resulting spaces and lift the corrections by rational sections. The slow coordinates are bounded in $u/H, \beta - A(u)$, where $A = C_{[j]}$; the rational ones have controlled common denominator in raw arguments. There are only r grades. Polynomial families in $\hat{Y}$ are coordinates in a fixed controlled rational group, so this is the same finite grade induction, not an induction over their individual coefficients.

Raise the long threshold after these outputs, and let $z_{\text{all}} = (z, z_{\text{new}})$ be all short coordinates, with remaining variables v_L . The whole off- K polynomial has the form

$$\text{proj}_{K^\perp} \beta(\psi(v_L, z_{\text{all}})) = o(z_{\text{all}}). \quad (8.6)$$

Its coefficients and common denominator have pre bounds. Indeed each coefficient involving a long variable is rational with pre denominator and is smaller than its reciprocal denominator by the fully slow path bound, hence is zero. Pigeonhole the remaining coefficient list. Define

$$\begin{aligned} p(X, z_{\text{all}}) &= (u, \text{proj}_K \beta + o(z_{\text{all}})), \\ P_F^*(X, z_{\text{all}}) &= P_F(p(X, z_{\text{all}})). \end{aligned} \quad (8.7)$$

On path data $p(X_\psi, z_{\text{all}}) = X_\psi$ polynomially. The same equality holds on the forecasting locus (8.4). To see this, choose pre-height integer annihilating rows for K . They vanish on W and pair integrally with the projected lattice basis vectors. Their values on active sheet vectors are zero: those vectors have size $O(1/K_i)$, with $k_i \gg (AL)^j$, and the threshold exceeds the row height. The row values therefore depend only on the inactive sheet coordinates, which match exactly. The short labels are actual entries of the same t_I .

Refilter in the inverse images H, H_F of $\mathfrak{v}, \mathfrak{v}_F$, using Lemma 7.6. Lift the matched symbols to full factors

$$P_F^* = E_F P_H R_F, \quad P_H \in H_F. \quad (8.8)$$

In the slow factor use u/H and $\text{proj}_K \beta - (C - c)(u)$; in the rational factor use $u, \text{proj}_K \beta$. Their middle symbol belongs to $\mathfrak{v}_F$, so positive-weight coefficients are adapted to the new filtration. In the vector quotient $F/H_{F,1}$, only the constant in X remains. It is polynomial in z_{all} . Split its coefficients, in rational coordinates, into bounded and integral parts and insert lifts on the inside of the outer factors. This is a coefficientwise split, not taking fractional parts of a polynomial at varying labels. It preserves its degree. We obtain E_F fully bounded with pre coefficients in the indicated centered and normalized arguments and short labels, R_F with pre common denominator in raw integer arguments and short labels, and P_H weighted filtered in X . These factors are fixed across the retained paths.

Correcting the full family marking. We must correct the full family marking before making local factors. Put

$$X_a = X_\psi(v_L, z, z_{\text{new}}), \quad X_{z'} = X_\psi(v_L, z', z_{\text{new}}).$$

Let ρ^0 be a filtered logarithmic section of the pointwise projection π^0 . Before quotienting the family, replace its local map by

$$z' \longmapsto \rho^0 \left(P_Y(X_a; z') P_Y(X_{z'}; z')^{-1} \right) g_\psi(v_L, z', z_{\text{new}}), \quad (8.9)$$

and use the partners $Q_i(\psi(v_L, z, z_{\text{new}}))$. Call the resulting map into G $\mathbf{g}_\psi$. Its family value at $z' = z$ is unchanged. The inserted ratio has identity pure-long symbol, since X_a and $X_{z'}$ have the same pure-long leading input. Thus the former joint symbol splitting is retained as a whole family, independently of the values of z_{all} . But its full projection is now $P_F^*(X_a, z_{\text{all}})$, at one common actual input. Explicitly, the projected family product is

$$[P_Y(X_a; z')P_Y(X_{z'}; z')^{-1}]P_Y(X_{z'}; z') = P_Y(X_a; z').$$

No commutativity is needed. This correction is why we do not need evaluation to commute with refiltration.

Lifting the prescribed outer factors. We construct

$$\mathbf{g}_\psi = E_\psi h_\psi R_\psi, \quad \pi E_\psi = E_F, \quad \pi R_\psi = R_F, \quad \pi h_\psi = P_H, \quad (8.10)$$

where projected factors use X_a, z_{all} and h_ψ is adapted to H in v_L . More importantly, E_ψ, R_ψ have ambient-with-label versions with these exact formal projections. They are products of fixed logarithmic-section lifts of E_F, R_F and kernel discrepancies. The slow discrepancies have bounded polynomial coefficients in normalized long labels θ_L and short labels; the rational discrepancies have controlled common denominator in integer labels.

Here are the exact corrections. Write the local symbol splitting as ePd , with $P \in \exp \mathfrak{v}$. Let e_F, d_F be the symbols of the prescribed projected outer factors. By compatible splitting, on sufficiently long axes,

$$(\pi e)^{-1}e_F \in \exp \mathfrak{v}_F, \quad d_F(\pi d)^{-1} \in \exp \mathfrak{v}_F.$$

The first is slow and the second rational. Lift them through a controlled graded section inside $\mathfrak{v}$ and reset the local splitting. Its outer symbols now project exactly to e_F, d_F . Fix a filtered logarithmic section ρ of π , and denote the reset outer symbols by e', d' . The two ratios

$$e'(\text{Sym } \rho(E_F))^{-1}, \quad (\text{Sym } \rho(R_F))^{-1}d'$$

are kernel symbols. Strongness gives actual kernel lifts: if a representative in the i th layer projects into the $(i+1)$ st, subtract a section lift in that deeper layer. Coefficientwise lifts yield kernel polynomial maps on the left of $\rho(E_F)$ and on the right of $\rho(R_F)$, respectively, with the claimed bounds. Their residual has all positive-long-degree coefficients adapted to the layers of H .

For its constant in v_L , use the vector quotient G/H_1 . Its image belongs coefficientwise to the image of $\ker \pi$, because its projection is supplied by $P_H \in H_F$ and $\pi(H) = H_F$. Split that short-label polynomial coefficientwise into bounded and integral parts, lift within the actual kernel, and insert the factors on the inside of the two outer factors. This puts the remaining constant in H_1 , preserves all exact projections, and proves (8.10).

We choose the ambient versions after this constant adjustment in the definite order

$$\begin{aligned} E_\psi &= K_{E,\psi}(\theta_L, z_{\text{all}})\rho(E_F(X, z_{\text{all}})), \\ R_\psi &= \rho(R_F(X, z_{\text{all}}))K_{R,\psi}(\tilde{t}_L, z_{\text{all}}). \end{aligned} \quad (8.11)$$

The kernel polynomials here have no additional site argument. This order will let us compare path-dependent fields while the fixed site factors cancel: at a fixed site and labels,

$$R_\psi^{-1}R_{\psi'} = K_{R,\psi}^{-1}K_{R,\psi'}.$$

Thus unrestricted coefficients of R_F will not enter a mesh of the rational discrepancies. To preserve this order when absorbing an inside constant, first conjugate it past the common factor *on the path*, and express the conjugation using the pullback $\rho(E_F(X_\psi(t), z_{\text{all}}))$ or $\rho(R_F(X_\psi(t), z_{\text{all}}))$ as a polynomial in the labels. The first pullback is fully slow with controlled normalized coefficients. The second has controlled common denominator in integer labels, because X_ψ has integral polynomial coefficients. Extend these kernel label formulas off the path, while retaining the displayed common site factors. They still agree with the corrected local factors on paths and have the exact formal projections everywhere. This procedure bounds denominators independently of the unrestricted integral coefficients of X_ψ .

The threshold order here is important. Any final increases needed for separation compare the pure-long family symbols with the already constructed matched symbols. Those symbols have no cost from lengths of axes made short by this very increase. Likewise off-space vanishing uses full coefficient slowness of y . Subsequent polynomial estimates may charge the resulting short lengths. Thus there is no threshold iteration once per axis.

At this point the full factorization (8.10) has the exact projections required for reconstruction, and its middle map is adapted to the refiltered group H . The first stage proved injectivity at the top. We can now remove that top layer while retaining the two payoffs as functions on the lower quotient.

3. Degree reduction and restoration of the score

Reconstructing the payoff on the lower quotient. Read the L_0 -averaged original fields on G , evaluating a lift of the family quotient at its actual diagonal short label and ignoring partners. Denote them $\overline{F}_{\psi,u}^\pm$, with label arguments suppressed. They are well defined because L_0 is central and the family lattice evaluates into the pointwise lattice at integral labels. Their Lipschitz bounds have pre-cost, using bounded-label evaluation and controlled quotient sections. This averaging is nonexpansive in supremum norm at fixed labels. By (8.9), their local evaluations on $\mathbf{g}_\psi$ are exactly the retained averaged payoffs.

Consider, at each site and label, the function on H

$$h \mapsto \overline{F}_{\psi,u}^\pm(E_\psi h R_\psi),$$

using the ambient-with-label factors. It descends on a controlled deep source lattice. Top injectivity and Lemma 7.6 give a rational filtered isomorphism

$$H \simeq (H/H_r) \times_{H_F/H_{F,r}} H_F. \quad (8.12)$$

Choose sufficiently deep lattices in the two target groups so their inverse image lies in the source lattice. In logarithmic coordinates this follows by applying the rational linear inverse of the joint embedding to a sufficiently divisible product grid. The reconstruction argument of Proposition I.6 therefore gives a common controlled Lipschitz constant on the joint image. Extend each function by the same clipped infimum extension, and specialize the second coordinate at $P_H(X, z_{\text{all}})$. The extension agrees exactly on compatible pairs.

For clarity, a common source grid permits all right translates R_ψ : their logarithmic coordinates have one controlled common denominator after pigeonholing denominator values, and the adjoint formulas have bounded denominators independently of numerators. Left translation has controlled Lipschitz cost because E_ψ is fully bounded. These are the hypotheses required by the lattice reconstruction, not bounds on the middle orbit.

Preserving the two mesh conditions. The resulting fields on H/H_r satisfy Definition 8.1. For point-value meshes, the parameters are the old function value, bounded E_ψ , the right lattice coset of R_ψ , bounded-label evaluation, and the compact specialization value of P_H . The clipped infimum extension is nonexpansive in its input function and Lipschitz in specialization. Thus these are finite-dimensional parameter meshes; no net for all Lipschitz functions is used.

For field meshes in the path index, the marked functions P_H, E_F, R_F are fixed as functions of the site and short labels. Slow kernel factors have controlled coefficient meshes. Rational kernel factors have a pre number of right-coset fields despite unrestricted numerators. Indeed prescribe their polynomial coefficients modulo a sufficiently divisible pre grid. In the BCH expression for the difference, every noncancelled term contains a coefficient difference, and all other denominators are controlled; at integer labels the difference belongs to the prescribed deep lattice. This fixes the right coset uniformly over sites and integer labels. The same calculation supplies a pre period in long integer labels. Taking a common selected period preserves divisibility by the eventual $\mathfrak{M}$. The cancellation following (8.11) shows that this comparison involves only the kernel label factors. The common site factor $\rho(R_F)$ introduces no conjugation or coefficient bound.

Continuous long-label variation enters only the old fields and the controlled slow factors. It does not enter the possibly unbounded fixed site polynomial P_H , which uses the actual site and exact short labels. Additional short axes contribute only pre many label choices to the meshes. These observations verify all parts of the refreshed field contract without meshing any unrestricted orbit coefficient.

The lower-degree return and stochastic restoration. We have a lower-degree problem on $H/H_r \rightarrow H_F/H_{F,r}$ with actual local orbit $h_\psi \bmod H_r$, exact marking $P_H(X, z_{\text{all}}) \bmod H_{F,r}$, and the refreshed fields. Its score uses exactly the original slice and flag. Apply the induction hypothesis with the remaining error tolerance. Its stochastic output lifts jointly with the exact P_H to a polynomial $h^\sharp \in H$ by (8.12). Multiply $E_\psi h^\sharp R_\psi$ on the left by

$$\rho\left(P_F(X)P_F^*(X, z_{\text{all}})^{-1}\right). \quad (8.13)$$

The projection is now exactly $P_F(X)$ formally. The multiplier is the identity on (8.4), by (8.7). Project to the family quotient, lift by a logarithmic section, multiply by an independent uniform constant in $L_0/(L_0 \cap \widehat{\Gamma}^0)$, using bounded Borel representatives, and evaluate at $z' = z$. This defines $g^\sharp$ and proves its formal marking (8.3).

Fix any admissible site and labels on the exact-inactive locus. The reconstructed pair is compatible, so the extended lower-degree function agrees exactly with the translated averaged original function. The correction (8.13) is the identity there. Averaging the newest family translate realizes exactly the earlier L_0 -average after diagonal evaluation. Haar invariance removes any dependence on the selected family lift, and evaluation of its lattice lies in the original pointwise lattice. Consequently, for each label choice, the expectation of each restored payoff equals the corresponding lower-step payoff. Taking the remaining expectations gives representative independence by induction, separately for the two signs. It also preserves the $\mathcal{K}$ -score. This is an identity of expectations; an individual restored polynomial need not be periodic in the parallel integer labels.

All random parameters just introduced are constants independent of the arguments. Evaluation of a fixed random polynomial family at the variable short label remains polynomial. Group multiplication, logarithmic sections, the common-site correction, coefficient lifts, and family evaluation are bounded-degree operations. They are nested only a bounded number of times in this degree induction. Taking simultaneous partners, solving all component equations, or intersecting many subspaces does not iterate a degree increase once per component. Hence total degree depends only

on the input degree bounds, while dimensions and pre logarithmic costs have boundedly many successive polynomial increases.

Finally, forecast the flag truncation, modeling, symbol, and period bounds at each of those pre stages before choosing Z . Choose Z to include the required individual periods, and only then choose perturbation, range, rank, and side cutoffs. The scalar comparison uses individual moduli and witness products, not a Fourier expansion at the full possibly enormous $\mathfrak{M}$. All losses to the objective have been the prescribed additive parts of η ; short-label selection, partner discovery, and symbol restrictions reduced only productive path fractions. This proves the theorem. $\square$

Remark 8.3 (What the macro argument may use). In the initial instance of Theorem 8.2, the fields have no direct parameter-label dependence: the labels occur only in the replacement polynomial. All auxiliary label and period subdivisions used inside its proof have disappeared from the score's slice mask. The macro argument may therefore change parallel residue representatives pointwise, using the expected-payoff identity, before fixing the stochastic constants by averaging. It retains a warm full-mass score and the original warm slice, but must not assume bounded coefficients or periodicity of an individual replacement orbit.

9 Parameter extraction and exact restoration

We now turn a scalar forecasting surplus into a new pair of polynomial boxes on the parent, retaining every old determining equation. The forecasting measure has exact equations for the inactive integer parameters, real output constraints, and modular matching conditions. We solve these three kinds of constraints in turn. A coefficient-one column solves each nonomitted inactive equation over the integers; a real change of variables solves the spatial and lift constraints; and residue matching is either reduced to a small modulus or solved by affine sections over the full finite ring. Averaging then fixes the remaining parameters and leaves an actual polynomial formula on the parent.

Fix a layer j , a feasible center c , and a path retained by the appropriate micro replacement. We use the notation of Lemma 7.3. In particular,

$$X = (u, \beta), \quad y = C(u) - c - \beta, \quad b(X) = (u, w, z_{\text{act}}), \quad m = m_j.$$

Here $b(X)$ denotes the integer residue output, not the current raw determining block, which is $b_j = \beta + b_{j,0}$. Write I for the inactive sheet axes, including omissions, and $t_{\parallel}$ for all other columns. In the forecasting functional, the normalized real parallel labels and parallel residues are separate integration variables, with independent base measures before the incidence and output weights are imposed. Each inactive parameter has one actual integer value. The original slice mask has warm cost. The two unsigned payoffs are the original cutoff evaluations at this layer, including their site multipliers; thus they take values in $[0, 1]$. In particular the positive payoff includes the original pulled-back input, and the negative one does not. They have no direct dependence on auxiliary parameter labels once the micro replacement has returned to its original codomain.

At a passive layer $j > s$, Corollary 7.9 leaves a fixed polynomial orbit and a matching modulus M_* of warm log cost. At an active layer $j \leq s$, Theorem 8.2 leaves a stochastic polynomial $g^\#$ with the original determining marking exact formally. Its two expected payoff evaluations are pointwise independent of the representatives of parallel residues on the exact-inactive chart locus. Here the retained path is modular-good, as required by the comparison measure. We request an active replacement error at most one quarter of its input score. This error and the original slice have warm logarithmic bounds; short-label ranges and other auxiliary bounds may remain cold preliminary data. In either case the incoming integral has full-mass surplus at least δ , where δ^{-1} is

warm-bounded. “Full mass” means that a slice is inserted as an indicator rather than renormalized away.

Proposition 9.1 (One-layer extraction). *Under these hypotheses, after a prescribed small proportional discount of the target, one obtains a parent slice and a pair of cutoff boxes which add the current raw block b_j to the determining system. The inherited determining polynomials and integer arguments are restored exactly. The new slice and current-block congruence have warm log cost, and the current radius r'_j satisfies*

$$\log \frac{w_j}{r'_j} \leq \text{poly}(\text{warm log data}).$$

The logarithm of the reciprocal returned full-mass score has cold cost at layer j . At passive layers the absolute weights are unchanged. At active layers a fixed multiplicative increase of those weights suffices; determining weights remain exactly their original weights. All degree bounds are independent of the budget dimensions.

We prove the proposition in several steps. Constants described below as warm may be exponential in a fixed polynomial of the warm log data; their logarithms, rather than their values, are the quantities charged to the budget. The returned polynomial may have arbitrarily large coefficients. We therefore localize only its domain, in the joint normalized coordinates $(u/H, y/v)$, and keep the formula unchanged. The new radius will be $r'_j = \ell v$, with both $\log(1/\ell)$ and $\log(w_j/v)$ warm. The returned score may contain the chart mass m_j , whose inverse can depend on the current width. Thus the score may incur a current-width cost even though the relative width loss does not.

9.1 Removing flags and solving inactive equations

At an active layer insert the selected-plane weight W_* of Lemma 6.1 into the forecasting functional and remove the no-bad flag from the positive payoff. This only requires a proportional target discount. Indeed, if $P, N \geq 0$ denote the two payoffs and

$$W_* \leq 1 + \eta \quad \text{everywhere,} \quad W_* \geq 1 - \eta \quad \text{on the no-bad set,}$$

then, for $\lambda' = (1 - \eta)\lambda/(1 + \eta)$,

$$\int W_*(P - \lambda'N) \geq (1 - \eta) \int (1_{\text{no bad}}P - \lambda N). \quad (9.1)$$

Choose the constant in the flag construction small enough for the available target slack. This retains a fixed fraction of δ ; it does not require the flag constant to be smaller than δ . At a passive layer no such insertion is needed. Below we rename the discounted target λ and the retained surplus δ .

For a nonomitted inactive axis, let $r_i = t_{I,0i}$ be its extra linear column, with side length T_{0i} . Fixing its dedicated product columns, the exact inactive equation has the unique solution

$$r_i = z_i(X) - z_i(0) - \sum_{\text{blocks}} \prod_{\text{columns}} t_{I,\text{column}}. \quad (9.2)$$

We retain the actual range, interval, and stride conditions on this integer, and on omitted axes retain $z_i(X) = 0$. The coefficient of r_i is exactly one. Consequently its averaging factor $1/T_{0i}$, together with the inactive normalization in the functional, leaves

$$C_I = \prod_{i \text{ omitted}} k_i \prod_{\substack{i \in I \\ i \text{ nonomitted}}} \frac{k_i}{T_{0i}}. \quad (9.3)$$

This factor has warm log cost: omitted k_i are warm-bounded and T_{0i} is comparable to k_i within warm factors. There is no divisibility condition or rounding in Equation (9.2).

Each $z_i(X)$ is an integral linear form in β . On the chart, $z_i(X)/k_i$ is a linear function of y/v with warm norm. Thus r_i/T_{0i} has a warm Lipschitz bound in normalized lift coordinates after the other inactive columns are fixed, regardless of the size of the coefficients of C .

9.2 Real pivots and boundary losses

The real part of the functional is the density of the principal spatial and lift outputs, with their actual constant shifts. Pivot on one spatial kernel block and on one variable in a dedicated product block for each continuous or active lift direction. The remaining real variables are held fixed. The pivot values are affine functions of

$$(u/H, y_W/v, z_{\text{act}}/k). \quad (9.4)$$

The absolute inverse Jacobian is a factor independent of the site. The spatial kernel matrix has a warm-controlled inverse. Each lift pivot coefficient is its nonzero principal amplitude times a product of boundedly many of the other normalized variables in that block.

Lemma 9.2 (Exceptional domains). *One may discard small-determinant choices from both payoffs, and narrow the positive domain by removing small boundary strips for all real pivot interval tests. One may also remove positive-domain endpoint strips for the inactive integers in Equation (9.2) whenever T_{0i} exceeds a warm cutoff. The total loss is at most any prescribed small fraction of δ , using warm logarithmic cutoffs. On the retained domain the inverse determinant is warm-bounded and all these positive interval tests have inverse-warm margins. The remaining short inactive interval tests, and omitted-axis tests, can be handled by exact integer equality, with their k_i warm-bounded.*

Proof. First consider the real variables before solving the pivots. Exclude strips of sufficiently small warm width around zero in the nonpivot coordinates appearing in the chosen product coefficients. Off these strips each individual coefficient has an inverse-warm lower bound. Multiplying over polynomially many directions still gives a determinant whose inverse logarithm has warm polynomial cost. Equivalently, choose the determinant threshold below the resulting product lower bound. Its exceptional set is then contained in the union of these coordinate strips.

Small parameter volume alone does not bound the density contributed at a fixed output. Here unused independent spatial kernel blocks and unused dedicated real lift blocks provide a joint output density with a uniformly warm cap. For lift outputs this follows from the order-zero product-block estimate in Lemma G.4; enough independent product sums give bounded density in normalized units. For the spatial output the independent nonsingular blocks give the corresponding cap by the forecasting argument preceding Proposition H.1. Their parameters are disjoint from the chosen pivot groups. Thus convolving with these spare blocks bounds the output density coming from any exceptional set in a pivot group by a warm constant times that set's Lebesgue mass. The same argument applies when the pivot variable lies in a small strip around an endpoint of its allowed interval. One can discard the groups separately and sum the bounds. For an upper bound all other slice masks may be dropped, and all unsigned payoff factors may be replaced by one.

Here is why arbitrary site-dependent payoff factors cause no problem. For each fixed actual t_I , the modular output density of parallel residues is warm-bounded by modular goodness. This remains true for a retained subset of primes. In the active case W_* is uniformly bounded as well. Moreover,

$$\mathbb{E}_U 1_{\text{chart}}(u) 1_{z_I(X)=z_I(t_I)} \leq B \frac{m}{\prod_{i \in I} k_i}, \quad (9.5)$$

where B is warm. To see the Haar bound, use the lattice chart: fixing the inactive sheet values leaves continuous volume of order $v^{\dim W}$ and at most a dimension-controlled multiple of $\prod_{i \notin I} k_i$ active sheets, divided by the covolume. This is precisely the right-hand scale in Equation (9.5). Single-site high rank transfers that bound to the parent box. Buffered majorants for the exact inactive values may require late Fourier precision depending on $\log L$ and Q_j ; such a request is allowed. Their errors are chosen small relative to the displayed scale.

Multiplying Equation (9.5) by the functional's prefactor $m^{-1} \prod_{i \in I} k_i$ cancels the possible cold concentration. The preceding real-density and modular caps therefore show that every discarded real strip costs at most a warm constant times its parameter mass, even with arbitrary bounded site payoffs inserted. Choose these masses smaller than δ divided by a sufficiently large warm bound and by the number of strips.

The argument for an inactive endpoint strip uses the full real-density cap and Equation (9.5), then averages over the actual uniform integer r_i . A strip of normalized width ε contains a proportion $O(\varepsilon + T_{0i}^{-1})$ of its range. Use this estimate only when T_{0i} exceeds the required warm cutoff. Otherwise retain the interval test exactly; then k_i is warm-bounded because it is comparable to T_{0i} . Omissions already have this property.

Finally, the density used here is well-defined pointwise, including at true sites on exceptional output fibers. A pivot-group density, initially defined almost everywhere, is convolved with the independent bounded-Lipschitz spare-block density. This gives a continuous version of the full density, and the solved-pivot integral gives that same version. The same observation applies with rectangular restrictions. No arbitrary choice of an almost-everywhere density version is being evaluated at a site. Arbitrary bounded label-dependent payoffs are integrated against this explicit nonnegative pivot-fiber measure and dominated by its unweighted mass; continuity of their weighted output laws is neither asserted nor needed. $\square$

We henceforth impose these restrictions. Denote the narrowed positive domain by D_- and the full remaining negative domain by D_+ ; both also include the other range, exact-axis, and congruence conditions. The surviving surplus is still a fixed fraction of δ .

9.3 Eliminating the full active modulus

At a passive step we can fix the parallel residues modulo M_* , including the slice strides. The matching weight $M_*^{m_b}$ has warm log cost. The remaining tests, after Equation (9.2), are congruences in (u, β) at warm moduli.

At an active step the full modulus $M = \mathfrak{M}$ may not have affordable log cost. The following exact finite-ring identity is the reason it need not be imposed on the parent.

Lemma 9.3 (Affine-plane sections). *Fix all residue parameters except the pure-free vector $x \in (\mathbb{Z}/M\mathbb{Z})^{v_f}$, and fix all inactive parameters except their extra linear columns r . Put $m_b = \dim b$ and $l = m_b + \dim r$. The weighted matching measure is the image, under $x = a + V(B, r)$, of a finite nonnegative measure on selected origins (a, V) satisfying $b(a, 0) = 0 \pmod{M}$. Explicitly, for every test function F on the free residues,*

$$\begin{aligned} \mathbb{E}_x M^{m_b} 1_{b(x, r) = B \bmod M} W_*(x, r) F(x) \\ = M^{-v_f + m_b - v_f l + J} \sum_{(a, V) \text{ selected origins}} F(a + V(B, r)). \end{aligned} \quad (9.6)$$

Each origin has mass

$$M^{-v_f + m_b - v_f l + J}.$$

The origin set and its weights are independent of both B and r , and their total mass is at most $1 + O(c_*)$.

Proof. The selected matrices satisfy the formal polynomial identity

$$b(x + V(s, h), r + h) = b(x, r) + s \quad \text{over } \mathbb{Z}/M\mathbb{Z}. \quad (9.7)$$

By the prime-power lifting and CRT construction in Lemma 6.1, their number at (x, r) is $M^{v_f l - J} W_*(x, r)$. Their selected reductions remain selected when the basepoint moves along the same plane: the defining upper-count and nonsingularity conditions test every basepoint of its reduction.

Suppose $b(x, r) = B$. Set $a = x - V(B, r)$. Substituting $(s, h) = (-B, -r)$ in Equation (9.7) gives $b(a, 0) = 0$. Translating the same formal identity to this origin gives

$$b(a + V(s, h), h) = s.$$

Conversely the latter identity evaluated at (B, r) gives $x = a + V(B, r)$ and $b(x, r) = B$. These transformations are inverse and preserve selection. The calculation holds over the whole finite ring, not just over its residue fields; all identities are coefficientwise polynomial identities.

Expanding W_* into its normalized selected count gives Equation (9.6) and the asserted common mass. The origin description has no dependence on B or r . Taking $F = 1$ and averaging the left side over uniform B gives $\mathbb{E}_x W_*(x, r) \leq 1 + O(c_*)$. Since the section mass is independent of B , this bounds its total mass. Independence of r follows from the same origin description and is important when r is subsequently the site-dependent integer in Equation (9.2). $\square$

Choose ordinary integer representatives of each selected a, V and make the unrestricted integer substitution

$$x = a + V(b(X), r(X)). \quad (9.8)$$

Keep actual inactive values in this expression, not reduced representatives of them. The original stride masks give the same answers because their moduli divide M . The two *expected* micro payoffs also give the same answers by the pointwise representative-independence conclusion of Theorem 8.2, since the exact-inactive equations hold on this support.

This equality is used before freezing stochastic constants. An individual stochastic polynomial need not be periodic in its integer labels, and no such periodicity is used later. The real free labels remain separate normalized real integration variables. Substitution (9.8) concerns only the residue labels and does not create a real interval restriction with large coefficients.

The number of origins may be enormous, but it is their total measure, not their number, that enters the argument. The unrestricted coefficient convention allows the selected representatives in the polynomial orbit. Their size has no effect on degree. Thus this step does not charge $\log M$ to any budget.

9.4 Quantitative freezing

After the exact inactive substitution and the real pivots, apply the passive residue choice or Lemma 9.3. The remaining variables to be fixed are the inactive product parameters, the nonpivot real variables, the residue constants and sections, and, in the active case, the independent stochastic polynomial constants. All their selection measures are nonnegative and independent of the site. The inverse determinant off the removed set, C_I , and the passive residue factor if present have warm upper bounds. The active section measure has uniformly bounded total mass. Therefore the remaining measure, say μ , has total mass at most a warm number B .

The retained forecasting surplus has the form

$$m^{-1} \int \mathbb{E}_U[1_{D_-(\omega)} P_\omega(X) - \lambda 1_{D_+(\omega)} N_\omega(X)] d\mu(\omega) \geq c\delta, \quad (9.9)$$

where $c > 0$ is fixed and $0 \leq P_\omega, N_\omega \leq 1$. Fubini consequently supplies a fixed choice of ω for which

$$\mathbb{E}_U[1_{D_-} P(X) - \lambda 1_{D_+} N(X)] \geq c\delta m/B. \quad (9.10)$$

This is a quantitative full-site surplus. Neither an unknown support mass nor the cardinality of the section set is divided out. Real domains and compact torus constants can be assigned bounded Borel representatives, so the required integrals are ordinary measurable integrals.

The fixed payoffs now use an actual polynomial orbit in (u, β) . Its normalized real labels are the affine pivot formulas in Equation (9.4) and the exact inactive variables in Equation (9.2); its integer labels are those inactive variables, the affine expression (9.8), and constants. Since $y = C(u) - c - \beta$, $\deg C \leq j$, and β has weight j , these substitutions have bounded weighted degree depending only on j . The exact determining projection is preserved formally.

The remaining residue masks are determined by (u, β) modulo a common warm-cost integer. At active layers the full matching condition has been solved identically, so only the *original* slice strides remain. Substitution of integer matrices with arbitrary coefficients does not enlarge these moduli. At passive layers include M_* . Auxiliary periods and subdivisions used only in the micro proof are absent from this domain: we have returned to the original two function fields and original slice.

9.5 Localizing the domain in normalized coordinates

We next replace $D_- \subset D_+$ by a pair of cutoff boxes for the current determining slot, without approximating either polynomial payoff.

Lemma 9.4 (Warm domain localization). *There is a common normalized cell scale ℓ with warm $\log(1/\ell)$ such that a spatial residue subbox, a warm-cost congruence class for β , and a translated pair of current-slot cutoffs of radius $r'_j = \ell v$ preserve a quantitative positive score after a prescribed proportional target discount. Their larger support lies in the fixed padded chart of the current determining block, and the logarithm of the reciprocal score has cold cost at j .*

Proof. The real pivot interval arguments have warm Lipschitz bounds in the joint coordinates $(u/H, y/v)$ by the determinant restriction. The same is true of r_i/T_{0i} on the nonsmall inactive axes. Their positive-domain margins are inverse-warm. On a small exceptional axis the *unscaled* $z_i(X)$ has warm Lipschitz bound, because k_i is warm-bounded. On true lifts it is integer. Hence sufficiently close true lifts have equal small-axis values. This preserves all short interval and omitted-axis tests exactly. Congruences are preserved by fixing their common warm modulus.

All points of $D_\pm$ are well inside the buffered uniqueness chart and the fixed padded chart for the old current block. Indeed the principal real outputs and the exact integer sheet outputs have the prescribed narrow ranges. Choose ℓ small enough, at warm log cost, for all these buffers, Lipschitz bounds, and positive margins. Partition the spatial coordinates into subboxes of normalized size comparable to ℓ , with the required residue restrictions; lower side cutoffs absorb endpoint rounding. Also prescribe β in classes modulo the common mask modulus.

On each such discrete/spatial piece average the usual small and large smooth cutoffs at radius $r'_j = \ell v$ in

$$b_j - C_j = b_{j,0} - c - y.$$

Centers are uniform in a fixed sufficiently large *normalized* lift range about $b_{j,0} - c$. Retain a center only if there is an actual D_- point with this spatial and residue data and with lift distance at most $2r'_j$ from the center. Choose the ratio 2 with the harmless fixed buffers in the cutoff convention.

Every D_- point then receives the full small-cutoff coverage: every center contributing to its small cutoff is retained. Conversely, every true site and tuple contributing to the large cutoff is close in both joint coordinates to the witnessing D_- point. It lies in D_+ by the positive interval margins, the exact equality of the small-axis data, and the fixed congruences. It also lies in the uniqueness and padded charts. Conversely, any integral tuple counted by this cutoff is the true lift on that chart, so no extra tuple has entered the argument. This reasoning needs no derivative bound on C in physical coordinates: we localize separately in u/H and y/v .

Let a_- be the full normalized small coverage and a_+ the corresponding upper bound for large coverage. The standard product cutoffs give

$$a_+ \leq (1 + C\gamma)^{d_j} a_-, \quad \log(a_-^{-1}) \leq \text{poly}(\text{warm log data}).$$

The latter uses normalized center measure; it does not introduce a factor v^{d_j} . With $\lambda'' = \lambda a_- / a_+$, integrating the new signed scores over centers and summing spatial/residue pieces gives at least a_- times the score in Equation (9.10). The total number of discrete and spatial pieces has warm logarithm, and the center measure has warm total mass. One retained piece therefore has score at least

$$m \exp(-\text{poly}(\text{warm log data})).$$

The logarithms of m and m^{-1} have cold-pre bounds at this layer, so this is the claimed cold score bound. Finally,

$$\log(w_j/r'_j) = \log(w_j/v) + \log(1/\ell)$$

is warm. The center is bounded by feasibility and the old center convention. Keeping the slice as an indicator preserves the full-site score interpretation. $\square$

9.6 Exact slots and fixed weight inflation

Keep the two polynomial payoff formulas unchanged in the localization argument. Substitute $\beta = b_j - b_{j,0}$ in their restored orbit. For active determining slots, the exact formal marking returns the old polynomials and classes after undoing the fixed triangular integer normalizations used to form the group representation. Passive masks were evaluated using canonical padded lifts. They agree exactly with the corresponding old slots on the returned support: earlier arguments have been fixed at this parent, the new current slot is in its padded box, and every higher nonzero contribution likewise uses the unique padded lift. Thus no later polynomial is evaluated at an approximately correct integer argument. The original two absolute cutoffs, including the original bounded full-weight part S^0 , have also been kept. The current flag has been removed, whereas any earlier flag in the pulled-back input remains for the preceding ascent step to remove.

It remains to check weights. At a passive step $j > s$, the restored absolute orbit is a polynomial of degree at most s in u alone; no current lift argument is inserted, so its weights are unchanged. At an active step let the current absolute weights be their original weights multiplied by L_{wt} . Initially $L_{\text{wt}} = 1$; after the first active step choose it larger than s .

In the triangular transformation $S^0 g^\sharp$, consider the cost of a monomial counting only its absolute-slot arguments. It is at most the absolute destination weight. If equality holds, its coefficient is constant in all parameter, lift, and determining arguments: such a term comes from the unchanged full-weight part S^0 , since the group acts strictly below destination weight. Undoing determining

normalizations does not alter this property, because these normalizations do not involve absolute slots.

Every other absolute monomial has a positive gap, at least L_{wt} , between its absolute cost and its destination weight. The weighted degree in all other arguments is bounded by a fixed constant: it uses only the bounded polynomial degree of the restored formula, the current lift weight $j \leq s$, and determining weights at most s . Multiplying all absolute weights by a sufficiently large fixed integer makes the enlarged gap absorb that other cost for all monomials at once. A term involving an absolute source variable of the same weight as its destination is necessarily linear in that variable, with constant coefficient, and remains triangular. Full-weight monomials in strictly lower-weight absolute variables may of course be nonlinear. Determining weights have not changed.

There are only s active steps. Hence the required weight multiples and degree cutoffs are fixed by a recursion of depth s , independently of all dimensions, widths, and density rounds. Choose D in advance at least the maximal resulting absolute weight, and take all sampler and algebra degree bounds large enough for this finite recursion. Together with Lemma 9.4, this completes the proof of Proposition 9.1.

10 Triangular iteration

We finish by organizing the choices in one increment and then iterating them. The distinction between a large numerical budget and an actual increase in dimension is essential here. All constants, polynomial exponents, and depths of successive polynomial-budget compositions in this section depend only on k . We use the warm and cold budget conventions of Definition 2.3.

10.1 Returning to simultaneous residual boxes

At the root of the ascent, all flags have been removed. We have the original input on a root slice, the prepared determining slots, and one fresh absolute tuple with at most d_0 slots. The determining equations are the exact old equations after the prescribed changes of integer variables. In particular, there is no error to estimate in an argument of a higher determining polynomial.

Lemma 10.1 (Root compression). *The root return of Section 9 can be converted to the simultaneous two-box certificate of Theorem 2.1, with a further arbitrarily small proportional discount of its target. The conversion adds no slots. At weight i its additional logarithmic width loss and its congruence cost are warm at i . Before common-width compression, the fresh absolute widths have logarithmic cost polynomial in p and the dimension bounds.*

Proof. Replace each small smooth cutoff by its strict box indicator and each large smooth cutoff by its enlarged box indicator. These replacements increase the positive payoff and decrease the magnitude of the negative payoff. Reindex the spatial slice and write each determining congruence in the form $b_i = M_i b'_i + \alpha_i$. The new residual target is obtained by this exact substitution and division by M_i ; its width is divided by the same number. The logarithm of M_i has the asserted warm cost.

Order the slots of one weight according to their inherited triangular order. A term involving an earlier slot of this same weight must be linear with constant coefficient: all variables have positive weights. By successive triangular integral changes of the integer slots, reduce those coefficients to a fixed bounded interval. Make the same changes exactly in every later target. The vector of residuals at this weight now has the form

$$Tb - C(u, b_{<i}),$$

where T is unitriangular, its entries are bounded, and $\log(2 + \|T\| + \|T^{-1}\|)$ is bounded polynomially in the block dimension. Thus, on putting $\tilde{C} = T^{-1}C$, the old residual vector is $T(b - \tilde{C})$.

Let P be the image under T^{-1} of the old strict residual box, and let $w_{\min}$ be the least coordinate radius of that old box. Choose

$$0 < \rho \leq \min \left\{ \frac{1}{32}, \frac{\gamma w_{\min}}{4 \max\{1, \|T\|_{\infty}\}} \right\},$$

with equality up to fixed factors. Average centers of simultaneous ρ -boxes in $b - \tilde{C}$ over $P + [-\rho, \rho]^m$. Every old strict point receives full coverage volume $(2\rho)^m$. Every contribution of a new enlarged box is within $(2 + \gamma)\rho$ of P in maximum norm. Its image under T is therefore inside the old enlarged box, since $\gamma \leq 1$. The upper enlarged coverage volume at any point is $(2(1 + \gamma)\rho)^m$. Nonnegativity and a proportional target discount paying their ratio $(1 + \gamma)^m$ select a positive certificate. This is the same coverage argument as in Section 4; it requires neither a lower bound on the original surplus nor regularity of the sites near a boundary. It also shows that the selected centers can be kept bounded.

The least of several radii contributes the maximum of their logarithmic inverse widths. For old coordinates these logs are the old Q_i plus warm losses, while for the fresh coordinates they are polynomial in p and dimensions. Their maximum is therefore at most Q_i plus an allowed warm loss, padding an empty block with harmless fixed width data. No new coordinate is introduced. We may now reset every value space to its full ambient space for the next increment. $\square$

10.2 Structural choices come first

Section 9 bounds the weights of a fresh absolute patch by a degree recursion with at most s active steps. Passive steps do not increase these weights, and dimensions do not enter the recursion. Choose D to dominate its output. We must also check that this same choice works in every density round.

Preparation preserves this choice. A cut at weight h retains a target of weighted degree at most h , introduces a target of degree at most $h - 1$, and replaces the old h -slot by an exact polynomial of weighted degree at most h . Substitution into a higher target consequently preserves its destination weight. Slots that were absolute in an earlier round are simply old determining slots in the next round; those above s are then passive. Fix all sampler, family-polynomial, and algebra degree cutoffs to cover these bounds. None will subsequently depend on a dimension or a numerical precision.

Proposition 10.2 (Triangular dimension bounds). *For one round, the increase in the dimension of weight i is bounded by d_0 plus a fixed polynomial in the incoming dimensions strictly above i . The number of preparation cuts and all structural variable counts are polynomial in the incoming dimensions and p . For a proposed run of $T = O_k(1 + p)$ rounds on an interval, starting without slots, all these quantities admit polynomial bounds in $2 + p$, chosen before any numerical precision.*

Proof. Write d_i for the incoming ambient dimension at weight i . Within one preparation round set

$$H_D = d_D, \quad H_i = d_i + H_{i+1}^2 \quad (1 \leq i < D).$$

These bound the largest ambient dimensions reached during preparation. Indeed every cut at i lowers $\dim W_i$ by one. Its dimension can be replenished only by incoming full copies from $i + 1$, whose dimensions are already included in H_i . There are therefore at most H_i cuts at i . Each exports a copy of dimension at most H_i to $i - 1$, so the total export is at most H_i^2 . This proves the recurrence downwards from D . Restarting a probability test does not reset its value space, so it does not affect this count.

The fresh absolute tuple adds at most d_0 coordinates in total. Consequently, with $H_{D+1} = 0$, the output dimensions satisfy

$$d_i^{\text{out}} \leq d_i + d_0 + H_{i+1}^2. \quad (10.1)$$

Here and below an expression with H_{D+1} is interpreted as zero. If a block is harmlessly relabeled to a still lower weight, replace the last term by the sum of the corresponding higher exports; the dependence is still strictly triangular. The total cut count is at most $\sum_i H_i$. Structural path-variable counts are fixed polynomials of the dimension data, including the root spatial dimension. Composing them through the fixed number D of levels preserves polynomial growth.

For completeness, let $P_i(p, d_{>i})$ be nondecreasing polynomials dominating the additions in (10.1), including d_0 . Define downward envelopes

$$B_D = d_D(0) + TP_D(p), \quad B_i = d_i(0) + TP_i(p, B_{>i}).$$

An induction first on i downwards and then on the round number gives $d_i(t) \leq B_i$ for $0 \leq t \leq T$. Starting with $d_i(0) = 0$ and $d_0 \leq (2+p)^C$, these envelopes are polynomial in $2+p$. For an arbitrary fixed root spatial dimension n , the corresponding structural forecasts are polynomial in $2+p+n$: the root slices retain that dimension, and the path-column counts are fixed-degree polynomials composed through only D layers within each round. Path dimensions do not replace the root dimension in the next round. In the interval application $n = 1$. Choose a polynomial bound d_* dominating the envelopes and all within-round structural counts, and a polynomial bound N_c for the cut count. This forecast does not use a width, rank, scale, or accuracy parameter. $\square$

10.3 The numerical schedule

Fix the structural bounds d_*, N_c . Choose the gaps and proportional discounts so that all preparation selections, the D ascent coverings, and root compression together consume an arbitrarily small fixed part of the absolute gain. For example, γ can be a sufficiently small fixed gain-dependent constant divided by a sufficiently large polynomial in $2+p+d_*+N_c$. Divide the permitted preparation discount among the N_c possible cuts. Thus $\log(1/\gamma)$ is polynomially bounded before the remaining choices. The fixed-depth forward and flag errors may be chosen as small fixed constants depending on k and the absolute gain.

Let q_j be the incoming width log at level j . We describe a descending schedule $j = D, D-1, \dots, 1$. Its record U_j will contain the logarithmic bounds for every request at level j , including its late rank and parent side requirements. At the time level j is scheduled, all $U_{>j}$ are already fixed. We shall bound each preparation cut's logarithmic width loss at this level by A_j , and the total loss from ascent and root compression by A'_j . The two required dependencies are

$$\begin{aligned} U_j &\leq F_j(p, d_*, N_c, \log(1/\gamma), q_j, U_{>j}), \\ N_c A_j + A'_j &\leq G_j(p, d_*, N_c, \log(1/\gamma), U_{>j}), \end{aligned}$$

for fixed nondecreasing polynomials F_j, G_j . The complete record may use the current width; the added width loss may not. We therefore choose the geometric scales and relative losses first, and only then use q_j to choose the remaining accuracies, rank thresholds, and side cutoffs. The following four steps verify this order.

1. **Normalized geometry and minimum sides.** Using only p , the structural and gap bounds, and $U_{>j}$, choose the spatial principal scale, the lift ratio v/w_j , and the minimum-side threshold below which an inactive axis is omitted. These are warm choices. The minimum sides must meet the parent-side requirements of the higher layers, or the absolute-return requirement at

the last layer. They must also permit the full-side separations used when lifting a rank failure from a strictly higher layer.

In that separation, the path-evaluated slow coefficient and common denominator bounds, including bases for the later tensor variables, use only higher-layer data and dimensions. The separate real and denominator bounds in Lemmas 4.7 and 4.2 are precisely what ensures this assertion. Auxiliary subproblems needing cold data at j temporarily set further pre-short columns to zero; they do not change this minimum-side requirement. A rank failure at j itself is lifted only through layers below j . Thus the present minimum-side choice uses neither the final rank at j , its value-space height, nor its own width loss.

2. **Preparation width forecast.** Choose a warm per-cut bound A_j , polynomial in the same data and independent of q_j . In a cut at $h > j$, subdividing the normalized deviation variables at level j charges the higher failure's coefficient and denominator bounds, the chosen ratio v/w_j , the higher spatial shrink factors, and the omission cutoff controlling the variation of omitted rows. The linear reduction in the incoming copy and the clearing of its residues charge only higher data in addition to these. If the lower width must be absolutely smaller than a higher width, the corresponding relative shrink factor suffices because every old width is less than one.

A cut at j requires only gap and dimension factors for its own residual width. Its large rational section vector is handled in the lower localizations and the incoming copy, not by conditioning the new j -residual. The copy can be assigned an arbitrarily small incoming width without demanding that the formula error be small at that width: the formula error only needs to be small relative to the gap times the cut-layer width. Cuts below j do not shrink level j . We may therefore budget all preparation widths at this level by

$$q_j + N_c A_j.$$

3. **Warm ascent forecast.** Next forecast the warm modular-product bound, inherited score and slice bounds, productive probabilities, and macro boundary and cell precisions. The passive ascent also uses its warm modeling and subslice bounds. These require only boundedly many polynomial increases in the warm data, and do not enlarge the preparation bound A_j .

At an active step, the micro argument preserves a warm score on the original warm slice. The extraction in Section 9 depends on that score, normalized pivot geometry, the full-law caps relative to chart mass, and the original slice stride. It does not subdivide according to the active micro period, the short-axis lengths, or the coefficients of the restored polynomial orbit. The affine-plane parameterization solves its large congruences identically. Hence the relative current-width loss and the returned current-block congruence cost remain warm. At passive steps the sharpened major bounds give the same conclusion. Include these losses and those of Lemma 10.1 in a warm bound A'_j .

4. **Cold-pre and late choices.** We may now charge $q_j + N_c A_j$, as well as the preceding data. Forecast the cold-pre detection, modeling, and major-refinement requests; the bounded nests of pure-long threshold changes and symbol splittings; the short-axis and exceptional-product bounds; and every flag-truncation accuracy required by those requests. Choose the cutoff for the modulus cover after the relevant model and period bounds. Only individual moduli and witness comparisons of the stated logarithmic size are required, as in Section 6.

After all these pre requirements, choose the perturbation, range, rank, and parent-side bounds. Choose the rank-stop probability small enough for the forward calculation using the current

prepared width and higher-layer center masses. Auxiliary cold computations that establish a sharpened warm conclusion do not enlarge that conclusion's output budget. In particular their own cold costs do not return to A_j or A'_j .

Let U_j dominate this entire schedule and record its rank-failure requirements for the next lower layer. Every call to the relative major refinement at an ancestor is forecast at that ancestor before its late choices. The height of W_j after repeated cuts has the simultaneous stacked-row bound of Proposition 4.1 in terms of dimensions, cut count, and $\log R_j$; it is not needed in choosing R_j . Earlier rows are extended by zeros on incoming copies, and uniform scalar residue changes preserve the value space. All rank tests can consequently restart after each transaction with these uniform schedules. A nonroot path box pays the specified later-layer parent-side requirements, not the root slicing costs of other transactions.

Proposition 10.3 (Triangular width bounds). *The choices above can be made with fixed polynomial logarithmic bounds. They give, in one round,*

$$Q_j^{\text{out}} \leq q_j + P_j(p, d_*, q_{>j}). \quad (10.2)$$

where P_j is a fixed polynomial independent of q_j and of all lower width logs. The total root logarithmic slice loss and the sufficient root log-side cutoff are polynomial in p, d_* and all incoming width logs. For the interval run of $O_k(1+p)$ rounds in Proposition 10.2, starting without slots and with trivial width data, these width logs and the sums of root costs are polynomial in $2+p$.

Proof. Every step in the descending schedule uses a bounded composition of polynomial logarithmic bounds, giving the two dependencies displayed above. Descending substitution through only D levels shows that $N_c A_j + A'_j$ is polynomial in p, d_* and $q_{>j}$. Preparation, ascent, and root compression give $Q_j^{\text{out}} \leq q_j + N_c A_j + A'_j$, proving the stated recurrence.

Each transaction's spatial slice loss has polynomial logarithmic cost in the full recorded budgets. Their sum over at most polynomially many transactions has the same form, as does the sufficient root log-side cutoff. This statement concerns logarithms: a numerical scale may be exponential in such a polynomial without violating the estimate.

For this interval run of T rounds starting without slots, Proposition 10.2 supplies a structural bound d_* polynomial in $2+p$. Define downward width envelopes by

$$V_D = Q_D(0) + TP_D(p, d_*), \quad V_j = Q_j(0) + TP_j(p, d_*, V_{>j}).$$

Then $Q_j(t) \leq V_j$ for every $t \leq T$, by the same descending induction as for dimensions. Trivial initial widths, $T = O_k(1+p)$, and the polynomial bound on d_* make every V_j polynomial in $2+p$. The recorded budgets and each round's root costs are therefore polynomial as well. Summing the logarithmic root losses over T rounds preserves this bound. $\square$

10.4 The increment and the density bound

Completion of the proof of Theorem 2.1. Preparation preserves a strict positive certificate with an arbitrarily small proportional target discount. Proposition 6.2 then supplies productive terminal paths on a fraction at least a fixed multiple of a , at a threshold differing from a by a sufficiently small fixed proportion. This use of the forward mass comparison is important: it does not charge the inverse of the incoming strict surplus, which may be arbitrarily small.

Apply Lemma 2.2 on a productive terminal box. Its sides meet the dimension-dependent requirement by the minimum-side choices above. Its fresh rank is bounded by d_0 independently of that dimension. If its hypotheses are impossible, the original certificate cannot exist. Otherwise it

gives the quantitative positive return at a fixed multiplicative gain. The passive and active ascents preserve enough of that gain, and restore all determining equations exactly. Apply Lemma 10.1 at the root.

Choose the fixed proportional losses sufficiently small that a net factor $1 + \eta_k$, with $\eta_k > 0$, remains. The dimension assertion of Theorem 2.1 is Proposition 10.2; its width assertion and root-cost assertion are Proposition 10.3. Thus all three required bounds hold simultaneously, with iteration-independent polynomial exponents. $\square$

Proposition 10.4 (Finite-horizon closure). *There are constants $C_k > 0$ and $A_k \geq 1$ such that a progression-free subset of $[N]$ of density $\alpha > 0$ cannot satisfy*

$$\log N \geq C_k(2 + \log(1/\alpha))^{A_k}.$$

Proof. Start with no slots, $f = 1_A$, and target $a_0 = \alpha/2$. Take one fixed parameter $p = C(2 + \log(1/\alpha))$, with C sufficiently large, and reserve

$$T = \left\lceil \frac{\log(2/\alpha)}{\log(1 + \eta_k)} \right\rceil + 1 = O_k(1 + p)$$

rounds. Use this horizon in the advance dimension and width forecasts. Every temporary application threshold is at least a fixed positive multiple of a_0 , so it is at least e^{-p} after enlarging C . The same parameter p , fresh-rank bound, and structural forecasts are therefore valid throughout the run.

Propositions 10.2 and 10.3 bound the sum of all root logarithmic slice losses, the maximal remaining log-side requirement, and the endpoint rounding margins by a fixed polynomial in $2 + p$. Taking the initial $\log N$ larger than their sum sustains every possible round. Only the root is spatially resliced from one round to the next; it remains an integer-affine slice of the original interval.

If no impossibility occurs, the targets satisfy $a_t \geq (\alpha/2)(1 + \eta_k)^t$, and hence $a_T \geq 1$. But a certificate at any target $a \geq 1$ is impossible, since

$$fB^- \leq B^+$$

pointwise. An absolute-rule impossibility gives the contradiction sooner. Enlarging constants to cover the polynomial side requirement proves the proposition. $\square$

Proof of Theorem 1.1. For $\alpha = r_k(N)/N > 0$, the contrapositive of Proposition 10.4 gives

$$\log N < C_k(2 + \log(1/\alpha))^{A_k}, \quad \alpha < e^2 \exp(-C_k^{-1/A_k}(\log N)^{1/A_k}).$$

If $\alpha = 0$ the density assertion is immediate. Renaming the positive constants gives Equation (1.1).

We also check the threshold formulation at its endpoints. Let C_k, c_k, ε_k now denote the constants in Equation (1.1), and put $\overline{C}_k = \max\{1, C_k\}$. For every $0 < \alpha \leq 1$, the condition

$$\log N \geq \left(\frac{\log(2\overline{C}_k/\alpha)}{c_k} \right)^{1/\varepsilon_k}$$

implies $r_k(N) \leq \alpha N/2 < \alpha N$. Thus a set of density *at least* α contains the required progression, including at equality in the density assumption. Enlarging a constant $A_k \geq \max\{1, 1/\varepsilon_k\}$ makes $A_k(2 + \log(1/\alpha))^{A_k}$ dominate the displayed threshold for the whole range $0 < \alpha \leq 1$. Conversely, a threshold of this form excludes an extremal progression-free set whenever it holds at $\alpha = r_k(N)/N$, and the preceding inversion gives a bound of the form Equation (1.1).

For an integer $r \geq 1$, a largest class in an r -coloring has density at least $1/r$. Substituting $\alpha = 1/r$ therefore gives a monochromatic nonconstant k -term progression whenever

$$N \geq \left\lceil \exp(A_k(2 + \log r)^{A_k}) \right\rceil.$$

The same constant covers $\alpha = 1$ and $r = 1$. All constants depend only on the fixed length k . $\square$

11 Weighted consequences

The density bound gives uniform bounds on harmonic sums and tails over progression-free sets, and permits weights larger than the reciprocal. It also implies progressions in every subset of the primes with positive relative upper density. We derive these consequences from dyadic summation and the resulting counting estimates.

Fix $k \geq 3$. In this section write $c = c_k$, $C = C_k$, and $\varepsilon = \varepsilon_k$ for constants such that

$$r_k(N) \leq CN \exp(-c(\log N)^\varepsilon) \quad (N \geq 2). \quad (11.1)$$

We may decrease ε and enlarge C so that $0 < \varepsilon < 1$. All bounds below are uniform over sets containing no nonconstant k -term arithmetic progression.

11.1 Dyadic summation

Proposition 11.1 (Dyadic weighted summation). *For $w : \mathbb{N} \rightarrow [0, \infty)$, set*

$$S_k(w) = \sum_{m=0}^{\infty} r_k(2^m) \max_{2^m \leq n < 2^{m+1}} w(n).$$

If $A \subseteq \mathbb{N}$ contains no nonconstant k -term progression, then

$$\sum_{a \in A} w(a) \leq S_k(w).$$

Consequently, whenever $S_k(w) < \infty$, divergence of the weighted sum forces a nonconstant k -term progression.

Proof. Each interval $[2^m, 2^{m+1}) \cap \mathbb{Z}$ has length 2^m , so translation invariance gives at most $r_k(2^m)$ elements of A there. Bound each weight by the maximum on that interval and sum. All terms are nonnegative, so passage to infinite sums is valid. $\square$

Corollary 11.2 (Uniform harmonic bound). *There is $H_k < \infty$ such that every k -term-progression-free $A \subseteq \mathbb{N}$ satisfies*

$$\sum_{a \in A} \frac{1}{a} \leq H_k.$$

One may take $H_k = \sum_{m \geq 0} 2^{-m} r_k(2^m)$.

Proof. Apply Proposition 11.1 with $w(n) = 1/n$. By Equation (11.1), its terms for $m \geq 1$ are at most $C \exp(-c(m \log 2)^\varepsilon)$, whose sum converges. $\square$

11.2 Weights larger than the reciprocal

The following weights are defined by their formulas on a sufficiently large tail and set to zero before that tail. Changing finitely many values does not affect a divergence criterion.

Corollary 11.3 (Stretched-exponential weights). *Every k -term-progression-free set has a uniformly bounded sum for each of the weights*

$$\begin{aligned} w_a(n) &= \frac{\exp(a(\log n)^\varepsilon)}{n}, & 0 < a < c, \\ w_{*,\eta}(n) &= \frac{\exp(c(\log n)^\varepsilon)}{n \log n (\log \log n)^{1+\eta}}, & \eta > 0. \end{aligned}$$

The bound may depend on k and the displayed parameters. Divergence for either weight therefore forces a nonconstant k -term progression.

Proof. Both weights are eventually decreasing. Indeed, with $y = \log n$, the derivatives of their logarithms with respect to y are

$$-1 + a\varepsilon y^{\varepsilon-1}, \quad -1 + c\varepsilon y^{\varepsilon-1} - y^{-1} - (1 + \eta)(y \log y)^{-1},$$

respectively. Since $\varepsilon < 1$, both are negative for large y . On every sufficiently late dyadic interval the maximum is therefore its left-endpoint value. For w_a , the corresponding summand in Proposition 11.1 is at most

$$C e^{-(c-a)(m \log 2)^\varepsilon},$$

which is summable. For $w_{*,\eta}$ the exponential factors cancel at that same endpoint, leaving

$$\frac{C}{m \log 2 [\log(m \log 2)]^{1+\eta}},$$

also summable. The omitted initial intervals contribute a finite amount. $\square$

11.3 Logarithmic weights and dense subsets of the primes

The exponent ε in the preceding two weights depends on the fixed length k . Applying Theorem 1.1 separately for each fixed length gives the following consequence with a single logarithmic weight.

Corollary 11.4 (Logarithmic weights and every length). *For every fixed $B \geq 0$, if $A \subseteq \mathbb{N}$ satisfies*

$$\sum_{a \in A} \frac{(\log(2+a))^B}{a} = \infty,$$

then A contains nonconstant arithmetic progressions of every finite length.

Proof. Fix $k \geq 3$. On the m th dyadic interval the weight is at most $C_B(1+m)^B 2^{-m}$. Hence the terms in Proposition 11.1 are bounded, apart from finitely many, by

$$C_{k,B}(1+m)^B e^{-c_k(m \log 2)^{\varepsilon_k}}.$$

This sequence is summable because $m^{\varepsilon_k}/\log m \rightarrow \infty$. Thus a set with the stated divergence cannot be k -term-progression-free. Lengths one and two follow from infinitude. $\square$

The same comparison shows that, for each fixed $B > 0$, there is a constant $C_{k,B}$ with

$$r_k(N) \leq C_{k,B} N (\log N)^{-B} \quad (N \geq 3).$$

Dense subsets of the primes. Theorem 1.1 also recovers the established dense-primes theorem of Green and Tao [21, Theorem 1.2]. If A is a subset of the primes with $\limsup_{N \rightarrow \infty} |A \cap [N]|/\pi(N) > 0$, where $\pi(N)$ counts the primes at most N , then A contains infinitely many nonconstant k -term progressions for every fixed $k \geq 3$. Indeed, the preceding estimate with $B = 2$ gives $r_k(N) = o(N/\log N)$. Positive relative upper density and the prime number theorem give $c > 0$ and an unbounded sequence of N with $|A \cap [N]| \geq cN/\log N$. For each fixed M , deleting $A \cap [M]$ leaves at least $(c/2)N/\log N$ elements along all sufficiently large members of this sequence. This exceeds $r_k(N)$, so there is a progression entirely above M . Since M is arbitrary, infinitely many occur. Lengths one and two follow from infinitude.

11.4 Uniform harmonic tails

Corollary 11.5 (Harmonic tails). *There is $C'_k > 0$ such that every k -term-progression-free set $A \subseteq \mathbb{N}$ satisfies, for all sufficiently large real x ,*

$$\sum_{\substack{a \in A \\ a > x}} \frac{1}{a} \leq C'_k (\log x)^{1-\varepsilon_k} e^{-c_k (\log x)^{\varepsilon_k}}.$$

In particular, for every $0 < a < c_k$, this tail is at most $C_{k,a} e^{-a(\log x)^{\varepsilon_k}}$.

Proof. Let $A(t) = |A \cap [1, t]|$. The function $te^{-c(\log t)^{\varepsilon}}$ is eventually increasing, since $\varepsilon < 1$. Applying Equation (11.1) at $\lfloor t \rfloor$ therefore gives $A(t) \leq Cte^{-c(\log t)^{\varepsilon}}$ for all sufficiently large real t . Partial summation and $A(t)/t \rightarrow 0$ yield

$$\sum_{\substack{a \in A \\ a > x}} \frac{1}{a} \leq C \int_x^\infty e^{-c(\log t)^{\varepsilon}} \frac{dt}{t} = \frac{C\varepsilon^{-1/\varepsilon}}{\varepsilon} \int_{c(\log x)^{\varepsilon}}^\infty v^{1/\varepsilon-1} e^{-v} dv.$$

Put $b = 1/\varepsilon - 1 > 0$. If $v_0 \geq 2b$, then $(v/v_0)^b \leq e^{(v-v_0)/2}$ for $v \geq v_0$, by $\log(1+t) \leq t$. Thus

$$\int_{v_0}^\infty v^b e^{-v} dv \leq 2v_0^b e^{-v_0}.$$

Substitution proves the first claim, with constant $2C/(c\varepsilon)$ on a sufficiently large tail. Finally the polynomial factor in $\log x$ is absorbed by $e^{(c-a)(\log x)^{\varepsilon}}$, proving the second claim. $\square$

The absolute rule used in the main proof requires a fresh-coordinate bound independent of the dimension of the box. This uniformity requires more than positive counting. Rational and symbol calculus in Appendices A and B supports the shift comparison of Appendix C; positive counting in Appendix D then gives an absolute increment in fixed dimension. Its extension to arbitrary dimension is the final relative-lifting argument.

Sampling supplies the connection. After scalar transfer on ordinary affine samples in Appendix E, Appendix F constructs one ensemble carrying the required polynomial lifts. It has two distinct uses. Cube comparison in Appendix G transfers detection and yields finite linear models. Those models need not have positive coefficients, so they alone do not return a positive score. The nonnegative comparison weights of Appendix H provide that return and recover the required integer slots.

Appendix I uses detection and approximation to replace local orbits by an ambient orbit with an exact marked projection. It combines this globalization with positive score transfer to prove relative lifting with rank at most $d + d_0$. Applying it to the constant old patch, of rank zero, gives the dimension-independent fresh rank in Lemma 2.2. This argument permits the stated late costs in the returned patch. The main triangular return has the separate task of retaining the old determining equations with width losses that can be iterated. Table 1 records the precise analytic inputs and their proofs.

Throughout the appendices, degree parameters are fixed, while dimensions are included in the stated logarithmic budgets. The polynomial notation of Section 2 is repeated where needed. The refined width estimates required for the triangular increment are proved in the main text; they are not hypotheses of these analytic results.

Input	Result used	Full statement and proof
A bounded function with large cube norm	Correlation with a controlled niltest; interval input and local box/product-cyclic reductions	Theorem A.7
A biased top-degree nil-test	Slow-structured-rational symbol decomposition, with all parameter dimensions included in the budget	Lemma B.1
Positive comparison against degree- d tests	Uniform comparison against shifted degree- $(d - 1)$ tests outside a quantitatively small exceptional set	Theorem C.1
A progression-free input on fixed-dimensional prime boxes	An absolute increment and conversion between niltests and patches	Proposition D.3; Lemma D.5
High formal rank, with structural choices fixed before precision	One ensemble satisfying geometry, detection and transfer for the prescribed inputs and accuracies	Proposition F.2; proofs in Appendices G and H
A fixed strong projection and exact marked polynomial	Globalization preserving the marked identity with early output bounds	Proposition I.6
An absolute rule and an old patch of rank d	A relative increment with rank at most $d + d_0$, including its separate impossibility clause	Proposition D.7; proof in Theorem I.15

Table 1: Analytic inputs to the main proof. Each reference specifies the complete hypotheses and quantitative bounds; the table records the object passed between the statements.

A Quantitative conventions and symbols

The analytic argument uses polynomial niltests both to detect structured correlation and to compare positive functions. This section fixes their quantitative meaning and develops the operations needed to reduce their degree. The rational calculus keeps its costs polynomial in logarithmic complexity when dimensions grow; the symbol calculus separates normalized real control from rational denominators. We then prove the local box versions of the inverse and converse inverse theorems and the normalized square construction used in the step-drop argument.

For a positive integer M , write $[M] = \{1, \dots, M\}$. If Q is a nonempty finite set, $\mathbb{E}_{x \in Q}$ denotes its uniform probability average. All correlations on such a set are normalized: “ f correlates with T by at least ρ ” means

$$|\langle f, T \rangle_Q| \geq \rho, \quad \langle f, T \rangle_Q = \mathbb{E}_{x \in Q} f(x) \overline{T(x)}.$$

Real nonnegative tests require no conjugation. Slices and subboxes always carry their own uniform probability measures unless another measure is specified.

Throughout this section the degree s is fixed. An expression $\mathcal{P}(p)$ denotes $(2+p)^C$, and $\mathcal{B}(p)$ denotes $\exp((2+p)^C)$, for a constant C depending only on the fixed degrees. The constant can increase from one occurrence to the next. These are existence bounds. In particular, “ $T \geq \mathcal{B}(p)$ sufficiently large” means that some bound of the displayed form suffices. A polynomial number of products, or a bounded number of compositions of these bounds, is permitted. An unbounded number of successive polynomial changes of the budget is not implicit in this notation.

For $T = (T_1, \dots, T_n)$ and a multi-index α , put $T^\alpha = \prod_i T_i^{\alpha_i}$. Every coefficient estimate below uses these actual side lengths. No bound on $\max T_i / \min T_i$ is assumed. The height of a rational array bounds its numerators and positive denominators. A *denominator bound* does not bound numerators. The letter n counts integer parameters. In the nilmanifold discussion d denotes the group dimension; later uses of d as a propagation degree, polynomial weight, or patch rank are specified locally. Variable dimensions and ranks enter the displayed budgets, whereas constants may depend on fixed degrees.

A.1 Filtered nilmanifolds

Definition A.1 (Niltests). A rational filtered nilmanifold consists of a connected, simply connected nilpotent Lie group G , a cocompact lattice Γ , and connected rational subgroups

$$G = G_1 \supseteq G_2 \supseteq \dots \supseteq G_{s+1} = \{1\}, \quad [G_i, G_j] \subseteq G_{i+j}.$$

Put $d = \dim G$. We fix rational exponential coordinates on its Lie algebra $\mathfrak{g}$, with rational bases of the filtered subspaces $\mathfrak{g}_i = \log G_i$. A polynomial map $g : \mathbb{Z}^n \rightarrow G$ is adapted to this filtration when $\log g$ modulo $\mathfrak{g}_{i+1}$ has ordinary degree at most i . With explicitly assigned variable weights, ordinary degree is replaced by weighted degree.

A degree- s niltest is $t \mapsto F(g(t)\Gamma)$, where F is bounded and Lipschitz for the quotient of the right-invariant Riemannian metric that makes the declared coordinate basis orthonormal at the identity. A different metric is permitted only with a comparison constant whose logarithm is included in the complexity budget. Its complexity is at most p if the dimension, logarithmic heights of the structural constants and filtered bases, and $\log(2 + \|F\|_\infty + \text{Lip } F)$ are at most p , and if

$$l\mathbb{Z}^d \subseteq \log \Gamma \subseteq l^{-1}\mathbb{Z}^d \quad \text{for some integer } 1 \leq l \leq e^p. \quad (\text{A.1})$$

There is no coefficient bound on g . On $\mathbb{Z}/N\mathbb{Z}$ the evaluation uses representatives $0, \dots, N-1$. Degree-zero tests are constants. The group of dimension zero is allowed.

Here and below one can enlarge the budget by $\mathcal{P}(p)$ when changing rational coordinates. The following elementary facts explain the uniformity in a variable dimension.

Lemma A.2 (Rational operations). *Suppose a polynomial number of rational matrices have polynomially many rows and columns and entries of height $\mathcal{B}(p)$. Their kernels, images, intersections, quotient maps and sections of surjections have rational bases or matrices of height $\mathcal{B}(p)$, provided these operations are performed jointly, or at a bounded depth.*

The analogous assertions hold for rational filtered subgroups, quotients by connected rational normal subgroups, and coordinate formulae obtained from the Baker–Campbell–Hausdorff formula at fixed step. The resulting lattices have grid bounds of the form (A.1) with $l \leq \mathcal{B}(p)$.

Proof. Clear all denominators by multiplying them. The logarithm of this product is polynomially bounded. For an $r \times r$ integer matrix whose entries have magnitude at most H , the determinant has magnitude at most $r!H^r$. Cramer’s rule applied to a nonzero maximal minor gives the asserted bases and sections. An intersection is a single kernel calculation on the combined defining equations; performing a separate elimination for every subspace is unnecessary.

At fixed step the Baker–Campbell–Hausdorff formula has bounded commutator length. In dimension d , each resulting coordinate has only $d^{O_s(1)}$ terms. The same remains true, with the number of factors included in the polynomial, for a polynomial number of multiplications. Apply the preceding rational estimates to their coefficients. Restricting or projecting the grids in (A.1), and clearing the denominators of the chosen rational bases, proves the subgroup and quotient grid assertions. $\square$

The log-polynomial definition in Definition A.1 agrees with the Taylor/binomial definition of an adapted polynomial sequence. Indeed, modulo each successive layer, subtract the homogeneous terms using factors $\exp(X \binom{t}{\alpha})$ with $X \in \mathfrak{g}_{|\alpha|}$. The finite Baker–Campbell–Hausdorff formula proves the converse and closure under products. This also proves that a rational linear section respecting every filtered layer lifts polynomial maps. The section need not be a Lie algebra homomorphism: lifting in logarithmic coordinates is sufficient.

We shall use finite covers with the same quantitative convention. Choose an ordered rational basis adapted to the filtration. After scaling every basis vector by a sufficiently divisible integer of size $\mathcal{B}(p)$, its integral ordered products form a subgroup of Γ ; this follows directly from the triangular multiplication and inverse formulae. This supplies controlled Mal’cev coordinates and bounded fundamental representatives. It also supplies common congruence covers for a polynomial number of rational requirements. In particular, if z has bounded denominator but arbitrary numerator, the formula $\text{Ad}_z = \exp(\text{ad } \log z)$ shows that a fixed sufficiently deep integral grid is contained in $\log(z\Gamma z^{-1})$. The depth depends on denominators, not numerators. A bounded-degree rational polynomial map with common denominator $\mathcal{B}(p)$ is periodic modulo such a cover with period $\mathcal{B}(p)$: subtract its values at arguments congruent modulo a sufficiently divisible period and use the same polynomial formulae.

When a cover normal in Γ is needed, one can obtain it without the factorial loss from taking a permutation-group core. Let $\Lambda_0 \leq \Gamma$ be the controlled cover already constructed, and choose $m_0 \leq \mathcal{B}(p)$ with $m_0\mathbb{Z}^d \subseteq \log \Lambda_0$. Choose an integer m divisible by lm_0 , and let

$$\Lambda = \langle \gamma^m : \gamma \in \Gamma \rangle.$$

This subgroup is characteristic in Γ . Since $\log(\gamma^m) = m \log \gamma \in m_0\mathbb{Z}^d$, every generator belongs to Λ_0 , and hence $\Lambda \subseteq \Lambda_0$. A Mal’cev normal form with d integral exponents shows that $[\Gamma : \Lambda] \leq m^d$: in the quotient each exponent can be reduced modulo m , using normality. Moreover

$$m\mathbb{Z}^d \subseteq \log \Lambda \subseteq \log \Gamma \subseteq l^{-1}\mathbb{Z}^d.$$

Thus normality, index and grid bounds all have the same quantitative cost. This construction does not assert that an arbitrary uniform exponential grid is itself invariant under conjugation.

On bounded fundamental representatives these coordinate formulae also give Lipschitz bounds $\mathcal{B}(p)$ for bounded rational group homomorphisms and bounded left translations. Fixed right translation is an isometry before passing to a suitable cover. For varying left translations, right invariance gives the useful exact estimate

$$d_G(ag, a'g) = d_G(a, a').$$

Comparing small coordinate balls with metric balls, and constructing Lipschitz partitions at precision $\mathcal{B}(p)^{-1}$, therefore costs only $\mathcal{B}(p)$.

Lemma A.3 (Reconstruction from rational images). *Let $\phi : H \rightarrow L$ be a rational homomorphism between controlled filtered nilpotent groups, and suppose the matrix of its differential in the declared rational coordinates has height $\mathcal{B}(p)$. Assume its kernel is connected and rational. A controlled Lipschitz function on H/Γ_H invariant under left translation by $\ker \phi$ can be represented, on $\phi(H)$, by a controlled Lipschitz function of its image in a sufficiently fine lattice cover of L . The cost is $\mathcal{B}(p)$. The same assertion applies to a polynomial number of homomorphisms by taking their product.*

Proof. Write $J = \phi(H)$. Rational linear algebra and the grid bounds allow a target lattice Γ'_L with $J \cap \Gamma'_L \subseteq \phi(\Gamma_H)$. Thus the function is well defined on $J/(J \cap \Gamma'_L)$: changing a source lift changes it by the connected kernel, and changing an image representative changes it by the image of a source lattice element.

For the Lipschitz estimate take bounded representatives in J . A target lattice element identifying two sufficiently close image points is itself bounded. In logarithmic coordinates its distance from the rational subspace $\log J$ is either zero or at least $\mathcal{B}(p)^{-1}$. This follows by applying a bounded rational matrix defining that subspace to the bounded-denominator lattice coordinates. At a smaller distance it consequently belongs to J . The small displacement then lifts by a bounded rational linear section of $\log H \rightarrow \log J$, which proves the local Lipschitz bound. Boundedness of the function gives the global bound for larger distances. Real and imaginary parts can be extended separately from the image by the Lipschitz extension theorem; clipping preserves a specified real interval when the function is real valued. $\square$

Lemma A.4 (Vertical decomposition). *Put $e(x) = \exp(2\pi ix)$. A controlled Lipschitz function can be approximated uniformly, to error $\mathcal{B}(p)^{-1}$ with any prescribed polynomial logarithmic accuracy, by a sum of $\mathcal{B}(p)$ functions F_η satisfying*

$$F_\eta(zx) = e(\eta(\log z))F_\eta(x), \quad z \in G_s.$$

The frequencies and the Lipschitz complexities have size $\mathcal{B}(p)$. If the initial function has a specified character on a subtorus, every nonzero summand has that character there.

Proof. The quotient $G_s/(G_s \cap \Gamma)$ is a central torus with a rational lattice basis of controlled height. Convolve in that torus with a product of sufficiently high-order Fejér kernels, then expand the resulting trigonometric polynomial. If the torus has dimension $d \leq \mathcal{P}(p)$, Lipschitz constant $L \leq \mathcal{B}(p)$, and required error $\delta \geq \mathcal{B}(p)^{-1}$, the cutoff in each coordinate can be taken of size $(1 + dL/\delta)^{O(1)}$. The number of terms and all frequency heights are thus $\mathcal{B}(p)$. Translation by the central torus commutes with the relevant metric comparisons, so the summands retain the asserted Lipschitz bounds. The last assertion follows by taking the Fourier transform along the subtorus. $\square$

A.2 Symbols and separation

Let $\mathfrak{n} = \bigoplus_{j=1}^s \mathfrak{g}_j / \mathfrak{g}_{j+1}$ be the associated graded Lie algebra. The *symbol* of g is the map X into its simply connected group for which $(\log X)_j$ is the degree- j homogeneous part of $\log g$ projected to grade j . Symbol extraction respects products, is unchanged by parameter translation, and commutes with integer dilation. These statements follow by retaining the terms whose parameter degree equals their Lie layer in the Baker–Campbell–Hausdorff formula.

A symbol is *slow* if its coefficient of t^α has size at most $\mathcal{B}(p)T^{-\alpha}$. A symbol is *rational* if all its logarithmic coefficients have a common denominator at most $\mathcal{B}(p)$; their numerators are unrestricted. Both classes are closed under products and inverses at bounded depth.

We distinguish this from a *fully slow* polynomial map on the box. In coordinates translated so that the box starts at the origin, a scalar or vector polynomial $q(t) = \sum_\alpha c_\alpha t^\alpha$ is fully slow at budget p if

$$\|c_\alpha\| \leq \mathcal{B}(p)T^{-\alpha} \quad \text{for every coefficient, including } \alpha = 0.$$

A group-valued polynomial a is fully slow when its polynomial $\log a$ satisfies these bounds in the declared logarithmic coordinates. Thus its constant logarithm is bounded by $\mathcal{B}(p)$, and the condition controls every lower-degree coefficient as well as the homogeneous symbol. For a translated box the definition is applied after translating the parameter coordinates; it always uses that box's actual side lengths.

Lemma A.5 (Separation). *Let U be a rational subspace of controlled height. Suppose $a + b \in U$, where $|a| \leq \mathcal{B}(p)T^{-\alpha}$, $|\alpha| > 0$, and b has bounded denominator. If $\min T_i \geq \mathcal{B}(p)$ is sufficiently large, then $a, b \in U$.*

Proof. Choose a rational matrix Q of controlled height with kernel U . The vector $Qb = -Qa$ has bounded denominator. A nonzero coordinate therefore has absolute value at least $\mathcal{B}(p)^{-1}$, whereas $|Qa| \leq \mathcal{B}(p) / \min T_i$. Taking the lower length cutoff sufficiently large forces $Qa = Qb = 0$. $\square$

Lemma A.6 (Compatible symbol splittings). *Suppose*

$$X = EPR = E'P'R',$$

where E, E' are slow, R, R' are rational, and P, P' take values in the same graded rational Lie subalgebra U of controlled height, with its logarithmic height included in p . Then $E^{-1}E'$ and $R'R^{-1}$ take values in U , for sufficiently large sides.

If a polynomial number of controlled graded rational Lie subalgebras $U^{(b)}$ each admit a splitting of X of this kind, then their intersection admits one, with the same quantitative convention.

Proof. For the first assertion put $A = E^{-1}E'$ and $D = R'R^{-1}$, so that $P = AP'D$. Work upwards in grade. Once the lower-grade logarithmic coefficients of A, D belong to U , every lower-grade commutator in this identity belongs to U . Its current-grade coefficients therefore say that a slow coefficient plus a rational coefficient belongs to U . Lemma A.5 completes the induction.

For the second assertion construct common slow and rational factors one grade at a time. At a given grade, reset the lower grades of each individual splitting to the already chosen common factors. The first assertion, applied in the truncated group, puts the required ratios inside $U^{(b)}$. Lifting those ratios by setting the unused logarithmic layers to zero lets one absorb them into the middle factor.

Let V be the current coefficient space, and let $q : V \rightarrow \bigoplus_b V/U^{(b)}$ be the simultaneous quotient map. The remaining coefficient array v satisfies $q(v) = a + b$, with a slow and b rational. Modulo the rational subspace $q(V)$, Lemma A.5 puts a and b separately in $q(V)$. A bounded rational section of q gives common slow and rational corrections; their removal puts the current coefficients in $\ker q = \bigcap_b U^{(b)}$. All matrices are solved jointly at each grade, and there are only s grades. $\square$

These symbolic splittings lift to actual polynomial maps. Lift the homogeneous logarithmic coefficients of the slow and rational symbols through fixed rational sections, assigning no other logarithmic terms; call the lifts e_0, r_0 . Thus $e_0(0) = r_0(0) = 1$. Choose $\kappa \in \Gamma$ so that $c = g(0)\kappa^{-1}$ is a bounded representative, and put

$$E = e_0 c, \quad b = c^{-1} e_0^{-1} g \kappa^{-1} r_0^{-1}, \quad R = r_0.$$

Then $g = EbR\kappa$ exactly and $b(0) = 1$. Multiplication by constants does not change a symbol, so the symbol of b is the middle symbol of the original splitting. Fixed-step BCH shows that E is fully slow, including its constant coefficient, while R has a bounded common denominator. All these assertions use the coordinates translated to the box origin. If the middle symbol takes values in U , then b is adapted to the refiltration

$$\mathfrak{h}_j = \{v \in \mathfrak{g}_j : v \bmod \mathfrak{g}_{j+1} \in U_j\}.$$

Commutator closure follows from that of U , and the middle map takes values in $H_1 = \exp \mathfrak{h}_1$. This statement does not bound the lower-degree coefficients of that middle map.

A.3 Gowers norms and the normalized square

Here is the precise normalization of the Gowers norms. Write $Cz = \bar{z}$. On a finite abelian group H ,

$$\|f\|_{U^j(H)}^{2^j} = \mathbb{E}_{x, h_1, \dots, h_j \in H} \prod_{\omega \in \{0,1\}^j} C^{|\omega|} f(x + \omega \cdot h).$$

For a finite nonempty integer box $Q \subseteq \mathbb{Z}^n$, put

$$C_j(Q) = \#\{(x, h_1, \dots, h_j) \in (\mathbb{Z}^n)^{j+1} : x + \omega \cdot h \in Q \text{ for every } \omega\}.$$

Writing f_0 for the zero extension of f to $\mathbb{Z}^n$, we fix

$$\|f\|_{U^j(Q)}^{2^j} = \frac{1}{C_j(Q)} \sum_{x, h_1, \dots, h_j \in \mathbb{Z}^n} \prod_{\omega \in \{0,1\}^j} C^{|\omega|} f_0(x + \omega \cdot h). \quad (\text{A.2})$$

Thus $\|1\|_{U^j(Q)} = 1$ and $\|f\|_{U^1(Q)} = |\mathbb{E}_Q f|$. For fixed j there are constants $0 < c_j < C'_j$ such that

$$c_j^n |Q|^{j+1} \leq C_j(Q) \leq (C'_j)^n |Q|^{j+1}.$$

The upper bound follows by choosing x and then bounded coordinate differences; for the lower bound restrict x to a central subbox and each h_i to a sufficiently small box. Bounded side lengths are covered by taking all corresponding increments zero. Therefore normalizing the same sum by $|Q|^{j+1}$ changes the norm by at most $\exp(O_j(n))$.

Embed Q , after translation, in $H = \prod_{i=1}^n \mathbb{Z}/(4T_i)\mathbb{Z}$, where T_i are its side lengths. All cube identities with vertices in Q are ordinary integer identities: a face discrepancy cannot be a nonzero multiple of the corresponding period. If $\tilde{f}$ is the zero extension in H , the exact comparison is

$$\|\tilde{f}\|_{U^j(H)}^{2^j} = \frac{C_j(Q)}{|H|^{j+1}} \|f\|_{U^j(Q)}^{2^j}. \quad (\text{A.3})$$

In particular the triangle and mixed Cauchy-Schwarz inequalities transfer with this fixed normalization.

For clarity, define $\Delta_h f(x) = f(x)\overline{f(x+h)}$. The derivative identity on a group uses the uniform average

$$\|f\|_{U^{j+1}(H)}^{2^{j+1}} = \mathbb{E}_{h \in H} \|\Delta_h f\|_{U^j(H)}^{2^j}.$$

On a box let $Q_h = Q \cap (Q - h)$. The corresponding exact identity has cube-count weights:

$$C_{j+1}(Q) \|f\|_{U^{j+1}(Q)}^{2^{j+1}} = \sum_{h: Q_h \neq \emptyset} C_j(Q_h) \|\Delta_h f\|_{U^j(Q_h)}^{2^j}. \quad (\text{A.4})$$

This is obtained by fixing the last increment in the unnormalized cube sum; in particular $\sum_h C_j(Q_h) = C_{j+1}(Q)$. We use either the ambient group identity or this weighted box identity, never an unweighted average of separately normalized overlap norms.

Theorem A.7 (Quasipolynomial inverse theorem). *Fix an integer $s \geq 1$. Let $p \geq 2$. A unit-bounded function on an interval whose U^{s+1} norm is at least e^{-p} correlates by at least $\mathcal{B}(p)^{-1}$ with a unit-bounded degree- s niltest of complexity $\mathcal{P}(p)$. The same conclusion holds on integer boxes and on products of cyclic groups, with the following dimension convention: for every fixed $A \geq 1$, dimension $n \leq (2+p)^A$ is permitted, and the constants and polynomial exponents may depend only on s, A . On a product of cyclic groups the niltest is evaluated in the coordinatewise integer representatives.*

The interval assertion is the inverse theorem of Leng, Sah and Sawhney [35, Theorem 1.2]; for $s = 1$ it follows from Fourier inversion and Parseval. The cited Lipschitz norm includes the supremum norm. Dividing its correlator by $\max(1, \|F\|_\infty)$ therefore makes the correlator unit-bounded at an admissible cost.

We first compare the coordinate conventions. In the same adapted Mal'cev basis of dimension d , let d_c be the source's right-invariant chain metric and let d_R be our right-invariant Riemannian metric. For $d \geq 1$,

$$d_c(x, y) \leq d_R(x, y) \leq d d_c(x, y).$$

For the second inequality, write xy^{-1} as an ordered product of coordinate exponentials. Traversing its factors from right to left gives a path of length at most the sum of the absolute coordinate values. Apply this to every link in a chain. For the first inequality, subdivide a smooth path into small displacements; the first-order coordinate sup norm is bounded by its Riemannian speed. Take successively the mesh limit and the infimum over paths. Both comparisons pass to quotient metrics. Fixed-step BCH gives polynomially many rational coefficients for the changes between ordered and exponential coordinates. Clearing their denominators in both directions supplies the lattice grids of Definition A.1 with polynomial logarithmic cost. No bound on the polynomial orbit coefficients is required.

Here are the local extensions of the interval theorem. Put $j = s + 1$, translate an integer box to $Q = \prod_{i=1}^n \{0, \dots, T_i - 1\}$, and set

$$b_i = 4T_i, \quad a_1 = 1, \quad a_i = \prod_{r < i} b_r, \quad L = \prod_i b_i, \quad \Phi(x) = 1 + \sum_i a_i x_i.$$

Then $\Phi(Q) \subseteq [L]$ and $|Q|/L = 4^{-n}$. If $\Phi(u) + \Phi(v) = \Phi(w) + \Phi(z)$, the coordinate discrepancies $c_i = u_i + v_i - w_i - z_i$ satisfy $|c_i| < b_i$ and $\sum_i a_i c_i = 0$. Reduction modulo b_1 , followed by division and repetition, gives $c_i = 0$ for every i . Thus this affine map preserves and reflects every two-face identity, and hence all additive cube identities. If $\tilde{f}$ equals f on $\Phi(Q)$ and is zero elsewhere in $[L]$, the unnormalized cube sums agree exactly. Consequently

$$\|\tilde{f}\|_{U^j([L])}^{2^j} = \frac{C_j(Q)}{C_j([L])} \|f\|_{U^j(Q)}^{2^j}.$$

The ratio is at least $\exp(-O_j(n))$ by the cube-count bounds. The interval input therefore has norm at least $\exp(-p - O_j(n))$. Pull its correlating orbit back along Φ . The correlation on Q is the interval correlation multiplied by $L/|Q|$, and the degree is unchanged. This proves the box assertion, including anisotropic boxes.

Finally let $H = \prod_{i=1}^n \mathbb{Z}/N_i\mathbb{Z}$. Partition each representative interval into $\min(4, N_i)$ balanced nonempty intervals. Each piece has diameter less than $N_i/3$ and length at least $N_i/8$. There are at most 4^n product boxes. The Gowers triangle inequality selects one, call it Q , such that

$$\|f1_Q\|_{U^j(H)} \geq 4^{-n} \|f\|_{U^j(H)}.$$

A two-face discrepancy with all vertices in this box has absolute value less than $2N_i/3$ in coordinate i . If it is zero modulo N_i , it is therefore zero as an integer. The group and integer cube sums agree, so

$$\|f1_Q\|_{U^j(H)}^{2^j} = \frac{C_j(Q)}{|H|^{j+1}} \|f|_Q\|_{U^j(Q)}^{2^j}.$$

In particular the restricted box norm is at least $e^{-p}4^{-n}$. The box result gives a cap-one representative-coordinate niltest T with $|\mathbb{E}_Q f \overline{T}| \geq \eta$, where $\eta \geq \exp(-\text{poly}_s(p+n))$. Put $\lambda = (|Q|/|H|)\eta \geq 8^{-n}\eta$.

We may insert a smooth selector without a lower bound on the cyclic moduli. For each nonsingleton cyclic coordinate, take the circle arc whose endpoints lie halfway between the grid points just inside and outside its selected interval. A $[0, 1]$ -valued Lipschitz approximation W_i with transition radius θ and Lipschitz constant $O(\theta^{-1})$ has discrete L^1 error $O(\theta)$. Indeed, if $\theta < 1/(2N_i)$ it agrees with the indicator on the grid; otherwise each transition interval contains $O(N_i\theta)$ grid points. For $N_i = 1$ take $W_i = 1$. The product selector $W = \prod_i W_i$ has error $O(n\theta)$. Choose θ so that this is at most $\lambda/4$. Then

$$|\mathbb{E}_H f \overline{WT}| \geq 3\lambda/4.$$

The selector is a degree-one niltest on an n -dimensional torus, and its logarithmic Lipschitz cost is polynomial in $p+n$. Thus WT has degree at most s , cap one, and the required complexity and correlation. This completes the local extension.

For a subgroup $H \leq G$, write $H^\Delta = \{(h, h) : h \in H\}$ for its diagonal subgroup in $G \times G$.

The relative-square construction and its diagonal quotient occur in Green and Tao [23, Proposition 7.2 and Lemma 7.4]. We record the normalization and its quantitative coordinate proof here so that the number of parameters and the group dimension remain explicit.

Lemma A.8 (Normalized square). *Let F be vertical of frequency η on G_s and let g be an adapted polynomial map. For a fixed integer shift h , after a constant right lattice translation and a bounded constant left translation of $g(t+h)$, the pair with $g(t)$ is polynomial in*

$$G_j^\square = G_j^\Delta(G_{j+1} \times G_{j+1}).$$

The function given by the product of the translated F and $\overline{F}$ descends by G_s^Δ . It is consequently a degree- $(s-1)$ niltest, of controlled complexity independent of the coefficients of g and of h .

Proof. The horizontal difference $g(t+h)g(t)^{-1} \bmod G_2$ is constant in t . Reduce it modulo the projected lattice. Thus choose $\gamma_h \in \Gamma$ and bounded $\epsilon_h \in G$ whose horizontal logarithms are m_h, e_h , respectively, and whose sum is this difference. The first path $\epsilon_h^{-1}g(t+h)\gamma_h^{-1}$ now has the same horizontal coordinate as the second path.

Put $d_h(t) = \epsilon_h^{-1}g(t+h)\gamma_h^{-1}g(t)^{-1}$. The logarithm of d_h , in original Lie layer j , has parameter degree at most $j-1$. In the finite Baker–Campbell–Hausdorff expansion, a term without a shift

difference or inserted constant cancels, while either operation lowers parameter degree relative to Lie layer by at least one. The factorization

$$(\epsilon_h^{-1}g(t+h)\gamma_h^{-1}, g(t)) = (d_h(t), 1)(g(t), g(t))$$

therefore proves polynomiality in $G_{\bullet}^{\square}$. Right multiplication by γ_h^{-1} leaves the original coset unchanged. The new function is $F(\epsilon_h x)\overline{F(y)}$; its Lipschitz bound depends on the bounded ϵ_h alone. Central equivariance cancels on G_s^{Δ} . The final filtration layer is that diagonal group, so quotienting reduces the degree. All structural and metric bounds follow from the preceding rational operations; only the coefficients of the new orbit can be large. $\square$

Lemma A.9 (Niltests detect Gowers uniformity). *If a unit-bounded function on a box of dimension $\mathcal{P}(p)$ correlates by at least e^{-p} with a degree- s niltest of complexity $\mathcal{P}(p)$, then its U^{s+1} norm is at least $\mathcal{B}(p)^{-1}$. The assertion also holds on cyclic groups, including products of a bounded number of translated tests in representative coordinates.*

The interval case is the converse inverse theorem of Leng, Sah and Sawhney [35, Lemma B.5]. We give the box argument to keep its dependence on the number of parameters explicit.

Proof. Induct on s , the case $s = 0$ being the definition of U^1 . Lemma A.4 selects a top mode T , which we may rescale to be unit-bounded, with $\rho := |\langle f, T \rangle_Q| \geq \mathcal{B}(p)^{-1}$. Put $a = f\overline{T}$ and $c_h = \mathbb{E}_{Q_h} \Delta_h a$. The exact identity is

$$\rho^2 = \sum_{h: Q_h \neq \emptyset} \frac{|Q_h|}{|Q|^2} c_h. \quad (\text{A.5})$$

There are at most $2^n|Q|$ shifts. With $\tau = 2^{-n-2}\rho^2$, overlaps of volume below $\tau|Q|$ contribute at most $\rho^2/4$ in absolute value. Discarding also $|c_h| < \rho^2/4$ leaves a total absolute contribution at least $\rho^2/2$. Each weight is at most $1/|Q|$, so at least $\rho^2|Q|/2$ shifts remain. Each retained overlap has volume at least $\tau|Q|$ and hence every side at least τ times its original side.

The product of the two mode values is the lower-degree niltest in Lemma A.8. Induction gives a lower bound $\mathcal{B}(p)^{-1}$ for $\|\Delta_h f\|_{U^s(Q_h)}$ on every retained overlap. Inserting their count and volume bounds into (A.4), and using the bounds for $C_s(Q_h)$, proves the assertion. All dimension factors are $\mathcal{B}(p)$.

In the cyclic case cut at every wrap point of the bounded number of translates and subdivide further into intervals of diameter less than $N/3$. This gives a bounded number of ordinary intervals. Select a piece retaining a controlled fraction of the correlation. Every cyclic cube supported on that piece has ordinary integer face identities, since a face discrepancy has magnitude less than $2N/3$ and is a multiple of N . Thus the integer and cyclic cube sums of its zero extension coincide, with the stated normalization factors. Finally a cutoff of an interval cannot convert a sufficiently small U^j norm, $j \geq 2$, into a large one at the present costs: smooth the cutoff with sufficiently small $\mathcal{B}(p)^{-1} L^1$ error and approximate it by a Fourier sum of $\mathcal{B}(p)$ total coefficient mass. Use character invariance and the triangle inequality, and bound the U^j norm of a unit-bounded error by its L^1 norm to the power $1/2^j$. This transfers the conclusion back to the uncut cyclic function. $\square$

B Dropping the top frequency of a biased symbol

A biased top-frequency niltest need not have a slowly varying orbit: large rational coefficients may remain. We separate these contributions at the level of the homogeneous symbol, leaving a structured factor whose top layer is annihilated by the tested frequency.

The antecedents are Leng's efficient equidistribution theorem [32, Theorem 4] and the degree-reduction results of Leng, Sah and Sawhney [35, Theorem 5.4 and Corollary 5.5]. Unequal side lengths already occur in Leng's theorem. The proof here supplies the symbol formulation with bounds uniform in a growing number of parameters, retaining each side length throughout.

Lemma B.1 (Step drop for symbols). *Fix $s \geq 1$. Let $F(g(t)\Gamma)$ be a degree- s niltest of complexity at most p , on a box with $n \leq p$ variables and sides T_i . Suppose F is vertical of top frequency η of height at most e^p , and*

$$|\mathbb{E}_t F(g(t)\Gamma)| \geq e^{-p}.$$

For $\min_i T_i \geq \mathcal{B}(p)$ sufficiently large, the symbol admits a factorization $X = EPR$, where E is slow, R is rational, and $\log P$ takes values in a graded rational Lie subalgebra W of height $\mathcal{B}(p)$ such that $\eta(W_s) = 0$.

The proof inducts on the fixed degree rather than the group dimension. Differentiation gives many biased normalized squares, to which the lower-degree statement applies. After fixing their rational data, we obtain slow-plus-rational relations for derivatives of one common symbol. A lattice argument recovers such relations for each coordinate derivative; the remaining argument removes their slow and rational parts grade by grade.

B.1 Recovering coordinate derivatives

The recovery step below involves the adjoint action of a symbol whose coefficients need not be bounded. Its identity diagonal blocks nevertheless force bounded vertical lattice directions to be determined by their horizontal coordinates. The following lemma makes this assertion quantitative at the actual, possibly unequal, side lengths.

Lemma B.2 (Lattice extraction). *Let $1 \leq n, m \leq \mathcal{P}(p)$, $0 \leq a \leq m$, $T_i > 0$, and $l, R, \delta^{-1} \leq \mathcal{B}(p)$, where l is a positive integer, $R \geq 1$ and $0 < \delta \leq 1$. Put $D_T = \text{diag}(T_1, \dots, T_n)$, and let L be a positive diagonal $m \times m$ matrix. Its first a entries equal 1, and every remaining entry is at least $T_* := \min_i T_i$. Let M be lower unitriangular, with identity on its first $a \times a$ block, and let $Y : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be any linear map. Suppose the lattice*

$$\Lambda = \{(D_T^{-1}h, L(Yh - Mr)) : h \in \mathbb{Z}^n, r \in l^{-1}\mathbb{Z}^m\} \quad (\text{B.1})$$

contains at least $\delta \prod_i T_i$ vectors of norm at most R with distinct h -coordinates. No size bound is imposed on M or Y . If $T_ \geq \mathcal{B}(p)$ is sufficiently large, there are a rational subspace $K \subseteq \mathbb{R}^a$ of height $\mathcal{B}(p)$ and linear maps $S, \mathcal{R} : K \rightarrow \mathbb{R}^m$ with*

$$S = MR, \quad S(k)_{\text{hor}} = \mathcal{R}(k)_{\text{hor}} = k, \quad |LS(k)| \leq \mathcal{B}(p)|k|. \quad (\text{B.2})$$

On a controlled rational basis of K , $\mathcal{R}$ has a common denominator $\mathcal{B}(p)$, with no numerator bound. Moreover

$$Ye_i = s_i + Mr_i + S(k_i), \quad |Ls_i| \leq \mathcal{B}(p)/T_i, \quad (\text{B.3})$$

where $k_i \in K$ and the r_i have a common denominator $\mathcal{B}(p)$.

Proof. The map defining (B.1) is invertible on $\mathbb{R}^{n+m}$, so Λ is a lattice. Let Z be the span of its vectors of norm at most R , and put $\Lambda_Z = \Lambda \cap Z$. If their h -projections lay in a proper subspace, one nonzero real linear equation would determine one coordinate from the others. The number of integer possibilities would be at most

$$(3R)^{n-1} \frac{\prod_i T_i}{T_*},$$

contrary to the hypothesis when T_* is sufficiently large. Thus Z projects onto $\mathbb{R}^n$.

Write $d = \dim Z$. There are d independent lattice vectors of norm at most R spanning Z . Their integral span has covering radius at most $dR/2$, by rounding real coefficients to integers. The larger lattice Λ_Z has no larger covering radius. Its Voronoi fundamental region is consequently contained in a ball of radius $dR/2$. Translate that region by the given bounded lattice vectors. The translates are disjoint up to null sets and are contained in a ball of radius $(1 + d/2)R$. Therefore

$$\text{covol}(\Lambda_Z) \leq C_* / \prod_i T_i, \quad C_* := \delta^{-1}(C(n+m)R)^{C(n+m)}, \quad (\text{B.4})$$

for an absolute constant C .

Let $V = Z \cap (\{0\} \times \mathbb{R}^m)$ and $\Lambda_V = \Lambda \cap V$. The projection in unscaled h -coordinates is a full-rank subgroup $H \leq \mathbb{Z}^n$; put $I = [\mathbb{Z}^n : H]$. The lattice Λ_V has full rank in V : the projection of a lattice basis of Λ_Z is an integer matrix of rank n in the unscaled h -coordinates, and its real kernel is spanned by rational, hence integral, kernel vectors. Orthogonally to V , the space Z is the graph of a linear map $A : \mathbb{R}^n \rightarrow V^\perp \subseteq \mathbb{R}^m$, whose input is the scaled coordinate $D_T^{-1}h$. Lift a basis of $D_T^{-1}H$ to Λ_Z and subtract its V -components when calculating the exterior product with a basis of Λ_V . This gives the exact identity

$$\text{covol}(\Lambda_Z) = \text{covol}(\Lambda_V) \frac{I}{\prod_i T_i} \sqrt{\det(\text{id} + A^*A)}.$$

Writing $E_A = \sqrt{\det(\text{id} + A^*A)}$, we obtain

$$\text{covol}(\Lambda_V) I E_A \leq C_*. \quad (\text{B.5})$$

Every singular value of $x \mapsto (x, Ax)$ is at least one, so $|(x, Ax)| \leq E_A |x|$.

Put $k = \dim V$, and write a basis of the vertical lattice, after changing signs if necessary, as the columns of LMR_b , where $R_b \in (l^{-1}\mathbb{Z})^{m \times k}$. Scan the rows of R_b from top to bottom, retaining a row exactly when it increases the span. Let J be the resulting k pivot rows. Every earlier nonpivot row is a linear combination of earlier pivot rows. Hence multiplication by the lower unitriangular matrix M changes the selected rows by a lower unitriangular row operation. In particular,

$$\det(MR_b)_J = \det(R_b)_J.$$

If w_j denotes the j th diagonal entry of L , Cauchy–Binet therefore gives

$$\text{covol}(\Lambda_V) \geq |\det(LMR_b)_J| = \left(\prod_{j \in J} w_j \right) |\det(R_b)_J| \geq l^{-k} \prod_{j \in J} w_j. \quad (\text{B.6})$$

If any pivot were nonhorizontal, the last expression would be at least $l^{-k}T_*$. This contradicts (B.5) once $T_* > C_* l^m$. All pivots are horizontal. Thus horizontal projection is injective on V and its image is a rational subspace $K \subseteq \mathbb{R}^a$. Equations (B.5)–(B.6) give

$$I, E_A \leq C_* l^k, \quad l^{-k} \leq \text{covol}(\Lambda_V) \leq C_*.$$

Let H_b be the horizontal rows of R_b . They are also the horizontal rows of LMR_b . Every maximal minor of H_b is bounded in absolute value by C_* , and belongs to $l^{-k}\mathbb{Z}$. Choose a nonzero maximal minor $(H_b)_{J_0}$. The columns of $H_b((H_b)_{J_0})^{-1}$ form a rational basis of K . Its entries are ratios of maximal minors, hence have controlled height. No short basis for Λ_V is required here.

The unique graph lift from K to V has volume expansion

$$\frac{\text{covol}(\Lambda_V)}{\text{covol}(H_b \mathbb{Z}^k)} \leq C_* l^k.$$

As its horizontal part is the identity, its singular values are at least one and this also bounds its norm. Denote the lift by $k_0 \mapsto LS(k_0)$, and define

$$\mathcal{R}(H_b((H_b)_{J_0})^{-1}u) = R_b((H_b)_{J_0})^{-1}u.$$

Then $S = M\mathcal{R}$. The matrix on the right has denominator dividing the nonzero integer $\det(l(H_b)_{J_0})$, whose magnitude is at most $C_* l^k$. This proves (B.2).

Finally $I\mathbb{Z}^n \subseteq H$. Lift Ie_i to Λ_Z and divide the lift by I . The resulting vector is

$$(T_i^{-1}e_i, L(Ye_i - Mr'_i/I)), \quad r'_i \in l^{-1}\mathbb{Z}^m.$$

Its component orthogonal to V has norm at most E_A/T_i . Write that component's vertical coordinate as LS_i , and the V -component as $LS(k_i)$. Taking $r_i = r'_i/I$ proves (B.3). Its denominator divides lI . Zero-dimensional lattices have covolume one; the same argument, with empty bases, covers $k = 0$. $\square$

B.2 Proof of the step-drop lemma

Proof. If $\eta = 0$, take $W = \mathfrak{n}$ and $E = R = 1$. A zero-variable symbol is also trivial, so assume that the box has at least one variable and that $\eta \neq 0$. We induct on s . If $s = 1$, the group is a vector group and verticality gives $F(x) = ce(\eta(x))$ with $|c| \leq e^p$. Choose a rational vector v of controlled height with $\eta(v) = 1$. Write $\log X(t) = \sum_i a_i t_i$. The product formula for the mean of a linear phase and the geometric-sum bound give integers q_i such that

$$|\eta(a_i) - q_i| \leq e^{2p}/(2T_i).$$

Indeed the product of the n normalized geometric sums has magnitude at least e^{-2p} , so each individual magnitude is at least e^{-2p} . Take slow coefficients $(\eta(a_i) - q_i)v$, rational coefficients $q_i v$, and middle coefficients in $\ker \eta$. This proves the base case.

The differentiated symbol. Assume $s \geq 2$. Apply the weighted-shift selection in the proof of Lemma A.9, after rescaling F to be unit-bounded. Together with Lemma A.8, it gives at least $\mathcal{B}(p)^{-1} \prod_i T_i$ distinct shifts h , with $|h_i| \leq T_i$, for which the normalized square test has a mean of magnitude at least $\mathcal{B}(p)^{-1}$ on its overlap, and whose individual side lengths are at least $\mathcal{B}(p)^{-1}T_i$.

Write $\mathfrak{n}_j = \mathfrak{g}_j/\mathfrak{g}_{j+1}$. The associated graded Lie algebra of the quotient square has diagonal layers $\mathfrak{n}_j$, $1 \leq j < s$, and relative layers $\mathfrak{n}_{j+1}$ in square degree j . The relative part is abelian in this associated graded algebra: the bracket of two relative inputs has one extra filtration degree. The diagonal acts on it by the ordinary adjoint action.

Use the normalization in Lemma A.8, writing

$$(\log X)_1(h) = e_h + m_h,$$

where e_h has controlled size and m_h lies in a fixed rational grid. For a symbol C write $Y_h(C) = (\partial_h C)C^{-1}$ for its right logarithmic derivative. In relative-then-diagonal semidirect coordinates, the square symbol is

$$(X_{\leq s-1}, Y_h(X) - e_h - \text{Ad}_X m_h). \tag{B.7}$$

Here the relative component in original layer j is its homogeneous parameter part of degree $j - 1$, and the horizontal part is zero. Here is a formal verification, including the case where the original filtration has no grading. Choose complements E_j of $\mathfrak{g}_{j+1}$ in $\mathfrak{g}_j$ and let δ_z act by z^j on E_j . The brackets

$$[u, v]_z = \delta_z[\delta_z^{-1}u, \delta_z^{-1}v]$$

are polynomial in z : a component in E_k of a bracket of E_i, E_j is multiplied by z^{k-i-j} , with $k \geq i + j$. At $z = 0$ this is the associated graded bracket. Write $\exp_z, \log_z$ for the finite Baker–Campbell–Hausdorff group with this bracket. The polynomial

$$P_z(t) = \delta_z \log g(t/z)$$

has constant term $P_0(t) = \log X(t)$. Similarly the scaled normalizing constants are $ze_h + O(z^2)$ and $zm_h + O(z^2)$. Here $O(z^2)$ denotes terms divisible by z^2 in a formal polynomial, without any bound on their coefficients. The scaled relative product is therefore

$$\exp_z(-ze_h + O(z^2)) \exp_z(P_z(t + zh)) \exp_z(-zm_h + O(z^2)) \exp_z(-P_z(t)).$$

When the shift and inserted constants are zero, the middle product equals the identity for every z . Consequently the first-order variations of the bracket itself and of the subleading coefficients in P_z cancel identically. The coefficient of z depends only on the displayed shift and insertions, evaluated in the graded group at $z = 0$. It can thus be calculated over the dual numbers $\mathbb{R}[\varepsilon]/(\varepsilon^2)$, where the exact identity is

$$\exp(-\varepsilon e_h)X(t + \varepsilon h) \exp(-\varepsilon m_h)X(t)^{-1} = \exp(\varepsilon(Y_h(X) - e_h - \text{Ad}_X m_h)).$$

Indeed $X(t + \varepsilon h)X(t)^{-1} = \exp(\varepsilon Y_h(X))$, conjugation of the right insertion gives $\text{Ad}_X m_h$, and brackets of two ε terms vanish. In original layer j , the coefficient of z is exactly the homogeneous parameter part of degree $j - 1$ of the relative logarithm, since δ_z multiplies that monomial by $z^{j-(j-1)}$. This proves (B.7). The same calculation gives the semidirect multiplication rule $y + \text{Ad}_X y'$ under a preceding diagonal coordinate X .

A common fast subgroup. Apply Lemma A.4 to the quotient square. One top mode retains bias $\mathcal{B}(p)^{-1}$, and its restriction to the relative copy of $\mathfrak{n}_s$ is η . The induction hypothesis factors its symbol into slow, fast and rational factors. Every resulting fast subalgebra has polynomial dimension and rational height $\mathcal{B}(p)$. There are only $\mathcal{B}(p)$ possibilities, since a rational basis is specified by polynomially many bounded-height entries. Retain a set of $\mathcal{B}(p)^{-1} \prod_i T_i$ shifts with the same fast subalgebra $\tilde{U}$.

Let U be its diagonal projection, with $U_s = \mathfrak{n}_s$, and let $V_j \subseteq \mathfrak{n}_j$ be its kernel in the relative part, setting $V_1 = 0$. Then

$$\eta(V_s) = 0, \quad [U_i, V_j] \subseteq V_{i+j}. \quad (\text{B.8})$$

All these are controlled rational spaces. For each retained shift, denote the diagonal splitting by $X = A_h B_h D_h$ through grade $s - 1$. Fix one retained shift and denote its outside diagonal factors by A, D . Lemma A.6 puts $A^{-1}A_h$ and $D_h D^{-1}$ in U . Lift their logarithms to $\tilde{U}$ by a bounded rational section of the diagonal projection. The first lift is slow, and the second is rational. Absorbing them into the middle factor makes its diagonal equal to

$$B = A^{-1} X D^{-1}$$

for every retained shift. Extend A, D to original grade s with zero logarithmic component there; the same formula defines the full symbol B and puts it in U .

In a subgroup of a semidirect product with abelian kernel, two elements with the same diagonal have relative coordinates equal modulo the kernel. The fast relative coordinate, denoted Z_B , is therefore common modulo V . Comparing the relative coordinates in (B.7) and using

$$Y_h(ABD) = Y_h(A) + \text{Ad}_A Y_h(B) + \text{Ad}_{AB} Y_h(D)$$

gives

$$Y_h(B) = s_h + \text{Ad}_B r_h + Z_B \pmod{V}. \quad (\text{B.9})$$

For clarity, if u_h, Z_h, v_h are the three relative factors, the two sides before solving for $Y_h(B)$ are

$$Y_h(X) - e_h - \text{Ad}_X m_h = u_h + \text{Ad}_A Z_h + \text{Ad}_{AB} v_h.$$

Thus one can take

$$\begin{aligned} s_h &= \text{Ad}_{A^{-1}}(e_h + u_h - Y_h(A)), \\ r_h &= v_h + \text{Ad}_D m_h - Y_h(D). \end{aligned}$$

The coefficients of s_h in original layer j , parameter degree $j - 1$, satisfy $\mathcal{B}(p)T^{-\alpha}$ bounds. This uses $|h_i| \leq T_i$ when differentiating the slow symbol A . The coefficients of r_h have bounded denominator; integrality of h is used for $Y_h(D)$. These claims include the horizontal constant terms. Pigeonhole the denominators and subtract one retained instance of (B.9) from every other instance. The term Z_B disappears, the number of distinct shift differences is unchanged up to one, and the side and coefficient bounds only change by controlled factors.

Extraction of a horizontal kernel. Let $\mathcal{M}$ be the direct sum, over $j = 1, \dots, s$, of the coefficient spaces of degree- $(j - 1)$ homogeneous polynomials with values in $\mathfrak{n}_j/V_j$. Choose controlled rational coordinates. The operator $M = \text{Ad}_B$ on $\mathcal{M}$ is block lower unitriangular by Lie layer, and is well defined by (B.8). The linear map $h \mapsto Y_h(B)$ takes values in $\mathcal{M}$. Let L_T multiply a coefficient of t^α by T^α . The horizontal scales are one and every other scale is at least $\min_i T_i$.

The subtracted identities supply the bounded lattice vectors required by Lemma B.2, with a common grid for the r_h ; changing from the chosen rational coordinates to a rectangular grid merely enlarges its denominator. Consequently there are a controlled rational $K \subseteq \mathfrak{n}_1$, linear maps S, R on K , and, for every coordinate derivative,

$$\begin{aligned} S &= \text{Ad}_B R \pmod{V}, & S(k)_1 &= R(k)_1 = k, \\ |L_T S(k)| &\leq \mathcal{B}(p)|k|, \\ Y_i(B) &= s_i + \text{Ad}_B r_i + S(k_i) \pmod{V}, & |L_T s_i| &\leq \mathcal{B}(p)/T_i. \end{aligned} \quad (\text{B.10})$$

Here R on a controlled rational basis and all the r_i have bounded common denominator, and $k_i \in K$ need not be rational. We henceforth use these identities in the quotient coefficient space, so equalities of components are always interpreted modulo V_j .

The target subalgebra. We have obtained the derivative system (B.10), with B taking values in U . Define

$$W_1 = \{v \in K \cap U_1 : [v, K] \subseteq V_2\}, \quad W_j = V_j \cap U_j \quad (2 \leq j \leq s).$$

These controlled rational spaces form a graded Lie subalgebra. Indeed $[W_1, W_1] \subseteq V_2 \cap U_2$, while every other required bracket lies in the appropriate U layer and in V by (B.8). Also $\eta(W_s) = 0$. It remains to remove slow and rational factors from B until its logarithmic coefficients lie in W .

Write $P_j = (\log B)_j$, updating this notation after each removal. First we put P_1 in $U_1 \cap K$. At stage $j \geq 2$ we arrange $[P_{j-1}, K] \subseteq V_j$ and then put P_j in $U_j \cap V_j$. The accompanying derivative terms must vanish at the same time so that the next layer reduces to an ordinary coefficient equation. We begin with the two operations that preserve the derivative system.

Removing slow and rational derivatives. The system (B.10) is preserved when $B = aB'b$, where a is slow and b rational, both in U . The correct transformed quantities are

$$\begin{aligned} S' &= \text{Ad}_{a^{-1}} S, & R' &= \text{Ad}_b R, \\ s'_i &= \text{Ad}_{a^{-1}}(s_i - Y_i(a)), & r'_i &= \text{Ad}_b r_i - Y_i(b). \end{aligned} \quad (\text{B.11})$$

The identities follow from the product rule for Y_i . Their quantitative bounds persist: after coefficient scaling by T^α , multiplication by a slow coefficient only multiplies norms by a controlled factor; rational products preserve bounded denominators. Also, for $k_s, k_r \in K$, the replacement

$$s_i \leftarrow s_i - S(k_s), \quad r_i \leftarrow r_i - R(k_r), \quad k_i \leftarrow k_i + k_s + k_r \quad (\text{B.12})$$

preserves the system. Its bounds persist if $|k_s| \leq \mathcal{B}(p)/T_i$ and k_r has bounded denominator.

The coefficient of t_i in P_1 is $s_{i,1} + r_{i,1} + k_i \in U_1$. Modulo the rational space $U_1 + K$, separation puts $s_{i,1}$ and $r_{i,1}$ separately in that space. Use a controlled rational decomposition into a U_1 part and a K part. Absorb the K parts by (B.12), and remove the U_1 parts by factors a, b whose logarithms have only layer one. We have arranged

$$P_1 \in U_1 \cap K, \quad s_{i,1} = r_{i,1} = 0. \quad (\text{B.13})$$

We now process $j = 2, \dots, s$. At the start of stage j , suppose the following hold:

$$\begin{aligned} P_d &\in U_d \cap V_d & (2 \leq d < j), \\ [P_d, K] &\subseteq V_{d+1} & (1 \leq d \leq j-2), \\ S_d &= R_d = 0 & (2 \leq d < j), \\ s_{i,d} &= r_{i,d} = 0 & (1 \leq d < j). \end{aligned}$$

The last two lines are statements in the quotient by V . Expanding $S = \text{Ad}_B R$ in layer j gives

$$[P_{j-1}, k] = S_j(k) - R_j(k) \pmod{V_j}, \quad k \in K. \quad (\text{B.14})$$

Indeed all lower terms of R except its horizontal part vanish modulo V , and every term in $\exp(\text{ad } P)k$ with an inner bracket from a layer at most $j-2$ belongs to V .

Let $E = U_{j-1} \cap V_{j-1}$, or $E = U_1 \cap K$ when $j = 2$. On a fixed controlled rational basis (k_b) of K , define the rational linear map

$$Q_j : E \longrightarrow \bigoplus_b \mathfrak{n}_j / V_j, \quad v \longmapsto ([v, k_b])_b.$$

The right side of (B.14) is a slow coefficient array minus a rational one. Project modulo $Q_j(E)$, and apply Lemma A.5 coefficient by coefficient. Each part lies separately in the image. A controlled rational section provides homogeneous polynomials a_{j-1}, b_{j-1} with values in E such that

$$Q_j(a_{j-1}) = (S_j(k_b))_b, \quad Q_j(b_{j-1}) = -(R_j(k_b))_b.$$

The first is slow and the second rational. Remove their exponentials from the left and right of B . Formula (B.11) now gives

$$[P_{j-1}, K] \subseteq V_j, \quad S_j = R_j = 0.$$

Lower vanishing conditions persist. When $j = 2$, the operation creates horizontal parts $-\partial_i a_1$ and $-\partial_i b_1$ in s_i, r_i ; both lie in K . Absorb them by (B.12). Since the new S_2, R_2 are already zero modulo V_2 , this restores (B.13) without changing their vanishing. For $j > 2$, the analogous new parts lie in V_{j-1} and are already zero in the quotient.

At this point the right logarithmic derivative satisfies

$$Y_i(B)_j = \partial_i P_j \pmod{V_j}.$$

To verify this, expand it as a finite sum of nested brackets of P and $\partial_i P$. A derivative in a preceding layer at least two lies in V . A horizontal derivative lies in K , and its innermost bracket lies in V by the bracket conditions just established. Subsequent brackets preserve V by (B.8). Likewise $(\text{Ad}_B r_i)_j = r_{i,j}$ modulo V_j , since all its lower components are already zero there. Thus (B.10) reduces to

$$\partial_i P_j = s_{i,j} + r_{i,j} \pmod{V_j}.$$

For every monomial t^β of degree j , choose one i with $\beta_i > 0$ and divide the corresponding derivative coefficient by β_i . This expresses its coefficient, modulo V_j , as a slow part of size $\mathcal{B}(p)T^{-\beta}$ and a rational part. Because P_j takes values in U_j , separation modulo $U_j + V_j$ and a bounded rational section allow both parts to be lifted into U_j . Remove their exponential lifts at layer j . We have obtained $P_j \in U_j \cap V_j$. The derivative identity then says $s_{i,j} + r_{i,j} = 0$ modulo V_j ; separation, now at scale $T_i T^\alpha$, makes the two terms vanish separately. This completes stage j .

The final logarithmic coefficients now lie in the target algebra W , whose top layer is annihilated by η . There were only s stages, with joint rational linear algebra at each stage. All corrections therefore combine into slow and rational outside factors with the stated bounds. Together with the original A, D , these give the required factorization of X . $\square$

C Shift comparison

Shift comparison lowers the degree of a positive test after multiplying by a translate of an arbitrary nonnegative function. The degree-one case uses Bohr-set almost-periodicity. At higher degree we recover the test from two fixed finite lists and apply the induction outside one exceptional set of shifts. The lower-degree test may then depend on the shift.

Put $G = \mathbb{Z}/N\mathbb{Z}$, with its uniform probability measure, and evaluate nilsequences on the representatives $0, \dots, N-1$. For $d \geq 0$ and $q \geq 2$, let $\mathcal{T}_d(q)$ denote the $[0, 1]$ -valued niltests of degree at most d and complexity at most q . Thus $\mathcal{T}_0(q)$ consists of constants in $[0, 1]$.

Theorem C.1 (Shift comparison). *For fixed $d \geq 1$ and $0 < \epsilon < 1$, there is a constant $C = C(d, \epsilon)$ such that the following holds. Suppose that $p \geq 2$, $N \geq \exp((2+p)^C)$ is odd, and $0 \leq f, g, J \leq e^p$. Set $P = (2+p)^C$. If*

$$\mathbb{E}_n(f - g)(n)T(n) \leq e^{-P} \quad (T \in \mathcal{T}_d(P)), \tag{C.1}$$

then there is a set $E \subseteq G$, depending on f, g, J , with $|E| \leq e^{-P}N$, such that

$$\mathbb{E}_n(f - (1 + \epsilon)g)(n)J(n+h)T'(n) \leq e^{-P} \quad (h \notin E, T' \in \mathcal{T}_{d-1}(p)). \tag{C.2}$$

The exceptional set is chosen before T' .

The proof is given in Proposition C.6 for $d = 1$ and Proposition C.12 for higher degrees. The latter uses the structural results of Section A and Lemma B.1.

C.1 The degree-one argument

For a finite set Ω of characters of G and $0 < w \leq 1$, put

$$B(\Omega, w) = \{x \in G : |\chi(x) - 1| \leq w \text{ for every } \chi \in \Omega\}.$$

We use these symmetric common-width Bohr sets, retaining Ω and $w(B) = w$ as part of the data; their rank is $D = |\Omega|$. The whole group has the presentation with one trivial character and width one. A regular Bohr set B of rank $D \geq 1$ satisfies

$$|B_{1+t}| = (1 + O(D|t|))|B| \quad (|t| \leq c/D).$$

Here dilating a Bohr set means dilating its defining widths. The usual covering argument gives $|B|/N \geq (w(B)/10)^D$ and $|B_\rho| \geq (\rho/C)^{CD}|B|$ for $0 < \rho < 1$; a regular dilate exists between any two scales whose ratio is two. Also, if $v \in B_\rho$, regularity gives

$$\|\mu_B(\cdot - v) - \mu_B\|_1 \ll D\rho, \quad (\text{C.3})$$

Here $\mu_B(x) = 1_B(x)/|B|$ is the probability mass function on B , and the norm in (C.3) is the unnormalized sum over G . Regular dilation and the following almost-periodicity input preserve the common-width convention. Widths will be tracked explicitly; adding characters does not permit a width-independent estimate for the relative size of the resulting Bohr set.

The convolution almost-periodicity method originates with Croot and Sisask [12]. We use the following radius-sensitive Bohr-set form of Schoen and Sisask.

Lemma C.2 (Almost-periodicity with a width bound). *Fix $0 < \eta < 1/2$. If S, S' are regular of rank $D \geq 1$, $A_1 \subseteq S$, $A_2 \subseteq s_0 + S'$, both relative densities are at least e^{-q} with $q \geq 1$, and $|K| \leq 2|S|$, then, for independent uniform $a_i \in A_i$, the probability that $a_1 - a_2$ belongs to K changes by at most η after adding an independent uniform element of a regular Bohr set $D_* \subseteq S'$ with*

$$\text{rank } D_* \leq D + C_\eta(1 + q)^4, \quad w(D_*) \geq w(S') \exp(-C_\eta(1 + q + \log(2 + D))). \quad (\text{C.4})$$

Proof. We apply Schoen–Sisask [43, Theorem 5.4], which uses unnormalized counting convolution. In its notation, a regular Bohr set of rank d and radius ρ , a subset of relative density greater than σ , a sumset bound $|A + S_0| \leq K_1|A|$, and $\beta = |M|/|L| \leq 1$ give a regular sub-Bohr set of rank at most $d + d'$ and radius at least $c\rho\eta\sqrt{\beta}/(d^2d')$, where

$$d' \leq C\eta^{-2} \log^2(2/(\eta\beta)) \log(2/\beta) \log(2K_1) + C \log(1/\sigma).$$

The error in its three-fold convolution is at most $\eta|A||M|$.

Write α_i for the two relative densities, and take a regular $B_0 = S'_\tau$ with $\tau \asymp 1/D$ sufficiently small. Use B_0 as the ambient Bohr set, and set $A = -A_2$, $S_0 = B_0$. The increment set S_0 has density one in B_0 , so $\sigma = 1/2$ is valid; regularity gives $|A + S_0| \leq 2|S'| \leq (2/\alpha_2)|A|$. If $|K| \geq |A_1|$, take $M = A_1$, $L = -K$; then $\alpha_1/2 \leq \beta \leq 1$. If $\eta|A_1| \leq |K| < |A_1|$, interchange M and L ; then $\eta \leq \beta \leq 1$, and the error is still at most $\eta|A_1||A_2|$. In either case $d' \leq C_\eta(1 + q)^4$, and the radius lower bound implies (C.4). The counting convolution at zero, divided by $|A_1||A_2|$, is the probability in the statement; symmetry of D_* gives the asserted smoothing estimate. Finally, if $|K| < \eta|A_1|$, every translate of the difference law assigns K probability at most $|K|/|A_1| < \eta$. Taking $D_* = B_0$ proves this case as well. $\square$

Only fixed $\eta > 0$ is used here. All convolution normalizations in the following local calculation are stated explicitly.

Lemma C.3 (A bilinear reduction). *Let $a = f - (1 + \epsilon)g$. In degree one it suffices to prove*

$$\mathbb{E}_{x,y} H(x)W(y)a(x+y) \leq e^{-4p} \quad (0 \leq H, W \leq 1). \quad (\text{C.5})$$

Proof. If the conclusion fails on a set E of more than $e^{-p}N$ shifts, then $\mathbb{E}_n a(n)J(n+h) > e^{-p}$ for every $h \in E$: a constant test cannot create a positive violation where this inequality fails. Take $H = e^{-p}J$ and $W(y) = 1_E(-y)$. The left side of (C.5) equals $e^{-p}\mathbb{E}_h 1_E(h)\mathbb{E}_n a(n)J(n+h) > e^{-3p}$, a contradiction. $\square$

The refinement to be proved. To prove the bilinear estimate, we decompose its two independent averages into smaller Bohr cells. The difficulty is that a restriction can concentrate either weight. We measure both weights by a concave potential, so that concentration and genuine smoothing each pay for the further restrictions they require.

Write $a = f - (1 + \epsilon)g$. A node consists of a regular Bohr set B , two translates $x_B + B, y_B + B$, and functions $H, W \in [0, 1]$ supported on these translates. Define

$$I = \mathbb{E}_{x \in x_B + B, y \in y_B + B} H(x)W(y)a(x+y), \quad u = \mathbb{E}_{x \in x_B + B} H(x), \quad v = \mathbb{E}_{y \in y_B + B} W(y), \quad \Phi = (uv)^{1/4}.$$

Every node average uses the full size of its cell, even after a weight has been restricted and extended by zero. In particular, $I \leq e^p uv \leq e^p \Phi$. The original bilinear average is the node with $B = G$.

We shall construct a refinement with successor nodes indexed by ν and nonnegative coefficients c_ν such that

$$I \leq \sum_{\nu} c_{\nu} I_{\nu} + \varepsilon_{\text{err}} \Phi, \quad \sum_{\nu} c_{\nu} \Phi_{\nu} \leq (1 - \gamma) \Phi. \quad (\text{C.6})$$

Here $\gamma > 0$ depends only on ϵ , and ε_{err} can be any sufficiently small fixed multiple of e^{-10p} . Terms which need no further refinement are included in the error in the first inequality. The same construction controls the geometry: if D and w are the parent's stored rank and width, every successor has

$$D_{\nu} \leq D + (2 + p)^{A\epsilon}, \quad \log \frac{w}{w_{\nu}} \leq (2 + p + D)^{B\epsilon}. \quad (\text{C.7})$$

We need this refinement only for descendants of the root during $R = \lceil c_{\text{depth}} p \rceil$ rounds. The geometric bounds above forecast polynomial upper bounds for the ranks and $\log(1/w)$ of these nodes and their intermediate dilates. Choose P to cover those bounds; the refinement is then uniform over every center and pair of weights encountered in the run.

There are two reasons to stop part of a node. If either cell mean is tiny, its entire positive contribution is small relative to Φ . If a localized convolution is close to 1 in a sufficiently high moment, positive comparison with a smooth Bohr test makes that localized contribution small. The other parts become successors. A highly concentrated restriction is paid for by the potential lost when that restriction is removed. In the absence of such concentration, a large convolution moment yields almost-periods and forces a fixed decrease of the average potential. The following two elementary estimates quantify these last two mechanisms.

Lemma C.4 (Potential estimates). *Put $\theta = 1/4$. The following estimates hold.*

1. *If $u, v > 0$, $0 \leq bu_1 \leq u$, and $u_1 \geq Ku$, then*

$$(uv)^{\theta} - ((u - bu_1)v)^{\theta} \geq \theta K^{1-\theta} b(u_1 v)^{\theta}. \quad (\text{C.8})$$

2. *If $0 \leq X \leq M$, $\mathbb{E}X \leq 1$, and $\mathbb{E}X^2 \geq 1 + c$, where $M \geq 1$, then*

$$\mathbb{E}\sqrt{X} \leq 1 - \frac{c}{2(1 + \sqrt{M})^2}. \quad (\text{C.9})$$

Consequently, if $X, Y \geq 0$, $\mathbb{E}X, \mathbb{E}Y \leq 1$, and X satisfies these additional bounds, then $\mathbb{E}(XY)^{1/4} \leq 1 - c/[4(1 + \sqrt{M})^2]$.

Proof. Concavity gives $1 - (1 - t)^\theta \geq \theta t$ for $0 \leq t \leq 1$. Apply this with $t = bu_1/u$, and use $(u_1/u)^{1-\theta} \geq K^{1-\theta}$. For the second assertion,

$$\frac{1+x}{2} - \sqrt{x} = \frac{(x-1)^2}{2(1+\sqrt{x})^2} \geq \frac{(x-1)^2}{2(1+\sqrt{M})^2}.$$

Moreover $\mathbb{E}(X-1)^2 \geq c$ and $(1 + \mathbb{E}X)/2 \leq 1$. The final assertion follows from Cauchy–Schwarz and $\mathbb{E}\sqrt{Y} \leq 1$. $\square$

The local moment argument and the weighted dependent-random-choice step below adapt the moment and sifting methods of Kelley and Meka [30, Sections 4–5], as developed in Bloom and Sisask [8, Sections 2.1–2.2]. Here they are used inside the refinement of two weighted Bohr cells.

Lemma C.5 (Local convolution alternatives). *Let L, S be symmetric regular Bohr sets, with S a sufficiently small dilate of L . Let $F, G \geq 0$ be supported on L , have $|L|^{-1} \sum F = |L|^{-1} \sum G = 1$, and satisfy $F, G \leq M$. Define*

$$(F * G)(t) = \frac{1}{|L|} \sum_x F(x)G(t-x), \quad (F \circ G)(t) = \frac{1}{|L|} \sum_x F(x+t)G(x).$$

Write ν_+ for the law of $s_1 + s_2$ and ν_- for the law of $s_1 - s_2$, where s_1, s_2 are independent and uniform on S . For fixed $\delta > 0$, and integer $m \geq 1$, either

$$\|F * G - 1\|_{L^{2m}(\nu_+)} \leq \delta, \tag{C.10}$$

or, for one of $D = F, G$ and an integer $m' \leq C_\delta m$,

$$\|D \circ D\|_{L^{2m'}(\nu_-)} \geq 1 + c_\delta. \tag{C.11}$$

It is enough that the regularity errors, multiplied by M , be sufficiently small in terms of δ .

Proof. For real A, B and every integer $j \geq 1$, expansion and Cauchy–Schwarz give

$$\|A * B\|_{L^{2j}(\nu_+)} \leq \|A \circ A\|_{L^{2j}(\nu_-)}^{1/2} \|B \circ B\|_{L^{2j}(\nu_-)}^{1/2}. \tag{C.12}$$

Indeed, after expanding the $2j$ th power, apply Cauchy–Schwarz to the two factors $\mathbb{E}_{s \in S} \prod_{i=1}^{2j} A(s - t_i)$ and $\mathbb{E}_{s \in S} \prod_{i=1}^{2j} B(s + t_i)$, summing each t_i over the group with weight $|L|^{-1}$. The same expansion shows

$$\mathbb{E}_{\nu_-} (A \circ A)^j \geq 0 \quad (j \geq 1), \tag{C.13}$$

because the resulting expression is a sum of squares.

Put $F_1 = F - 1_L$. Regularity and the cap M show, uniformly on $S + S$ and $S - S$, respectively, that

$$1_L * G = 1 + o_\delta(1), \quad F \circ F = 1 + F_1 \circ F_1 + o_\delta(1).$$

If (C.10) fails, then $\|F_1 * G\|_{L^{2m}(\nu_+)} \geq \delta/2$. Either $\|G \circ G\|_{L^{2m}(\nu_-)} \geq 2$, or (C.12) gives $\|F_1 \circ F_1\|_{L^{2m}(\nu_-)} \geq \delta^2/8$. In the latter case put $X = F_1 \circ F_1$ and choose $m' = Rm$ with a sufficiently large integer $R = R(\delta)$. All terms in the binomial expansion have nonnegative expectations by (C.13); hence

$$\mathbb{E}(1 + X)^{2m'} \geq \binom{2m'}{2m} \mathbb{E}X^{2m} \geq (R\delta^2/8)^{2m}.$$

Choose R so that $(R\delta^2/8)^{1/R} > 1 + 2c_\delta$, and absorb the uniform regularity error using the triangle inequality. $\square$

Proposition C.6 (Degree-one shift comparison). *Theorem C.1 holds for $d = 1$.*

Proof. We construct the refinement in (C.6), with the geometry in (C.7) and the full-cell averaging convention fixed above. Put $\theta = 1/4$.

Fix K so large that $\theta K^{1-\theta} > 2$, and choose $c_0 > 0$ with $(c_0 K)^\theta \leq 1/2$. Choose $\kappa = e^{-c_{\text{cut}} p}$ and $\zeta = \exp(-(2+p)C'_\epsilon)$ sufficiently small. Choose $c_{\text{cut}} = c_{\text{cut}}(\epsilon)$ first to control the tiny-node error below. The moment constant and the number of rounds will be chosen afterwards; all are independent of the node rank. If $\min(u, v) < \kappa$, stop the node, since its positive contribution is at most

$$e^p uv \leq e^p \kappa^{1-\theta} \Phi. \quad (\text{C.14})$$

Here and throughout this proof, the required scale choices are regular dilates whose logarithmic width losses are polynomial in $p + D$, where $D = \text{rank } B$. Successor ranks are at most $D + (2+p)C''_\epsilon$. To make the set of admissible successor shapes precise, retain $w = w(B)$ and allow all regular presentations C satisfying

$$C \subseteq B_{c\zeta/(2+D)}, \quad \text{rank } C \leq D + (2+p)A_\epsilon, \quad w(C) \geq w \exp(-(2+p+D)B_\epsilon).$$

Choose A_ϵ from (C.4), which will be applied with $q = O_\epsilon(p^2)$. Choose B_ϵ after the exponent defining ζ , large enough to cover that almost-periodicity width loss and the bounded number of regular dilations within one refinement. These choices precede peeling. The exponent defining P is chosen last.

Removing concentration. While a translate of an admissible C has mean $u_1 > Ku$ of H , remove the whole slice from H . Set $b = |C|/|B|$. The removed contribution is represented, up to $O(e^p \zeta b)$, by nodes of shape C with coefficient b : keep the first center fixed, average the second center in its parent cell, and add an independent uniform element of C on that side. This error bound is (C.3), applied to the second parent cell. The second means average to at most v . Consequently their total potential is at most $b(u_1 v)^\theta$. Inequality (C.8) pays a factor greater than two for this potential. Perform the same operation if W has such a slice.

Stop immediately if either current mean becomes smaller than κ . At every other removal, the removed mass is at least $K\kappa b_{\min} > 0$, where the volume bound and admissibility give

$$b \geq |C|/N \geq b_{\min} := \left(\frac{w \exp(-(2+p+D)B_\epsilon)}{10} \right)^{D+(2+p)A_\epsilon}.$$

Thus peeling terminates in finitely many steps. Moreover its errors are summable without multiplying by the number of steps: since $u_1, v \geq \kappa$, each b is at most $\kappa^{-2\theta} b(u_1 v)^\theta$, and (C.8) telescopes. It remains to treat a non-tiny remainder for which every admissible translate has H -mean at most Ku and W -mean at most Kv .

Matching two local scales. Choose tiny regular dilates $S \ll L \ll B$. Choose x_0, y_0 independently and uniformly in the two parent cells, take $r \in L$ and $s = s_1 + s_2$ with $s_1, s_2 \in S$ independently uniform, and put

$$x = x_0 + r, \quad y = y_0 - r + s.$$

The resulting average differs from the parent average by $O(e^p \zeta)$, by (C.3) twice. Restrict both weights to $x_0 + L$ and $y_0 + L$; the further error is $O(e^p \zeta)$ because s is on a much smaller scale than L . Let u_0, v_0 be the resulting means on L . The absence of concentration gives $u_0 \leq Ku$, $v_0 \leq Kv$; independence of x_0, y_0 and concavity give

$$\mathbb{E}_{x_0, y_0} (u_0 v_0)^\theta \leq (uv)^\theta. \quad (\text{C.15})$$

One can return to independent cells of any sufficiently small common shape C by adding independent $t_1, t_2 \in C$ to x, y . In (r, s) coordinates this translates the law by $(t_1, t_1 + t_2)$, so its total variation changes by $O(\zeta)$. If U, V are the new cell means, then, exactly,

$$\mathbb{E}_{r,s}U \leq u_0, \quad \mathbb{E}_{r,s}V \leq v_0, \quad \mathbb{E}_{r,s}(UV)^\theta \leq (u_0v_0)^\theta. \quad (\text{C.16})$$

For example, for each fixed s, t_2 , summing the zero-extended second weight over $-L + s + t_2$ is at most its sum over the whole group, namely $|L|v_0$. This proves the second inequality; the first is identical and the last follows by Cauchy–Schwarz and concavity.

If $u_0 < c_0u$ or $v_0 < c_0v$, return to independent cells in this way. The total potential of these centers is at most $(c_0K)^\theta \Phi \leq \Phi/2$. For the other centers $u_0, v_0 \geq c_0\kappa$. Fix one and normalize its truncated weights to functions F, G on L of mean one. Their caps are at most $M = (c_0\kappa)^{-1}$. Choose $m = \lceil c_{\text{mom}}p \rceil$ with $c_{\text{mom}} = c_{\text{mom}}(\epsilon, c_{\text{cut}})$ large enough, and apply Lemma C.5 with sufficiently small fixed $\delta = \delta(\epsilon)$.

The flat alternative terminates a matched term. Its integral, divided by u_0v_0 , is $\mathbb{E}_{s \sim \nu_+} a(x_0 + y_0 + s)(F * G)(s)$. Outside a set of ν_+ -measure at most 2^{-2m} , $|F * G - 1| \leq 2\delta$. Since $0 \leq F * G \leq M$, the exceptional part costs at most $O(e^p(M+1)2^{-2m})$. On the complementary set use

$$a(F * G) \leq (1 + 2\delta)f - (1 + \epsilon)(1 - 2\delta)g.$$

Choose δ so that $(1 + \epsilon)(1 - 2\delta) \geq 1 + 2\delta$.

Hypothesis (C.1) gives $\mathbb{E}_{s \sim \nu_+} (f - g)(x_0 + y_0 + s) \leq \eta$ for any prescribed $\eta = \exp(-(2+p)^{O_\epsilon(1)})$, after taking P sufficiently large. Here is the needed approximation justification. Approximate a translated regular Bohr indicator from above and below by $[0, 1]$ -valued Lipschitz functions on its defining torus, with transition width $\rho w(S)$. Their difference has normalized group mean at most $O(D\rho)|S|/N$, and their Lipschitz constants are at most $O(D/(\rho w(S)))$. These are degree-one niltests: the torus has lattice $\mathbb{Z}^D$ and the orbit is given by the defining characters. With $\alpha = |S|/N$, the hypothesis and the signed comparison yield

$$\mathbb{E}_{s \in S} (f - g)(z + s) \leq \alpha^{-1}e^{-P} + O(e^p D\rho) \quad (z \in G).$$

The width and rank bookkeeping below bounds both $\log(1/w(S))$ and $\log(1/\alpha)$ polynomially in p . Thus ρ can be chosen with polynomial logarithmic loss to make the last display at most η , with all test complexities at most P . Averaging over the second copy of S proves the assertion for ν_+ . Thus this node has positive contribution at most an arbitrarily small multiple of $e^{-10p}(u_0v_0)^\theta$.

The unbalanced alternative contracts the potential. Let Q be the function supplied by (C.11), and write $q = 2m' = O_\epsilon(p)$. Choose a tiny regular shrink S' of S . Regularity, with error small after multiplication by M^{2q} , supplies an $s_0 \in S$ such that

$$\mathbb{E}_{a \in S, b \in s_0 + S'} (Q \circ Q)(a - b)^q \geq (1 + c)^q$$

for a fixed $c > 0$. Set $K_0 = \{Q \circ Q \geq 1 + c/2\} \cap (S - s_0 - S')$. Regularity gives $|K_0| \leq 2|S|$.

For dependent random choice with these real weights, let L^+ be a slight dilate of L containing $L - a$ for every $a \in S \cup (s_0 + S')$. The scale separation and regularity ensure $|L^+| \leq 2|L|$. Independently sample $t_1, \dots, t_q$ uniformly from L^+ . Given these samples, include each $a \in S$ in A_1 , and each $b \in s_0 + S'$ in A_2 , independently, with probabilities

$$\prod_{j=1}^q \frac{Q(t_j + a)}{M}, \quad \prod_{j=1}^q \frac{Q(t_j + b)}{M}.$$

The expected pair weight is $(|L|/(|L^+|M^2))^q(Q \circ Q)(a-b)^q$. The contribution of pairs outside K_0 , relative to the lower bound for all pairs, is at most $((1+c/2)/(1+c))^q$. By increasing m by a fixed factor, this is smaller than any chosen fixed constant. To justify the simultaneous density and bad-pair bounds, write $Z = |A_1||A_2|/(|S||S'|)$ and let Z_{bad} be the same normalized count restricted to differences outside K_0 . The preceding estimates give $\mathbb{E}Z \geq \exp(-O_\epsilon(p^2))$ and $\mathbb{E}Z_{\text{bad}} \leq \delta_* \mathbb{E}Z$, where δ_* can be made any sufficiently small fixed constant by increasing the fixed multiple defining q . Given a sufficiently large fixed C_* , take $C_*\delta_* \leq 1/2$. Some outcome then has

$$Z - C_*Z_{\text{bad}} \geq \frac{1}{2}\mathbb{E}Z.$$

It follows that $Z \geq \exp(-O_\epsilon(p^2))$ and $Z_{\text{bad}}/Z \leq C_*^{-1}$. Each relative density is at least Z , since both are at most one. Thus both densities have the required lower bound and an arbitrarily high fixed fraction of the differences lie in K_0 .

Apply (C.4) to these sets. Choose a tiny regular shrink C of D_* so that adding independent opposite shifts from C changes the uniform law on D_* by $O(\zeta)$. If d is uniform on D_* and c_1, c_2 are independent and uniform on C , the almost-periodicity estimate therefore gives

$$\mathbb{E}_{a \in A_1, b \in A_2, d, c_1, c_2} 1_{K_0}(a - b + d + c_1 - c_2) \geq 1 - \eta_0,$$

where $\eta_0 > 0$ is any sufficiently small fixed constant. For a finite set A , write $P_A v(x) = |A|^{-1} \sum_{t \in A} v(x+t)$; these averaging operators do not use the $|L|^{-1}$ normalization of function-function convolution. Since $Q \circ Q \geq 1 + c/2$ on K_0 and is nonnegative everywhere, the last display implies

$$\max_x P_C P_{-C}(Q \circ Q)(x) \geq (1 + c/2)(1 - \eta_0) \geq 1 + c'$$

for a fixed $c' > 0$. This function is the autocorrelation of $P_C Q$, with sums normalized by $|L|$. Cauchy-Schwarz puts its maximum at zero. Its mass outside L contributes $O(M^2\zeta)$ to the second moment, so $\mathbb{E}_{r \in L}(P_C Q)(r)^2 \geq 1 + c'/2$. The same estimate holds on the reflected and slightly translated law used by the second weight, by regularity.

Return to independent cells of shape C , with centers $x_0 + r$ and $y_0 - r + s$, where r is uniform on L and $s \sim \nu_+$ independently. Their means satisfy

$$U/u_0 = P_C F(r), \quad V/v_0 = P_C G(-r + s).$$

This shape is admissible: its rank increase is polynomial in p , and its logarithmic width loss relative to B is polynomial in $p + D$, by (C.4) and the intervening regular dilations. The absence of concentration gives $U/u_0, V/v_0 \leq K/c_0$. Equations (C.16) give means at most one, and one of these variables has second moment at least $1 + c'/4$. Lemma C.4 therefore contracts the local potential by a fixed factor $q_0 < 1$.

Summing the refinements. Let A be the matched potential from small-ratio centers and B_1 that from the other centers. We have $A \leq \Phi/2$ and $A + B_1 \leq \Phi$, so the surviving potential is at most $A + q_0 B_1 \leq (1 + q_0)\Phi/2$. For completeness, let S_{peel} be the total weighted potential already extracted by peeling, and R_{rem} the potential of its remainder. Telescoping gives $2S_{\text{peel}} + R_{\text{rem}} \leq \Phi$. The matching argument contracts the surviving remainder potential by $\lambda = (1 + q_0)/2 < 1$; a tiny remainder has no successors. Hence the total successor potential is at most

$$S_{\text{peel}} + \lambda R_{\text{rem}} \leq \max(1/2, \lambda)\Phi.$$

This gives a fixed $\gamma = \gamma(\epsilon) > 0$ such that all successors of a node have total weighted potential at most $(1 - \gamma)\Phi$. Their integrals bound its integral from above up to $c_\epsilon e^{-10p}\Phi$, where c_ϵ can be made

arbitrarily small. Indeed all errors before a tiny-mean cutoff are bounded by fixed powers of $e^p \kappa^{-1}$ times ζ ; the extracted errors have already been summed by (C.8).

This proves (C.6) with $\varepsilon_{\text{err}} = c_\epsilon e^{-10p}$; admissibility of the successor shapes gives (C.7).

Apply the refinement for $R = \lceil c_{\text{depth}} p \rceil$ rounds. After r rounds, the total weighted potential of the remaining nodes is at most $(1 - \gamma)^r \Phi_{\text{root}}$. Hence the sum of all refinement errors is at most $\varepsilon_{\text{err}} \Phi_{\text{root}} / \gamma$, and the remaining positive integral is at most

$$e^p (1 - \gamma)^R \Phi_{\text{root}}.$$

Since $\Phi_{\text{root}} \leq 1$, choosing c_{depth} sufficiently large and then the permitted error constant sufficiently small makes their sum less than e^{-4p} .

It remains to justify one comparison budget for the whole refinement. Along every branch, (C.7) gives

$$D_{r+1} \leq D_r + (2 + p)^{A_\epsilon}, \quad \log(1/w_{r+1}) \leq \log(1/w_r) + (2 + p + D_r)^{B_\epsilon}.$$

Starting with $D_0 = 1, w_0 = 1$, both D_r and $\log(1/w_r)$ are polynomial in p for $r \leq R$. The same is true of the intermediate dilates used inside a round. The Bohr volume bound $|B|/N \geq (w(B)/10)^D$ consequently gives a polynomial bound for $\log(N/|B|)$. One choice $P = (2 + p)^C$ therefore covers the smooth-test approximations, their required errors, and the size cutoff at every node. This proves the bilinear estimate and, by Lemma C.3, degree-one shift comparison. $\square$

C.2 The finite-list reduction

For the induction from degree r to degree $r + 1$, a violating positive test of degree r may be chosen separately at every shift. We cannot take a union bound over all these tests. Instead we will recover them, on a sufficiently large set of shifts, from two fixed finite lists of positive tests: one observed at n , the other at $n + h$. The remaining factor may depend on h , but has degree at most $r - 1$. Degree- r shift comparison is uniform over this remaining factor, so only the two fixed lists enter the union bound. The next lemma gives the precise reduction. The rest of the proof constructs its factorization.

Lemma C.7 (Why fixed lists suffice). *Assume shift comparison at degree $r \geq 1$. Let $0 < \epsilon < 1$, $0 \leq f, g, J \leq e^p$, and suppose that N is odd and $N \geq \exp(P)$ for the sufficiently large polynomial budget P in the conclusion. Suppose, for a set $H \subseteq G$ of density at least σ , that positive tests T_h satisfy*

$$\mathbb{E}_n(f - (1 + \epsilon)g)(n)J(n + h)T_h(n) > \delta.$$

Suppose there are fixed families $A_i, B_j \in \mathcal{T}_d(Q)$, $d = r + 1$, of sizes at most F , and, for each $h \in H$, an approximation

$$T_h(n) = \sum_{\ell=1}^M A_{i(\ell, h)}(n) B_{j(\ell, h)}(n + h) C_{\ell, h}(n) + e_h(n), \quad C_{\ell, h} \in \mathcal{T}_{r-1}(Q),$$

with $\mathbb{E}|e_h| \leq \delta e^{-2p}/6$. Here all factors take values in $[0, 1]$. If

$$Q + \log(F + M + 2) + \log(1/\sigma) + \log(1/\delta) \leq (2 + p)^{O_{d, \epsilon}(1)},$$

then (C.1), with P a sufficiently large polynomial in p , rules out these violations.

Proof. Choose $q \geq \max(p, Q)$ polynomially large so that $F^2 e^{-q} < \sigma$ and $M e^{-q} < \delta/2$. For each fixed i, j , apply degree- r shift comparison to fA_i, gA_i, JB_j , with multiplicative slack $\epsilon/2$ and budget q . Its hypothesis follows from (C.1): multiplying a degree- r test by A_i gives a degree- d positive test, of complexity bounded polynomially in the required budgets. There are at most F^2 fixed pairs, so some $h \in H$ lies outside every exceptional set. Uniformity over all lower tests then bounds each summand by e^{-q} . Replacing $f - (1 + \epsilon/2)g$ by $f - (1 + \epsilon)g$ only decreases the integral. Finally $|(f - (1 + \epsilon)g)J| \leq 3e^{2p}$, so the approximation error costs at most $\delta/2$. This contradicts the displayed violation. $\square$

C.3 Modeling and a common anchor

To construct the fixed lists, we first approximate each input by a finite linear combination of niltests in a seminorm adapted to the shifted comparisons. The separation argument is a form of the Hahn–Banach method for deriving decompositions from inverse theorems [19, Section 3.2]; the explicit finite list is then obtained by sampling. On a finite probability space, the extended dual of a seminorm X is

$$X^*(\psi) = \sup_{X(v) \leq 1} |\langle v, \psi \rangle| \in [0, \infty].$$

It is finite precisely when ψ annihilates $\ker X$. A family is *balanced* if it is closed under multiplication by complex scalars of modulus at most one.

Lemma C.8 (Modeling from a seminorm detector). *Let Ω be a finite probability space, let X be a seminorm on $\mathbb{C}^\Omega$ with $X(v) \leq K\|v\|_1$, and let $\mathcal{S}$ be a nonempty compact, balanced collection of functions of supremum at most one. Fix $M, K \geq 1$, $\beta > 0$, and $0 < \tau < 1$. Suppose that every ψ satisfying*

$$\|\psi\|_\infty \leq K/\tau, \quad X(\psi) \geq \tau/M^2$$

has $|\langle \psi, Q \rangle| \geq \beta$ for some $Q \in \mathcal{S}$. Then every b with $\|b\|_\infty \leq M$ has a representation

$$b = \sum_{i=1}^{\ell} c_i Q_i + e, \quad \sum_i |c_i| \leq 2/\beta, \quad X(e) \leq 2\tau,$$

with $\ell \leq 1 + 4K^2/(\beta^2\tau^2)$.

Proof. Write $R = 2/\beta$. Suppose that b does not belong to $R \operatorname{conv}(\mathcal{S}) + \{e : X(e) \leq \tau\}$. Separation in the underlying finite-dimensional real vector space gives a nonzero ψ with

$$\operatorname{Re}\langle b, \psi \rangle > R \sup_{Q \in \mathcal{S}} |\langle Q, \psi \rangle| + \tau X^*(\psi).$$

The right side is positive; normalize it to one. Then $X^*(\psi) \leq 1/\tau$. Since $X(v) \leq K\|v\|_1$, testing on point masses yields $\|\psi\|_\infty \leq K/\tau$. Also $\|\psi\|_2 > 1/M$, by the strict lower bound on $\operatorname{Re}\langle b, \psi \rangle$. Duality gives $X(\psi)X^*(\psi) \geq \|\psi\|_2^2$ and consequently $X(\psi) > \tau/M^2$. The detection hypothesis contradicts $R \sup_Q |\langle Q, \psi \rangle| \leq 1$. The indicated convex sum therefore exists with $X(e) \leq \tau$. Its structural part has ℓ^1 coefficient norm at most R . Sample ℓ independent terms with probabilities proportional to their coefficient magnitudes, absorbing their phases into $\mathcal{S}$. The expected squared L^2 error of the empirical average is at most R^2/ℓ . Take $\ell \geq R^2 K^2/\tau^2$ and use $X \leq K\|\cdot\|_2$. $\square$

In the application to shift comparison,

$$X(e) = \mathbb{E}_h \sup_L |\mathbb{E}_n e(n) B(n+h) L(n)|, \tag{C.17}$$

where L ranges over complex, unit-bounded degree- r niltests of controlled complexity and all their cyclic translates. Thus the family includes $n \mapsto L_0(n+t)$, with addition taken in G , even when its representative path wraps. The supremum allows a different test and translation at each h . It is a seminorm and $X(e) \leq \|B\|_\infty \|e\|_1$. Choosing an approximately maximizing test and aligning its phase, then applying Cauchy–Schwarz at $n+h$, gives

$$X(e)^2 \leq \|B\|_\infty^2 \mathbb{E}_{h,k} \left| \mathbb{E}_n e(n) \overline{e(n+h-k)} L_h(n) \overline{L_k(n+h-k)} \right|. \quad (\text{C.18})$$

The product in (C.18) contains at most two translated representative niltests. Lemma A.9 therefore applies uniformly, and the derivative identity gives a quantitative U^{r+2} detector. The quasipolynomial inverse theorem [35] then supplies the structural tests for Lemma C.8. These structural tests are ordinary degree- $(r+1)$ niltests; only the testing family has been enlarged. The translated family permits the change of variable $m = n+h$ when modeling the second input: $L_h(n)$ becomes $L_h(m-h)$. The supremum handles these translations directly, without a union bound over them. All cutoff errors in the converse estimate are chosen below the detection threshold.

Lemma C.9 (A common anchor). *Let H be a finite nonempty set. For $1 \leq j \leq m$ and $h \in H$, let $v_{h,j}$ be complex functions of supremum at most one on finite probability spaces Ω_j . Suppose $|\mathbb{E} v_{h,j}| \geq \delta > 0$ for all h, j . There exist $h_0 \in H$ and $H' \subseteq H$ with $|H'| \geq \delta^{2m} |H|/2$ such that*

$$|\mathbb{E} v_{h,j} \overline{v_{h_0,j}}| \geq \delta^{2m}/2 \quad (h \in H', 1 \leq j \leq m).$$

Proof. Multiply each $v_{h,j}$ by a unit scalar so its mean is real and nonnegative, and tensor the functions in their independent variables to form V_h on $\prod_j \Omega_j$. Then $\mathbb{E} V_h \geq \delta^m$ and $\|V_h\|_2 \leq 1$. Consequently

$$\mathbb{E}_{h,k} \text{Re} \langle V_h, V_k \rangle = \|\mathbb{E}_h V_h\|_2^2 \geq \delta^{2m}.$$

Choose h_0 with the same lower bound after averaging only h . At least a $\delta^{2m}/2$ fraction have real inner product at least $\delta^{2m}/2$. For these h , the absolute inner product is a product of m numbers in $[0, 1]$, so each factor is at least $\delta^{2m}/2$. Removing the aligning phases changes no absolute value. $\square$

C.4 Recovering tests from two fixed observations

The next lemma records precisely what positive discretization can do after the requisite geometry has been established.

Lemma C.10 (Positive discretization). *Fix finite families of nonnegative functions $\{A_i\}_{i \in I}$ and $\{B_j\}_{j \in J}$ on G , each summing to one. For each shift h , suppose that, outside a set E_h ,*

$$T_h(n) = F_h(z_h(n), \xi_h(n)),$$

where $0 \leq F_h \leq 1$ and F_h is L -Lipschitz in its second variable. Suppose that whenever $A_i(n)B_j(n+h)$ and $A_i(n')B_j(n'+h)$ are positive, with $n, n' \notin E_h$, one has $d(\xi_h(n), \xi_h(n')) \leq \rho$. Then there are functions $C_{ij,h}(n) = F_h(z_h(n), \xi_{ij,h}(n))$ such that

$$\left| T_h(n) - \sum_{i,j} A_i(n) B_j(n+h) C_{ij,h}(n) \right| \leq L\rho \quad (n \notin E_h). \quad (\text{C.19})$$

An empty cell pair may be assigned $C_{ij,h} = 0$. If the first-variable sections of F_h are bounded Lipschitz functions on a degree- $(r-1)$ nilmanifold and z_h is a polynomial path on it, then every $C_{ij,h}$ is a degree- $(r-1)$ niltest with the corresponding uniform complexity.

Proof. For every nonempty cell pair select a point $n_{ij,h} \notin E_h$ where its weight is positive and put $\xi_{ij,h} = \xi_h(n_{ij,h})$. For each contributing pair the replacement error is at most $L\rho$. Sum against the nonnegative weights, whose sum is one. The final assertion follows by restricting F_h to the chosen second coordinate. $\square$

Lemma C.11 (Uniform recovery after right normalization). *Fix controlled rational nilpotent groups, lattices, and a rational subgroup V of their product. Fix a controlled lattice Λ_V on V . There are fixed fine covers of the input lattices with the following property. Fix bounded left translations and arbitrary right translations by elements of the original input lattices, and apply these same translations to both of two tuples. Suppose that both normalized tuples belong to V . If the two original tuples are within ρ on the fixed input covers, then their normalized images are within $\mathcal{B}(p)\rho$ on V/Λ_V , provided $\rho \leq \mathcal{B}(p)^{-1}$. The covers and constants do not depend on the right translations. The left translations may instead differ by at most ρ in each coordinate; the same conclusion holds after enlarging its constant. One may additionally require compatibility with right translation by rational elements having a fixed common denominator $q \leq \mathcal{B}(p)$.*

Proof. First choose a fine ambient product lattice Λ_0 such that $V \cap \Lambda_0 \subseteq \Lambda_V$. The restriction of the ambient quotient metric has a controlled local inverse on $V/(V \cap \Lambda_0)$. To see this directly, reduce both points using $V \cap \Lambda_0$ to bounded representatives. An ambient lattice element witnessing closeness is then bounded. Its logarithm lies in a controlled rational grid; if its distance from the rational subspace $\log V$ is below the rational separation bound, it lies in that subspace exactly. It therefore belongs to $V \cap \Lambda_0$. Controlled coordinate changes compare the ambient and intrinsic small distances. This is also the metric argument in Lemma A.3.

Choose each input cover Λ inside the corresponding factor of Λ_0 , normal in the original lattice Γ . Here is an explicit construction. If $\log \Gamma \subseteq l^{-1}\mathbb{Z}^D$ and $m_0\mathbb{Z}^D \subseteq \log \Lambda_0$, choose a controlled integer m divisible by lm_0 , and set

$$\Lambda = \langle \gamma^m : \gamma \in \Gamma \rangle.$$

Every generator lies in Λ_0 because $\log(\gamma^m) = m \log \gamma \in m_0\mathbb{Z}^D$. This subgroup is characteristic in Γ . A Mal'cev normal form with D coordinates gives $[\Gamma : \Lambda] \leq m^D$ by reducing each exponent modulo m ; powers of the Mal'cev generators give controlled grids for Λ . Thus all logarithmic index and coordinate costs remain polynomial.

For one coordinate, let $a = \epsilon^{-1}x\gamma^{-1}$ and $a' = \epsilon'^{-1}x'\gamma^{-1}$, with ϵ bounded and $\gamma \in \Gamma$. If $d_G(x\lambda, x') \leq \rho$ for $\lambda \in \Lambda$, then

$$a(\gamma\lambda\gamma^{-1}) = \epsilon^{-1}x\lambda\gamma^{-1}, \quad d_G(a(\gamma\lambda\gamma^{-1}), a') \leq \text{Lip}(L_{\epsilon^{-1}})\rho.$$

Normality puts $\gamma\lambda\gamma^{-1}$ in $\Lambda \subseteq \Lambda_0$. Thus normalized ambient distances are controlled uniformly in γ . Apply the preceding local inverse on V . If the bounded left normalizers are ϵ and ϵ' , right invariance gives

$$d_G(\epsilon^{-1}x'\gamma^{-1}, (\epsilon')^{-1}x'\gamma^{-1}) = d_G(\epsilon^{-1}, (\epsilon')^{-1}).$$

Inversion is controlled on the bounded set of normalizers, so their variation adds at most $\mathcal{B}(p)\rho$ before applying the local inverse. No orbit representative enters this error bound.

For rational right translations, the adjoint formula has a common denominator bounded in terms of q and the structural data; its numerators need not be bounded. Choose the earlier congruence strides sufficiently divisible to force the required conjugates into Λ_0 . This requires no bound on those numerators. Again normal refinement inside Γ handles arbitrary lattice right translations at the same time. $\square$

Proposition C.12 (Higher-degree propagation). *Lemmas A.3, A.6, A.8, A.9, and B.1, together with the quasipolynomial inverse theorem, imply the propagation principle in Theorem C.1 for every fixed degree.*

Proof. Induct on d , using Proposition C.6 for the base case. Set $r = d - 1 \geq 1$ and $a = f - (1 + \epsilon)g$. Suppose that a set of more than $e^{-p}N$ shifts has a violating positive degree- r test T_h . All losses in this proof have logarithm polynomial in p ; the degree and the multiplicative slack are fixed. A precision called “sufficiently small” below is at least $\exp(-\mathcal{P}(p))$. No list is obtained by discretizing the functions f, g, J or the coefficients of a polynomial orbit.

Structural models at the two sites. For the seminorm (C.17), (C.18) implies the needed detector as follows. A lower bound $X(e) \geq \tau$, after normalizing its cap, gives a proportion $\mathcal{B}(p)^{-1}$ of pairs (h, k) with derivative correlation at least $\mathcal{B}(p)^{-1}$. The product of translated tests in that correlation is covered by Lemma A.9, including representative wraps. Thus a proportion $\mathcal{B}(p)^{-1}$ of differences $h - k$ has derivative U^{r+1} norm at least $\mathcal{B}(p)^{-1}$: the difference is uniform, and each fixed difference has exactly N pairs. The Gowers derivative identity gives $\|e\|_{U^{r+2}} \geq \mathcal{B}(p)^{-1}$. The quasipolynomial inverse theorem therefore gives a degree- d correlator of polynomial logarithmic complexity and correlation $\mathcal{B}(p)^{-1}$.

Apply Lemma C.8 with these detectors. To fit its compactness hypothesis, take the closure in the finite-dimensional space of values on G of the balanced family of allowed niltests. The detector still supplies an actual niltest. After obtaining the finite model, approximate its finitely many closure points by actual niltests to any prescribed further L^2 accuracy. The cap bound $X \leq \|B\|_\infty \cdot \|\cdot\|_2$ controls this additional error. Thus a bounded function at either site has, in the corresponding seminorm, a model with list length and ℓ^1 coefficient sum at most $\mathcal{B}(p)$.

Positive Fourier projection and fixed partners. Write $T_h = F_h(P_h(n)\Gamma_h)$. Truncate its top-torus Fourier expansion at sufficiently high controlled precision. Declare a frequency significant if the corresponding vertical term has correlation against $a(n)J(n+h)$ larger in absolute value than a sufficiently small controlled threshold. Average F_h over the connected annihilator of the rational span of the significant frequencies. This is a positive averaging operator, so the resulting test still lies in $[0, 1]$. It changes the tested integral by at most the Fourier truncation error plus the number of discarded truncated frequencies times the significance threshold. Hence a positive violation of size $\mathcal{B}(p)^{-1}$ remains. Replace F_h and T_h by these averages and retain their notation. Each significant vertical mode is unchanged by this replacement.

Choose a basis $\eta_{h,1}, \dots, \eta_{h,m}$ from the significant frequencies, with corresponding vertical terms $U_{h,j}$. After pigeonholing, m is fixed and at most $\mathcal{P}(p)$. If $m = 0$, the averaged test factors through the degree- $(r-1)$ quotient, and degree- r shift comparison already rules out the violations after a polynomial enlargement of its budget. Thus assume $m \geq 1$. Model a with other input J . Next model J for each fixed structural term of that first model. Choose the second precision after the first list and its coefficient bound have been fixed. For the latter application, write $u = n + h$ and use the shift $-h$: the fixed structural factor is evaluated at $u - h$, and the varying test $U_{h,j}(u - h)$ belongs to the translated testing family in (C.17). Markov’s inequality applied to the modeling seminorm errors removes only a negligible proportion of shifts, uniformly for every selected $U_{h,j}$. Expanding the two models shows that each remaining significant term has partners Q_j, Q'_j with

$$\left| \mathbb{E}_n U_{h,j}(n) Q_j(n) Q'_j(n+h) \right| \geq \mathcal{B}(p)^{-1}. \quad (\text{C.20})$$

The factors can be scaled to have cap one. Take vertical modes in the top-torus expansions of both partners, and pigeonhole these choices for all j simultaneously. The number of choices is $\mathcal{B}(p)^{\mathcal{P}(p)} = \mathcal{B}(p)$. We retain a set of shifts of density $\mathcal{B}(p)^{-1}$ on which all partner maps and functions are fixed.

An anchor and a common rational factorization. Apply Lemma C.9 to the functions in (C.20), tensoring in j . For one fixed h_0 , a further set of density $\mathcal{B}(p)^{-1}$ satisfies, for every j ,

$$\left| \mathbb{E}_n U_{h,j}(n) \overline{U_{h_0,j}(n)} |Q_j(n)|^2 Q'_j(n+h) \overline{Q'_j(n+h_0)} \right| \geq \mathcal{B}(p)^{-1}. \quad (\text{C.21})$$

For each j , divide the representative interval at the two wrap points in the shifted partners. Intervals shorter than a sufficiently small $\mathcal{B}(p)^{-1}N$ have negligible total contribution. One remaining interval has both relative length and normalized bias at least $\mathcal{B}(p)^{-1}$. On it, extend the partners using their corresponding ordinary integer shifts. Pigeonhole the finitely many anchor branches.

Write $Q_j(n) = V_j(g_j(n)\Gamma_j)$ on G_j/Γ_j . Since Q_j is vertical at degree d , $|Q_j|^2$ is a function of the degree- r quotient path $a_j(n) = g_j(n)(G_j)_d$ in $\overline{G_j} = G_j/(G_j)_d$. Let $s_{h,j}(n)$ be the normalized square path supplied by Lemma A.8 for the last two factors in (C.21), and let S_j be its degree- r quotient group. Taking top Fourier modes of the functions on $\overline{G_j}$ and S_j selects a pair that retains bias $\mathcal{B}(p)^{-1}$ on the selected interval. Their frequencies, denoted by $\alpha_{h,j}$ and $\beta_{h,j}$, give a unit-bounded degree- r test on

$$D_{h,j}(n) = (P_h(n), P_{h_0}(n), a_j(n), s_{h,j}(n))$$

with combined top frequency

$$\Lambda_{h,j} = (\eta_{h,j}, -\eta_{h_0,j}, \alpha_{h,j}, \beta_{h,j}).$$

Apply Lemma B.1 to this biased test, obtaining a graded rational subalgebra $W_{h,j}$ whose top layer is annihilated by $\Lambda_{h,j}$.

Writing G_h for the ambient group of P_h , use the common degree- r product map

$$(P_h, Z_h) = (P_h, P_{h_0}, (a_j, s_{h,j})_{j=1}^m) \quad \text{into} \quad G_h \times G_{h_0} \times \prod_{j=1}^m (\overline{G_j} \times S_j).$$

Let ρ_j be its coordinate projection to the four factors of $D_{h,j}$. The inverse image of $W_{h,j}$ under the induced graded projection $\text{gr}(\rho_j)$ admits a splitting of the total symbol: lift the slow and rational symbols by placing identities in the omitted coordinates. The selected intervals may differ with j , but parameter translation leaves the symbol unchanged, and their lengths are at least $\mathcal{B}(p)^{-1}N$. Thus all these splittings have slow bounds at scale N , after enlarging the budget. Lemma A.6 gives a simultaneous splitting through the intersection U_h of these inverse images. Its lift is an actual factorization

$$(P_h, Z_h) = E_h b_h R_h \kappa_h, \quad b_h(n) \in H_h, \quad (\text{C.22})$$

with its refiltration $(H_h)_i$; E_h is fully slow, R_h is rational with bounded common denominator, and κ_h is a constant lattice element.

Write π for the projection to Z_h . The total filtration has degree r , so $(H_h)_r = \exp((U_h)_r)$, and each $\Lambda_{h,j} \circ d\rho_j$ annihilates $(U_h)_r$. On

$$K_h = \ker((H_h)_r \longrightarrow \pi((H_h)_r)), \quad (\text{C.23})$$

all coordinates except the G_h coordinate are the identity. The frequency equation therefore says that every $\eta_{h,j}$ annihilates the logarithm of the first-coordinate projection of K_h . All frequencies of the projected positive function belong to their rational span, so the function is invariant under K_h . This kernel is the exponential of a rational subspace of the central top layer, hence is connected, rational, and central in the total group.

Fixing rational data before choosing covers. The partner groups and their square quotients are fixed. The controlled rational group data for the first factor, the subgroups H_h , and their filtrations

and quotient maps range over at most $\mathcal{B}(p)$ possibilities: bases use polynomially many rational entries, each of bounded height. The same finite bound applies to controlled lattice choices; one may work with the integral congruence covers constructed in the preliminaries. Pigeonhole all these rational choices, the common denominators of R_h , and any required congruence moduli. The retained shift density is still $\mathcal{B}(p)^{-1}$. We now write H for the fixed subgroup and regard all rational diagrams as fixed. The polynomial orbits, bounded real constants, and Lipschitz functions may still depend on h . We have not taken a common multiple over exponentially many alternative moduli.

Freezing the slow and rational factors. Choose a fixed fine partition of the coordinate n/N into cells of diameter ρ . On one such cell the slow factor $E_h(n)$ varies by at most $\mathcal{B}(p)\rho$; its values and the values of its inverse are bounded by $\mathcal{B}(p)$. By rational polynomial periodicity there is a fixed $q \leq \mathcal{B}(p)$ such that, on each class $n \bmod q$, the right lattice class needed to evaluate the first component of $R_h(n)$ is constant. Use a bounded representative of that class. After freezing these factors, the original test is, up to $\mathcal{B}(p)\rho$, a controlled positive Lipschitz function of $b_h(n)$ on a fine lattice cover of H . Its invariance under (C.23) persists, since that subgroup is central.

Let G_Z be the fixed full product group containing the other inputs Z_h . Its dimension is polynomially bounded and its filtration has degree at most r : its factors are the anchor degree- r paths, the degree- r quotients of the unshifted partners, and the normalized square quotients. Use the homomorphism

$$H \longrightarrow H/H_r \times G_Z, \quad b \longmapsto (bH_r, \pi(b)),$$

whose kernel is $K_h = K$, now fixed. Lemma A.3 represents the frozen function as a positive bounded Lipschitz function of these two image coordinates on fine product covers, at cost $\mathcal{B}(p)$. Write $\Lambda \leq G_Z$ for the full ambient lattice in the second cover. Using G_Z , rather than only $\pi(H)$, is necessary because the other factors in the projected factorization need not lie in $\pi(H)$. Extend from the image to the product metric space by Lipschitz extension and clip to $[0, 1]$. The first coordinate is a polynomial path of degree $r - 1$. The remaining task is to recover the second coordinate from fixed observations at n and $n + h$.

For this recovery, also freeze the *left* lattice class of $\pi(R_h(n))$ for the full ambient target lattice Λ . Its cocompactness supplies bounded rational representatives of these classes. Rational periodicity, whose period depends only on the fixed structural data and common denominator, permits a single refinement of q by a controlled amount, uniformly over all retained h and simultaneously with the earlier right-class requirement. On a residue class write $\pi(R_h(n)) = \lambda_n z_{h,c}$, with $\lambda_n \in \Lambda$ and $z_{h,c}$ a bounded rational representative. Projecting (C.22) then gives the exact identity

$$\pi(b_h(n))\Lambda = \pi(E_h(n))^{-1}Z_h(n)\pi(\kappa_h)^{-1}z_{h,c}^{-1}\Lambda. \quad (\text{C.24})$$

The order of these factors is essential. A right lattice class of R_h alone would not justify this identity.

Recovering squares from fixed observations. The anchor paths and unshifted quotient paths in Z_h are fixed paths at n . A square path is obtained from two original partner paths by bounded left normalization and a lattice right normalization, followed by the fixed subgroup inclusion and square quotient. Lemma C.11 gives fixed input covers on which this operation is uniformly Lipschitz. In particular, the arbitrary magnitude of its lattice normalizer does not enter the metric estimate. The same lemma makes the right factors $\pi(\kappa_h)^{-1}z_{h,c}^{-1}$ in (C.24) compatible with fixed finer covers. For fixed h , residue class and wrap branches, both locations being compared use the same square normalizers and the same right factors. The left slow inverses can differ by $\mathcal{B}(p)\rho$; the varying-left-normalizer estimate in Lemma C.11 controls the resulting error independently of the orbit representatives.

The ordinary extension of an anchored partner is a fixed path $n \mapsto g'_j(n + h_0 - b_j N)$, with its

branch b_j already fixed. An ordinary extension at the shifted site is one of the fixed paths

$$m \mapsto g'_j(m + tN), \quad t \in \{-1, 0, 1\}, \quad m = \text{rep}(n + h).$$

Which branch is used is determined by h and the coarse n/N cell except when that cell meets the wrap point. Discard a neighborhood of width $O(\rho N)$ around that point and the endpoint of the interval-coordinate chart. This has relative size $O(\rho + 1/N)$, uniformly in h . There are only polynomially many fixed paths, including every branch required for all partners.

Positive partitions and the uniform oscillation estimate. At the first site measure those fixed paths on their chosen covers, the coordinate n/N , and the residue $n \bmod q$. At the second site measure all the fixed shifted-path alternatives. Take positive Lipschitz partitions of unity into cells of diameter at most ρ on these two product spaces. The residue partition can be realized exactly on the grid by a circle path $n \mapsto n/q \bmod 1$, with Lipschitz cost $O(q)$. The interval-coordinate partition is realized similarly away from the discarded endpoint neighborhood. Since the total dimension and $\log(1/\rho)$ are polynomial in p , both families have at most $\mathcal{B}(p)$ members and complexity at most $\mathcal{P}(p)$. Denote them by $A_i(n)$ and $B_j(n + h)$; these are fixed degree- d niltests.

For a fixed h , consider two good locations contributing to the same pair of cells. Their frozen residue classes and wrap branches agree; their slow factors differ by at most $\mathcal{B}(p)\rho$. Their fixed orbit observations are within $O(\rho)$ on the chosen covers. Lemma C.11, followed by (C.24), therefore gives

$$d(\pi(b_h(n))\Lambda, \pi(b_h(n'))\Lambda) \leq \mathcal{B}(p)\rho. \quad (\text{C.25})$$

This is a uniform estimate over all retained h , rather than a continuity assertion whose constant may depend on h .

Freeze the first-site slow value separately for each cell i . For every nonempty pair (i, j) choose a good point and replace the second image coordinate in the extended reconstruction function by its value there. The resulting function of $b_h(n)H_r$ is a positive degree- $(r - 1)$ test $C_{ij,h}$. The proof of Lemma C.10, allowing the frozen function to be chosen separately for each first-site cell, gives

$$T_h(n) = \sum_{i,j} A_i(n)B_j(n + h)C_{ij,h}(n) + e_h(n), \quad \mathbb{E}|e_h| \leq \mathcal{B}(p)\rho + O(1/N).$$

No sum of errors over all cell pairs is necessary, since their nonnegative weights sum to one. Choose ρ small enough, and the length cutoff large enough, to make this error adequate for Lemma C.7. All costs remain within $\mathcal{P}(p), \mathcal{B}(p)$.

The retained shift set and its positive violations both have controlled inverse-exponential lower bounds. Lemma C.7 now contradicts them by degree- r shift comparison. Its finitely many fixed pairs are chosen before that application; the lower tests $C_{ij,h}$ may depend arbitrarily on h because the induction is uniform in those tests. Choosing budgets backwards through the fixed degree depth gives a single polynomial $P = (2 + p)^{C(d,\epsilon)}$. This proves shift comparison in every fixed degree. $\square$

D Positive counting and absolute increments

A function on an integer box is called progression-free when its support contains no tuple $u, u + v, \dots, u + (k - 1)v$ with $v \in \mathbb{Z}^n \setminus \{0\}$ and all vertices in the box. The length k will always be fixed in the statement using this term.

Positive comparison first gives an absolute increment on a box of fixed dimension. We convert its niltest to a patch and then state the relative lifting theorem, proved in Theorem I.15. That theorem samples fixed-dimensional boxes and returns their increments with rank at most $d + d_0$.

Applied to the constant old patch, of rank $d = 0$, it gives the starting rule of Lemma 2.2: the fresh rank d_0 is independent of the ambient dimension. The triangular argument in the main proof must additionally control the losses in the old widths; that assertion is not part of the relative lifting theorem.

D.1 From positive comparison to counting

We use Theorem C.1 as an input. Its uniformity in the lower-degree test is part of the input. In particular, the exceptional set may depend on the two positive functions but must not depend on that test. Throughout this subsection the ambient group is $H = \mathbb{Z}/N\mathbb{Z}$. For positive counting at length k we assume $\gcd(N, (k-1)!) = 1$, so every nonzero slope difference of absolute value at most $k-1$ is invertible in H . The grid lemma needs only that N be odd, as required by shift comparison.

Lemma D.1 (Positive grids). *Fix s , $0 < \delta < 1$, and $C_0 \geq 1$. There exist C and $\xi_0 > 0$, depending only on these parameters, with the following property. Let $p \geq 2$, let $1 \leq q \leq C_0 p$ be an integer, let N be odd, and fix $0 < \xi \leq \xi_0$. Suppose that a finite collection of functions $0 \leq f \leq e^p$ on $H = \mathbb{Z}/N\mathbb{Z}$ satisfies*

$$\mathbb{E}_H(f - (1 + \xi))T \leq e^{-(2+p)^C}$$

for every $[0, 1]$ -valued niltest of degree at most s and complexity at most $(2+p)^C$. If $N \geq \exp((2+p)^C)$, then

$$\mathbb{E} \prod_{\beta \in [a_1] \times \cdots \times [a_j]} f_\beta(t_{1,\beta_1} + \cdots + t_{j,\beta_j}) \leq (1 + \delta)^{a_1 \cdots a_j} \quad (\text{D.1})$$

whenever $1 \leq j \leq s+1$, $1 \leq a_i \leq q$, and each f_β is one of the given functions or the constant 1. All the $t_{i,b}$ are independent and uniform on H .

Proof. We use precisely Theorem C.1, with its exceptional set chosen before the lower-degree test. Fix a grid, put $V = \prod_i a_i$ and $K = q^{s+1}$, and note that $V \leq K$ and K is polynomial in p . The case $j = 1$ follows from independence and the comparison with the constant test, so suppose $j \geq 2$.

Here is one axis compression. Let the current functions h_a have an upper comparison with a constant g : initially $g = 1 + \xi$, and after normalization below $g = 1$. Fix a small constant $\varepsilon > 0$ and put $c = (1 + \varepsilon)g$. The telescoping identity is

$$\prod_{a=1}^A h_a(y + t_a) - c^A = \sum_{a=1}^A c^{a-1} (h_a(y + t_a) - c) \prod_{b>a} h_b(y + t_b).$$

In insertion a , condition on every shift except t_a . The function $J_a(y) = \prod_{b>a} h_b(y + t_b)$ is then fixed and nonnegative. Against a lower-degree test $T(y)$ the corresponding term, after $n = y + t_a$, is

$$c^{a-1} \mathbb{E}_n(h_a(n) - (1 + \varepsilon)g) J_a(n - t_a) T(n - t_a).$$

This has exactly the form of (C.2), with shift $-t_a$. The exceptional set may depend on the other shifts through J_a , which causes no difficulty: condition on them first, then average in t_a . Its uniformity in the test is indispensable, since the translated test depends on t_a . A union bound over all fibers and insertions costs at most K exceptional probabilities at each compression; no union over the entire original collection is needed.

At positive output degree, a cyclically translated test is represented by its two ordinary translates, on the intervals separated by the wrap point, multiplied by smooth nonnegative interval cutoffs. Ordinary translation changes only the polynomial orbit, whose coefficients are unrestricted in Definition A.1. For transition width ρ , the cutoff logarithmic complexity is polynomial in $\log(1/\rho)$

and its discrete error is $O(\rho + N^{-1})$. Thus the error in the displayed insertion is at most this proportion times the supremum of the *complete* multiplier $c^{a-1}(h_a - c)J_a$. The tests retain the same degree, since a positive output degree permits a degree-one cutoff factor. At degree zero they are constants and no split is needed.

For explicit precision bookkeeping, every intermediate normalized fiber is a product of at most K original inputs divided by a constant at least one. All fiber, insertion, and telescoping multiplier logarithmic caps are therefore at most $B_0 = C_s K(p + 1)$, after fixing $\xi, \varepsilon < 1/4$ and enlarging C_s . Suppose the requested output test complexity is L and the requested comparison error is e^{-L} . Choose

$$\rho = \frac{e^{-L-2B_0}}{10K}, \quad R = (2 + L + B_0)^{A_s},$$

where A_s is large enough to contain the translated-test complexity and all the displayed caps and error sums. Apply shift comparison with parameter R , requesting its input complexity and precision at $P = (2 + R)^{C(d, \varepsilon)}$, where d is the current degree. The two translated tests and the at most K insertions then give the requested output error after division by c^A . The condition $N \geq e^P$ absorbs the N^{-1} cutoff error as well. Choosing ρ before R avoids requiring a cutoff of complexity exceeding the very parameter used in that call.

Starting with the last compression, choose its output budget L large enough that all mean errors are a sufficiently small fixed constant, and so that

$$L \geq pK + \log \frac{4(s+1)K}{\delta} + B_0.$$

Recursively request the input budget P just described as the preceding compression's output budget. There are at most s compressions, so this backward recursion is a bounded composition of polynomials in p . One initial budget $(2 + p)^C$ dominates all of them. At each stage the conditional probability of failure is at most Ke^{-R} ; the total failure contribution to the original grid average is at most

$$e^{pV} \sum_{r=1}^{j-1} Ke^{-R_r} < \delta/4.$$

It remains to record the multiplicative loss. Put $A_r = \prod_{i=1}^r a_i$, and let M_r be the constant divided out of each fiber after compressing its first r axes. The preceding construction gives

$$M_1 = [(1 + \xi)(1 + \varepsilon)]^{a_1}, \quad M_r = M_{r-1}^{a_r} (1 + \varepsilon)^{a_r} \quad (r \geq 2),$$

and every normalized fiber has comparison with 1 at degree $s - r$. Consequently

$$\frac{\log M_r}{A_r} = \log(1 + \xi) + \log(1 + \varepsilon) \left(1 + \sum_{h=1}^{r-1} A_h^{-1} \right) \leq \log(1 + \xi) + r \log(1 + \varepsilon).$$

After $j - 1$ compressions, the last-axis evaluations are independent; their mean upper bounds finish the estimate. Choose ξ_0 and ε so that $(1 + \xi_0)(1 + \varepsilon)^s < 1 + \delta/4$, and make the final mean errors small enough to bound the good-event contribution by $(1 + \delta/2)^V$. Since $(1 + \delta)^V - (1 + \delta/2)^V \geq \delta/2$, the exceptional contribution fits the remaining gap. This proves (D.1). $\square$

Proposition D.2 (Positive counting). *Let $k \geq 3$, $s = k - 2$, and $p \geq 2$. Given $\tau > 0$, there exist $\xi_0 > 0$ and C , depending only on k, τ , such that the following holds. Let $H = \mathbb{Z}/N\mathbb{Z}$ with $\gcd(N, (k - 1)!) = 1$, and fix $0 < \xi \leq \xi_0$. Suppose $0 \leq \mu_i \leq e^p$ on H for $1 \leq i \leq k$,*

$$|\mathbb{E}\mu_i - 1| \leq 2\xi, \quad \mathbb{E}(\mu_i - (1 + \xi))T \leq e^{-(2+p)^C},$$

for all $[0, 1]$ -valued niltests of degree at most s and complexity at most $(2+p)^C$. If $N \geq \exp((2+p)^C)$, then, simultaneously for every subset $I \subseteq [k]$,

$$\left| \mathbb{E}_{x,d} \prod_{i \in I} \mu_i(x + id) - 1 \right| \leq \tau. \quad (\text{D.2})$$

Proof. Induct on $|I|$, requesting smaller fixed tolerances for proper subsets. The cases $|I| \leq 2$ follow from the mean estimates and invertibility of the difference of two slopes. For the induction step choose two slopes as the final slopes and call the others pivots.

We first turn a count deficit into an upper-bound problem. The two final factors will be coupled so that the coefficient of their product can be chosen negative. After Cauchy–Schwarz separates the two final sites, each resulting square is a positive mixture of grids. Thus the same grid upper bound can treat both a surplus and a deficit. Let $e_0 = 1$ and let e_1 take the values $1, -1$ with equal probability. For $0 \leq b \leq 1$, put

$$C_1 = be_0 + \sqrt{b(1-b)}e_1, \quad C_2 = be_0 - \sqrt{b(1-b)}e_1, \quad \nu_i(u) = e_0 + C_i(u-1).$$

Their two kernels are

$$B_b(u, v) := \mathbb{E}_\sigma \nu_1(u) \nu_2(v) = 1 + b(u+v-2) + (2b^2 - b)(u-1)(v-1), \quad (\text{D.3})$$

$$\mathbb{E}_\sigma \nu_i(u) \nu_i(u') = 1 - b + buu'. \quad (\text{D.4})$$

For a surplus use $b = 1$, when $B_b(u, v) = uv$. For a deficit use $b = 1/4$, when

$$B_{1/4}(u, v) = \frac{3}{8} + \frac{3}{8}u + \frac{3}{8}v - \frac{1}{8}uv.$$

If every proper-subset error is at most $\tau/18$, a deficit of τ makes the count with $B_{1/4}$ at least $1 + \tau/16$. A surplus is detected directly by B_1 . Thus failure of (D.2) makes the count I_0 with the final two factors replaced by B_b exceed $1 + c_\tau$ for a fixed $c_\tau > 0$.

Choose an even integer q that is a sufficiently large fixed multiple of p . Each pivot elimination uses a q th moment. After taking the successive roots, the total cap loss will be $\exp(lp/q)$ for l pivots, whereas the largest grid has only $2q^l$ vertices. Thus the cap loss can be made a fixed small factor while all grids remain of polynomial size, since $l \leq k-2$ is fixed. After eliminating l pivots, denote their slopes by $p_1, \dots, p_l$ and set $m = q^l$. The argument of the factor at slope i and cell $\beta \in [q]^l$ is

$$x + id + \sum_{a=1}^l (i - p_a) z_{a, \beta_a}.$$

All displayed variables are independent and uniform. The integrand contains the remaining pivot factors and one copy of B_b in each cell. For the next pivot p_{l+1} let D be the product of its m factors. It depends only on $w = x + p_{l+1}d$ and the previously introduced z 's. Write the rest of the integrand as F . Weighted Hölder and $D \leq e^{pm}$ give

$$|I_l| \leq (\mathbb{E}D)^{1-1/q} e^{pm/q} I_{l+1}^{1/q}, \quad I_{l+1} := \mathbb{E}_{w,z} |\mathbb{E}_d F|^q. \quad (\text{D.5})$$

Expansion of the even moment adds a new axis of q independent shifts along the kernel of the pivot form. Explicitly, write its independent free variables as $d + z_{l+1,b}$ and the corresponding first coordinates as $x - p_{l+1}z_{l+1,b}$. This gives exactly the displayed cell formula with l replaced by $l+1$.

The product D is a grid with the l shift axes of size q and a root axis of size 1: the coefficient on shift axis a is $p_{l+1} - p_a$, which is invertible in H . Thus Lemma D.1 gives $\mathbb{E}D \leq (1 + \delta)^m$.

When all $l = |I| - 2$ pivots have been eliminated, condition on the shifts and change (x, d) into the independent roots at the two final slopes. Use an independent σ for each cell in (D.3), and apply Cauchy–Schwarz between the two root-averaged factors. Each square is a product of the kernels (D.4). Expanding it gives a positive convex combination of grids with l axes of size q and one axis of size 2. Each grid has at most $2q^l$ vertices, so

$$|I_l| \leq (1 + \delta)^{2q^l}.$$

Iterating (D.5) therefore yields

$$|I_0| \leq e^{lp/q}(1 + \delta)^{l(1-1/q)+2} \leq e^{lp/q}(1 + \delta)^{l+2}.$$

First choose the multiple defining q large, and then δ small, so that this is less than $1 + c_\tau$. This contradicts the detection inequality and completes the induction. $\square$

Proposition D.3 (Absolute increment on a fixed-dimensional box). *Assume Theorem C.1. Fix $k \geq 3$ and $n \geq 1$. There are $\xi > 0$ and C , depending only on k, n , with the following property. Let $Q = \prod_{j=1}^n \{0, \dots, P_j - 1\}$, where the P_j are distinct primes, and put $p = \max(2, \log(1/\alpha))$. Suppose that $\min P_j \geq \exp((2 + p)^C)$ and that $f : Q \rightarrow [0, 1]$ has mean $\alpha > 0$ and support containing no nonconstant integer k -term progression. In the cyclic variable supplied by the Chinese remainder theorem there is a $[0, 1]$ -valued niltest T of degree $k - 2$ and complexity at most $(2 + p)^C$ such that*

$$\mathbb{E}_Q(f - (1 + \xi)\alpha)T \geq \exp(-(2 + p)^C). \quad (\text{D.6})$$

Proof. We reserve separate internal score and external side exponents $C_{\text{sc}} < C$, depending only on k, n . Suppose that every permitted test of complexity at most $(2 + p)^{C_{\text{sc}}}$ has score less than $\exp(-(2 + p)^{C_{\text{sc}}})$, and write $\mu = f/\alpha$. Enlarge the side cutoff so that every $P_j > k - 1$. Put $N = \prod_{j=1}^n P_j$; then $\gcd(N, (k - 1)!) = 1$. Identify Q with $\mathbb{Z}/N\mathbb{Z}$ by the Chinese remainder theorem. All cyclic averages in this proof use this identification. The exponents will be fixed after the internal estimates below. For every bounded-complexity smooth nonnegative linear mask $0 \leq \phi \leq 1$, the function

$$\mu_\phi = 1 + \phi(\mu - 1)$$

is between 0 and $\max(1, \mu) \leq e^p$. If η is the normalized comparison error, applying the upper comparison to ϕ and $1 - \phi$, and using $\mathbb{E}(\mu - 1) = 0$, gives

$$|\mathbb{E}\mu_\phi - 1| \leq \xi + \eta.$$

For a positive test T and $\xi' = 2\xi$, one also has

$$\mathbb{E}(\mu_\phi - (1 + \xi'))T = \mathbb{E}(\mu - (1 + \xi))\phi T + \mathbb{E}(\xi\phi - \xi')T \leq \eta.$$

Taking $\eta \leq \xi$ and choosing the original ξ sufficiently small therefore supplies all the hypotheses of Proposition D.2, with slack ξ' . Products of these tests still have the same degree, since $k - 2 \geq 1$, and incur only polynomial complexity losses.

By inclusion–exclusion applied to the subset counts, for every nonempty $I \subseteq [k]$ we consequently have

$$\left| \mathbb{E}_{x,d} \prod_{i \in I} \phi_i(x + id)(\mu(x + id) - 1) \right| \leq \varepsilon, \quad (\text{D.7})$$

where $\varepsilon > 0$ is any sufficiently small fixed constant. This also holds, up to a fixed multiplicative constant, for character masks: write real and imaginary parts of a character as linear combinations of $[0, 1]$ -valued smooth masks.

Choose a fixed smooth nonnegative function W of the $2n$ torus coordinates of (x, d) , supported where, for every j ,

$$\frac{1}{4} < \frac{x_j}{P_j} < \frac{1}{3}, \quad \frac{1}{20k} < \frac{d_j}{P_j} < \frac{1}{10k}.$$

Its discrete mean tends to a positive constant as the primes grow. On its support every $x + id$, $1 \leq i \leq k$, is represented without wraparound, and d is nonzero. Hence

$$\mathbb{E}_{x,d} W(x, d) \prod_{i=1}^k \mu(x + id) = 0. \quad (\text{D.8})$$

On the other hand, approximate W uniformly by a fixed finite trigonometric polynomial, and then expand the product as $\prod_i (1 + (\mu(x + id) - 1))$. The approximation error is bounded using $\prod_{i \in I} |\mu(x + id) - 1| \leq \prod_{i \in I} (\mu(x + id) + 1)$; expansion of the right side and Proposition D.2 bound its mean by a constant depending only on k . For a term with at least two centered factors, express each configuration character as a product of characters at two of those sites; invertibility of the difference of the two slopes justifies this change of variables. Equation (D.7) then bounds that term. With exactly one centered factor, averaging on the fibers of its form either annihilates the configuration character or leaves a single character at that site, to which the one-factor case applies. Thus the left side of (D.8) is arbitrarily close to $\mathbb{E}W > 0$, a contradiction. To fix the budgets, let $(2 + p)^D$ dominate every internal counting and test-product budget. Choose $C_{\text{sc}} > D$ so that $(2 + p)^{C_{\text{sc}}} \geq (2 + p)^D + p + 1$. The assumed failure of the internal score bound, divided by $\alpha \geq e^{-p}$, gives normalized comparison error at most

$$\exp(-(2 + p)^{C_{\text{sc}}} + p) \leq \exp(-(2 + p)^D),$$

as required. The resulting test therefore has the stronger score $\sigma \geq \exp(-(2 + p)^{C_{\text{sc}}})$ and complexity at most $(2 + p)^{C_{\text{sc}}}$. Finally choose the displayed side exponent $C > C_{\text{sc}}$ large enough to imply every internal size cutoff and

$$N \geq 100(1 + \xi) \exp((2 + p)^{C_{\text{sc}}}).$$

This proves the stated, weaker score and complexity bounds. Retaining the internal score exponent also gives $N^{-1} \leq \sigma/(100(1 + \xi))$, which we will use in patch conversion. $\square$

D.2 Patches and the lifting statement

Definition D.4 (Patch). A degree- s patch on integer parameters u is a function

$$B(u) = \sum_{b \in \mathbb{Z}^d} \Phi(S_u(b)), \quad S_u(b)_i = b_i - A_i(u, b_1, \dots, b_{i-1}). \quad (\text{D.9})$$

The slots b_i have nondecreasing integer weights $1 \leq w_i \leq s$; parameters have weight 1; and A_i has weighted degree at most w_i . The function $\Phi : \mathbb{R}^d \rightarrow [0, 1]$ is Lipschitz and supported in $[-1/4, 1/4]^d$. The rank is d . A logarithmic complexity bound bounds the number of parameters, the rank, and $\log(2 + \text{Lip}(\Phi))$, but does not bound the coefficients of the A_i . Rank zero includes $B = 1$.

At most one summand in (D.9) is nonzero: first the condition on slot 1 determines its possible integer uniquely, and then one continues in increasing slot order. In particular $0 \leq B \leq 1$. Figure 1 shows a two-slot example.

For $B \geq 0$, a positive score $\mathbb{E}(f - a)B$ says that the mean of f under the probability weight proportional to B exceeds a . Its explicit lower bound also records how much mass is available for

later restrictions and approximation errors. Rank and complexity have different roles. The relative lifting theorem below bounds the returned rank by $d + d_0$. Its rank-zero application therefore introduces at most d_0 slots even when the resource budget is enlarged to include the ambient dimension. The main iteration uses this fresh-rank bound together with its additive triangular dimension recurrence. Other logarithmic costs may grow polynomially, with their compositions accounted for explicitly.

For the qualitative correspondence between generalized-polynomial coordinates and polynomial nilmanifold orbits, see Bergelson and Leibman [3, Theorem A*]. The conversion below gives the quantitative bounds needed for our Lipschitz patches.

Lemma D.5 (Conversion of patches). *For fixed s , a patch of rank d and logarithmic complexity p is a niltest of degree at most s , nilmanifold dimension at most $(1 + d)^{C_s}$, and logarithmic complexity at most $(2 + p)^{C_s}$. Conversely, the test in Proposition D.3 can be replaced by a single patch on the original box, of rank and logarithmic complexity $(2 + p)^{C_{s,n}}$, with score at least $\exp(-(2 + p)^{C_{s,n}})$.*

Proof. For the rational Mal'cev-coordinate calculus used here, see Green and Tao [23, Appendix A]. We give the triangular shear realization and its lattice bounds explicitly.

From patches to niltests. Integer polynomial changes of the summation variables normalize the coefficients of the A_i modulo integers, in increasing slot order. Indeed, replace b_i by $b_i + R_i(u, b_{<i})$ for an integer-coefficient polynomial R_i and substitute this change in the later slots. This is a bijection of $\mathbb{Z}^d$ for each integer u , preserves the residual vector, and preserves weighted degree. Thus every coefficient can be put in $[-1/2, 1/2]$ without changing Φ . A later polynomial may acquire large coefficients during an earlier substitution; these are reduced when its own slot is reached, so the final bound is independent of every original coefficient.

Let S^0 be the part of S_u homogeneous in the b variables of weight w_i in slot i . It is independent of u . Write $S_u = S^0 \circ Y_u$. Then Y_u belongs to the group $\mathcal{Y}$ of triangular polynomial transformations whose addition in slot i has weight strictly less than w_i . Filter this group by weight deficit. It is a simply connected nilpotent group of step at most s and of dimension at most

$$d \binom{d + s}{s} \leq (1 + d)^{C_s}.$$

For completeness, its Lie algebra has the monomial shears $b^\beta \partial / \partial b_i$ with $\text{wt}(\beta) < w_i$ as a basis. Such a shear has deficit $w_i - \text{wt}(\beta)$; a bracket has the sum of the deficits. The exponential of each shear is its elementary triangular substitution. Ordering these shears by increasing deficit gives triangular polynomial multiplication and inversion, with rational coefficients of bounded logarithmic complexity. The integer-coefficient transformations form a cocompact lattice Γ ; successive reduction of their triangular coefficients gives bounded fundamental representatives. Equivalently, the ordered elementary shears give integral Mal'cev coordinates.

We verify both the logarithmic lattice grids and the precise polynomial condition in Definition A.1. On the finite-dimensional space of polynomials in b of weight at most s , let Y^* denote substitution by Y and put $U = Y^* - I$. Since U lowers weight, $U^{s+1} = 0$ and

$$\log Y^* = \sum_{h=1}^s \frac{(-1)^{h+1}}{h} U^h.$$

Applied to b_i , this gives the coefficients of the logarithmic shear field. For an integer-coefficient transformation their denominators divide $s!$. Conversely, if a shear field X has all coefficients in $s!\mathbb{Z}$,

the finite formula $\exp(X)b_i = \sum_{h=0}^s X^h b_i / h!$ has integer coefficients. Writing $D = \dim \mathcal{Y}$, we obtain the explicit grids

$$s! \mathbb{Z}^D \subseteq \log \Gamma \subseteq (s!)^{-1} \mathbb{Z}^D.$$

The combined weight bound for Y_u says that its coefficient of b^β in slot i has parameter degree at most $w_i - \text{wt}(\beta)$. Each substitution in the finite logarithm preserves this combined bound. Thus the logarithmic coefficient of a shear of deficit h has parameter degree at most h , which proves the required filtered polynomial condition.

The function

$$F(Y\Gamma) = \sum_{b \in \mathbb{Z}^d} \Phi(S^0 Y(b))$$

is well-defined, since a lattice transformation permutes $\mathbb{Z}^d$. To bound its Lipschitz constant, use bounded representatives for Y . The coefficients of S^0 are bounded after normalization. Within a fixed weight, inversion of S^0 requires a unitriangular linear system of size at most d , whose inverse has coefficients at most $\exp((1+d)^{C_s})$. There are only s weight layers. It follows that S^0 , its inverse, and their derivatives on the relevant bounded sets have this bound. A contributing integer b is correspondingly bounded; there are at most $\exp((1+d)^{C_s})$ such b on any bounded coordinate chart. Summing their Lipschitz estimates proves the asserted bound. The function is well-defined on the quotient, and these estimates on overlapping coordinate neighborhoods give its global Lipschitz bound. Finally $B(u) = F(Y_u \Gamma)$.

From the absolute increment to a patch. For the converse, retain the score σ and the internal exponent C_{sc} from the proof of Proposition D.3. In particular $N^{-1} \leq \sigma/(100(1+\xi))$; choosing the external side exponent after the internal score exponent supplies this rounding allowance. Choose $\eta = \sigma/(100(1+\xi))$ and a smooth function $0 \leq \chi \leq 1$ supported in $(\eta, 1-\eta)$ and equal to one on $[2\eta, 1-2\eta]$. The set where $\chi(n/N) \neq 1$ has discrete measure at most $4\eta + 2/N \leq 6\sigma/(100(1+\xi))$. Multiplying the test by $\chi(n/N)$ therefore loses at most $\sigma/4$ of its score, since $|f - (1+\xi)\alpha| \leq 1+\xi$ and CRT makes n uniform. Split χ into a smooth nonnegative partition with $L_0 \leq \exp((2+p)^{C_s})$ pieces. Every piece is supported in $(0, 1)$ and in an interval of radius strictly less than $1/4$. The truncated score is at least $3\sigma/4$, so one localized score is at least $3\sigma/(4L_0)$. If $A(u)$ is the integer linear Chinese-remainder representative and the selected interval is centered at $c \in (0, 1)$, introduce a weight-1 slot b_0 with residual

$$b_0 - A(u)/N + c.$$

On its localized support $b_0 = \lfloor A(u)/N \rfloor$ and $n = A(u) - Nb_0$ exactly. This removes the representative wrap.

Here are the quantitative chart details needed for the next localization. Choose a rational basis adapted to the filtration and multiply every axis by a sufficiently divisible integer $L \leq \exp((2+p)^{C_s})$. The common denominators of the brackets, the lattice-grid denominators, and the fixed BCH denominators can all be cleared by such an L . In the scaled axes, integral ordered products form a subgroup $\Gamma' \subseteq \Gamma$: ordered-coordinate multiplication and inversion have integral polynomial coefficients after this scaling. These ordered products have a compact fundamental region with ordered coordinates in $[0, 1]^D$. Thus Γ' is a lattice of finite index, and its logarithmic complexity remains polynomial in p . Lift the orbit to $g(n)\Gamma'$ and pull the original test back under the projection $G/\Gamma' \rightarrow G/\Gamma$; its evaluation is unchanged.

For some $K = \exp((2+p)^{C_s})$, this quotient has representatives z with $|\log z| \leq K$, its nontrivial lattice elements have log-coordinate distance at least K^{-1} from zero, and coordinate maps and their inverses on bounded regions have derivative bounds K^{C_s} . These statements follow respectively by reducing ordered coordinates, by the lattice grid, and by the finite BCH formulas. Consequently the

maps $v \mapsto z \exp(v) \Gamma'$ are injective for $|v| \leq K^{-C_s}$ after enlarging C_s . Indeed equality of two such points would put $\exp(-w) \exp(v)$ in Γ' , contradicting lattice separation unless $v = w$. A Euclidean grid in the bounded representative region supplies at most $\exp((2+p)^{C_s})$ such charts whose smaller concentric charts cover the quotient. Smooth coordinate cutoffs equal to one on those smaller charts have the same exponential Lipschitz bound. Dividing them by their sum gives a nonnegative partition of unity, still with that bound, since the sum is at least one. Multiplying the test by this partition and selecting one term retains an inverse-exponential score. If the atlas has L_1 charts, the selected score is at least $3\sigma/(4L_0L_1)$. On a chart centered at a bounded representative z , write the small coordinates of $z^{-1}g(A(u) - Nb_0)\gamma$ by BCH, with γ parametrized by an ordered integral lattice product. In increasing Lie-layer order, each coordinate is a nonzero fixed multiple of its new integer slot plus a polynomial in u, b_0 , and the earlier slots. Dividing by that multiple gives exactly (D.9); BCH bounds the weighted degree by the slot weight. Choose the chart small enough that all residuals lie in $[-1/4, 1/4]$. More explicitly, let ψ be the selected CRT cutoff, let $K_z(v)$ be the localized test in the chart, extended by zero, and let D_z be the diagonal matrix of the fixed slot multipliers. The new kernel is

$$\Phi(r_0, r) = \psi(c - r_0) K_z(D_z r).$$

It takes values in $[0, 1]$, has the required support and Lipschitz bound, and has no remaining dependence on u or the integer slots. There is one slot per Lie coordinate and the additional slot b_0 , and all chart and score losses have the asserted bounds. $\square$

Definition D.6 (Parameter slice). A parameter slice of an integer box is a subbox intersected with one coordinatewise residue class modulo a common positive integer, followed by affine reindexing as a box. Logarithmic cost P means that each new side contains at least an e^{-P} proportion of the corresponding old side, counting the modulus. Successive slice costs add. The affine map is injective and sends a nonconstant integer progression to a nonconstant integer progression.

We now state the lifting theorem used to obtain the absolute starting rule, ahead of its sampling-based derivation in Theorem I.15. The word “rule” below quantifies over *all* boxes and functions in its stated class; it does not mean a choice of test for one function.

Proposition D.7 (Relative lifting). *Fix $s \geq 1$, put $k = s + 2$, and fix a sufficiently large dimension $n_0 = n_0(s)$ and $\varepsilon_0 > 0$. There is an exponent E , depending only on these fixed parameters, with the following property.*

Fix the side-comparability constant $R = 2^{n_0+1}$. Let $p \geq 2$, $e^{-p} \leq a \leq \Lambda \leq 1$, and suppose an absolute rule holds on every sufficiently large prime-sided box of dimension n_0 , with distinct sides and largest side at most R times the smallest: every progression-free input of mean at least a has, on a parameter slice of cost at most p , a degree- s patch of rank at most d_0 , complexity at most p , and score at least e^{-p} against input minus Λ . The cutoff on every prime side is at most e^p .

Let Q be any box of dimension between 1 and p , with all sides at least $\exp((2+p)^E)$. Suppose $0 \leq f \leq 1$ is progression-free on Q , and an old degree- s patch B of rank d and complexity at most p satisfies

$$\mathbb{E}_Q(f - a)B \geq e^{-p}.$$

Then some parameter slice of Q , of cost at most $(2+p)^E$, carries a degree- s patch B' satisfying

$$\begin{aligned} \text{rank } B' &\leq d + d_0, \\ \text{complexity}(B') &\leq (2+p)^E, \\ \mathbb{E}(f - (1 - \varepsilon_0)\Lambda)B' &\geq \exp(-(2+p)^E). \end{aligned}$$

There is also a separate impossibility alternative: if $e^{-p} \leq a \leq 1$ and the absolute rule states that no such input of mean at least a exists, the old positive score is impossible. This alternative has no target parameter Λ . The exponent is independent of the absolute rule, of d, d_0 , and of every variable dimension included in p .

Remark D.8 (The role of additive rank). Theorem I.15 proves Proposition D.7, including its impossibility alternative and the additive rank bound $d_0 + d$. For the main proof, its application with $d = 0$ supplies the dimension-independent fresh-rank bound in Lemma 2.2. The triangular increment theorem then retains the old coordinates and controls their relative widths by a separate argument. Its iteration uses that triangular dimension recurrence, rather than repeated applications of the present theorem to the growing old patch.

E Unconstrained sampling and scalar transfer

We compare averages on the parent box with averages on random affine boxes. Proposition E.1 transfers a scalar comparison against a fixed niltest. Its proof first stabilizes the niltest under conditioning on small-prime residues, then uses contraction on the remaining residue coordinates.

Section F constructs the affine sampling law when polynomial constraints are imposed. Sections G and H then treat path-dependent tests and positive score recovery. Here the structured test is fixed, and the purpose is to preserve a scalar comparison.

All path probabilities and expected testing suprema refer to the ensemble constructed before productive paths are selected. A subsequent discard is estimated in that original probability space. Conditioning on productivity does not supply a new residue or equidistribution law.

E.1 Scalar comparison without constraints

Proposition E.1 (Scalar transfer). *Fix the degree, an integer $\ell \geq 2$, and $\varepsilon > 0$. There are constants $C, C' > 0$ with the following property; C' depends only on ℓ , and one may take $C' = \ell + 1$. Let $p \geq 2$, let $\mathcal{I}$ be an integer box of dimension $n \leq p$ and side lengths H_i , and let $g : \mathcal{I} \rightarrow [0, 1]$ be a niltest of complexity at most p . Let $0 \leq f \leq 1$ and $e^{-p} \leq \lambda \leq 2$. Suppose, for every parameter slice $B \subseteq \mathcal{I}$ of log cost at most $P = (2 + p)^C$, that*

$$\mathbb{E}_B fg \leq \lambda \mathbb{E}_B g + e^{-P}. \quad (\text{E.1})$$

Extend fg and g by zero. Choose integer x, V uniformly from product boxes at scales H_i and H_i/L , respectively, with V having ℓ columns. Sample t uniformly from an integer box in $[-e^p L, e^p L]^\ell$, possibly restricted to one residue in each coordinate modulo a common $D \leq e^p$, with every restricted side at least Le^{-2p} . If $L \geq \mathcal{B}(p)$ and $\min_i H_i \geq \mathcal{B}(p)L^{C'}$, then

$$\Pr_{x,V} \{ \mathbb{E}_t (fg - (1 + \varepsilon)\lambda g)(x + Vt) > e^{-p} \} \leq e^{-p}. \quad (\text{E.2})$$

The same conclusion holds when the outer law of (x, V) is dominated by these product laws by e^p , and for scale changes by $e^{O(p)}$, after polynomial changes of the log budgets. All complexity and precision requirements are prescribed before choosing L . The conditional law of t remains the stated uniform law on the prescribed parameter slice; arbitrary reweighting of t after observing (x, V) is not included.

E.1.1 Stability of the structured baseline

The no-increment hypothesis compares the mass of fg with the mass of g on each slice. Conditioning on a few residue coordinates can change both masses, so the reference mass must be controlled before we use contraction on the remaining coordinates. The next proposition does this for the structured function g . It gives a partition on which every further restriction involving a prescribed number of prime coordinates changes the mean of g by at most δ .

Proposition E.2 (Stability of baseline marginals). *Let g be as in Proposition E.1. Select a finite collection of distinct primes l , with one power $q_l = l^{a_l}$ per prime, where $\log q_l$ and the logarithm of the number of primes have prescribed polynomial bounds. Regard $u \bmod q_l \in (\mathbb{Z}/q_l\mathbb{Z})^n$ as a single coordinate. Prescribe $\delta > 0$, an integer r_0 , and an initial set of mandatory coordinates, with $\log \delta^{-1}$, r_0 , and the number of mandatory coordinates polynomially bounded.*

On any prescribed box of polynomial log cost, there is a decision tree whose nodes fix entire prime coordinates, with every leaf fixing all the prescribed mandatory coordinates, whose depth on every branch is polynomially bounded, and such that every nonempty leaf B satisfies

$$|\mathbb{E}[g \mid B, u_J = z] - \mathbb{E}[g \mid B]| \leq \delta \quad (|J| \leq r_0) \quad (\text{E.3})$$

for every set J of still unfixed coordinates and every nonempty refinement $u_J = z$. The depth and the sufficient lower bound for the original side lengths have polynomial log bounds in the prescribed inputs, independently of the coefficients of the polynomial orbit defining g .

Proof. We use Lemmas A.4, B.1, A.6, and A.3, together with the actual-factor lifting and cover constructions in Section A. Let $A \geq 2$ dominate the original complexity, $\log \delta^{-1}$, r_0 , the number of mandatory primes, $\max_l \log q_l$, the logarithm of the number of selected primes, and the initial slice cost. Replace δ by $\min(\delta, 1)$ if necessary. First split on all mandatory coordinates. If the starting box is a parameter slice with residue modulus M_{init} , also fix every selected prime dividing M_{init} ; this adds at most $\log_2 M_{\text{init}}$ coordinates. The initial residue modulus is then coprime to every set of still-unfixed prime coordinates.

We first describe how a failure of stability produces new rational information. At a positive degree j , the comparison concerns a fixed polynomial map a into a filtered group G . Retain a graded subalgebra V , initially the full associated graded algebra, obtained by intersecting the fast algebras supplied by earlier failures. Its role is to record the part of the symbol not yet eliminated. There are two possibilities: the present discrepancy forces a strict decrease of V , or it passes to a map of smaller degree. Since a degree-zero map cannot distinguish the two laws, some positive-degree subalgebra must eventually decrease.

Here is the reduction. Suppose the two laws at a node and a residue refinement differ on the incoming test by $d_* > 0$. Approximate that test uniformly to error $d_*/100$ by top-vertical terms. Let M bound both their number and total coefficient mass; $\log M$ is polynomial in the current comparison budget. Call a term significant when its average on either law has modulus at least $d_*/(100M)$. If a significant frequency η does not annihilate V_j , step drop on that law's residue box gives a fast algebra U with $\eta(U_j) = 0$. Consequently

$$\dim(V \cap U) < \dim V.$$

Record this splitting and replace V by $V \cap U$. Fix the witness coordinates J to their full selected prime powers, splitting into all assignments rather than retaining only the assignment that witnessed the discrepancy. Translation does not change the symbol, so the same record applies on every sibling after transport to its coordinates. We call this strict decrease an event.

Otherwise every significant frequency annihilates V_j . Averaging over the central torus generated by this top subspace loses at most $d_*/100$ on each law, apart from the Fourier approximation error. After factoring the map and freezing its controlled slow and rational factors, the surviving discrepancy passes to a map of degree at most $j-1$. The constructions below justify this descent with one fixed lower-degree map. Repeating the alternatives must reach an event, since the discrepancy cannot persist at degree zero.

It remains to bound the cost of repeating this procedure. Refactoring the map produced by each successive violation would compose polynomial complexity bounds once per violation, potentially more than a fixed number of times. We instead retain the map and combine its recorded symbol restrictions in one simultaneous splitting, so that a new polynomial is composed into the budget only when the degree decreases. A new violation starts from its own node and refinement, but uses the same retained maps until a recorded subalgebra changes. In particular, the subboxes used to freeze a slow factor are chosen afresh from that factor; they are not intersections of the subboxes selected for earlier violations. This controls the accumulated costs; the strict dimension decreases will bound the number of events. We now record the quantitative invariant and verify the factorization, transport, and descent used above.

The fixed-map invariant. Maintain at most s levels, of strictly decreasing filtered degree. An epoch at degree j has a fixed polynomial map a into a fixed filtered group G , expressed in the affine integer coordinates of a fixed ancestor residue box. Its coefficients are unrestricted. The physical box underlying this ancestor has sides W_i . At a node, a comparison at this level evaluates this same map under the node and a refinement by still-unfixed primes J , possibly on an additional subbox and one auxiliary residue class. The subbox has physical sides at least $e^{-B_j}W_i$, the auxiliary modulus has logarithm at most B_j and is coprime to the primes in J , and the test complexity and inverse discrepancy logarithm are at most B_j . The bounds depend on the parent epoch, not on the number of visits to this level.

At this level retain the intersection V of the recorded graded fast subalgebras. For each new value of V , retain one factorization and one child map. An event at this level resets the lower levels; an event above it resets this entire epoch. Each new violation of (E.3) starts again from its node/refinement pair and descends through the retained maps. Auxiliary subboxes from a previous violation are not successively intersected.

Transport and joint intersection. Write the old piece and current node in physical coordinates as

$$u = a_0 + M_0 d_0 t, \quad u = a_1 + M_1 v, \quad M_0 \mid M_1,$$

where M_0, M_1 are the accumulated residue moduli and d_0 is the old controlled auxiliary modulus. Thus $t = c + hv$, with $h = (M_1/M_0)/d_0$. The homogeneous-polynomial identity for symbols removes c and multiplies a coefficient of multidegree α by $h^{|\alpha|}$; the identity holds for this rational h as well as integer dilations. If an old rational symbol has common denominator D_0 , its transported common denominator divides $D_0 d_0^s$. The integer M_1/M_0 affects only numerators, and the fast subalgebra's rational orientation is unchanged.

Let T_i, T'_i be old and current parameter sides. If the old physical piece has sides at least ρW_i and the current physical span is contained in the epoch box, then

$$|c_\alpha h^{|\alpha|}|(T')^\alpha \leq \mathcal{B}(B_j) \prod_i \left(\frac{M_1 T'_i}{M_0 d_0 T_i} \right)^{\alpha_i} \leq \mathcal{B}(B_j) \rho^{-s}.$$

Thus every recorded slow symbol is controlled on the current full residue box, with no accumulated-modulus or aspect-ratio loss. There are at most $\dim G$ records in this epoch. Multiply their common denominators if needed and apply the simultaneous assertion of Lemma A.6 jointly. The resulting common splitting has polynomial logarithmic cost. No repeated powering of a complexity bound at the same level is used.

The actual factorization and fixed child. After each event, on each resulting full residue box write the current map as a in its affine coordinates and let $X = E_{\text{sym}}PR_{\text{sym}}$ be its common symbol splitting, with P taking values in V . Lift the homogeneous logarithmic coefficients of $E_{\text{sym}}, R_{\text{sym}}$ through fixed rational sections, adding no other logarithmic terms; call the lifts $E_{\text{hom}}, R_{\text{hom}}$. Choose $\gamma \in \Gamma$ such that $c = a(0)\gamma$ is a bounded representative, and define

$$E = cE_{\text{hom}}, \quad R = R_{\text{hom}}\gamma^{-1}, \quad b = E^{-1}aR^{-1}.$$

Then $a = EbR$, $b(0) = 1$, and the symbol of b lies in V . The finite BCH formula shows that E is fully slow: its constant is bounded, and every product of nonconstant slow coefficients has the corresponding product of inverse-side factors. It also shows that R has one controlled common denominator, since $\log \Gamma \subset l^{-1}\mathbb{Z}^d$, even though the numerators of γ may be arbitrarily large.

Set

$$\mathfrak{h}_i = \{v \in \mathfrak{g}_i : v \bmod \mathfrak{g}_{i+1} \in V_i\}.$$

Graded bracket closure makes these a filtration. The degree- i coefficients of $\log b$ lie in $\mathfrak{h}_i$ by its symbol condition, while coefficients of larger degree already lie in $\mathfrak{g}_{i+1} \subset \mathfrak{h}_i$. Hence b takes values in $H = \exp \mathfrak{h}_1$ and is adapted to this filtration. The map $b \bmod H_j$ has degree at most $j - 1$. No bound on the lower-degree coefficients of b is asserted or used.

Let D_R be the actual common denominator of this fixed polynomial R , enlarged by the fixed structural and lattice denominators. At integer arguments, its values and their adjoint matrices have denominators dividing a fixed controlled power of D_R . The cover construction in Section A therefore gives one controlled lattice Λ contained in every required $R(t)\Gamma R(t)^{-1}$. Its cost depends on D_R , not on the numerators or on t . We use the actual common denominator here; we do not take a common multiple of every integer below its bound. The source lattice $H \cap \Lambda$ and the quotient data for $H \rightarrow H/H_j$ are now fixed.

The same construction gives a controlled period D for R in right lattice class. Include every selected prime dividing D , to its full selected power, keeping the child map and cover unchanged while splitting these assignments. At most $\log_2 D$ coordinates are added. The selected power may be smaller than the corresponding prime power in D , so this need not make R constant. It does ensure that every remaining auxiliary period is coprime to every future J : its selected primes have been removed from the unfixed collection. Period primes absent from the collection can never belong to J . When V is full, the fixed child is simply the ordinary top quotient.

Descent when there is no event. Average the actual incoming test over the central torus associated with H_j , obtaining $\bar{F}$. This removes exactly the vertical frequencies not annihilating V_j . If none of them is significant, their total average on each law is at most $d_*/100$. Including the Fourier approximation errors, $\bar{F}$ retains at least a fixed fraction of d_* as discrepancy. It remains nonnegative, bounded and Lipschitz.

Use the retained factorization to freeze E on small subboxes and R exactly in right lattice class on residue pieces. The freezing error is independent of b , by right invariance:

$$d_G(E(t)b(t)R(t), E_0b(t)R(t)) = d_G(E(t), E_0).$$

A fully slow polynomial remains fully slow on later affine restrictions within its creation box: expansion of a translated monomial uses bounded normalized offsets, and dilation cancels against the new sides. Thus subboxes of controlled relative physical sides suffice throughout this epoch. They may be chosen with comparable lengths; rounding or boundary remainders have arbitrarily small aggregate mass once the sides are large enough. At each visit these pieces are obtained afresh from the retained factors, which proves the fixed relative-side part of the invariant.

For a frozen value R_0 , the function

$$h \longmapsto \bar{F}(E_0 h R_0 \Gamma)$$

is well defined on $H/(H \cap \Lambda)$. Fixed right translation is an isometry, bounded left translation by E_0 has controlled Lipschitz norm, and the rational subgroup metric comparison is controlled. Since $H_j \subset G_j$ is central, this function is invariant under left H_j . Lemma A.3 applies to the rational quotient $H \rightarrow H/H_j$, whose kernel is connected and rational and whose differential is controlled. It gives a controlled nonnegative test of the already fixed child map. The target cover can be chosen from the fixed source lattice and quotient map, independently of the frozen values.

It remains to compare the weights of the pieces. If the current selected-prime modulus is M , the pending refinement modulus is m_J , and the auxiliary piece modulus is d_0 , then $\gcd(m_J, M d_0) = 1$. The M - and d_0 -residues may overlap, but their compatibility is unchanged by the J -refinement. Counting integers in each physical side modulo the appropriate least common multiple gives its length divided by that modulus, with error at most one. Sufficiently large sides therefore make the two normalized piece laws arbitrarily close in total variation, uniformly in their residue assignments. After the freezing and weight errors, some piece has normalized discrepancy at least $d_*/2$, after harmless adjustment of the precision constants. There is no loss by the number of pieces: the discrepancy is a weighted average of their discrepancies.

The new test, auxiliary modulus, inverse relative sides and inverse discrepancy logarithm are polynomially bounded in the current epoch's bounds. This is the invariant at the next level. At degree zero the map is constant, so a positive discrepancy is impossible. Every original violation therefore forces an event at some positive degree.

Budgets, depth and termination. Choose $\Phi(x) = (2 + x)^{C_s}$ to dominate the fixed number of Fourier, step-drop, joint-intersection, lifting, cover and freezing costs just used. Before constructing the tree set

$$B_s = \Phi(A), \quad B_{j-1} = \Phi(B_j).$$

There are only s compositions. At degree j , all step-drop inputs depend on the fixed incoming bound B_j , not on the common splitting assembled from that level's previous records. Only the child budget depends on this common splitting. Consequently all B_j are polynomial in A .

Every child is a subgroup quotient and covers preserve dimension. If the original dimension is d , the number N_j of events through a degree- j stack satisfies the coarse recurrence

$$N_0 = 0, \quad N_j \leq d + (d + 1)N_{j-1}, \quad N_s \leq (d + 1)^s - 1.$$

Indeed a level has at most d strict dimension decreases and therefore at most $d + 1$ successive child epochs. Each event adds at most r_0 witness coordinates and polynomially many period coordinates. Including the mandatory coordinates, branch depth is polynomial in A . Each selected $\log q_l$ is bounded by A , so the accumulated branch modulus also has polynomial logarithm.

All auxiliary moduli, inverse relative sides and required inverse errors have polynomial logarithms. Thus an original side bound $\min_i H_i \geq \exp(\text{poly}_s(A))$ suffices simultaneously for every step-drop,

separation and residue-counting use, including all refinements of size at most r_0 . There is no circular choice: choose this bound for a construction truncated after the displayed maximum number of events; the dimension recurrence rules out the next event. The resulting finite tree terminates, and a leaf cannot violate (E.3), since a violation would force another event. $\square$

Remark E.3 (Uniformity of the stability estimates). Lemma B.1 gives bounds independent of polynomial-orbit coefficients, Lemma A.6 performs simultaneous splitting by joint linear algebra, and Lemma A.3 controls the reconstructed test on rational images. The fixed-map and cover constructions in the proof above preserve these bounds for the frozen right-translate test independently of the accumulated prime modulus.

E.1.2 Residue contraction and low-degree projections

The residue operator contracts a component once for every prime on which it depends. Components involving many primes are therefore small. For the components involving only a few primes, we need bounds from their conditional marginals; stability and the no-increment hypothesis will supply those bounds. The lemmas below separate these two estimates.

Residues of an integer point are close to product measure on each small collection of prime coordinates, but not necessarily on the entire collection. We therefore also prove versions of the low-degree estimates that use only these small marginals.

Lemma E.4 (A single-prime contraction). *Let $q = l^a$, let $X \in (\mathbb{Z}/q\mathbb{Z})^n$ and $V \in (\mathbb{Z}/q\mathbb{Z})^{n \times \ell}$ be independent uniform variables, and let $t \in (\mathbb{Z}/q\mathbb{Z})^\ell$ be independent and uniform. The operator*

$$(T\varphi)(X, V) = \mathbb{E}_t \varphi(X + Vt)$$

from functions on $(\mathbb{Z}/q\mathbb{Z})^n$ to functions of (X, V) has norm at most $l^{-\ell/2}$ on the subspace of mean-zero functions.

Proof. Expand in the orthonormal characters $\chi_\xi(u) = e(\xi \cdot u/q)$. For $\xi \neq 0$,

$$T\chi_\xi(X, V) = \chi_\xi(X) 1_{\{\xi \cdot V_j = 0 \text{ for every } j\}}.$$

The image of the homomorphism $v \mapsto \xi \cdot v$ has order at least l . Thus $\|T\chi_\xi\|_2^2 \leq l^{-\ell}$. The X average makes the images of different characters orthogonal, so Parseval proves the claim. $\square$

The low-degree global-function strategy here is motivated by global hypercontractivity; see [28] for background. The bounded-marginal estimate below and its approximate-product extension are proved here.

A stronger bound for an exact product law follows from Keller, Lifshitz and Marcus [29, Theorem 5.4]. We give a moment proof of the bound needed here, because the same argument will also handle approximately product laws.

Lemma E.5 (Low-degree mass under bounded marginals). *Let $\Omega = \prod_{i=1}^m \Omega_i$ carry a product probability measure μ , where the coordinate spaces are finite. Let $F : \Omega \rightarrow [0, M]$ and $K \geq 1$. Suppose*

$$\mathbb{E}[F \mid x_J] \leq K^{|J|} \quad (J \subseteq [m], |J| \leq r),$$

pointwise on positive-mass atoms, including $\mathbb{E}F \leq 1$ when $J = \emptyset$. Define the orthogonal product components (also called ANOVA components) by

$$F_S = \sum_{T \subseteq S} (-1)^{|S|-|T|} \mathbb{E}[F \mid x_T].$$

Thus $F = \sum_S F_S$, and each nonconstant component is centered in every coordinate on which it depends. Put

$$U_k^2 = \sum_{|S|=k} \|F_S\|_2^2.$$

If $p_1 \geq \max\{r, \log(2 + M)\}$, then, with $R = 4(1 + K)(p_1 + 2)$,

$$U_k \leq R^{2k} \quad (0 \leq k \leq r).$$

Proof. We first record an elementary mixed-norm fact. Suppose a function A of k variables, on arbitrary measure spaces, has L^2 norm at most E_0 , and every section obtained by fixing any j variables has L^2 norm at most E_j . In particular $\|A\|_\infty \leq E_k$. If $b_1 \geq \dots \geq b_k \geq 2$, with variable 1 outermost in the mixed norm, then

$$\|A\|_{L^{b_1} \dots L^{b_k}} \leq E_0^{2/b_1} \prod_{j=1}^{k-1} E_j^{2/b_{j+1} - 2/b_j} E_k^{1-2/b_k}. \quad (\text{E.4})$$

To see this, interpolate the innermost norm between L^2 and L^∞ . For $B = (\int |A|^2 dx_k)^{1/2}$ the remaining norm is bounded by E_k^{1-2/b_k} times the mixed norm of B , with exponents $2b_1/b_k, \dots, 2b_{k-1}/b_k$, raised to the power $2/b_k$. The section bounds for B are $E_0, \dots, E_{k-1}$, by Fubini. Induction on k proves (E.4). Zero bounds follow by approximation or by observing that the relevant section vanishes.

We prove the lemma by induction on k , simultaneously over all finite product spaces and all densities satisfying the hypotheses. For $k = 0$, $U_0 = \mathbb{E}F \leq 1$. For the induction step, represent the array of F_S , $|S| = k$, by the tensor

$$A((i_1, z_1), \dots, (i_k, z_k)) = \begin{cases} F_{\{i_1, \dots, i_k\}}(z_1, \dots, z_k), & i_1 < \dots < i_k, \\ 0, & \text{otherwise,} \end{cases}$$

on the disjoint union of the coordinate spaces, with measure $\nu = \sum_i \mu_i$. Its L^2 norm is exactly U_k .

Fix any $j > 0$ tensor variables. If their coordinate indices are repeated or incompatible with the ordering, the section vanishes. Otherwise call their set T , and fix their values z_T . For $A_0 \subseteq T$, define a density on the coordinates outside T by holding $x_{A_0} = z_{A_0}$ and averaging F in $x_{T \setminus A_0}$; denote this density by F^{A_0} . The density $F^{A_0}/K^{|A_0|}$ has cap at most M and satisfies the marginal hypotheses through level $r - |A_0|$. Consequently its level- $(k-j)$ norm is at most $R^{2(k-j)}$ by induction. Expanding the centering operators in the fixed variables gives

$$F_{T \cup U}(z_T, x_U) = \sum_{A_0 \subseteq T} (-1)^{j-|A_0|} (F^{A_0})_U(x_U), \quad |U| = k-j, \quad U \cap T = \emptyset.$$

The ordering restriction merely omits some U from the sum of squared norms. The triangle inequality in that direct-sum Hilbert space therefore gives the uniform section bound

$$E_j = (1 + K)^j R^{2(k-j)} \leq R^{2k-j} \leq R^{2k-1} \quad (1 \leq j \leq k).$$

Use $E_0 = U_k$.

Put $P_k = \sum_{|S|=k} F_S$, and choose an even integer q with $p_1 \leq q \leq p_1 + 2$. In the expansion of $\mathbb{E}P_k^q$, a term vanishes if a coordinate index occurs exactly once, because each ANOVA component is separately centered. Group the remaining terms by the partition of the qk slots into blocks of equal coordinate indices. There are at most $(qk)^{qk}$ such partitions, and every block has multiplicity

at least two. A partition repeating a block within one tensor copy contributes zero, by the strict ordering of that copy's indices; discard these partitions. For each remaining partition, take absolute values and drop the requirement that distinct blocks receive distinct indices. Its contribution is then bounded by an integral of $|A|$ over independent variables in the disjoint-union measure ν .

Order these block variables by decreasing multiplicity. Successive Hölder inequalities, starting with the innermost variable, give one mixed norm of A per tensor copy; in each norm the exponent of a variable is its block multiplicity. Indeed, at a block of multiplicity b , exactly b tensor copies involve that variable, so Hölder uses exponent b for each of those copies. The exponents in each resulting tensor norm are nonincreasing from outermost to innermost. By the mixed-norm bound, each norm is at most $\max_{0 \leq j \leq k} E_j$, since the displayed powers in the mixed-norm bound are nonnegative and sum to one. It follows that

$$\|P_k\|_q \leq (qk)^k \max\{U_k, R^{2k-1}\}.$$

Orthogonality gives $U_k^2 = \mathbb{E} F P_k$. Also $\|F\|_{q/(q-1)} \leq M^{1/q} (\mathbb{E} F)^{1-1/q} \leq e$. Since $qk \leq (p_1 + 2)^2$ and $k \geq 1$, our choice of R implies $e(qk)^k \leq R^{2k}$. Thus the moment bound yields

$$U_k^2 \leq R^{2k} \max\{U_k, R^{2k-1}\}.$$

If the maximum is U_k , divide by U_k when it is nonzero. Otherwise $U_k \leq R^{2k-1/2}$. In either case $U_k \leq R^{2k}$, completing the induction. $\square$

Lemma E.6 (Truncated projections for an approximately product law). *Let $\Omega = \prod_{a \in \mathcal{A}} \Omega_a$ have a product probability law μ , and let ν be another probability law on Ω . Suppose that for every $S \subseteq \mathcal{A}$ with $|S| \leq 2b$, the density $d\nu_S/d\mu_S$ differs from 1 by at most η pointwise. Let $0 \leq F \leq M$ and let $h = d(F\nu)/d\mu$. Let h_S be its orthogonal product projections, and put $P = \sum_{|S| \leq b} h_S$ and $A = \sum_{|S| \leq b} \|h_S\|_{L^2(\mu)}^2$. If $N_b = \sum_{j=0}^b \binom{|\mathcal{A}|}{j}$, then*

$$A \leq 2M\mathbb{E}_\nu F + 2\eta N_b^2 4^b M^2 (1 + \eta)^2.$$

Proof. The marginal density of $F\nu$ on any S of size at most b is bounded by $M(1 + \eta)$. Inclusion–exclusion therefore gives $\|h_S\|_\infty \leq 2^{|S|} M(1 + \eta)$. Orthogonality for μ gives $\int h P d\mu = A = \int P^2 d\mu$. Each term in P^2 uses at most $2b$ coordinates, so

$$\left| \mathbb{E}_\nu P^2 - A \right| \leq \eta N_b^2 4^b M^2 (1 + \eta)^2 =: E.$$

Consequently Cauchy–Schwarz and $F^2 \leq MF$ give $A^2 \leq M\mathbb{E}_\nu F(A + E)$. Writing $B = M\mathbb{E}_\nu F$, this implies $A \leq B + \sqrt{BE} \leq 2B + 2E$, as asserted. The same proof applies to a finite positive base measure, as will be needed after conditioning by reference atom masses. $\square$

Lemma E.7 (Low levels for an approximately product base measure). *Let μ be a product probability measure on N finite spaces, and let ν be a finite positive measure on the same product. Let $0 \leq F \leq M$, $K \geq 1$, and $h = d(F\nu)/d\mu$. Suppose*

$$\mathbb{E}_\mu[h \mid x_J] \leq K^{|J|} \quad (|J| \leq r),$$

including $\int h d\mu \leq 1$. Let $p_1 \geq \max\{r, \log(2 + M)\}$, and choose an even integer q with $p_1 \leq q \leq p_1 + 2$ and $q \geq 2$. Assume every marginal of ν on at most $r(q + 1)$ coordinates has density within η of 1 relative to μ , where

$$\eta \leq \frac{1}{2} (2^{r+1} (2 + N)^r (2 + M))^{-q}.$$

Then, with $R = 8(1 + K)(p_1 + 2)$,

$$\left(\sum_{|S|=k} \|h_S\|_{L^2(\mu)}^2 \right)^{1/2} \leq R^{2k} \quad (0 \leq k \leq r).$$

The assertion remains uniform after conditioning or marginalizing coordinates, provided their number plus the needed marginal order is at most the original cutoff. Here conditioning a measure on $x_A = z$ means restriction divided by its reference atom mass $\mu_A(z)$; the resulting base measure need not have mass exactly one. The weighted marginal hypotheses are renormalized by $K^{|A|}$ after fixing A .

Proof. Repeat the induction in Lemma E.5 for the product projections of h . We give the changes explicitly. Restricting a set A of coordinates to fixed values and marginalizing any other coordinates produces a base measure whose low-order densities are the corresponding marginal densities of ν divided by the reference atom mass. They still lie between $1 - \eta$ and $1 + \eta$. Its weight is a conditional expectation of F , hence remains between zero and M . The weighted marginal hypotheses, divided by $K^{|A|}$, remain exact. Thus the section induction has the same bounds $E_j = (1 + K)^j R^{2(k-j)} \leq R^{2k-j}$ for $j > 0$, and $E_0 = U_k$. Along a branch at most r coordinates are fixed; the assumed marginal order leaves at least qk free coordinates for every moment calculation.

The product moment estimate is therefore unchanged: $\|P_k\|_{L^q(\mu)} \leq (qk)^k \max\{U_k, R^{2k-1}\}$. Each raw marginal of h on at most k coordinates is bounded by $M(1 + \eta)$, so $\|h_S\|_\infty \leq 2^k M(1 + \eta)$. Every term of P_k^q depends on at most qk coordinates. Expanding it termwise gives

$$\left| \int P_k^q d\nu - \int P_k^q d\mu \right| \leq \eta(2^k N^k M(1 + \eta))^q \leq 1.$$

Orthogonality still gives the exact identity $U_k^2 = \int h P_k d\mu = \int F P_k d\nu$. Hölder with respect to ν , the cap on F , and $\int F d\nu \leq 1$ now yields

$$U_k^2 \leq e \left((qk)^k \max\{U_k, R^{2k-1}\} + 1 \right).$$

Since $e(qk)^k \leq R^{2k}/2$, this inequality rules out $U_k > R^{2k}$: in that case the maximum is U_k , and the right-hand side is strictly less than U_k^2 . This proves the induction, including for the restricted base measures just described. $\square$

For residues of a uniform integer point, the hypothesis of Lemma E.6 follows directly by counting residues modulo the product of at most $2b$ of the selected prime powers. The full residue law need not be close to product measure. This distinction prevents an invalid use of the full product L^2 norm of its density.

E.1.3 Proof of scalar transfer

Proof of Proposition E.1. Apply Proposition E.2 with its uniform bounds. Increase polynomial log budgets when necessary. We may first work with independent uniform product laws for x, V : domination transfers a sufficiently small failure probability to any of the allowed sampling laws. Let $0 \leq W(x, V) \leq 1$ be the indicator of failure. If the failure probability exceeds e^{-p} , then

$$\mathbb{E}_{x,V,t} W(x, V) (fg - (1 + \varepsilon)\lambda g)(x + Vt) > e^{-2p}. \quad (\text{E.5})$$

We will make the upper bound for this expression smaller than e^{-2p} . We first replace the two site functions by their low-degree residue projections. We then compare the resulting product-coordinate pairing by conditioning both sides and applying the no-increment hypothesis.

Physical approximation. Choose $Q = \exp(\mathcal{P}(p))$ sufficiently large, at least e^p . For every prime $l \leq Q$ use the largest power $q_l \leq Q$ as its residue coordinate. This collection captures every modulus at most Q , and in particular D . Partition $\mathcal{I}$ into a fine grid of boxes with comparable relative side lengths and polynomial log cost, keeping the boundary of $\mathcal{I}$ among the grid boundaries. All site weights and their approximations are zero outside $\mathcal{I}$. Thus stability will only be applied to restrictions of the original niltest to interior cells. Extend the grid over a fixed enclosing box for the sampled sites, assigning zero weights to its exterior cells; its side ratios and log cost have the initial polynomial bounds. For $h = g$ or $h = fg$, let $h^\#$ be the density, with respect to the product reference law of residue coordinates, of the actual h -weighted law on one such cell. Set, on that cell,

$$h^{\text{lo}}(u) = \sum_{|S| \leq b} h_S^\#(u_S).$$

The cutoff $b \geq \log_2 Q$ is polynomial and will be chosen after the conditioning costs below. The number of terms, each term's cap, and the sum of caps are $\mathcal{B}(p)$ once b is fixed. Products of a polynomial number of selected moduli have polynomial logarithms. More explicitly, let N be the number of selected primes, let $N_b = \sum_{j \leq b} \binom{N}{j}$, and put $C_b = N_b 2^b (1 + \eta)$. If a cell has minimum side J , every marginal on at most $2b$ prime coordinates has relative density error at most $O(nQ^{2b}/J)$. Prescribe this error to be at most η , with $\eta C_b^2 \leq 1/10$. Lemma E.6, followed by the same termwise comparison for the square, gives

$$\mathbb{E}_C(h^{\text{lo}})^2 \leq 2 + 3\eta C_b^2, \quad \mathbb{E}_C|h - h^{\text{lo}}|^2 \leq 7.$$

For $m \leq Q$ its prime support has size at most $\log_2 Q \leq b$. The product projection identity, compared with the actual cell law on the union of that support and each projection support, therefore gives

$$\left| \mathbb{E}_C(h - h^{\text{lo}}) 1_{u \equiv z \pmod{m}} \right| \leq \eta C_b m^{-n}. \quad (\text{E.6})$$

Indeed the product projection error against this residue indicator is exactly zero, and the pointwise relative marginal error contributes its reference atom mass m^{-n} times η and the sum of projection caps. Only marginals on at most $2b$ primes are used. The accuracy η and the corresponding side-length cutoff will be chosen after b .

We next verify that this replacement is harmless for the averaged sample in $L^2(x, V)$. For this estimate alone dominate the uniform spatial laws by smooth product laws at comparable scales. In the expansion of the square, omit the pairs t, t' for which

$$\gcd(t' - t) > Q \quad \text{or} \quad |t' - t| < L/B_0,$$

where $B_0 = \exp(\mathcal{P}(p))$ is chosen before b . After removing the fixed residue modulo D , divisibility counting gives an upper bound $e^{O_\ell(p)}(Q^{1-\ell} + L^{1-\ell})$ for the first set of pairs, with an additional negligible diagonal term and an allowed factor from D . Explicitly, for a nonzero integer difference of size $O(e^p L)$, the probability that a positive integer a divides all ℓ coordinates is at most $e^{O_\ell(p)}(a^{-1} + L^{-1})^\ell$; summing over $Q/D < a \leq e^{O(p)} L$ gives the assertion. We later require $L \geq Q$, which absorbs the finite-length term. The second set has mass at most $e^{O_\ell(p)}(B_0^{-1} + L^{-1})^\ell$. Write $e_h = h - h^{\text{lo}}$. Summing the cell bounds with their counting-volume weights gives $\mathbb{E}_\mathcal{I}|e_h|^2 \leq 7$, independently of the number and sizes of the cells. Conditional on t, t', V , the translate by x has point mass at most $e^{\mathcal{P}(p)}/|\mathcal{I}|$. Cauchy-Schwarz therefore bounds its absolute $e_h(u)e_h(u')$ average by $e^{\mathcal{P}(p)}$, with a constant depending only on the initial spatial scales, not on Q, B_0, b or the grid. Each discarded set costs this bound times its probability. Choose Q, B_0 to make these contributions negligible.

For a remaining pair put $m = \gcd(t' - t) \leq Q$. The joint location law of $u = x + Vt$ and $u' = x + Vt'$ is zero unless $u \equiv u' \pmod{m}$ coordinatewise. On that congruence its probabilities are, up to an arbitrarily small normalized error, of the form

$$\frac{m^n}{\prod_i H_i^2} \Phi_{t,t'}(u/H, u'/H), \quad (\text{E.7})$$

where $\Phi_{t,t'}$ has polynomially bounded log cap and Lipschitz constant, independently of b . Here is a rowwise derivation, retaining the error needed for the scale bound. Use smooth compactly supported row weights $\rho_x(x_i/H_i) \prod_j \rho_v(LV_{ij}/H_i)$, with bounded normalized derivatives; the allowed scale changes contribute only $e^{\mathcal{P}(p)}$. Put $\theta = t/L$, $\Delta = (t' - t)/L$, $s = u_i/H_i$, and $s' = u'_i/H_i$. Choose k with $|\Delta_k| \geq (\sqrt{\ell}B_0)^{-1}$. Eliminating

$$z_k = \frac{s' - s - \sum_{j \neq k} \Delta_j z_j}{\Delta_k}$$

gives the continuous row density

$$\Phi_i(s, s') = \frac{1}{|\Delta_k|} \int \rho_x(s - \theta \cdot z) \prod_j \rho_v(z_j) dz_{j \neq k}.$$

Its cap is at most $e^{O_\ell(p)}B_0$ and its Lipschitz constant at most $e^{O_\ell(p)}(1 + B_0)^2$, by differentiation of this formula.

For $d = t' - t$, the integer solutions of $d \cdot V_i = u'_i - u_i$ exist exactly on the displayed congruence. Their kernel lattice has covolume $|d|/m$ in its real hyperplane. Successive two-entry Bezout column operations transform d to $(m, 0, \dots, 0)$; at most $\ell - 1$ operations with entries bounded by $O_\ell(1 + |d|)$ give a kernel basis and a fundamental region of diameter $O_\ell((1 + |d|)^{\ell-1})$. At the normalized scale LV_i/H_i this diameter is at most $e^{O_\ell(p)}L^\ell/H_i$. Partition the hyperplane into these regions. The error between the smooth weight at a lattice point and its integral over the region is bounded by its derivative times the diameter; only regions meeting a fixed bounded enlargement of the support contribute. After normalization, coarea and the covolume give the row factor m/H_i^2 and error $e^{\mathcal{P}(p)}L^\ell/H_i$ in its normalized density. Taking the product over $n \leq p$ rows gives the pointwise law

$$1_{u \equiv u' \pmod{m}} \frac{m^n}{\prod_i H_i^2} (\Phi_{t,t'}(u/H, u'/H) + O(\epsilon_{\text{fib}})), \quad \epsilon_{\text{fib}} \leq e^{\mathcal{P}(p)} \frac{L^\ell}{\min_i H_i}. \quad (\text{E.8})$$

The product changes only polynomial log constants, since each row uses its own scale H_i . Thus $C' = \ell + 1$ suffices; this exponent has no dependence on n , b , or the ratios of spatial side lengths.

Choose the first physical grid so that every $\Phi_{t,t'}$ differs by at most a prescribed ζ from a constant on each pair of cells. There is no count loss in using this approximation. Indeed, CRT gives $|\{u \in \mathcal{I} : u \equiv z \pmod{m}\}| \leq 2|\mathcal{I}|m^{-n}$, so residue-wise Cauchy-Schwarz gives

$$\frac{m^n}{|\mathcal{I}|^2} \sum_z \left(\sum_{u \equiv z \pmod{m}} |e_h(u)| \right)^2 \leq 2\mathbb{E}_{\mathcal{I}} |e_h|^2 \leq 14.$$

Consequently the grid error and the error in Equation (E.8) cost $O(\zeta + \epsilon_{\text{fib}})$. For the cell-pair constants, Equation (E.6) in each cell bounds the sum by $\|\Phi_{t,t'}\|_\infty (\eta C_b)^2$: the two factors m^{-n} cancel the fiber factor and the number of residue assignments, and cell volumes sum as probability weights. Combining these estimates with the bad-pair bound yields

$$\|\mathbb{E}_t e_h(x + Vt)\|_2^2 \leq e^{\mathcal{P}(p)} \left(\Pr\{\text{discarded pair}\} + \zeta + \epsilon_{\text{fib}} + \sup_{\text{retained } t,t'} \|\Phi_{t,t'}\|_\infty (\eta C_b)^2 \right).$$

Thus

$$\left\| \mathbb{E}_t(h - h^{\text{lo}})(x + Vt) \right\|_{L^2(x, V)} \leq \text{prescribed small error.}$$

Here Q, B_0 , then ζ and the first physical grid, are chosen before b . The fiber-counting and Chinese remainder accuracies can use side lengths chosen after b .

The product-coordinate calculation. The physical replacement is now established. Finely mesh the normalized x, V, t variables so that $x + Vt$ remains in one physical cell on every retained triple of mesh boxes. Triples meeting cell boundaries have small total measure, since x varies at the full H_i scales; their cost is bounded by the individual mode caps, which are now available. On a retained triple, sufficiently large side lengths allow the t average to be evaluated modulo every prime support involved, with negligible Chinese remainder error. Let $W^\#$ be the product-reference density of the actual W -weighted residue law of $Y = (x, V)$ on its mesh box. The expressions to compare are

$$\sum_{|S| \leq b} \langle T_S h_S^\#, W_S^\# \rangle, \quad h = fg, g. \quad (\text{E.9})$$

For a prime coordinate, the reference coupling is $X_l = x_l + V_l t_l$, with $Y_l = (x_l, V_l)$ uniform and t_l uniform subject to the imposed D -residue. Both marginals are uniform. Outside the primes of D , Lemma E.4 gives centered operator norm at most $l^{-\ell/2}$. Distinct primes couple independently.

Conditioning the two sides. Take $\tau = \exp(-C_0(2+p))$, with C_0 sufficiently large. Choose $r = C_2(2+p)$ next, with C_2 large enough to absorb the $e^{O(2+p)}$ normalized cap losses below using the tail bound 2^{-r} . Then choose C_1 sufficiently large for the low-level estimates below. These are choices of fixed constants before any conditioning. Declare all prime coordinates of D and all $l \leq (2+p)^{C_1}$ mandatory. Their number is polynomially bounded and each has $\log q_l \leq \log Q$. Apply Proposition E.2 on the first physical cell, with these mandatory coordinates, tolerance τ , and further-coordinate allowance $r + j_*$, where $j_* = \lceil \log_K(2/\tau) \rceil$ for a fixed $K > 1$. The meaning of mandatory is that every leaf fixes those coordinates. On a leaf let A be the fixed coordinates and let v be the conditional baseline mean. Also condition on Y_A with the true reference coupling. The no-increment hypothesis applies on each resulting site cylinder: its product modulus, physical subbox cost, and hence total log cost are polynomially bounded. Indeed these are precisely the subbox and common modulus slices of Definition D.6; the sum of their logarithmic side losses is polynomial in p since $n \leq p$. Actual conditional means and the corresponding product marginal densities differ by arbitrarily small relative normalization errors, by Chinese remainder counting. Fixing a rare reference atom does not enlarge this relative error: divide the restricted measure by its reference atom mass, as in Lemma E.7.

Decompose the remaining W density by disjoint cylinder removal. All means in this removal are exact reference conditional masses of the weighted density; the underlying physical residual weight stays in $[0, 1]$. If its unconditioned mass is at most τ , retain it as a discard. Otherwise, among cylinders fixing at most $j_* + r$ coordinates, maximize their conditional residual mass times K^{-j} , where j is the number fixed. Each such mass is at most $1 + \eta \leq 2$, whereas the empty cylinder has mass above τ . A maximizer therefore has $j \leq j_*$. Any refinement on at most r further coordinates is then included in the search; maximality implies its conditional mean is at most K^a times the selected mean, where a coordinates were added. Remove the entire residual weight on this cylinder and repeat. A removed cylinder cannot recur, and the number of available cylinders is $\mathcal{B}(p)$. Each retained piece has conditional mean $m \geq \tau$ and, after division by m , satisfies the low-level marginal hypothesis of Lemma E.7 through level r . The marginal inequalities here are

exact, and the normalized physical cap is at most $1/m$. Condition also on every possible matching X assignment on those selected primes. The already prescribed stability allowance $j_* + r$ includes these coordinates and r more. The number of available cylinders and their inverse reference atom masses are $\mathcal{B}(p)$: each uses at most $j_* + r$ prime alphabets of size at most $Q^{n(\ell+1)}$, from at most Q selected primes.

We justify using the full coordinate coupling on these fixed coordinates although Equation (E.9) was truncated. By linearity split both densities into parts supported on assignments of a common coordinate set I . A part has the form $1_{X_I=z} F'(X_{I^c})$ or $1_{Y_I=y} W'(Y_{I^c})$. For a fixed outside support S , summing all inside supports gives exactly the true coupling weight $\Pr(X_I = z, Y_I = y)$ times the outside pairing. All inside supports are present when $|S| \leq b - |I|$. The remaining shell, $b - |I| < |S| \leq b$, has total absolute cost at most

$$\mathcal{B}(p) 2^{-(b-|I|)}. \quad (\text{E.10})$$

Indeed the centered operator norm on each outside coordinate is at most $2^{-\ell/2} \leq 1/2$. Cauchy–Schwarz over outside supports, with Lemma E.6 for the normalized conditional densities, bounds their squared-mass sums up to the outside cutoff b . Chinese remainder estimates to order $2b + |I|$ suffice for these uses; the conditional base measure is divided by its reference atom mass. The finitely many partial inside sums, assignment counts, number of pieces, and inverse cylinder probabilities have polynomial logarithms and are charged to the prefactor. All those bounds, including $|I|$, were set before b . Choose b last, at least $r + \max |I|$ and sufficiently large to make Equation (E.10) negligible. For discard pieces only A is conditioned. No bound on the full product-reference L^2 density has been used.

Completing the comparison. On a retained piece, stability bounds every baseline mean obtained by fixing at most r further prime coordinates by $v + O(\tau)$. The no-increment hypothesis bounds the corresponding fg means by $\lambda(v + O(\tau))$, with an additive error at most $\lambda\tau$. Thus the natural scales for the two site densities are $v + \tau$ and $\lambda(v + \tau)$. Dividing by fixed multiples of these quantities gives the bounded-marginal hypotheses of Lemma E.7. The fixed multiples also absorb the relative CRT normalization errors and make the zeroth masses at most one. The normalized logarithmic caps, including that of W/m , are $O(2+p)$: reference conditioning keeps the unnormalized physical weights bounded by one, while $m, v + \tau \geq \tau$ and $\lambda \geq e^{-p}$.

For the levels $1, \dots, r$, the centered contraction $\kappa = ((2+p)^{-C_1})^{\ell/2}$ and Lemma E.7 bound the pairing of either site density with the removed cylinder by

$$\text{scale} \cdot m \sum_{k=1}^r \kappa^k (C(2+p))^{Ck}.$$

Choosing C_1 after C_2 makes this as small a fixed multiple of $\text{scale} \cdot m$ as required. For the complete levels $r < |S| \leq b - |I|$, Cauchy–Schwarz and Lemma E.6 give κ^r times the product of the two truncated norms. Their normalized cap losses are $e^{O(2+p)}$. The preceding choice of C_2 already controls this tail using $\kappa \leq 1/2$, so r need not be increased after choosing C_1 .

The zeroth level explains why this comparison suffices. Its contribution to the signed pairing is at most

$$-\varepsilon \lambda v m + O(\lambda \tau m),$$

by (E.1) and stability. Choose the contraction errors above to total at most $\varepsilon \lambda(v + \tau) m / 4$ for the two signed terms. Their sum with the zeroth level is then at most $O_\varepsilon(\lambda \tau m)$. This estimate also covers a vanishing or very small baseline mean; the normalization always uses $v + \tau$, never v alone.

Discard pieces have cost $O(\sqrt{\tau})$ by Cauchy–Schwarz and Lemma E.6. Their truncated squared masses are $O(\tau)$, since their physical caps are bounded and their means are at most τ , whereas the unnormalized site truncated squared masses are $O(1)$. The prescribed Gram errors are chosen smaller than these bounds.

These errors aggregate without multiplying by the number of cylinders. Leaf probabilities in the adaptive tree sum to one; Y_A is averaged conditionally on X_A ; disjoint removal ensures that the selected Y -cylinder probabilities times their means m sum to a bounded quantity; and on those coordinates the X assignments use their true conditional probabilities. The real mesh triples likewise carry their original probability weights. The second, finer mesh only partitions the sample variables x, V, t ; the site densities $h^\#$ and their stability trees still use the original physical u cells, chosen before b . Thus neither the stability budget nor the no-increment slice cost depends on that finer mesh. Its smaller sample boxes require only later Chinese remainder accuracy and side-length cutoffs. All cylinder counts are uniform per mesh triple, and averaging their probability weights incurs no count loss. Hence, after all approximation errors are included,

$$\mathbb{E}W\mathbb{E}_t(fg - (1 + \varepsilon)\lambda g)(x + Vt) \leq O(\sqrt{\tau}) + \text{prescribed approximation errors.}$$

Choose C_0 large and all finite approximation errors sufficiently small to contradict Equation (E.5). This proves Proposition E.1. The choice order is: Q, B_0 and the first physical mesh; stability and cylinder budgets; b ; final meshes and accuracies; then sufficiently large L and H_i . Every log cost is polynomial in the original input budget, and the power $C' = \ell + 1$ of L comes only from the ℓ -dimensional row lattice calculation; increasing the spatial dimension changes its exponential prefactor only. Every later CRT requirement involves a modulus with polynomial logarithm, so it is met by the final $\mathcal{B}(p)$ cutoffs for L and $\min_i H_i$, without enlarging this exponent. $\square$

F Constrained affine sampling

We now require sampled affine boxes to carry small real lifts of specified polynomials, with integral polynomial complements. An independent reference law on coefficient arrays describes the desired lifts. Evaluating its density on the coefficients of $C_h(x + Vt)$ then tilts the spatial law of (x, V) . These are distinct laws: the actual polynomial coefficients are determined by the sampled affine map, not sampled independently of it.

The construction will support two later arguments. Cube comparison in Section G transfers path-dependent detection and gives finite approximation with pre bounds. Positive score recovery in Section H uses a nonnegative comparison weight for the same ensemble; only its final patch may incur the specified late costs. We state both outputs before constructing the charts and coefficient tilt, and prove here the normalization and residue estimates that those two arguments share.

F.1 The constrained sampler and its parameter order

Let $U = \prod_{i=1}^n \{0, \dots, H_i - 1\}$, with $n \geq 1$. For $1 \leq h \leq s$, let $W_h \subseteq \mathbb{R}^{d_h}$ be a rational subspace and let $C_h : \mathbb{Z}^n \rightarrow W_h$ be an ordinary real polynomial of degree at most h . Neither its coefficients nor the height of W_h is bounded at this stage. Write $\Lambda_h = W_h \cap \mathbb{Z}^{d_h}$ and fix $c_h \in W_h/\Lambda_h$. Choose a representative in W_h , also denoted c_h . Within the chart used below, write

$$C_h(u) - c_h = y_h(u) + \beta_h(u), \quad \beta_h(u) \in \mathbb{Z}^{d_h}, \quad |y_h(u)|_\infty < \frac{1}{4}. \quad (\text{F.1})$$

The chart makes this decomposition unique whenever it exists.

Definition F.1 (Rank at scale H). The system has rank at least R if, for every h and every integer row $a \in \mathbb{Z}^{d_h}$ with $\|a\| \leq R$ and $a|_{W_h} \neq 0$, the homogeneous degree- h part of aC_h cannot be written as a polynomial with coefficient bounds R/H^ν plus a rational polynomial whose coefficients have a common denominator at most R .

The sampler has three stages of parameters. First fix the degree and structural dimensions $n, \sum_h d_h$. The sampling dimension m and every block count are fixed polynomials of these dimensions, independent of testing accuracy, recursive output budgets, and chart widths.

Next choose a budget $b_{\text{pre}} \geq 2$ for the fixed dimensions, testing complexities, logarithms of inverse chart widths, and requested logarithmic inverse accuracies. A *pre bound* means a bound of the following size, with C depending only on the fixed degrees:

quantity	allowed bound
counts, caps, and coefficient masses	$\exp((2 + b_{\text{pre}})^C)$
logarithmic complexity and slice cost	$(2 + b_{\text{pre}})^C$
positive fractions, scores, and errors	$\exp(-(2 + b_{\text{pre}})^C)$ from below.

We call a positive quantity with the last bound *pre-large*. None of these bounds may involve $\log L$, $\log R$, or the heights of the W_h . In particular, a pre-cost slice has polynomial *logarithmic* cost.

Finally choose the remaining parameters in the order

$$\text{pre requirements ; } \sigma \downarrow ; \quad A \uparrow ; \quad R \uparrow ; \quad \min_i H_i \uparrow . \quad (\text{F.2})$$

Here $0 < \sigma < 1$ is the perturbation parameter for the coefficient boxes in the construction below. The integer L is selected using W_h within $A \leq L \leq A^{O_s(1 + \sum_h d_h)}$. The thresholds for R and H_i must work throughout that range, independently of the subspace heights and of all polynomial coefficients. Only a bounded number of successive enlargements of the pre budgets is allowed.

Proposition F.2 (Constrained sampling). *Prescribe a finite collection of input caps, testing complexities, slice bounds, and requested accuracies at the pre stage. Assume the system has rank at least R in Definition F.1, with R and the side lengths chosen sufficiently large in the order (F.2). For every tuple of centers, the construction gives one ensemble satisfying all the prescribed requirements, with bounds uniform in the centers, the inputs, and the path-dependent slice and test choices within those requirements. All later choices are made for this prescribed collection; changing the requested accuracies permits a new construction. The ensemble consists of integer affine maps $\psi(t) = x + Vt$ from a box $\mathcal{T} = \prod_{j=1}^m \{0, \dots, T_j - 1\}$, where m was fixed as a polynomial of $1 + n + \sum_h d_h$ before all testing and accuracy budgets, $T_j \leq L$, and some sides have length L . The ensemble has the following properties.*

1. *Each $\psi(\mathcal{T})$ lies in U , and the functions $y_h \circ \psi$ are fully slow polynomial lifts while $\beta_h \circ \psi$ have integer ordinary coefficients, all of degrees at most h . Noninjective maps have negligible probability. If the centers are mixed independently with Haar measure, the one-site law, with t uniform, approximates the uniform law on U at the prescribed forward accuracy against every unit-bounded input.*
2. *A permitted site twist is a bounded Lipschitz function of $u/H, (y_h)_h$, multiplied by buffered chart cutoffs and by characters or residue indicators of $(u, \beta_h)_h$ of a pre-bounded common modulus. For fixed centers, suppose a pre-large fraction of sampled maps detects an arbitrary pre-bounded function $\phi : U \rightarrow \mathbb{C}$, on a pre-cost slice of $\mathcal{T}$, against a bounded local niltest of degree r , with $0 \leq r \leq s$, and pre complexity. The slices and niltests may be chosen separately*

for each map. Then ϕ has a pre-large correlation on U with a permitted twist times a bounded ambient degree- r niltest of pre complexity.

More precisely, write

$$X(\phi) = \mathbb{E}_\psi \sup_{B,F} |\mathbb{E}_{t \in B} \phi(\psi(t)) F(t)|,$$

where B, F range over the prescribed slices and tests. Every fixed unit-bounded input admits an approximation in X , to any prescribed pre accuracy, by a linear combination of the indicated ambient twisted niltests. The number of terms and the sum of the absolute coefficients have pre bounds.

3. Let $T_*(u, \beta)$ be a fixed positive patch formula with d_* further slots, in which u has weight 1 and β_h has weight h , and with pre-bounded complexity. If $0 \leq f \leq 1$, $0 \leq \lambda \leq 1$, and a pre-large fraction of maps have a pre-cost slice on which

$$\mathbb{E}_t(f(\psi(t)) - \lambda)T_*(\psi(t), \beta(\psi(t))) \geq e^{-b_{\text{pre}}},$$

then, for fixed $\varepsilon > 0$, a slice of U supports a positive patch of rank at most $d_* + \sum_h \dim W_h$ and an inverse-exponential-in-polynomial score against $f - (1 - \varepsilon)\lambda$. The output complexity, slice cost, and logarithmic inverse score in this item alone may charge the logarithmic heights of bases of W_h and $\log R, \log L$.

There is also an unconstrained version of items 1 and 2: for every fixed $\ell \geq \ell_0(s)$ one may prescribe the ℓ integer side lengths in advance, provided they are sufficiently large and pairwise comparable. In particular they may be prescribed distinct comparable primes. Its integer spatial law is dominated by a uniform product-box law at scales H_i and H_i/L by a pre-bounded factor, as required for Proposition E.1. The choice of ℓ is fixed before all testing and accuracy budgets; all constants may depend on it.

Remark F.3 (Proof and uniformity of the sampler). Section H.4 assembles all three properties for one ensemble. Corollary F.9 supplies the geometric and one-site conclusions in item 1. Lemmas G.10 and G.11 give detection and approximation in item 2; Proposition H.1 gives item 3. In item 2, the output complexity is independent of $\log L$, and the supremum permits tests and slices chosen separately on each path.

F.2 The geometry of small lifts

Lemma F.4 (Coordinates with bounds independent of subspace height). *Let $W \subseteq \mathbb{R}^d$ be rational, let $\Lambda = W \cap \mathbb{Z}^d$, and let $\Lambda' = \text{proj}_{W^\perp} \mathbb{Z}^d$. Choose a basis v_i of Λ' with orthogonality defect and lengths at most $e^{O(d^2)}$, and put $K_i = \max(1, \lceil |v_i|^{-1} \rceil$). In a small lift y with $\text{proj}_{W^\perp} y = \sum_i v_i z_i$, the coordinates consisting of the orthogonal W coordinates and the numbers z_i/K_i give forward and inverse linear maps of norm $e^{O(d^3)}$. Moreover*

$$e^{-O(d^3)} \leq \frac{\text{covol}(\Lambda)}{\prod_i K_i} \leq e^{O(d^3)}.$$

In these charts Haar measure on W/Λ is Lebesgue measure in W times counting measure in z , divided by $\text{covol}(\Lambda)$.

Proof. The lattice Λ' is generated by the projections of the standard basis, all of length at most one; lattice reduction gives the displayed bounds for a basis. The vectors $K_i v_i$ have lengths between one

and $e^{O(d^2)}$, and the same orthogonality defect as the v_i . Determinant and singular-value bounds therefore give the claimed bounds for the linear coordinate maps.

The subgroup Λ is primitive in $\mathbb{Z}^d$, and the quotient $\mathbb{Z}^d/\Lambda$ identifies with Λ' . Lifting a basis of Λ' and adjoining a basis of Λ therefore gives a basis of $\mathbb{Z}^d$. Its determinant, evaluated in the orthogonal splitting $W \oplus W^\perp$, proves $\text{covol}(\Lambda) \text{covol}(\Lambda') = 1$. The asserted covolume bound now follows from the determinant of the matrix with columns $K_i v_i$. Finally each local sheet is a translate of W ; translation invariance and the volume of a fundamental region of Λ give the Haar formula. $\square$

Choose integer lifts of the v_i and let $l(z)$ be their integer linear combination. In Equation (F.1), projection gives $\text{proj } \beta = -\sum_i v_i z_i$, so $\beta + l(z) \in \Lambda$. Let w be its integer coordinates in a basis of Λ . The maps $\beta \mapsto (z, w)$ are integer linear maps. Consequently conditions on z, w modulo M can be expressed by characters whose coefficients are reduced modulo M , without paying for the heights of the chosen lattice bases. This observation does not bound their real coordinate norms, and no such bound is needed for this modular statement.

F.3 The coefficient tilt

We now construct a product density on the polynomial coefficient arrays and use it to weight the spatial affine maps.

Choose one L within the allowed range so that, simultaneously in all layers, no $K_i^{1/h}$ lies strictly between L and AL . There are only $\sum_h d_h$ relevant numbers, so a collection of more than that many successive disjoint multiplicative gaps supplies such a choice, after rounding endpoints with a fixed buffer. Call a projection axis inactive if $K_i \leq L^h$, moderate if $L^h < K_i \leq L^{C_s}$, and enormous otherwise. An *active* integer axis means a moderate or enormous projection axis.

Give each continuous or projection axis a number J_0 , fixed as a polynomial of $1 + n + \sum_h d_h$ before all accuracy budgets, of separate blocks of h parameter columns. Use side length L in these blocks, except for an inactive axis, where it is $\max(1, \lfloor cK_i^{1/h} \rfloor)$. Include further disjoint kernel blocks, each with n columns. In normalized spatial coordinates the kernel matrices lie in small neighborhoods of fixed invertible matrices, with pre-bounded inverse norms. The number of kernel blocks is likewise structural. Choose these counts to dominate the structural dimension multiples in the modular-rank estimate. Choose J_0 large enough for Lemma G.4; its proof gives a threshold polynomial in the structural dimensions alone. No requested precision is allowed to increase these block counts or m ; precisions are handled afterwards by choosing widths, perturbation size, and the later scales.

Subtract the chosen representative of each center c_h only from the constant coefficient. The polynomial lift has an independent small constant shift on each normalized axis, with widths chosen at the pre stage and independent of σ, L . Its main nonconstant part is a sum of the products of the h parameters in the dedicated blocks. The coefficient is 1 on an inactive integer axis (or 0 for a bounded axis), an integer in a small interval of length comparable to K_i/L^h on an active integer axis, and a real number in an interval of length comparable to L^{-h} on a continuous axis. All remaining nonconstant coefficients vanish on inactive and moderate axes. On continuous and enormous axes give these remaining coefficients smooth perturbations of normalized width comparable to σ/T^ν . Constants and widths are chosen, after the dimensions, so the sum of all monomial contributions stays well inside the prescribed chart throughout the parameter box.

All principal coefficients in different dedicated blocks are independent. On continuous and active integer axes choose their normalized supports inside $[\gamma_h, 2\gamma_h]$, with $\gamma_h > 0$ fixed after the structural block counts and chart widths. The normalization is multiplication by L^h on a continuous axis and

by L^h/K_i on an active integer axis. The constants γ_h are small enough for the preceding chart-containment requirement. Thus even after these coefficients are fixed, their normalized magnitudes are bounded below by a pre-large constant. Continuous choices have fixed smooth buffered densities after this normalization.

For an integer choice with center a and scale S , use the probability mass

$$p(k) = \frac{\psi((k-a)/S)}{\sum_{j \in \mathbb{Z}} \psi((j-a)/S)},$$

where ψ is a fixed nonnegative, nonzero smooth bump supported in $(-1, 1)$, and a, S are chosen so its rescaled support lies strictly inside the selected interval. This convention also applies to the small perturbation coefficients on enormous axes. When $S = K_i \delta$ and $K_i \delta$ is sufficiently large, the denominator is comparable to $K_i \delta$, and

$$q(u) = \frac{K_i \psi((K_i u - a)/(K_i \delta))}{\sum_{j \in \mathbb{Z}} \psi((j - a)/(K_i \delta))}$$

interpolates $K_i p(k)$ at $u = k/K_i$, with derivatives of order r bounded by $C_r \delta^{-r-1}$ independently of K_i . Every enormous-axis scale satisfies this lower bound by the later choice of L . On inactive or moderate axes the permitted singleton interpolation instead costs $O(\log K_i) = O_s(\log L)$. For constant projection coordinates use this law centered at zero at a fixed pre normalized width whenever K_i times that width is sufficiently large; otherwise choose the constant zero. In the latter case K_i is pre-bounded. A pre-bounded inactive axis may likewise have all its nonconstant coefficients zero. In particular when $W_h = \{0\}$ every coefficient is zero. Constant continuous coordinates have independent smooth densities of fixed pre width about zero. These constant choices are independent of all nonconstant coefficients. All remaining continuous perturbations are sampled as their center plus their stated width times an independent variable with a fixed smooth buffered density. This specifies the probability laws, not only their supports.

To verify normalization, for one coefficient write $u = y_W$ and let $f(u)$ be its normalized continuous density and $p_i(z_i)$ its independent projection masses. Lemma F.4 gives its density relative to Haar explicitly as

$$\text{covol}(\Lambda_h) f(u) \prod_i p_i(z_i), \quad d\text{Haar} = \frac{du d\#z}{\text{covol}(\Lambda_h)}.$$

Its integral is exactly one. For a constant coefficient, write this density as

$$\frac{\text{covol}(\Lambda_h)}{\prod_i K_i} f(u) \prod_i (K_i p_i(z_i)).$$

Every factor has a pre cap: the smooth constant widths are pre, and a singleton constant was used only when K_i is pre-bounded. Thus the constant-coefficient density has a pre cap independently of subspace height. The supports were chosen so their full Cartesian product lies inside the buffered chart, so no truncation or further normalization is required. Singleton projection masses select discrete local sheets and do not destroy absolute continuity. Multiplying these independent laws over coefficients defines a density $\mathcal{D}$ of integral one on the product coefficient torus

$$\prod_{h=1}^s \prod_{|\nu| \leq h} W_h / \Lambda_h.$$

Let $\rho(x, V)$ be a smooth product law on integer choices, with x nearly uniform on an interior box. Its kernel-block matrices have the pre-bounded neighborhoods and inverse bounds specified above.

Every nonkernel entry is instead supported in

$$|V_{ij}| \leq \frac{c\sigma H_i}{mT_j}, \quad (\text{F.3})$$

with a fixed smooth profile after rescaling by this width, independently of x , the kernel entries, and all other nonkernel entries. Consequently the total normalized displacement from all nonkernel columns is $O(\sigma)$ on the parameter box. The laws of x and the kernel entries do not depend on σ . Normalized derivative and cap logarithms for the whole spatial law are polynomial in the pre budget and $\log \sigma^{-1}$; these are allowed later costs. The actual sampling law is

$$\frac{1}{Z} \rho(x, V) \mathcal{D}(\text{coeff}(C_h(x + Vt))_h), \quad Z = \mathbb{E}_\rho \mathcal{D}(\text{coeff}(C_h(x + Vt))_h). \quad (\text{F.4})$$

For the unconstrained version set $\mathcal{D} = 1$, omit all lift and dedicated-block conditions, and use any fixed number $\ell \geq \ell_0(s)$ of columns. Choose independent smooth integer entries of V in small but nondegenerate intervals at scales H_i/T_j . Their normalized widths are chosen after the testing budgets and have pre-bounded inverse logarithms. Together with the similarly chosen x law, this is dominated by a uniform product-box law by a pre-bounded factor. The bounds use only comparability of the prescribed T_j to L , not equality or compositeness; hence prescribed distinct prime sides are allowed. All columns serve as the free columns in the subsequent cube argument. Taking ℓ above its fixed degree-dependent threshold makes the exceptional dependent parameter cubes negligible, as in that argument. No coefficient tilt or special axis lengths need to be selected in this version.

The identity $\int \mathcal{D} = 1$ on the abstract coefficient torus does *not* imply $Z = 1$; the latter requires equidistribution of the integer spatial choices. On the support of the tilt, subtracting c_h from the constant coefficient and then the prescribed small coefficient lifts leaves integer ordinary coefficients. Their polynomial sum defines $\beta_h \circ \psi$, and the sum of the small lifted monomials defines $y_h \circ \psi$. The latter stays inside the uniqueness chart at every parameter site and has every coefficient bounded at its actual T^ν scale. Thus, whenever $Z > 0$, these polynomials agree with the sitewise lifts in Equation (F.1) and give the asserted fully slow and integer-coefficient identities exactly.

Lemma F.4 gives the bound for ambient Fourier approximations. After multiplying an integer-axis probability mass by K_i , its normalized density has height-independent bounds. A singleton restriction on an inactive or moderate axis can be extended by a smooth window of width less than $1/(2K_i)$, with logarithmic cost $O(\log L)$. On an enormous axis every prescribed interval contains many grid points once $C_s > s$ and L is sufficiently large in terms of σ^{-1} . Its smooth interpolation then has costs independent of K_i . Smooth windows supported strictly within the ambient charts can be periodized and approximated by product Fourier kernels. This gives costs polynomial in pre budgets, $\log \sigma^{-1}$, $\log L$, and inverse error logs. The pre bounds in item 2 of Proposition F.2 follow from the cube-image comparison in Proposition G.1 and the approximation argument in Lemma G.11.

F.4 Fourier modes and Boolean cube evaluation

Lemma F.5 (The ideal of an affine Boolean cube). *Let $t_\omega = t_0 + \sum_{j=1}^q \omega_j e_j \in \mathbb{Z}^m$, where the e_j are linearly independent, and put $T_\omega = (1, t_\omega)$. For $h \geq 1$, the vector space of homogeneous degree- h real polynomials vanishing at every T_ω is spanned by products of h rational linear forms such that, at each site T_ω , at least one factor vanishes. The vanishing factor may depend on ω . If q, h are bounded and $|t_\omega| \leq L$, the forms can be chosen with heights bounded by $(2 + m + L)^{O_{q,h}(1)}$.*

Proof. A rational change of coordinates sends the points to $(a_0, a_1, \dots, a_q, z) = (1, \omega_1, \dots, \omega_q, 0)$. One can obtain this change using a nonzero $q \times q$ minor of the matrix of the e_j ; Cramer's rule gives the asserted height bound.

Reduce a homogeneous monomial by the relations $z_i = 0$ and $a_i^2 = a_0 a_i$. Its remainder is a linear combination of $a_0^{h-|S|} \prod_{i \in S} a_i$, where $|S| \leq h$. The functions $\prod_{i \in S} \omega_i$ on $\{0, 1\}^q$ are linearly independent, by successive evaluation on indicator vectors of subsets. Therefore a polynomial vanishing on the cube has zero remainder, and lies in the degree- h part of the ideal generated by

$$z_i, \quad a_i(a_i - a_0) \quad (1 \leq i \leq q).$$

Multiplying each generator by monomials gives the required spanning products. The height bound is preserved under this bounded-degree construction and the inverse coordinate change. $\square$

Lemma F.6 (Algebraic reduction of non-site modes). *Let $W \subseteq \mathbb{R}^d$ and let λ be a row functional on the coefficients of a homogeneous W -valued degree- h polynomial in T . If λ does not factor through evaluation at the points in Lemma F.5, there are rational linear forms $a_i \cdot T$, $1 \leq i \leq h$, whose product vanishes at every site, such that*

$$\lambda'(w) = \lambda \left(w \prod_{i=1}^h (a_i \cdot T) \right)$$

is a nonzero functional on W . Its coefficient bounds depend only on those of λ and the parameter tuples, and not on the height of W .

Proof. A functional factors through a linear map if and only if it vanishes on its kernel. The kernel of site evaluation on W -valued polynomials is W tensored with the scalar kernel described in Lemma F.5. If λ does not vanish there, it is nonzero on wP for one of the spanning products P and some $w \in W$. This gives the assertion. The construction of P involved only the parameter tuples, which proves the last statement. $\square$

Lemma F.7 (Quantitative removal of non-site modes). *Assume Lemma B.1 and that the polynomial system has rank at least R in Definition F.1. Prescribe a Fourier coefficient and frequency budget, a common cover modulus, spatial smoothness bounds, and an inverse-error log. Include $m, n, \log L$ in a budget b containing these inputs; for Equation (F.3) include $\log \sigma^{-1}$ as well. The spatial law may also be restricted to bounded-modulus congruences. For a fixed cube of bounded order with independent parameter differences, insert arbitrary unit-bounded functions of its sites in a spatial expectation. Every coefficient Fourier mode that fails to factor through real site evaluation on at least one W_h has expectation at most the prescribed error, provided $R \geq \mathcal{B}(b)$ and the parent sides are sufficiently large $\mathcal{B}(b)$. Neither bound charges the heights of W_h . The same conclusion holds for one site.*

Proof. It suffices to treat one mode and one spatial congruence; the specified Fourier coefficient mass absorbs their sums. Choose the highest layer h at which the mode does not factor. Every higher-layer phase is a real linear combination of the values $C_j(vT_\omega)$ and can therefore be absorbed into the arbitrary bounded site functions. Such real linear coefficients can be chosen by a rational right inverse of the parameter evaluation matrix, with the prescribed height bounds: the kernel condition holds on W_j , so composing with that right inverse agrees with the mode there. No basis of W_j is needed.

By Lemma F.6, choose rational forms $a_i \cdot T$ whose product vanishes at every site and whose contraction gives a row λ' nonzero on W_h . Clear their denominators. Replacing the forms by integer

multiples changes λ' by a bounded nonzero rational factor; all numerator and denominator logs remain polynomial in b . Write the remaining phase as $P(v)$, of degree at most h in $v = (x, V)$.

Take independent uniform integer vector shifts w_i in boxes of side lengths $\tilde{H}_a$, with a common stride divisible by the spatial congruence modulus. Choose $\tilde{H}_a$ to be the original H_a divided by a sufficiently large $\mathcal{B}(b)$ factor. This factor also dominates the reciprocal normalized widths, all $T_j \leq L$, the integer forms, and the congruence stride. Every displacement $\sum_i w_i a_i^\top$ is therefore small compared with each actual spatial width, including $c\sigma H_a/(mT_j)$ on nonkernel columns. The total variation distance between the smooth spatial law and each such translate is smaller than the prescribed error. This follows by integrating its normalized derivative bound; integer rounding and the fixed stride cause no loss when the sides are sufficiently large. Thus if the original mode expectation has modulus at least δ , averaging these translations changes it by at most $\delta/2$. The triangle inequality then supplies one base tuple v for which

$$\left| \mathbb{E}_{w_1, \dots, w_h} e \left(P \left(v + \sum_i w_i a_i^\top \right) \right) \prod_\omega F_\omega \left(v T_\omega + \sum_i w_i (a_i \cdot T_\omega) \right) \right| \geq \delta/2.$$

Extend the bounded site functions outside U by zero. Each site factor misses at least one w_i , because $a_i \cdot T_\omega = 0$ for at least one i ; assign it to one such variable. Applying Cauchy–Schwarz once in each independent variable gives an absolute phase average at least $(\delta/2)^{2^h}$ on independent pairs w_i, w'_i . All terms of degree less than h vanish. The degree- h term is precisely

$$h! \text{Pol}(\lambda' C_h^{\text{top}})(w_1 - w'_1, \dots, w_h - w'_h),$$

up to sign. This identity follows by taking the coefficient of the product of the h shift scalars in $C_h^{\text{top}}(\sum_i w_i (a_i \cdot T))$ and applying the mode to its T coefficients. In particular the resulting phase is independent of the chosen base tuple v .

Apply Lemma B.1 to this polynomial on the last-layer torus: take $G = \mathbb{R}$, $\Gamma = \mathbb{Z}$, and $G_1 = \dots = G_h = \mathbb{R}$, $G_{h+1} = 0$, with character $e(u)$ of frequency one. The associated graded algebra has only grade h , so the fast algebra killed by that character is zero. There are $2hn$ integer variables, its inverse bias log is $O_h(1 + \log \delta^{-1})$, and all individual sides are sufficiently large after the preceding shrinkage and stride. Thus the exact hypotheses of the lemma hold without a bound on the phase coefficients. Hence the homogeneous polynomial just displayed is a sum of a slow polynomial and a rational polynomial with common denominator $\mathcal{B}(b)$. Set $w'_i = 0$ in the homogeneous identity and identify all $w_i = w$. There are only polynomially many coefficients, so this restriction preserves slow bounds at the $\tilde{H}_a$ scales, up to $\mathcal{B}(b)$. The diagonal left side is $h! \lambda' C_h^{\text{top}}(w)$. Clear the denominator of λ' and the shift strides, absorbing $h!$ into the integer row and any resulting rational denominator, and restore the original H_a scales. We obtain an integer row a of norm $\mathcal{B}(b)$, nonzero on W_h , for which $(aC_h)^{\text{top}}$ is slow plus rational with these same permitted bounds. Choosing R larger than all these bounds contradicts Definition F.1. This proves the removal estimate uniformly for each mode before its Fourier coefficient is summed. $\square$

Remark F.8 (Real factorization and torus covers). Factoring through evaluation as a real linear map is not necessarily the same as factoring through evaluation on a torus. For example, the character $x \mapsto e(x)$ does not factor through $x \mapsto 2x$ on $\mathbb{R}/\mathbb{Z}$. For rational evaluation matrices one may clear the denominators of a rational right inverse by a finite common cover. Surviving real-factoring modes must be read on that cover, and the coefficient density must be lifted to the same cover. The required denominators depend on the parameter matrix, not on W . Ignoring this distinction would invalidate the mode removal argument.

Corollary F.9 (Coefficient equidistribution and one-site consequences). *Assume the hypotheses and symbol estimate of Lemma F.7. The coefficient arrays of $C_h(x + Vt)$ are equidistributed to any prescribed accuracy against the bounded-cost Fourier approximations used in the tilt, jointly with the spatial smooth weights and bounded-modulus congruences. Consequently $Z = 1 + o(1)$ in Equation (F.4). Mixed centers give the one-site approximation in item 1 of Proposition F.2; with fixed centers the one-site law has negligible excess mass above a pre-bounded density relative to uniform measure on U . Noninjectivity has negligible probability.*

Proof. Without site insertions, the proof of Lemma F.7 removes every nontrivial coefficient mode: choose the highest nonzero layer and a product of coordinate forms on which its coefficient functional is nonzero. The vanishing-at-sites condition is unnecessary because there are no site factors to eliminate. Only the trivial characters on each W_h/Λ_h remain, exactly as in product Haar integration. This proves the stated equidistribution and, using the Fourier approximation of $\mathcal{D}$, the normalization claim. This estimate is uniform in the centers: translating a constant-coefficient law changes Fourier coefficients only by phases, preserving all budgets. It therefore also permits averaging the normalized laws $\rho\mathcal{D}/Z$ over those centers. The same reasoning works on a common integer cover. It also proves Haar equidistribution of the coefficient arrays obtained by expanding $C_h(x + \sum_{i=1}^q \omega_i d_i)$ on $\omega \in \{0, 1\}^q$ and reducing $\omega_i^2 = \omega_i$. These squarefree coefficient arrays are the *Boolean jets*; their dimensions are recorded in Equation (G.1). In ambient averages over the base x and independent differences d_i , the preceding highest-layer argument applies to every nonzero character of these arrays. Polytope restrictions that the cube lie in U can be approximated by spatial smooth cutoffs, with boundary errors bounded by elementary box counting.

For one site, Lemma F.7 keeps only modes factoring through that evaluation. Averaging the centers removes every nonzero frequency on the constant coefficient of each layer. A mode factoring through one evaluation has its site row also as its constant coefficient row, so the only surviving mode is trivial. The unweighted one-site spatial law approximates uniform measure on U : choose x nearly uniform on a box omitting a prescribed small boundary fraction and choose V so that the maximal site displacement is a still smaller fraction of each H_i . The corresponding translation and boundary errors are arbitrarily small at the forward pre accuracy. This proves the mixed-center assertion against arbitrary bounded site functions.

With fixed centers, the surviving weight is the conditional Haar average of the coefficient density given the site value. Its constant coefficients are independent of the other coefficients and have a pre-bounded density relative to Haar, by the fixed pre widths of the constant laws and Lemma F.4; convolution with the remaining coefficients cannot increase that cap. The unweighted spatial pushforward has a pre-bounded density because of the full-scale x shift, even after conditioning on all of V . In particular this cap does not use the σ -dependent joint spatial density or the late supremum bound of $\mathcal{D}$. Thus for every measurable set $E \subseteq U$ the one-site law obeys $\Pr(E) \leq C \Pr_U(E) + \eta$, where C is pre-bounded and η is arbitrarily small in the later choice order. Apply this to the set where its density exceeds $2C$ to bound the total excess mass there by 2η .

Finally, for a fixed nonzero difference $a = t - t'$ a collision requires $Va = 0$. In one row, after all but a column with $a_j \neq 0$ are fixed, at most one value of V_{ij} is possible. Its probability is at most $\mathcal{B}(b)L/(\sigma H_i)$ under the smooth spatial law, allowing a nonkernel pivot in Equation (F.3). A union bound over at most L^{2m} pairs, followed by the available supremum bound on the tilt and $Z = 1 + o(1)$, makes the collision probability arbitrarily small when the parent sides are sufficiently large. All losses have polynomial logarithms in the already chosen budgets. $\square$

Corollary F.10 (Finite residues under the actual tilted law). *Fix centers and an error $\eta > 0$. Prescribe a finite family of joint residue vectors, each consisting of polynomially many spatial residues*

and integer coefficient lifts β_h or their integer-linear coordinates z, w , modulo a common modulus M for that vector. The maximum of $\log M$ and the logarithm of the family size have prescribed bounds. Include these bounds in the Fourier budget of Lemma F.7. For every vector in the family, its joint distribution under Equation (F.4) is within η in total variation of the following comparison law: sample the coefficient-chart residues on the Haar cover with density $\mathcal{D}$, and independently sample the corresponding spatial residues under the unweighted law ρ . Exact independence in this comparison law is not an assertion of exact independence under the tilted law.

This permits a family of separate tests modulo prime powers at most Q and modulo prescribed bounded products of witness prime powers. It does not assert a joint approximation modulo the product of all primes at most Q , whose logarithm need not have the permitted bound. The rank threshold and parent side cutoff may depend on the maximum individual $\log M$, the log family count, and $\log \eta^{-1}$; previously fixed bounds for the chart maps and coefficient-law parameters do not.

In particular this is a joint finite-residue comparison. It does not assert conditional uniformity under the actual law after fixing an exact real lift value. A bounded insertion depending on the site is handled by Lemma F.7 with that insertion retained; the finite-residue conclusion alone does not remove it.

Proof. Treat one joint residue vector at a time. Lift every coefficient torus to an integer cover divisible by its modulus M and by any evaluation denominators. Under Haar measure on $W_h/M\Lambda_h$, the deck coordinate in $\Lambda_h/M\Lambda_h$ is uniform and independent of its image in W_h/Λ_h . In the chart coordinates this says that the $w_{h,\nu} \bmod M$ coordinates are independent uniform variables across coefficients, while the $z_{h,\nu}$ retain their prescribed projection-grid laws under the tilt. Fixing centers and any prescribed finite collection of chart pieces depending only on the small lifts and projection coordinates does not change this deck-coordinate statement. The Fourier approximation below concerns $\mathcal{D}$ times residue indicators; any further continuous chart restriction is taken with a smooth buffered cutoff. Within a buffered coefficient chart, the integer lift residue is locally constant. The functions $\mathcal{D}$ times its residue indicators consequently admit ambient smooth Fourier approximations on the cover with the same permitted costs, enlarged polynomially in the logarithm of its modulus. For z, w this uses that they are integer-linear functions of β ; the character coefficients can be reduced modulo the common modulus. There is no dependence on the heights of the lattice bases. Here is an explicit ambient extension. The map $W/M\Lambda \rightarrow \mathbb{R}^d/M\mathbb{Z}^d$ is injective, since $W \cap M\mathbb{Z}^d = M\Lambda$. Around each $c + \beta$, $\beta \in (\mathbb{Z}/M\mathbb{Z})^d$, use the disjoint buffered chart $c + \beta + y$, $|y|_\infty < 1/4$. On that chart the density is a product of smooth functions of $\text{proj}_W y$ and the normalized projection coordinates z_i/K_i , multiplied by $\text{covol}(\Lambda)/\prod_i K_i$. Lemma F.4 bounds both these linear maps and the last factor independently of the height. Interpolate discrete masses after multiplying them by K_i : a singleton window costs at most a power of L on a moderate or inactive axis, while an enormous axis has many grid points in every interval, so its interpolation costs only inverse normalized interval widths. Multiply the whole chart by the required residue mask of its index β . Disjoint buffered supports preserve the supremum and Lipschitz bounds when these functions are periodized on $\mathbb{R}^d/M\mathbb{Z}^d$. The same construction in the product of coefficient charts has polynomial dimension. Fourier smoothing on this ambient torus therefore gives uniform approximation, with frequency and coefficient-mass logarithms polynomial in the permitted budgets and $\log M$, independently of subspace heights.

Apply coefficient equidistribution with each imposed spatial residue class. The shortest spatial entry width is at least $\sigma H_i/\mathcal{B}(b)L$. Choosing the parent sides after the individual modulus M makes counting in its residue classes arbitrarily accurate, also for these nonkernel entries. This uses only the permitted late inverse widths and $\log M$, and preserves the product spatial law. It compares the unnormalized expectation of every joint residue indicator to its product Haar-cover integral times

the unweighted spatial probability, with an arbitrarily prescribed absolute error. The number of atoms of the finite joint residue space has bounded logarithm: it is at most the polynomial number of coordinates times $\log M$. Choose the error per atom smaller than η divided by this number of atoms and by a fixed constant. Summing absolute errors gives total variation at most $\eta/2$ for the unnormalized laws. The normalization $Z = 1 + o(1)$ contributes at most the remaining $\eta/2$. This proves the assertion. In particular it compares the actual integer-spatial model with the abstract tilted model. The estimates are uniform over the prescribed family by using the maximum individual budget; to sum errors over the family, first divide the desired error by its size. One never forms the possibly enormous common modulus of all family members. Haar approximation on continuous variables alone would not imply the stated finite-residue conclusion without this argument. $\square$

More explicitly, if b_{pre} is the prescribed initial budget and η the desired comparison error, the Fourier approximation budget can be taken as

$$b_0 = \mathcal{P}(b_{\text{pre}} + \log \sigma^{-1} + \log L + \log \eta^{-1} + \max \log M + \log(2 + \text{family size})).$$

The rank cutoff is $R_* = \exp(\mathcal{P}(b_0))$. Replace $\log L$ here by its upper bound $O_s(1 + \sum_h d_h) \log A$ to choose R_* uniformly before the subspaces select L .

The Fourier and equidistribution costs in this subsection are allowed to use $\log L$ and $\log \sigma^{-1}$. Section G supplies the pre bounds for the pathwise testing seminorm in item 2 of Proposition F.2, through Proposition G.1 and Lemma G.11. Section H supplies fixed-patch transfer in item 3, through Proposition H.1.

G Cube comparison and pathwise detection

The coefficient tilt of Section F produces integer-affine maps $\psi(t) = x + Vt$ on which the constrained polynomials have small real lifts and integral polynomial parts. We now show how a correlation detected on many sampled boxes can be detected on the parent box. The local slices and niltests may depend on the sampled map.

The first step compares the full sampled cube functional with a controlled mixture of parent cube functionals. Local detection then gives an averaged cube norm, and the ambient inverse theorem gives a parent niltest. Separation converts this detector into finite approximation in the pathwise testing seminorm.

All these output bounds are fixed before the later sampling scales. Section H uses the same ensemble to recover positive scores, with additional late costs permitted only in the final patch. Its concluding proof of Proposition F.2 makes the parameter choices once for all prescribed requests. Until then, the later scales are chosen to meet the finite requirements of each estimate.

G.1 The cube functional to be compared

For an integer box or parameter slice B , let $\text{Cub}_q(B)$ be the set of tuples $\mathbf{t} = (t_0, t_1, \dots, t_q)$ for which

$$t_\omega = t_0 + \sum_{i=1}^q \omega_i t_i \in B \quad (\omega \in \{0, 1\}^q).$$

Every cube average uses uniform probability on this set, including degenerate tuples. For $q = 0$ it is the one-site average. An affine map sends such a tuple to an ambient cube. Nevertheless, the distribution of that ambient cube need not be uniform: the coefficient tilt and the number of affine

maps producing a given cube both affect its weight. The following proposition expresses that weight through functions of the individual vertices.

Proposition G.1 (Cube comparison). *Use the coefficient law of Section F, with the numbers of dedicated product blocks and kernel columns chosen sufficiently large as polynomials of the initial dimensions for the fixed cube-order and derivative ranges. Assume the hypotheses of Lemma F.7. Fix a pre-cost parameter slice B , a bounded cube order q , and a pre accuracy $\varepsilon > 0$. The averaged sampled-cube functional*

$$(P_\omega)_\omega \mapsto \mathbb{E}_\psi \mathbb{E}_{\mathbf{t} \in \text{Cub}_q(B)} \prod_\omega P_\omega(\psi(t_\omega))$$

is, uniformly for arbitrary $|P_\omega| \leq 1$, within ε of a mixture of ambient cube functionals

$$\sum_\alpha c_\alpha \mathbb{E}_{\mathbf{u} \in \text{Cub}_q(U)} \prod_\omega P_\omega(u_\omega) J_{\alpha, \omega}(u_\omega).$$

The $J_{\alpha, \omega}$ are allowed site twists of pre complexity, and $\sum_\alpha |c_\alpha|$ is pre-bounded. Mixtures can be replaced by finite sums to any prescribed accuracy. The bounds do not use σ, L, R , or the heights of the subspaces W_h .

The functions P_ω may be arbitrary bounded inputs. Thus equidistribution of the coefficient arrays alone does not prove the proposition. We use Lemma F.7 with these vertex functions retained: only coefficient modes determined by evaluation at the cube vertices survive. Counting the spatial fibers then gives the weight of the resulting ambient cube. The remaining problem is to describe the distribution of the evaluated small lifts.

There are two kinds of lift coordinates. Coordinates parallel to W_h have real-valued laws, while projected integer coordinates have lattice laws at scales K_i . In both cases the principal polynomial is a sum of products on disjoint parameter blocks. Those products supply a smooth distribution of the evaluated values. For the real coordinates we must also justify removing the small nonprincipal coefficients. For the lattice coordinates we retain the congruence factors as well as the smooth dependence on z_i/K_i .

We first prove the product-block estimate, then count the spatial and coefficient fibers, and finally compare the real image laws. These three estimates are assembled in the proof of Proposition G.1. The lift estimates act on the fixed number of outputs associated with one lift axis. The number of axes and input parameters may grow. Their counts enter the bounds polynomially; caps and inverse precisions enter through their logarithms.

G.2 Evaluating the product blocks

Fix a degree $h \geq 1$ and a cube order $q \geq 0$, both in the prescribed fixed range. Evaluation of a degree- h polynomial P on a parameter cube has the form

$$P\left(t_0 + \sum_{i=1}^q \omega_i t_i\right) = \sum_{\substack{S \subseteq [q] \\ |S| \leq h}} P_S(t_0, \dots, t_q) \prod_{i \in S} \omega_i, \quad \omega \in \{0, 1\}^q.$$

Powers of each ω_i are reduced by $\omega_i^2 = \omega_i$. The vector $(P_S)_S$, called the Boolean jet, determines all the vertex values. It has

$$j(h, q) = \sum_{a=0}^{\min(h, q)} \binom{q}{a} \tag{G.1}$$

coordinates. This dimension is independent of the number of parameter columns and lift axes. At $q = 0$ the jet consists of a single value; the same estimate will therefore control the one-site weights used in score recovery.

For a sum of independent product blocks, the Fourier coefficients of its jet law multiply. We will show that a single block can have substantial Fourier bias only at a bounded rational frequency plus a bounded frequency in normalized coordinates. Taking a sufficiently large structural number of blocks then gives an absolutely summable Fourier expansion. The required block count is fixed before the eventual truncation accuracy. The next two elementary estimates establish the single-block bias assertion.

Lemma G.2 (A many-multiples estimate). *There are absolute constants $C > 1$ and $0 < c < 1$ with the following property. Let $0 < \eta \leq 1$, $0 < \delta < c\eta$, and $N > C\eta^{-2}$. If at least ηN integers $v \in [-N, N]$ satisfy $\|v\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \delta$, then there is an integer $1 \leq d \leq C\eta^{-1}$ such that*

$$\|d\theta\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{C\delta}{\eta^2 N}.$$

The same assertion, with changed absolute constants, holds for any interval of length comparable to N .

Proof. Two successive good integers have positive difference $d \leq C/\eta$, so $\gamma = \|d\theta\| \leq 2\delta$. There is nothing to prove if $\gamma = 0$. Split $[-N, N]$ into progressions of common difference d . On each progression, the successive points on the circle are spaced by γ , in either orientation. A progression of length M therefore visits an arc of length 2δ at most

$$C((\delta + \gamma)M + 1 + \delta/\gamma)$$

times: split the lifted interval of length $M\gamma$ at the integers, and count at most $1 + 2\delta/\gamma$ points in each resulting visit. Summing over the at most d progressions gives

$$\eta N \leq C((\delta + \gamma)N + d + d\delta/\gamma).$$

For a sufficiently small absolute c and a sufficiently large constant in the lower bound for N , the first two terms are at most $\eta N/2$. Thus $\gamma \leq Cd\delta/(\eta N) \leq C\delta/(\eta^2 N)$. $\square$

Lemma G.3 (Dense products and multilinear bias). *For every fixed $e \geq 1$ there are constants $C_e, c_e > 0$ with the following properties.*

First, suppose a set of at least an η fraction of the integer tuples $v_i \in [-N_i, N_i]$, $1 \leq i \leq e$, satisfies $\|\theta v_1 \cdots v_e\| \leq \delta$. If

$$\min N_i \geq C_e \eta^{-C_e}, \quad \delta \leq c_e \eta^{C_e},$$

then some integer $1 \leq d \leq C_e \eta^{-C_e}$ satisfies

$$\|d\theta\| \leq \frac{C_e \eta^{-C_e} \delta}{N_1 \cdots N_e}.$$

Second, let $F(x_1, \dots, x_e)$ be a real polynomial of degree at most one in each variable, with coefficient θ on $x_1 \cdots x_e$. Let the independent integer variables x_i range uniformly over intervals of lengths N_i . If

$$|\mathbb{E}e(F(x))| \geq \zeta > 0, \quad \min N_i \geq C_e \zeta^{-C_e},$$

then θ is within $C_e \zeta^{-C_e}/(N_1 \cdots N_e)$ of a rational with denominator at most $C_e \zeta^{-C_e}$.

Proof. For the first assertion use induction on e , the case $e = 1$ being Lemma G.2. A fraction at least $c\eta$ of the outer tuples $(v_2, \dots, v_e)$ have a fiber with at least $c\eta N_1$ good values of v_1 . On each such fiber the preceding Lemma gives an integer $d_1 \leq C/\eta$ such that

$$\|d_1 \theta v_2 \cdots v_e\| \leq \frac{C\delta}{\eta^2 N_1}.$$

One value of d_1 works on a fraction at least $c\eta^2$ of the outer tuples. Apply the induction hypothesis with density $c\eta^2$, tolerance $C\delta/(\eta^2 N_1)$, and coefficient $d_1 \theta$. Increasing C_e and decreasing c_e in terms of C_{e-1}, c_{e-1} makes all of its hypotheses hold. Its denominator multiplied by d_1 , and its error multiplied by C/η^2 , have the asserted form. For example a recurrence with C_e greater than $4C_{e-1} + 10$, together with sufficiently small c_e , suffices. The number of iterations is fixed.

For the second assertion the case $e = 1$ is the geometric-sum bound. For larger e , apply box Cauchy–Schwarz in the first $e - 1$ variables, using independent pairs x_i, x'_i . All terms missing one of these variables disappear after differencing. The only term linear in x_e that remains has coefficient $\theta \prod_{i < e} (x_i - x'_i)$; the remaining terms are constant in x_e and disappear on taking its absolute mean. Hence, with $a = \zeta^{2^{e-1}}$,

$$a \leq \mathbb{E}_{x_i, x'_i, i < e} \min \left(1, \frac{2}{N_e \|\theta \prod_{i < e} (x_i - x'_i)\|} \right).$$

For a fraction at least $a/2$ of these pairs the norm in the denominator is at most $4/(aN_e)$. Each difference value has probability at most $1/N_i$. Thus the corresponding difference tuples occupy a fraction at least $c_e a$ of the full product of integer intervals $[-N_i, N_i]$, $i < e$. Apply the first assertion with $\eta = c_e a$ and $\delta = 4/(aN_e)$. The prescribed sufficiently large power lower bound on all the N_i ensures its size and tolerance hypotheses. Divide the resulting estimate for $\|d\theta\|$ by d ; increasing C_e one last time gives the claimed rational approximation. $\square$

The laws to be smoothed. Fix h, q and write $j = j(h, q)$. In one projection direction the principal polynomial is a sum of J_0 products, each using its own h parameter coordinates. Its constant term is fixed or is an independent integer shift. There are two discrete laws. On an inactive axis, each product has coefficient 1 and its parameter sides are comparable to $cK^{1/h}$. On a moderate axis, its parameter sides are L and its coefficient has an independent normalized smooth integer law on an interval of length comparable to $c'K/L^h$. The constants c, c' have pre bounds. A zero principal polynomial is allowed only when K is pre-bounded; that case will be enumerated.

Cube parameters on different axes are independent. On a single axis their normalized domain is the polytope on which all cube vertices belong to its designated interval. We allow coordinatewise slices and conditioning modulo pre-bounded moduli. These restrictions preserve independence between axes and between the dedicated blocks. The moderate block coefficients are also independent of all parameter tuples.

For the continuous law, use real normalized cube parameters on these polytopes. Each principal coefficient is either fixed with magnitude in $[B^{-1}, B]$ or sampled independently with the specified normalized smooth density supported in that range. The bound B also includes all inverse normalized volumes, coefficient widths, smooth-density bounds and residue moduli.

An allowed slice of relative size at least δ has stride at most $2/\delta$ on a sufficiently long axis. Its normalized scalar cube polytope has dimension $q + 1$, at most 2^{q+1} faces and volume at least $c_q \delta^{q+1}$. Its density is therefore at most $C_q \delta^{-(q+1)}$. Conditioning modulo M multiplies the stride by at most M . We include these pre losses in B . If an inactive axis is too short for these comparisons, its K is pre-bounded and is enumerated instead.

We use a finite Fourier expansion for the discrete law. The normalized jet support lies in a fixed box. Choose a fixed positive integer R_{tor} so this box is contained in a fundamental box for $\mathbb{R}^j/R_{\text{tor}}\mathbb{Z}^j$. If Y is the integer jet, set

$$D_K(z) = K^j \Pr(Y = z), \quad z \in (\mathbb{Z}/(R_{\text{tor}}K)\mathbb{Z})^j,$$

using the genuine integer representatives on its support. This is density relative to grid volume K^{-j} . Fourier inversion reads

$$D_K(z) = R_{\text{tor}}^{-j} \sum_{\xi} \mathbb{E} e\left(-\frac{\xi \cdot Y}{R_{\text{tor}}K}\right) e\left(\frac{\xi \cdot z}{R_{\text{tor}}K}\right).$$

The sum is over the displayed finite group. The corresponding continuous convention is Fourier series on $\mathbb{R}^j/R_{\text{tor}}\mathbb{Z}^j$, with a smooth cutoff outside the small support.

Lemma G.4 (Uniform product-block Fourier estimate). *For the preceding laws, choose the number J_0 of disjoint blocks sufficiently large as a polynomial in the initial dimensions. In the moderate case use the sampling scale range*

$$K \leq L^{C_s}, \quad K/L^h \geq A^h, \quad \log L \leq C_0 d \log A, \quad d = 1 + \sum_h d_h,$$

where C_s, C_0 depend only on the fixed degree range.

The Fourier coefficients of D_K have a pre-bounded absolute sum, uniformly in K, L and in the permitted slices and residue conditions. For every pre error $\varepsilon > 0$, retaining pre-boundedly many characters gives pointwise error at most ε . Their coefficient sum remains pre-bounded, and every retained character can be written as

$$e((a/d' + \xi/K) \cdot z), \quad d', |\xi| \leq \mathcal{B}(b_{\text{pre}}), \tag{G.2}$$

with $a \in \mathbb{Z}^j$, $\xi \in \mathbb{R}^j$, and a positive integer denominator d' . Thus one factor is a bounded-modulus character and the other is smooth in z/K .

The continuous jet law has a density with pre-bounded derivatives through any fixed order chosen before J_0 . The choice of J_0 uses this derivative order and the initial dimensions, but not the eventual truncation accuracy. Fixed constant shifts preserve all these bounds.

Proof. We first locate the frequencies at which one block can have large bias. We then use the scale range to make this estimate uniform down to a power of K^{-1} , and multiply the estimates over the independent blocks.

Testing one product block against a jet character gives a multilinear phase in h vectors of $q+1$ cube coordinates. A sampled product coefficient supplies one further scalar variable. Every jet frequency is an entry of this multilinear form, up to the fixed principal coefficient: specialize its affine factors to 1 or to selected Boolean coordinates to isolate a squarefree monomial of degree at most h .

Suppose the bias has magnitude at least ζ . Subdivide each normalized cube polytope into boxes of mesh $(\zeta/B)^{C_{h,q}}$. A boundary strip of width δ has volume at most $C_{h,q}\delta$, since its dimension and number of faces are fixed. The corresponding discrete count has additional error at most $C_{h,q} \sum_i N_i^{-1}$. After division by the domain volume, which is at least B^{-1} , these boundary errors are small compared with ζ . The smooth weight varies by at most $C_{h,q}B\delta$ on a box. Consequently, if

$$\min_i N_i \geq (B/\zeta)^{C_{h,q}},$$

one retained box has unweighted bias at least $\zeta/4$. Its side lengths are at least a fixed power of ζ/B times the original lengths. Residue conditioning gives the same conclusion after reindexing by its pre-bounded stride; its counting error is controlled by the reciprocal reindexed side lengths.

Choose a tensor entry and freeze all other scalar coordinates at values retaining that bias. The chosen full multilinear coefficient is unchanged, because every monomial uses at most one scalar coordinate from each vector group. Apply Lemma G.3 to the remaining $e = h$ or $h + 1$ variables. It shows that this entry is within

$$\frac{(B/\zeta)^{C_{h,q}}}{N_1 \cdots N_e}$$

of a rational with denominator at most $(B/\zeta)^{C_{h,q}}$. Different entries may use different favorable frozen coordinates; their full multilinear coefficients do not depend on those choices. The product $N_1 \cdots N_e$ is comparable to K within pre factors. There are only j entries to control, so their denominators may be combined at the same type of pre cost.

We next check the side-length requirement uniformly in the sampling scale. Enlarge C_0 so that $C_0 \geq s$. On a moderate axis,

$$L \geq K^{1/C_s}, \quad K/L^h \geq A^h \geq L^{h/(C_0 d)} \geq K^{h/(C_s C_0 d)}.$$

Thus every scalar side, including the coefficient interval, is at least $B^{-1}K^\kappa$, with $\kappa = (C_s C_0 d)^{-1}$. Inactive sides satisfy the same weaker bound because they have size at least $B^{-1}K^{1/h}$. Let C be the exponent required by the single-block argument and choose

$$P_0 \geq 2C/\kappa, \quad B' \geq B^{1+1/C}.$$

At $\zeta = B'K^{-1/P_0}$ one has

$$(B/\zeta)^C \leq B^{-1}K^{\kappa/2},$$

so the side-length requirement holds. It also holds for larger ζ . The exponent P_0 depends polynomially on the initial dimensions and not on the requested accuracy. If this lower bias cutoff exceeds a desired retained level ζ_0 , then $K \leq (B'/\zeta_0)^{P_0}$ is pre-bounded; the whole grid can be enumerated.

At a bias level ζ , the possible major frequencies number at most $(B/\zeta)^{C_{h,q}}$. Indeed the denominator and numerator ranges have those bounds, and each interval of width $(B/\zeta)^{C_{h,q}}/K$ contains only that many points of the Fourier grid. The number j of coordinates is fixed. Choose common nested major sets at dyadic bias levels, using the uniform estimate for every block. Outside the major set at level ζ , each single-block Fourier coefficient has modulus at most ζ . Independence gives modulus at most ζ^{J_0} for the sum of the blocks.

Summing over the levels above the cutoff gives a convergent geometric sum when J_0 is sufficiently large. Below the cutoff, the trivial number $O(K^j)$ of grid frequencies gives the bound

$$O(K^j)(B'K^{-1/P_0})^{J_0}.$$

If the cutoff is below ζ_0 , this is at most a fixed multiple of

$$(B')^{P_0 j} \zeta_0^{J_0 - P_0 j}.$$

Take $J_0 > 2P_0 j + 2C_{h,q}$. The two discarded sums become as small as requested by decreasing ζ_0 at pre cost. Fixing ζ_0 once at a constant level gives the absolute Fourier cap independently of the final accuracy. The retained major frequencies have precisely the form (G.2).

For the continuous law, suppose a single-block Fourier coefficient at frequency ξ has magnitude at least $\beta > 0$. Approximate the normalized real variables by grids of mesh $1/m$. For all sufficiently

large m the discrete bias is at least $\beta/2$. A selected full multilinear coefficient becomes $t(\xi)/m^e$, where $t(\xi)$ is its original linear functional of the jet frequency. The discrete estimate supplies

$$1 \leq d' \leq H, \quad \left| \frac{t(\xi)}{m^e} - \frac{a}{d'} \right| \leq \frac{H}{m^e}, \quad H = (CB/\beta)^{C_{h,q}}.$$

The bound for H is independent of m . Taking $m^e > H(|t(\xi)| + H)$ forces $|a/d'| < 1/H$, hence $a/d' = 0$ and $|t(\xi)| \leq H$. The tensor entries include every jet frequency, up to a pre-bounded nonzero principal coefficient. It follows that $|\xi| \leq BH$, giving single-block decay $B'(1 + |\xi|)^{-c_{h,q}}$.

Independent blocks multiply this decay. For the derivative order r fixed initially, choose $c_{h,q}J_0 > j + r$. Fourier inversion with integrable weight $1 + |\xi|^r$ gives the claimed derivative bounds. Fixed shifts multiply Fourier coefficients by phases and leave every estimate unchanged. $\square$

G.3 Counting integer fibers

The spatial affine map sends integer inputs to the root and differences of the sampled cube. Its pushforward consists of a smooth real density multiplied by an integer-image mask. The next lemma identifies both factors. Applied one ambient coordinate or lift axis at a time, its mesh loss has an exponent depending on the fixed output rank rather than the growing number of input coordinates.

Lemma G.5 (Smooth counting on a linear fiber). *Let $A : \mathbb{Z}^e \rightarrow \mathbb{Z}^j$ have real rank j , integer entries bounded by CL^h , and image of index g in $\mathbb{Z}^j$. Let positive diagonal input and output scales be S_ν and P_i . Assume the rescaled matrix $\tilde{A}_{i\nu} = P_i^{-1}A_{i\nu}S_\nu$ has bounded entries and has a j -minor with bounded inverse. Let φ be a C^1 function supported in a fixed bounded set in normalized input coordinates. Give $z \in \mathbb{Z}^e$ mass $(\prod S_\nu)^{-1}\varphi((z_\nu/S_\nu)_\nu)$.*

At an output point $v \in A\mathbb{Z}^e$, its mass, multiplied by $\prod P_i$, differs from g times the continuous normalized pushforward density by at most

$$C_\varphi \frac{L^{hj}}{\min_\nu S_\nu}. \quad (\text{G.3})$$

Here C_φ depends on e, j, C , the support and C^1 bounds, and the normalized minor and index bounds. It is independent of the scales. The assertion is used only when the displayed mesh is small. Off $A\mathbb{Z}^e$ the discrete mass is zero.

Proof. Write $J(B) = \sqrt{\det(BB^\top)}$ for a full-row-rank matrix, and let S, P be the diagonal scale matrices. Choose a nonsingular j -column pivot A_0 . For each remaining column a_ν , the vector with pivot part $-\text{adj}(A_0)a_\nu$ and free coordinate $\det A_0$ is an integer kernel vector. These $e - j$ vectors are independent and their coordinates are bounded by $j!C^jL^{hj}$. Their fundamental parallelepiped bounds the covering radius of their span, and hence that of the full integer kernel, by $C_{e,j,C}L^{hj}$. Voronoi cells of the full kernel have diameter at most twice this bound. Thus the number of input variables enters the constant, not the exponent of L .

If $\Lambda = \ker(A) \cap \mathbb{Z}^e$, then

$$\text{covol}(\Lambda) = \frac{J(A)}{g}. \quad (\text{G.4})$$

Indeed Λ is primitive. Extend its basis by lifts of a basis of $A\mathbb{Z}^e$ to obtain a basis of $\mathbb{Z}^e$. Orthogonal projection onto $\ker(A)^\perp$, on which A has volume factor $J(A)$, gives $1 = \text{covol}(\Lambda)g/J(A)$.

The restriction of S from $\ker(\tilde{A})$ to $\ker(A)$ has Jacobian $\det(S)J(A)/(\det(P)J(\tilde{A}))$, by coarea applied to $AS = P\tilde{A}$. Consequently the normalized kernel lattice $\Lambda' = S^{-1}\Lambda$ satisfies the exact identity

$$\text{covol}(\Lambda') = \frac{\det(P)J(\tilde{A})}{g \det(S)}.$$

Transform the original kernel cells by S^{-1} . They tile each attained normalized fiber and have diameter at most $r = C_{e,j,C} L^{hj} / \min S_\nu$.

Suppose $\text{supp } \varphi \subset [-R_0, R_0]^e$, $r \leq 1$, and the inverse norm of the normalized pivot is at most K_0 . Solving for its j coordinates gives, uniformly in the output y ,

$$\frac{\mathcal{H}^{e-j}(\{\tilde{A}x = y\} \cap [-R_0 - 1, R_0 + 1]^e)}{J(\tilde{A})} \leq K_0^j (2R_0 + 2)^{e-j}.$$

The continuous normalized output density is $f(y) = J(\tilde{A})^{-1} \int_{\tilde{A}x=y} \varphi d\mathcal{H}^{e-j}$. The covolume identity shows that the normalized discrete mass is a cellwise Riemann sum for $gf(y)$. The oscillation on each cell is at most $\|\nabla\varphi\|_\infty r$; summing only cells meeting the support therefore bounds its error by

$$g\|\nabla\varphi\|_\infty r K_0^j (2R_0 + 2)^{e-j}.$$

This proves (G.3), with explicit dimensional growth. For a continuous input mass one, ordinary full-lattice cells also give a normalization error $O_{e,R_0}(\|\varphi\|_{C^1} / \min S_\nu)$. When this is small, division by the discrete total mass preserves the estimate, using the same uniform bound for f . $\square$

Lemma G.6 (Fixed-dimensional polynomial sublevels). *Let $N, d \geq 1$. If a real polynomial Q in N variables has degree at most d in each variable and has a coefficient of magnitude at least $c > 0$, then, for $0 < u \leq c$,*

$$|\{x \in [-1, 1]^N : |Q(x)| \leq u\}| \leq C_{N,d} (u/c)^{1/(Nd)}.$$

The same conclusion holds on any fixed containing box, with changed constants. A probability density bounded by H multiplies the right side by H .

Proof. In one variable, if the sublevel set has length $\lambda > 0$, choose $d + 1$ points in it separated by at least $\lambda/(4d + 4)$. Lagrange interpolation at these points bounds every coefficient by $C_d u \lambda^{-d}$. This gives the assertion for $N = 1$. For induction, expand in the last variable and choose a coefficient polynomial having a coefficient of magnitude at least c . Discard the base points where its magnitude is at most τ . The induction hypothesis and the one-variable estimate bound the total sublevel volume by

$$C_{N,d} \left((\tau/c)^{1/((N-1)d)} + (u/\tau)^{1/d} \right).$$

Set $\tau/c = (u/c)^{(N-1)/N}$. Rescaling a fixed box changes coefficient norms by fixed factors and proves the remaining claims. $\square$

G.4 Kernel tuples and finite evaluation covers

To compare a sampled cube with a parent cube, we will solve for spatial variables using the kernel columns. A retained parameter tuple must permit this operation over both the reals and the integers. A real minor controls the smooth spatial density; the integer cokernel determines a finite cover on which coefficient characters can be read as functions of the evaluated jet.

Lemma G.7 (Kernel tuples and evaluation covers). *Let the cube order q and the maximum polynomial degree s belong to the fixed ranges, and use the independent kernel-column laws with sides L and the prescribed coordinatewise slice and residue restrictions. Choose a structural number $J > q(q + 2)$ of kernel columns when $q \geq 1$. For every pre error $\eta > 0$, there is a pre bound B such that, for all sufficiently large later L , all but probability η of the parameter cube tuples have the following properties.*

Let D be the $q \times J$ matrix of their differences on the kernel columns. It has a real q -minor of magnitude at least L^q/B , and the finite cokernel $\mathbb{Z}^q/D\mathbb{Z}^J$ has exponent $a \leq B$. For such a tuple, let $E_{h,t}$ be the integer matrix taking ordinary coefficients of degree-at-most- h parameter polynomials to their Boolean jets, and let e_h be the number of these parameter monomials. For every $h \leq s$, this map has a rational right inverse with denominators dividing a^h .

Consequently $M = a^s$, or its product with a previously prescribed pre-bounded modulus, satisfies

$$E_{h,t}\mathbb{Z}^{e_h} \supset M\mathbb{Z}^{j(h,q)}.$$

Membership in the image lattice is determined modulo M , and its index is

$$[\mathbb{Z}^{j(h,q)} : E_{h,t}\mathbb{Z}^{e_h}] = \frac{M^{j(h,q)}}{|\text{im}(E_{h,t} \bmod M)|}.$$

Both its membership mask and its index are determined by the parameter tuple modulo M .

For $q = 0$ every tuple has these conclusions: the difference matrix has zero rows, its empty minor is 1, and its cokernel has exponent 1. Constant polynomials give an integral right inverse for the one-site evaluation map. One may take $M = 1$, or the previously prescribed modulus, without discarding a set.

Proof. Assume $q \geq 1$. Choose any fixed q kernel columns. Their normalized difference determinant is a polynomial of degree q with a coefficient of magnitude one. The normalized scalar cube domains have pre-bounded densities on fixed containing boxes. Lemma G.6 therefore bounds the probability that the determinant has magnitude at most $2/B$ by a pre constant times a fixed power of B^{-1} . For the lattice parameter law, majorize the bad event by a smooth function vanishing beyond $2/B$. Its Lipschitz norm has pre cost, so grid approximation adds an error tending to zero with L . This proves the real-minor assertion.

On the full-rank event, the cokernel exponent divides a nonzero q -minor, and is therefore at most $C_q L^q$. A character of exact order a on $\mathbb{Z}^q$ has some coordinate of order at least $a^{1/q}$. After fixing the other entries of one kernel column, annihilation by this character requires that selected difference entry to lie in at most one residue class of this order. Counting in a containing box and dividing by the scalar cube-domain mass bounds its probability by

$$B'(a^{-1/q} + L^{-1}).$$

The prescribed strides and residue conditions reduce the effective order by at most their pre-bounded modulus. Absorb this loss into B' . For $a \leq C_q L^q$ the displayed bound is at most $B'' a^{-1/q}$.

The kernel columns are independent before imposing any full-rank condition. If the matrix has full rank and cokernel exponent greater than B , a character of some exact order $B < a \leq C_q L^q$ annihilates every column. There are at most a^q such characters at order a . Thus

$$\Pr(\text{full rank and exponent} > B) \leq (B'')^J \sum_{a > B} a^{q-J/q} \leq C_{q,J} (B'')^J B^{q+1-J/q}.$$

The structural choice $J > q(q+2)$ makes this tail decrease as a fixed power of B . Choose B for both exceptional sets and then choose L for the grid errors. The number of columns has not been increased in response to η .

Fix a retained tuple, and write t_K for its kernel coordinates. For each $1 \leq r \leq q$, choose $n_r \in \mathbb{Z}^J$ with $Dn_r = ae_r$. The rational affine polynomial

$$\ell_r(t) = n_r \cdot (t_K - t_{0,K})/a$$

equals ω_r at the vertex t_ω . Products of at most h of these polynomials, together with the constant, give a right inverse for the Boolean-jet map with denominators dividing a^h . This proves the image inclusion with $M = a^s$; multiplying M by another prescribed modulus preserves it. Reduction modulo M now gives the asserted membership and index formulas.

If numerator bounds are needed, choose a nonzero pivot minor of D , reduce nonpivot coordinates of the solutions modulo its determinant, and solve for the pivot coordinates using its adjugate. This bounds the interpolation numerators by powers of L . The cover assertion itself needs only the bounded denominator. $\square$

The cover may depend on the retained kernel tuple. No common multiple over all possible exponents $a \leq B$ is taken. This is sufficient because the subsequent cube comparison first fixes that tuple and works on its individual cover.

G.5 Stability of nonsingular polynomial images

The real output here consists of the continuous lift axes and the enormous projection axes after their lattice laws have been replaced by continuous laws. We condition on all variables generating inactive and moderate grid outputs. On each remaining axis, independent principal product blocks supply a nonsingular minor. The following statement then allows us to remove the small nonprincipal coefficients jointly in the real outputs, uniformly in that conditioning. It does not assert total variation stability for perturbations of a discrete law. We use the total variation norm $\|\mu\|_{\text{TV}} = \sup_{|f| \leq 1} |\int f d\mu|$; thus differences of probability measures have norm at most 2.

Lemma G.8 (Perturbing a polynomial image). *Let $\Omega \subset \mathbb{R}^a$ be bounded, let ρdx be a probability measure on Ω , and let $F, G : \Omega \rightarrow \mathbb{R}^b$ be C^2 maps on a neighbourhood of $\bar{\Omega}$. Suppose that there is a function $\chi \in C_c^1(\Omega)$ with $0 \leq \chi \leq 1$ and*

$$\int_{\Omega} (1 - \chi) \rho \leq \eta, \quad w := \chi \rho \in C_c^1(\Omega).$$

Fix b input coordinates $i_1, \dots, i_b$, and write

$$A_t(x) = (\partial_{i_1} F_t(x), \dots, \partial_{i_b} F_t(x)), \quad F_t = F + t(G - F), \quad 0 \leq t \leq 1.$$

Assume A_t is invertible on a neighbourhood of $\text{supp } w$. For $1 \leq r \leq b$, let $V_{t,r}$ be the input vector field supported in these b coordinate directions whose selected components are $A_t^{-1} e_r$. Suppose

$$\sup_{t,r} \int_{\Omega} |\text{div}(w V_{t,r})| dx \leq B.$$

If $\|F - G\|_{L^\infty(\text{supp } w), \ell^\infty} \leq \varepsilon$, then, for every $\delta > 0$,

$$\|F_*(\rho dx) - G_*(\rho dx)\|_{\text{TV}} \leq 2\eta + 2bB\delta + Cb\varepsilon/\delta,$$

where C is an absolute constant. In particular, when $B \geq 1$ and $0 < \varepsilon \leq 1$, the right-hand side is at most $2\eta + C'b\sqrt{B\varepsilon}$.

Proof. Put $\mu_t = (F_t)_*(w dx)$. For a smooth compactly supported test function φ , the chain rule and integration by parts give

$$\begin{aligned} \int_{\mathbb{R}^b} \partial_r \varphi d\mu_t &= \int_{\Omega} w V_{t,r} \cdot \nabla(\varphi \circ F_t) dx \\ &= - \int_{\Omega} \varphi(F_t(x)) \text{div}(w V_{t,r})(x) dx. \end{aligned}$$

Consequently each distributional derivative of μ_t is a finite signed measure of total variation at most B . Integrating the corresponding identity along a coordinate translation and then summing over coordinates shows

$$\|\tau_z \mu_t - \mu_t\|_{\text{TV}} \leq B\|z\|_1.$$

One may first test this identity on smooth functions; regularity of finite Borel measures then gives the stated total variation bound.

Choose a nonnegative one-dimensional smooth mollifier ϑ , of integral 1 and support in $[-1, 1]$, and let $\vartheta_\delta^{\otimes b}$ be its product at scale δ . Averaging the translation estimate gives

$$\|\mu_t * \vartheta_\delta^{\otimes b} - \mu_t\|_{\text{TV}} \leq bB\delta.$$

In particular μ_t is absolutely continuous: it is a total variation limit of the absolutely continuous measures on the left. Its density belongs to the space BV of functions whose distributional first derivatives are finite measures, with variation at most B in each coordinate.

The measure $w \, dx$, of mass at most 1, couples μ_0 and μ_1 . The L^1 norm of each partial derivative of the product mollifier is at most C/δ . Hence this coupling gives

$$\|\mu_0 * \vartheta_\delta^{\otimes b} - \mu_1 * \vartheta_\delta^{\otimes b}\|_{\text{TV}} \leq Cb\varepsilon/\delta.$$

The triangle inequality proves the displayed estimate for the cutoff measures. The discarded measures each have mass at most η , giving 2η for the original measures. Take $\delta = \sqrt{\varepsilon/B}$ for the last assertion. $\square$

Remark G.9 (Checking the hypotheses and their costs). It is enough that, on the support of w , the selected minor of DF has inverse norm at most K , that F, G have C^2 norm at most H , and that $\|DF - DG\| \leq (2K)^{-1}$. A Neumann series gives $\|A_t^{-1}\| \leq 2K$, uniformly in t , and differentiating an inverse matrix gives $\|\partial_i A_t^{-1}\| \leq 4K^2 \|\partial_i A_t\|$. The product rule therefore bounds the required B by a polynomial in $a, b, K, H, \|w\|_{W^{1,1}}$ (with a harmless choice of matrix norms). For example, taking H to bound each selected second-derivative matrix in operator norm, one may use

$$B \leq 2K\|\nabla w\|_{L^1} + 4bK^2 H\|w\|_{L^1}.$$

All these estimates concern the good-domain cutoff only; no density bound is required on the discarded set.

There is also a sharper estimate if $\|F - G\|_{C^1} \leq \varepsilon$. Let U_t be the input vector field with selected components $A_t^{-1}(G - F)$. Differentiating in t and integrating by parts gives

$$\frac{d}{dt} \int \varphi(F_t(x))w(x) \, dx = - \int \varphi(F_t(x)) \operatorname{div}(wU_t)(x) \, dx.$$

The same inverse-matrix bounds show $\int |\operatorname{div}(wU_t)| \leq B'\varepsilon$, where B' has the same type of polynomial dependence. Integration in t yields the stronger bound $2\eta + B'\varepsilon$. The mollification proof above is useful because it needs only a uniform displacement bound once the first-derivative estimates for the two image measures have been supplied.

The required minors for product blocks. Fix integers $h \geq 1, q \geq 0$, and put

$$\mathcal{S} = \{S \subseteq \{1, \dots, q\} : |S| \leq h\}, \quad j = |S|.$$

For an affine factor write $\ell(\omega) = a_0 + \sum_{r=1}^q a_r \omega_r$. Let $\mathcal{J}$ send a polynomial on the Boolean cube to its coefficients in the squarefree monomial basis ω_S . Consider the map

$$F_{\text{ax}}(a) = \mathcal{J} \left(\sum_{B=1}^J c_B \prod_{v=1}^h \ell_{B,v}(\omega) \right), \quad 0 < c_0 \leq |c_B| \leq c_1,$$

where different blocks use disjoint input coordinates and $J \geq j$. Allocate a different block $B(S)$ to each $S \in \mathcal{S}$. If $|S| < h$, differentiate its first factor with respect to its constant coefficient; specialize $|S|$ of the remaining factors to the different variables ω_r , $r \in S$, and all other factors to 1. If $|S| = h$, differentiate its first factor with respect to a coefficient a_r with $r \in S$; specialize the other factors to the remaining $h - 1$ variables. In either case the resulting Jacobian column is exactly $c_{B(S)} e_S$. These specializations can be made simultaneously, because the allocated blocks are disjoint. The determinant of this selected j -minor is therefore a nonzero polynomial in a bounded number of input variables, and at a point with coordinates in $\{0, 1\}$ has absolute value at least c_0^j .

Its degree is at most $j(h-1)$, and it depends on at most $jh(q+1)$ variables; both numbers depend only on h, q . Its coefficients thus include one of magnitude at least $c_0^j / C_{h,q}$, because evaluation at a $\{0, 1\}$ point is a sum of at most $C_{h,q}$ coefficients. Lemma G.6 consequently gives

$$\mathbb{P}(|\det A_{\text{ax}}| \leq \kappa) \leq C \kappa^{\alpha_{h,q}},$$

for normalized cube-parameter laws on bounded polytopes of nonnegligible volume; C may depend polynomially on reciprocal coefficient bounds, normalized density bounds and domain-volume bounds. One applies that lemma on a fixed containing box, then uses the density bound or divides by the domain volume. For $h > 1$, an admissible exponent is $\alpha_{h,q} = 1/(jh(q+1)j(h-1))$; for $h = 1$ the determinant is a nonzero constant and the small sublevel sets are empty. The determinant need not be large at an interior point of the sampling polytope; its coefficient bound is sufficient. For random principal coefficients, condition on them first. The bound is uniform because their magnitudes lie between c_0 and c_1 .

Several axes and the order of the choices. Suppose there are D output axes with disjoint principal parameter blocks, and write d_ℓ for their selected determinants. Let $C_* u^\alpha$ bound their individual sublevel probabilities, with $\alpha > 0$ depending only on the fixed degree and cube-order bounds. Choose $\kappa \leq 1$ so that

$$C_* D (2\kappa)^\alpha \leq \eta/2.$$

The union of $\{|d_\ell| \leq 2\kappa\}$ has mass at most $\eta/2$. If H_0 bounds the entries of the selected Jacobian blocks, their inverse norms on $|d_\ell| \geq \kappa$ are at most $C_{h,q} H_0^{j-1} / \kappa$. The full selected ideal Jacobian is block diagonal, so its inverse norm is the maximum of these block norms. No determinant of growing degree is used.

For the product laws under consideration, normalize coefficient intervals to fixed boxes. The scalar cube factors have dimension $q + 1$, a bounded number of faces, and pre-bounded densities. Thus there are $F_0 = O_{h,q}(a)$ source faces, each with a width- $2r$ strip of probability at most $C_* r$, after enlarging C_* by the normalized domain bounds. Choose $r \leq 1$ with $C_* F_0 r \leq \eta/2$. Smooth product cutoffs can vanish on the width- r boundary strips and on $|d_\ell| \leq \kappa$, while equaling one outside the width- $2r$ strips and $|d_\ell| \leq 2\kappa$. For the latter factors use smooth functions of d_ℓ^2 / κ^2 . The resulting χ satisfies $\int (1 - \chi) \rho \leq \eta$ and

$$\|\nabla \chi\|_\infty \leq C \left(F_0 / r + \kappa^{-1} \sum_{\ell=1}^D \|\nabla d_\ell\|_\infty \right).$$

The determinant derivative bounds have fixed degree. Products of cutoff factors cost the sum of their derivative bounds, since each factor lies between zero and one. Similarly, normalized product coefficient densities have L^1 derivative bounds equal to the sums of their one-factor bounds. Hence

$$\|\chi\rho\|_{W^{1,1}} \leq \|\rho\|_{W^{1,1}} + \|\nabla\chi\|_\infty$$

has the claimed cost. Apply Lemma G.8 and Remark G.9 to this cutoff. In particular, if D , the input dimension, and all logarithmic structural costs are bounded by a fixed polynomial in a budget p , then specifying an error $\exp(-p^{O(1)})$ requires only

$$\log \kappa^{-1}, \quad \log B, \quad \log \delta^{-1}, \quad \log \varepsilon^{-1} \leq p^{O(1)}.$$

The order is: prescribe the discard error, choose the minor and boundary cutoffs, choose the mollification scale, and only then choose the size of the polynomial perturbation, also making its first derivative smaller than half the reciprocal ideal inverse bound. These choices use no later lattice scale. When the source has small perturbation coefficients, they must first be written as their width times unit-scale auxiliary variables; then the source norms above are independent of that width and its smallness appears in the image map itself.

These image estimates handle the continuous perturbation step. Lemma G.4 supplies the Fourier truncation on projection grids. The proof of Proposition G.1 combines them with integer fiber counting and transfers Haar L^1 errors to ambient polynomial cube averages using Corollary F.9. All these estimates use the independent blocks and normalized domains specified above.

G.6 Proof of the cube comparison

Proof of Proposition G.1. We first retain the coefficient modes that factor through cube evaluation, then count the spatial fibers and average the resulting jet law. Finally we remove the small perturbations and expand the ideal law into permitted site twists.

Discard the exceptional kernel tuples from Lemma G.7. Fix a retained tuple and let E_t be the evaluation map from polynomial coefficient arrays to Boolean jets. Its real right inverse alone would not suffice for torus characters: for example, $x \mapsto e(x)$ does not factor through $x \mapsto 2x$ on $\mathbb{R}/\mathbb{Z}$. The individual M -cover in that lemma clears the denominators of the right inverse. Work on this cover, pull the coefficient density $\mathcal{D}$ back to it, and normalize against Haar probability. Write

$$D_t = (E_t)_*\mathcal{D}.$$

This notation means the density of the pushed-forward measure. A Fourier character of the array that factors through the real jet map factors on this cover as well: compose with a rational right inverse having denominator dividing M . Every other character is removed by the assumed non-site-mode estimate, even with the arbitrary site functions present. Conditional Haar averaging therefore leaves precisely D_t .

Apply Lemma G.5 to the spatial map from (x, V) to the root and differences of the spatial cube, one ambient coordinate at a time. Its normalized pivot uses the kernel columns and the root. The resulting weight is a smooth real density multiplied by the index and the image-lattice mask. The original parent side lengths can be chosen to make the fiber error as small as needed, including the late sup bound of D_t .

Here is the spatial displacement estimate using the prescribed law. For a fixed good kernel tuple, the normalized image of (x, V_K) , where K denotes the kernel columns, has a density f_K with a pre C^1 bound. Indeed solve the output equations on the root and a controlled kernel pivot, integrate the other smooth variables, and differentiate the fixed normalized input density. The nonkernel

support condition $|V_{ij}| \leq c\sigma H_i/(mT_j)$ implies that its contribution Z_N to every normalized root or difference coordinate has magnitude at most $C_q\sigma$. Independence of the spatial entries gives the full continuous density

$$f_t = \mathbb{E}_N f_K(\cdot - Z_N), \quad \|f_t - f_K\|_\infty \leq C_{\text{pre}}\sigma.$$

The integer fiber still carries its full image index and membership mask, even after this replacement of its real density. These factors are pre-bounded. The resulting error in the functional is bounded by $C_{\text{pre}}\sigma$ times an ambient average of D_t . Its Haar mass is one, and Corollary F.9, with the available late Fourier approximation, transfers that mass bound to the ambient average. Thus this error is pre-small without requiring a pre supremum bound for D_t . Narrow nonkernel spatial profiles affect the earlier fiber error only through late powers of σ^{-1} , which the parent side lengths absorb.

Write $\mathbf{t}_K$ for the fixed kernel tuple and $\mathbf{t}_N$ for all remaining parameter coordinates of the cube tuple. Condition only on $\mathbf{t}_K$ and on $\mathbf{t}_N \bmod M$, not on the values of $\mathbf{t}_N$. The retained spatial weight g_{sp} is independent of the unconditioned tuple entries. To check this independence, notice that the spatial fiber has already been summed over; the parameter tuple is not conditioned on an equation $Vt = v$. After the nonkernel spatial displacement is removed, the real density uses only the fixed kernel tuple. The image lattice contains $M\mathbb{Z}^{q+1}$, so its membership and index are determined by the fixed tuple residues modulo M . The remaining cube coordinates retain their product law across parameter axes, including under the coordinatewise slice restrictions. Up to an arbitrarily small error, the original functional is the average over the retained kernel tuples and residue classes of expressions of the form

$$|U|^{-(q+1)} \sum_{\mathbf{u} \text{ cube in } U} g_{\text{sp}}(\mathbf{u}) \prod_{\omega} P_{\omega}(u_{\omega}) \overline{D}(\chi(\mathbf{u})), \quad \overline{D} = \mathbb{E}[D_t \mid \mathbf{t}_K, \mathbf{t}_N \bmod M], \quad (\text{G.5})$$

where $\chi(\mathbf{u})$ is the collection of the Boolean jets of $C_h(u_{\omega})$ on the M -cover. A finite Fourier expansion of the spatial residue mask and Fourier smoothing of the normalized real density express g_{sp} as a pre-cost mixture of products of single-site factors. The conversion from the normalization in (G.5) to uniform genuine box cubes costs at most an exponential in the ambient dimension.

It remains to approximate $\overline{D}$ in Haar L^1 . Use buffered site charts so narrow that every alternating sum of lifts has absolute components less than $1/2$. Higher Boolean differences vanish exactly: for $|S| > h$, the degree bound gives $\Delta_S C_h = 0$, while (F.1) gives $\Delta_S y_h = -\Delta_S \beta_h \in \mathbb{Z}^{d_h}$. The buffered chart makes $|\Delta_S y_h|_\infty < 1/2$, so this integer vector is zero. In these charts the lift jets are the actual real evaluations of their coefficient arrays. Reference Haar has Lebesgue measure in the continuous split coordinates and grid volume K_i^{-j} in a projection coordinate, with the previously computed covolume constants. The remaining integral lattice coordinates on the cover are independently uniform; their evaluated residues acquire only the image index and membership mask modulo M . Keep this mask as an external multiplier of the output density; it does not condition the independent principal coefficient laws. More explicitly, in the notation of Section F, the deck coordinates of $\beta + l(z)$ are independent of the small coefficient lifts. Since l is integer-linear, $E_t l(z_{\text{coeff}}) = l(E_t z_{\text{coeff}})$; evaluated deck residues therefore impose only the stated output mask and no additional condition on the input projection coefficients. This description introduces no carries and no new equidistribution hypothesis.

For an enormous projection axis, replace integer coefficient sampling by continuous sampling, keeping its index and residue mask. In Lemma G.5 the input scales are K/T^ν , the output scales are K , and the number of output rows is the fixed number $j(h, q)$. Here is why the normalized pivot has no hidden L loss. On the kernel columns all $T_i = L$. Divide the parameter coordinates by L . The retained real difference minor gives linear polynomials, with pre-bounded coefficients in these normalized coordinates, whose values are the individual Boolean coordinates. Their products,

together with the constant, give a pre-bounded real right inverse for the normalized degree- h jet matrix. This uses only kernel monomials. Cauchy–Binet then gives a j -column minor with pre-bounded inverse: the number of possible minors is polynomial in the number of input monomials, since j is fixed. The normalized entries are monomials in t_i/T_i and are bounded independently of L . The coefficient density on these coordinates may have widths of order σ ; these affect its cap and derivatives, or equivalently the inverse pivot after width rescaling, only by allowed powers of σ^{-1} .

Since $T^\nu \leq L^h$, the cell error is at most $B_{\text{late}} L^{hj}/(K/L^h)$. After accounting for coefficient widths and normalization, the error is thus bounded by

$$B_{\text{late}} L^{h(j+1)}/K.$$

The logarithm of B_{late} is polynomial in the pre budgets and $\log \sigma^{-1}$ and does not use $\log L$. The exponent of L is independent of the number of coefficient variables and of J_0 . Choose initially $C_s > \max_{h,q} h(j(h,q) + 1)$, over the fixed range of degrees and cube orders. Then the enormous cutoff $K > L^{C_s}$ and a later choice of A absorb the constant. These real image densities have late Lipschitz bounds, uniformly in the normalized nonkernel tuple entries. Continuous axes satisfy the same estimates without a lattice error. Normalized Riemann summation now makes the dedicated long parameter tuples of the real and enormous axes continuous. Its error is bounded by a late constant times M/L , including the boundary strips of their fixed-dimensional cube polytopes. Keep all parameter tuples and coefficients generating inactive and moderate grid outputs discrete.

Remove the σ -perturbations in the real polynomial images. First condition on all the variables generating those grid outputs. Write every small coefficient as its prescribed center plus σr in normalized coefficient coordinates. The auxiliary variable r has a fixed smooth density with pre bounds; its density does not have width σ . Thus the common continuous input law has pre derivative bounds. Uniformly in the conditioned grid variables, the true and ideal output maps F_σ, F_0 satisfy

$$\|F_\sigma - F_0\|_{C^2} \leq C_{\text{pre}} \sigma.$$

The normalized coordinates of the conditioned variables stay in a fixed box, which gives this uniformity even for short inactive axes.

Apply Lemma G.8 jointly to the real output axes. On each axis the disjoint principal blocks give the fixed-size nonzero Jacobian minor constructed above. The joint ideal Jacobian is block diagonal. Discard a pre-small sublevel set on each block and a pre-small boundary region; a union bound over the axes controls the discarded mass. The individual inverse-minor and cutoff estimates give the pre divergence bound in that lemma, and C^1 closeness preserves invertibility along the interpolation. Choose the discard error and cutoffs, then the mollification scale, and only then σ . The resulting total variation error is pre-small, uniformly under the conditioning. Integrating the conditional estimate proves a comparison of the joint real and grid law, not only of its real marginal. Grid summation on the enormous axes is justified by the late Lipschitz bound for the perturbed density and the pre bound for the ideal product-block density. The index and mask are pre-bounded and multiply absolute errors by at most a pre factor.

After this replacement all axes factor. Apply the product-block estimate to the inactive and moderate grid laws and to the continuous laws. Every retained character (G.2) is a smooth character in z/K times a bounded-modulus character of z . Boolean jets are linear combinations of site data, so these characters split into products of allowed site twists. Buffered splitting-coordinate cutoffs prevent periodic aliases and are equal to one on the ideal and true supports.

There are two different error transfers in this argument. The true and ideal image densities both live on the same M -cover for the fixed kernel tuple. Their remaining discrete grid components have $K \leq L^{C_s}$; the enormous components have the smooth interpolations just proved. They therefore

admit controlled late Fourier approximations. The absolute value of their difference has the same type of Lipschitz control. Corollary F.9 transfers its Haar integral, with the spatial weight, to the ambient cube average in (G.5). The arbitrary site insertions are bounded by one in this estimate. The final ideal-to-truncated error is instead pointwise on the site charts, so no joint cover for the many retained rational denominators is needed. Each retained term has only polynomially many such denominators, whose product is pre-bounded. Frequencies in the bounded continuous ranges can finally be placed on a pre-fine mesh; their characters change uniformly little on the bounded normalized supports. Averaging the corresponding coefficients makes the mixture finite, with the same type of coefficient-mass bound. $\square$

G.7 Detection and finite-dimensional approximation

Lemma G.10 (Detection on path-dependent slices). *Fix the centers and a degree $r \geq 0$, and assume the hypotheses and conclusion of Proposition G.1 at cube order $r + 1$. Suppose a pre-large fraction of sampled maps detects a pre-bounded function $\phi : U \rightarrow \mathbb{C}$ on permitted pre-cost slices against bounded degree- r local niltests of pre complexity. The slices and niltests may be chosen separately for each map. Then ϕ has a pre-large correlation on U with a permitted site twist times a bounded ambient degree- r niltest of pre complexity.*

Proof. Normalize the function and test caps first. Let $\rho > 0$ be a pre lower bound for the retained local correlations, and let δ be a pre lower bound for the relative side counts of the allowed slices. Mesh their normalized endpoints finely enough that passing to an inner mesh slice removes at most $\rho/4$ of its points. Its normalized correlation changes by at most $\rho/2$, regardless of how rapidly the test oscillates. On a sufficiently long axis every permitted stride is at most $2/\delta$; enumerate these strides, residues, and mesh endpoints. An axis too short for this comparison has pre-bounded length, so enumerate its resulting subsets exactly. If all axes are short, enumerate the actual slices rather than their possibly redundant large-modulus descriptions. The number of resulting slices is pre-bounded. Restricting a local niltest to the inner slice and reindexing changes its orbit coefficients, which are unrestricted in Definition A.1, and preserves its degree and complexity. Pigeonholing now gives one common slice supporting a pre-large correlation for a pre-large fraction of paths; their tests may still differ.

Lemma A.9 gives a pre-large U^{r+1} norm on this common reindexed box. Its normalization is precisely (A.2); the overlap steps in its proof use the cube-count weights in (A.4). Put $q = r + 1$. The cube power is nonnegative on every path. If the retained fraction is τ and their local U^q norm is at least a , then the average local cube power is at least τa^{2^q} . Apply Proposition G.1 with error smaller than half this number, taking $P_\omega = \mathcal{C}^{|\omega|}\phi$ at the vertices, where $\mathcal{C}$ denotes complex conjugation. If the coefficient mass of its mixture is M , one of the mixed ambient integrals has modulus at least $\tau a^{2^q}/(2M)$, by the triangle inequality; the coefficients need not be positive. Write $f_\omega = \phi \mathcal{C}^{|\omega|} J_\omega$ and embed U in $H = \prod_i \mathbb{Z}/(4H_i)\mathbb{Z}$, extending these functions by zero. If $\kappa = C_q(U)/|H|^{q+1}$, the mixed group average is κ times the mixed box average, while (A.3) multiplies each of the 2^q norms by $\kappa^{1/2^q}$. Gowers–Cauchy–Schwarz on H therefore gives, after exact cancellation of κ , the bound

$$\prod_{\omega \in \{0,1\}^q} \|\phi \mathcal{C}^{|\omega|} J_\omega\|_{U^q(U)}$$

for the modulus of that mixed integral. The caps are pre-bounded, so at least one of these norms is pre-large. Divide that function by its cap and apply Theorem A.7 directly on the integer box U . It supplies a degree- r ambient niltest correlating with ϕ times the selected twist. Moving that twist to the test side and using closure under conjugation gives the claimed twisted niltest. For $r = 0$, a

one-dimensional cube consists of two independent sites; its integral is a product of two scalar means, so a site twist alone detects ϕ . Restoring the original caps has only pre cost. This argument does not require one common local niltest across paths; the common slice and the lower bound for its cube norm are sufficient. $\square$

We use the Hahn–Banach inverse-to-decomposition argument of Gowers [19, Section 3.2] to turn detection into approximation. The version below also clips the dual witness against a sampling density and makes the number of atoms independent of the size of the ambient set. All pairings below are normalized on a finite set, and complex convex hulls are absolutely convex.

Lemma G.11 (Detection implies approximation). *Fix $\eta > 0$. Let U be finite, $m : U \rightarrow [0, \infty)$ with $\mathbb{E}_U m \leq C$, and let X be a seminorm satisfying*

$$X(g) \leq C\mathbb{E}_U m|g|.$$

Let $\mathcal{J}$ be a family of unit-bounded functions. Suppose that, for each specified cap B and threshold $a > 0$, every $|v| \leq B$ with $X(v) \geq a$ correlates by at least $\delta(B, a) > 0$ with some member of an allowed enlargement of $\mathcal{J}$. Assume also that a cap C_0 is fixed, and that the L^1 mass of m above C_0 is sufficiently small in terms of the target η , C , C_0 , and $\delta(CC_0/\eta, \eta/2)$. For a family of sampling ensembles, this means C_0 is selected first and the later ensemble scales are selected after this one required tail accuracy. It does not require a single fixed density to have arbitrarily small tails above C_0 . Then every $|f| \leq 1$ has an approximation by a bounded-coefficient-mass combination of members of the enlarged $\mathcal{J}$ with X -error at most η . The coefficient mass and the number of terms depend only on the quantitative data just stated.

Proof. Put $B = CC_0/\eta$ and $\delta_0 = \delta(B, \eta/2)$, and fix the single allowed enlargement for this cap and threshold, calling it $\mathcal{J}$ in the proof. It suffices to assume the explicit tail bound

$$\mathbb{E} m 1_{m > C_0} \leq \frac{\eta}{C} \min(1/8, \delta_0/4).$$

Choose $M > \max(1, 2/\delta_0)$. Suppose $(9/8)f$ is outside

$$M\overline{\text{aconv}}(\mathcal{J}) + \eta\{g : X(g) \leq 1\}.$$

The first summand is compact in this finite-dimensional space, so the sum with the closed seminorm ball is closed. Finite-dimensional separation, followed by normalization, gives a functional χ such that

$$\Re\langle (9/8)f, \chi \rangle = 1, \quad X^*(\chi) \leq \eta^{-1}, \quad \sup_{J \in \mathcal{J}} |\langle J, \chi \rangle| \leq M^{-1}.$$

The domination of X implies pointwise $|\chi| \leq Cm/\eta$: test the dual inequality on a function supported at one point. Clip χ along rays at B , obtaining v . The tail assumption gives

$$\mathbb{E} |\chi - v| \leq \frac{C}{\eta} \mathbb{E} m 1_{m > C_0} \leq \min(1/8, \delta_0/4).$$

Since $|f| \leq 1$, the normalization implies $\mathbb{E} |\chi| \geq 8/9$. Hence $\mathbb{E} |v| \geq 8/9 - 1/8 = 55/72$, and radial clipping gives

$$X(v) \geq \eta \Re\langle v, \chi \rangle \geq \eta \mathbb{E} |v|^2 \geq \eta(55/72)^2 > \eta/2.$$

Detection supplies $J \in \mathcal{J}$ with $|\langle v, J \rangle| \geq \delta_0$. Its cap is one, so $|\langle \chi, J \rangle| \geq 3\delta_0/4 > M^{-1}$, a contradiction. Dividing the resulting inclusion by $9/8$ gives

$$f \in (8M/9)\overline{\text{aconv}}(\mathcal{J}) + (8\eta/9)\{g : X(g) \leq 1\}.$$

There is now a margin $\eta/9$ for a finite approximation. First approximate the closed-hull term by a finite combination with coefficient mass at most M and X -error at most $\eta/18$; this is possible in finite dimension. Then sample its atoms independently, retaining their phases and total coefficient mass. For N samples the expected squared error in $L^2(1+m)$ is at most $(1+C)M^2/N$, since the atoms have cap one. Moreover

$$X(g) \leq C(\mathbb{E}m)^{1/2}(\mathbb{E}m|g|^2)^{1/2} \leq C^{3/2}\|g\|_{L^2(1+m)}.$$

A choice of N depending only on C, M, η gives a realization with X -error at most $\eta/18$. Its total error is at most $8\eta/9 + \eta/18 + \eta/18 = \eta$, and its number of terms is independent of $|U|$. The same sampling argument applies to a bounded mixture. $\square$

For the sampler take $X(\phi) = \mathbb{E}_\psi \sup_{\text{slice, test}} |\mathbb{E}_t \phi(\psi(t)) \text{test}(t)|$. Every allowed slice has at least a pre fraction of the full parameter mass, say δ_{sl} , and every test has been normalized to cap one. Before averaging over paths, its supremum is at most $\delta_{\text{sl}}^{-1} \mathbb{E}_{t \in \mathcal{T}} |\phi(\psi(t))|$. Consequently

$$X(\phi) \leq \delta_{\text{sl}}^{-1} \mathbb{E}_U m_U |\phi|, \quad \mathbb{E}_U m_U = 1,$$

where m_U is the full one-site density. No exchange of the supremum with a conditional expectation is involved. If $X(v)$ is pre-large for a pre-bounded v , boundedness of this pathwise supremum gives a pre-large fraction of detecting paths, to which Lemma G.10 applies. Corollary F.9 supplies the required tail above a fixed pre cap for m_U . Choose its one required tail accuracy after the detector cap and threshold in Lemma G.11, and then choose the later ensemble scales. That lemma gives the asserted approximation clause.

G.8 Cube comparison in normalized chart units

We verify Lemma 3.3 for the one-layer law constructed in Section 3. The parameter sides now satisfy $T_\ell = O(L^j)$; inactive sheet axes also have a coefficient-one linear column, and moderate axes have the specified pure-free perturbations. The general cube proof above identifies the image measures and covers needed for these changes.

Proof of Lemma 3.3. Retain a good spatial kernel tuple and its real-evaluation cover. Non-site removal leaves the Boolean-jet pushforward of the coefficient density. The spatial fiber still carries its full index and image-lattice mask. These are determined by the kernel tuple and parameter residues on its controlled cover. Neither the added free columns nor the inactive linear columns are required pivots, and both have small spatial displacement. Removing that displacement changes the real spatial density by $O(\sigma)$ in normalized units, without altering its integer image mask.

Condition on the parameters of the additional inactive linear columns. They translate the corresponding product-block jets. For moderate axes also condition on pure-free parameters and the coefficients used in their additional terms. They give bounded translations and use no dedicated smoothing column. The coefficient-mass estimate in Lemma G.4 is unchanged: a translated Fourier character is multiplied by a phase. Omitted axes permit exact enumeration at warm normalized cost. Singleton data before replacement require only the allowed late interpolation on axes with $k_i \leq L^{C_*}$.

For enormous axes the normalized real right inverse for Boolean jets still uses the original kernel columns. The output row count per axis depends only on fixed degrees and cube order. Replacing the original monomial bound by $T^\nu \leq L^{j^2}$ changes the fiber error to a fixed power of L divided by k_i , times a late factor independent of $\log L$. The choice of C_* and then the range scale makes this small. Crucially, this is a conditional normalized Haar L^1 estimate: the other grid-generating

variables and coefficients are fixed, and their laws have total mass one. There is no product of pointwise grid caps on those other axes for the enormous cutoff to overcome.

Riemann comparison makes the dedicated parameters on real and enormous axes continuous, conditionally also on every variable generating a remaining integer-grid output. The ideal real image has a nonsingular minor from the unchanged disjoint principal blocks. On normalized input variables its perturbed version is $O(\sigma)$ -close in C^2 , uniformly under this conditioning. First discard the prescribed small determinant sublevel and boundary sets, then choose the mollification precision, and only then choose σ . Lemma G.8 gives small total variation for the joint real/grid image. This order does not assert a pointwise density approximation on singular fibers.

Intermediate errors are transferred as Haar L^1 errors. Their late ambient Fourier approximations are available by the proof of Lemma 3.2; this includes absolute values of density differences and the full cover masks. Haar integration then bounds them in the ambient average before arbitrary bounded site functions are inserted. Product-block comparisons after these replacements remain conditional comparisons on each cover residue class. In particular, no equidistribution assertion for arbitrary site inputs has been substituted for mode removal. These observations prove the cold statement.

If $q \leq j$, the lift jets consist of all 2^q unrestricted site values. In normalized chart coordinates their reference Haar volume is $m_j^{2^q}$, up to warm dimension factors. The ideal/truncated product-block laws and all their Fourier coefficient-mass bounds are warm in these coordinates. The covolume factors cancel by Lemma 3.1. Each vertex therefore contributes precisely the scale m_j^{-1} in Equation (3.5). A pointwise error in normalized chart units, supported on the buffered charts, integrates to its warm bound by Haar jet counting. Powers of v^{-1} , K_i , and m_j^{-1} in physical-coordinate approximations are used only in late transfer errors. Buffered cutoffs equal one on true and ideal supports. Quantizing the bounded normalized Fourier ranges turns mixtures into finite sums at the requested accuracy, with the same warm coefficient mass. This proves the warm statement. $\square$

H Positive score recovery from constrained samples

Detection in Section G supplies finite linear models, but their coefficients can have either sign. The next proposition returns a positive score against a fixed patch to the parent box. The comparison weights used in its proof have pre bounds; only the final conversion of such a weight into a patch may charge later scales and the heights of bases of the constraining spaces.

Proposition H.1 (Transfer of a fixed scalar score). *Assume the uniform constrained-sampler inputs of Proposition G.1. Fix $\epsilon > 0$ and prescribe pre bounds for the input complexities and for ρ^{-1}, τ^{-1} , where $\rho, \tau > 0$. Include these data in the prescribed finite collection of sampler requests and choose the later parameters in the order of Section H.4. Fix a tuple of centers and sample $\psi : \mathcal{T} \rightarrow U$ by the normalized law (F.4). Suppose $f : U \rightarrow [0, 1]$, $\lambda \in [0, 1]$, and a fixed positive patch formula $T_*(u, \beta)$ of the prescribed complexity with d_* additional slots satisfy*

$$\Pr_{\psi} \left\{ \begin{array}{l} \text{there exists an allowed slice } B \subseteq \mathcal{T} \text{ such that} \\ \mathbb{E}_{t \in B} (f(\psi(t)) - \lambda) T_*(\psi(t), \beta(\psi(t))) \geq \rho \end{array} \right\} \geq \tau.$$

The inner average is uniform on B , which may be chosen separately for each path; the probability uses the fixed-center law without conditioning on this event. Then there is an ambient slice and a positive patch with at most

$$d_* + \sum_h \dim W_h$$

slots having positive score against $f - (1 - \epsilon)\lambda$. The output complexity and the logarithm of the reciprocal score may additionally use the allowed late parameters and the heights of bases of the W_h : both have polynomial bounds in the resulting augmented logarithmic budget, as does the cost of the output slice.

We first define a nonnegative comparison weight for a fixed path. Modular rank will give its uniform cap and its approximation by site twists, independently of the number of prime powers in the comparison law. These bounds let us forecast the finite model complexity before fixing the ensemble. We then compare the model atoms and recover the determining slots. The final subsection checks that this argument and the preceding cube comparison use one ensemble with one order of parameter choices.

H.1 A positive comparison weight for a fixed path

Fix a path and one allowed slice. Write $t = (t_{\parallel}, t_I)$, where t_I comprises the inactive blocks and $t_{\parallel}$ comprises all kernel, continuous, and active blocks. Every parallel side has length L before taking the slice. At a true site u , use the chart coordinates $y_W(u), z(u), w(u)$ of Section F, and put

$$b(u) = (u, w(u), z_{\text{act}}(u)).$$

On the fixed path these are integer polynomials $b(t) = (\psi(t), w(t), z_{\text{act}}(t))$. The spatial coordinates have degree tag 1 and the coordinates belonging to layer h have degree tag h . Let m_{out} be their total number.

A slice may have different strides D_i in its parallel coordinates. Put $D = \text{lcm}_i D_i$, so that $\log D \leq \sum_i \log D_i$. Partition each parallel progression into its nonempty residue classes modulo D , retaining their original probability weights. The inactive coordinates are not changed. Since D is pre-bounded and the parallel sides are chosen later, these pieces remain permitted pre-cost slices. We include these refinements in the prescribed slice family. There are at most $D^{\dim t_{\parallel}}$ pieces, a pre-bounded number. A productive original slice has a refined piece with at least the same score. Choosing that piece separately on each productive path loses no productive probability. Thus, for score transfer, we may work on one refined slice, on which all parallel congruences have modulus D .

There are three parts to the comparison law. The inactive parameters retain their actual integer values. Parallel parameters have separate real and residue copies, sampled independently conditional on that same inactive vector.

For the real part, the independent blocks give a continuous density with pre cap and Lipschitz bound. For each lift axis, apply the continuous product-block estimate with the principal coefficients fixed in their prescribed normalized intervals away from zero. For the spatial output use at least $n + 2$ independent nonsingular kernel blocks. If A is one normalized kernel matrix, its inverse bound ensures that some coordinate of $A^T \xi$ has magnitude at least a pre-large multiple of $|\xi|$. Integration in that parameter gives decay $B/(1 + |\xi|)$. The independent blocks multiply these bounds, giving integrable Fourier decay with a first moment in n dimensions. The kernel and lift blocks use disjoint parameters. Fourier inversion therefore gives a continuous joint density; we use this version at every output point. The same estimates hold after the allowed coordinatewise slice restrictions.

Write G for this density of

$$(u/H, y_W, z_{\text{act}}/K)$$

obtained from real normalized parallel parameters on the slice. Use the principal lift polynomials, including their actual constant shifts, and the kernel contribution to spatial displacement. These outputs depend only on $t_{\parallel}$. The small nonprincipal lift terms and nonkernel spatial displacement are omitted in this real density. Their smallness will be used only when comparing smooth test values.

For the residue part, choose a finite set of distinct primes and one positive power of each, and let N be their product. Given the actual t_I , sample $t_{\parallel} \bmod N$ uniformly subject to the slice congruences, independently of the real copy. Let $d_{\text{rat}}(b; t_I)$ be the density of the resulting true integer polynomial output $b(t) \bmod N$, relative to uniform measure on $(\mathbb{Z}/N\mathbb{Z})^{m_{\text{out}}}$. Every coefficient in this congruence law is the actual integer coefficient of the path. In particular, the small terms omitted from G remain in this law.

The comparison weight on the parent is

$$K_*(u) = \kappa G(u/H, y_W, z_{\text{act}}/K) \mathbb{E}_{t_I} \left[\left(\prod_{i \in I} K_i \right) 1_{z_I(u)=z_I(t_I)} d_{\text{rat}}(b(u); t_I) \right], \quad \kappa = \prod_h \frac{\text{covol}(\Lambda_{W_h})}{\prod_{i \text{ in layer } h} K_i}. \quad (\text{H.1})$$

Buffered chart cutoffs are included and equal one on the relevant supports. All factors are nonnegative. The prefactor converts Haar measure in the parent chart into real volume on the continuous axes, normalized active grid volume, and exact inactive counting measure. The exact equality on inactive coordinates uses the same t_I as the residue law.

The comparison required for each site-twist model atom J is

$$\mathbb{E}_U J K_* \approx \mathbb{E}_{t \text{ in the slice}} J(\psi(t)).$$

We prove it in Proposition H.1 by testing one bounded modulus at a time. The large product N defines a finite comparison law; it need not be an admissible equidistribution modulus.

Two uniform bounds are needed before that comparison can be used. The density d_{rat} must have a cap independent of the number of primes, and K_* must be approximable by site-twist combinations with bounds independent of the eventual prime cutoff. The real product-block estimate and its order-zero integer version already control G and the inactive law. The remaining task is arithmetic: we must control the Fourier coefficients of d_{rat} uniformly as more prime powers are included.

H.2 Modular rank and the bounds for the comparison weight

For each degree tag, a primitive row of the leading homogeneous polynomial can be differenced until its phase is linear. We measure modular rank by the probability that this final linear coefficient vanishes modulo a prime power. The next lemma turns a small vanishing probability into Fourier decay. It allows exceptional prime powers, provided that their product is bounded. The following probabilistic argument establishes that product bound for the coefficient law, with a bound chosen before the prime cutoff. We then return to (H.1) to prove its cap and site-twist approximation.

H.2.1 A modular Fourier estimate

Lemma H.3 and the coefficient-law calculation following it will verify the hypotheses below for the comparison weight just defined.

Lemma H.2 (Modular rank implies Fourier decay). *Fix $s \geq 1$, $C > 0$, integers $m_h \geq 0$, and put $m = \sum_{h=1}^s m_h$. For each parameter z , let*

$$F_h(t; z) \in \mathbb{Z}[t_1, \dots, t_v]^{m_h}, \quad \deg_t F_h \leq h.$$

Suppose the homogeneous degree- h part $H_h(t)$ is independent of z . Let $N = \prod_{p \in \mathcal{S}} p^{A_p}$, where $\mathcal{S}$ is a finite set of primes, and let $D \geq 1$ be an integer. Write $e_p = v_p(D)$. Choose integers $0 \leq b_p \leq A_p$,

and put $D_0 = \prod_{p \in \mathcal{S}} p^{b_p}$. Suppose that for every $p \in \mathcal{S}$, every $b_p < a \leq A_p$, every h with $m_h > 0$, and every row $\lambda \in (\mathbb{Z}/p^a\mathbb{Z})^{m_h}$ having a unit coordinate,

$$\mathbb{P}_{u_1, \dots, u_{h-1}} (\text{Lin}_t \Delta_{u_1} \cdots \Delta_{u_{h-1}} (\lambda \cdot H_h)(t) = 0 \pmod{p^a}) \leq p^{-Ca}. \quad (\text{H.2})$$

Here the u_i are independent and uniform in $(\mathbb{Z}/p^a\mathbb{Z})^v$, $\Delta_u P(t) = P(t+u) - P(t)$, and Lin_t means the vector of coefficients of the linear terms. For $h = 1$ there are no differences.

Independently at each prime, sample $t \bmod p^{A_p}$ uniformly subject to one prescribed coordinatewise congruence modulo $p^{\min(e_p, A_p)}$. Combine these samples by the Chinese remainder theorem. Let μ_z be the law of $(F_h(t; z))_{h=1}^s$ on $(\mathbb{Z}/N\mathbb{Z})^m$, and let d_z be its density relative to uniform measure. Set

$$\gamma_s = \frac{1}{2^s s!}, \quad P = \frac{C\gamma_s}{2^s}, \quad K_s = \frac{4s}{\gamma_s}, \quad P_1 = PK_s.$$

For every character χ of exact order q ,

$$|\hat{d}_z(\chi)| \leq (D_0 D)^{P_1} q^{-P}, \quad (\text{H.3})$$

uniformly in z and the prescribed residues. If $P > m + 1$, deleting all Fourier terms of order greater than an integer $T \geq 1$ gives a uniform error at most

$$\frac{(D_0 D)^{P_1}}{P - m - 1} T^{-(P-m-1)}. \quad (\text{H.4})$$

At a prime p with $b_p = e_p = 0$, the corresponding local density satisfies, provided $P \geq m + 2$,

$$\|d_{z,p} - 1\|_\infty \leq 2p^{-2}. \quad (\text{H.5})$$

Proof. We first work at one prime, suppressing its subscript. A nontrivial character has order p^a , with $1 \leq a \leq A$. Its restriction to the h th group of coordinates has order p^{a_h} , where $0 \leq a_h \leq a$ and $\max_h a_h = a$. An empty maximum of tag depths is interpreted as zero. Choose initially an index i with $a_i = a$. Whenever a larger index $j > i$ has $a_j \geq a_i/(2i)$, replace i by such a j . This process terminates at an index h satisfying

$$a_h \geq \gamma_s a, \quad \max_{j>h} a_j < \frac{a_h}{2h}. \quad (\text{H.6})$$

Indeed the indices strictly increase, and the product of the possible losses $2i$ is at most $2^{s-1}(s-1)!$.

Suppose first that $a > K_s(b+e)$. Put

$$k = \max\{e, \max_{j>h} a_j\}, \quad B = a_h - hk.$$

The definition of K_s and (H.6) imply

$$k < \frac{a_h}{2h}, \quad B > \frac{a_h}{2} \geq \frac{\gamma_s a}{2} > b.$$

Partition the prescribed residue class further into classes modulo p^k . In each class write $t = r + p^k x$, with x uniform modulo p^{a-k} . Every term from a tag $j > h$ is constant modulo its character denominator, since integer polynomials respect congruences.

The remaining phase has degree at most h in x . Its homogeneous part of degree h is

$$\frac{\lambda \cdot H_h(x)}{p^B} \pmod{1}$$

for a row λ primitive modulo p^B . This follows by expanding $F_h(r + p^k x; z)$: its top part is $p^{hk} H_h(x)$, and the character on tag h has exact order p^{ah} . Terms of smaller degree may depend on r and z and need no estimate.

Apply Cauchy–Schwarz $h - 1$ times on the finite group $(\mathbb{Z}/p^{a-k}\mathbb{Z})^v$. Averaging the resulting linear phase in x is zero unless its coefficient vector vanishes modulo p^B . The difference variables project uniformly modulo p^B , because $B \leq a - k$. Thus (H.2) gives, on every class,

$$|\mathbb{E}_x e(\text{phase}(x))|^{2^{h-1}} \leq p^{-CB}, \quad e(u) = \exp(2\pi i u).$$

Consequently the original character average has modulus at most

$$p^{-CB/2^{h-1}} \leq p^{-Pa}.$$

The same reasoning includes $h = 1$, when the average of a linear character is evaluated directly.

If instead $a \leq K_s(b + e)$, the trivial bound 1 gives

$$|\widehat{d}_{z,p}(\chi)| \leq p^{P_1(b+e)} p^{-Pa}.$$

This estimate therefore holds in both cases. If $b = e = 0$, only the first case occurs, so there is no prefactor. The local averages multiply by the Chinese remainder theorem; their orders also multiply. This proves (H.3).

The number of characters of order exactly q on $(\mathbb{Z}/N\mathbb{Z})^m$ is at most q^m : all such characters belong to the subgroup killed by q , which has that size. Hence the discarded Fourier terms have total absolute mass at most

$$(D_0 D)^{P_1} \sum_{q>T} q^{m-P} \leq \frac{(D_0 D)^{P_1}}{P - m - 1} T^{-(P-m-1)}.$$

Finally, at an uncharged prime the nonconstant Fourier mass is at most

$$\sum_{a \geq 1} p^{a(m-P)} = \frac{p^{m-P}}{1 - p^{m-P}} \leq 2p^{-2}.$$

□

H.2.2 Producing the modular rank hypotheses

We first give a probabilistic sufficient condition for (H.2), then verify its coefficient-law hypothesis using Corollary F.10.

Lemma H.3 (Independent coefficients and exceptional prime powers). *Retain s, m_h, m and C from Lemma H.2. Put*

$$c_s = 2^{1-s}, \quad J \geq \left\lceil \frac{2(C + m + 10)}{c_s} \right\rceil, \quad Q_s = \max\{2, s^{4/c_s}\}.$$

Consider random homogeneous vector polynomials H_h modulo prime powers. Assume that, for every component of H_h , there are J squarefree monomials of degree h , using pairwise disjoint sets of variables, whose coefficients are independent uniform elements of the coefficient ring, independently of all remaining coefficients. Assume also that the complete coefficient arrays at distinct primes are independent.

Call p^a bad if (H.2) fails for some tag and primitive row, or if $p^a < Q_s$. For any finite collection of tested prime powers, including all lower depths, let b_p be the largest bad depth, with $b_p = 0$ if none is bad. Define the degree-dependent integer

$$B_s = \prod_p p^{\max(\{a \geq 1: p^a < Q_s\} \cup \{0\})}.$$

For every integer $R \geq 2$,

$$\mathbb{P}\left(\prod_p p^{b_p} > B_s R\right) \leq \frac{2}{9R^9}. \quad (\text{H.7})$$

This bound is independent of the number and depths of the tested primes.

Proof. We use a simple sublevel estimate. If a multilinear polynomial of degree at most r has a unit coefficient on some squarefree monomial of degree r , then, for uniform independent inputs modulo p^a ,

$$\mathbb{P}(p^a \mid f) \leq r p^{-a/2^{r-1}} \quad (r \geq 1). \quad (\text{H.8})$$

For $r = 1$, condition on all but a variable having a unit coefficient. For the induction, write $f = xg + h$ using a variable in the distinguished monomial. The polynomial g has a unit coefficient on a monomial of degree $r - 1$. Where $p^{\lceil a/2 \rceil} \nmid g$, the conditional probability of $p^a \mid xg + h$ is at most $p^{\lceil a/2 \rceil - 1 - a} \leq p^{-a/2}$. On the complementary event apply the induction hypothesis at depth $\lceil a/2 \rceil$. Adding the two bounds proves (H.8).

Fix a primitive row on tag h modulo p^a , and choose one of its unit components. Use the J designated monomials of this component. From each monomial select one variable, and examine its linear coefficient after $h - 1$ differences. The designated random coefficient from that block occurs with multiplier equal to a permanent of an $(h - 1) \times (h - 1)$ matrix of uniform difference variables. No designated coefficient from another block enters this linear coordinate, because the variable sets of the designated monomials are disjoint.

Condition on the difference variables and all other polynomial coefficients. Each selected equation has conditional probability at most

$$p^{-a} \gcd(p^a, \text{perm}),$$

and these bounds multiply. The permanent has a unit coefficient on a squarefree monomial of degree $h - 1$. Splitting according to whether its valuation is at least $\lceil a/2 \rceil$ and using (H.8) bounds the expected cost of one equation by

$$h p^{-a/2^{h-1}} \leq s(p^a)^{-c_s}.$$

For $h = 1$ the multiplier is 1 and the cost is exactly p^{-a} . The shift variables used by the different permanents are disjoint, so these expected bounds also multiply. Thus, if ρ denotes the vanishing probability in (H.2),

$$\mathbb{E}\rho \leq (s(p^a)^{-c_s})^J.$$

Markov's inequality and a union bound over at most $s(p^a)^m$ rows and tags give

$$\mathbb{P}(p^a \text{ fails the rank test}) \leq s^{J+1} (p^a)^{C+m-c_s J} \leq (p^a)^{-10} \quad (p^a \geq Q_s).$$

The last inequality follows from the stated choices of J, Q_s . For a product d of powers of distinct primes, all at least Q_s , independence across primes therefore bounds the probability that all its chosen prime powers are bad by d^{-10} .

Put $W = \prod_{p: p^{b_p} \geq Q_s} p^{b_p}$, so that $\prod_p p^{b_p} \leq B_s W$. This discards whole small factors; it does not assert that dividing a bad prime power by a factor of B_s leaves a bad prime power. If $W > R$,

either a single factor exceeds R , or a subcollection has product in $(R, R^2]$ (multiply them one at a time until the product first exceeds R). Union-bounding over single-power witnesses and over these composite witnesses gives

$$2 \sum_{d > R} d^{-10} \leq \frac{2}{9R^9}.$$

The product of all largest bad powers is at most B_s times the product just considered, proving (H.7). $\square$

Remark H.4 (Approximate coefficient laws). Suppose every tested prime power is at most an integer Q , and the actual joint coefficient law modulo each witness modulus is within η in total variation of a model satisfying Lemma H.3. It is enough to test single prime powers at most Q and witness products in $(R, R^2]$; no modulus exceeds $\max(Q, R^2)$. The same union bound gives

$$\mathbb{P} \left(\prod_p p^{b_p} > B_s R \right) \leq \frac{2}{9R^9} + (Q + R^2)\eta.$$

Consequently the bound on the exceptional-prime product can be chosen before Q , provided the coefficient approximation accuracy can be chosen after Q . For the sampling law, Corollary F.10 supplies this comparison jointly with spatial residue conditions and the prescribed coefficient tilt, as verified next.

The probabilistic modular-rank requirement. Return to the integer output $b = (u, w, z_{\text{act}})$ in the comparison law, and let the slice stride be D . The pure degree- h part in the long variables is independent of t_I and of the translated origin of the slice. It is tested before dilation by D .

For a prime p , depth a , tag h , and primitive row on that tag modulo p^a , apply $h - 1$ independent complete differences to the corresponding top homogeneous polynomial. The desired good-depth condition says that the probability its remaining linear coefficient is zero modulo p^a is at most $p^{-C_* a}$. Write $m_{\text{out}} = n + \sum_h (\dim W_h + \#\{\text{active axes in layer } h\}) \leq n + \sum_h d_h$. Choose the fixed-degree multiple C_* so that $C_* \gamma_s / 2^s \geq m_{\text{out}} + 2$ uniformly over these choices. For an integer cutoff Q , test precisely the depths $p^a \leq Q$. Declare a tested depth bad if this condition fails or $p^a < Q_s$, and let b_p be the largest such depth, or zero if none. The required probabilistic conclusion is

$$\prod_{p \leq Q} p^{b_p} \leq D_0 \tag{H.9}$$

outside a prescribed small fraction of paths, with D_0 chosen before Q .

In the ideal independent-residue model, each component has the J disjoint monomials required by Lemma H.3: use dedicated blocks for active coordinates, free degree- h coefficient coordinates for w , and kernel linear coordinates for u . Their selected coefficients are uniform and independent of all remaining coefficients. Choose J by that Lemma with $C = C_*$, uniformly for $m = m_{\text{out}} \leq n + \sum_h d_h$. Thus the initial block count depends only on the structural dimensions, before the active axes are classified. Independence across primes then gives its uniform exceptional-product tail. Namely, for a target exceptional fraction δ , choose an integer R_{exc} with $2/(9R_{\text{exc}}^9) < \delta/2$ and put $D_0 = B_s R_{\text{exc}}$. The witnesses are single tested prime powers and distinct-prime products in $(R_{\text{exc}}, R_{\text{exc}}^2]$. Their number is at most $Q + R_{\text{exc}}^2$ and their moduli are at most $M_{\text{max}} = \max(Q, R_{\text{exc}}^2)$. No independence between different depths at one prime is required.

Here is the quantitative passage to the actual tilted law. For an integer probability mass obtained by sampling a smooth normalized interval density at scale ℓ , with normalized bounded-variation bound B , one has

$$\|\mathcal{L}(X \bmod M) - \text{Unif}(\mathbb{Z}/M\mathbb{Z})\|_{\text{TV}} \leq O(BM/\ell).$$

Indeed each progression of spacing M is a Riemann sum for $1/M$ of the total mass. Its error is controlled by the total variation of the density; summing over the M residues and normalizing proves the bound. For a product law the errors add over the coordinates.

Corollary F.10 compares the actual law, at each witness modulus M , to the product coefficient-chart law jointly with the unweighted spatial residue law. On its Haar covers the w coefficients are exactly uniform deck coordinates, independent of the projection coordinates and across coefficients. The active dedicated coefficients have interval lengths at least $c'A^h$. On enormous axes all other pure-long degree- h coefficient intervals have lengths comparable to $\sigma K_i/L^h$; these tend to infinity uniformly since $K_i > L^{C_s}$ and $C_s > h$. Include these remaining coefficients in the residue comparison as well. Moderate non-designated coefficients remain deterministic zeros. Finally, the spatial kernel coefficients have nondegenerate interval lengths comparable, up to pre factors, to H_i/L , and every other random spatial coefficient interval becomes broad at the later parent scales. Thus the displayed residue estimate replaces all random entries of the relevant homogeneous coefficient arrays by independent uniform residues modulo M , keeping deterministic zeros as zeros. CRT then makes the complete arrays independent across the primes dividing M .

The number of scalar coefficients is a fixed polynomial of the structural dimensions: the degree is fixed, the number of parameter variables is structural, and there are only polynomially many monomials. Choose the total comparison error at each witness to be at most $\eta = \delta/[2(Q + R_{\text{exc}}^2)]$, dividing it between the finite-residue corollary and the coordinatewise residue estimates. Choose A for the latter projection estimates after Q , and then the rank and parent scales for the corollary and spatial estimates. The logarithms of M_{max} , the witness count, and η^{-1} are permitted pre budgets. The approximate-law remark now gives an exceptional fraction below δ . In particular D_0 and all structural block counts remain fixed before Q .

H.2.3 Bounds for the comparison weight

For paths satisfying (H.9), summing Lemma H.2 with $P \geq m_{\text{out}} + 2$ gives a pre cap C_{rat} for d_{rat} independent of the finite prime collection. Let C_G bound G , and let C_I be the cap given by Lemma G.4 at cube order zero for the joint inactive law relative to its grid volume. Independence across inactive axes gives that joint cap by multiplication; bounded inactive axes are enumerated. Positivity therefore gives

$$0 \leq K_* \leq \kappa C_G C_{\text{rat}} \left(\prod_{i \in I} K_i \right) \Pr(z_I(t_I) = z_I) \leq \kappa C_G C_{\text{rat}} C_I.$$

Lemma F.4 bounds κ independently of subspace heights, so this is a pre cap.

Moreover K_* is uniformly approximable, to any pre accuracy, by a pre-cost sum of site twists. Truncate the congruence Fourier series to characters of individual order at most T , using its uniform absolute tail. For a character χ of order q , its coefficient is

$$c_\chi(t_I) = \mathbb{E}_{t_\parallel} \overline{\chi(b(t_\parallel, t_I))}.$$

Since b consists of integer polynomials, this depends only on $t_I \bmod q$ and has modulus at most one. Split into those at most $q^{\dim t_I}$ coordinatewise residue classes, weighted by their probabilities. The order-zero inactive product-block estimate on each class supplies the remaining smooth and residue site characters. If a conditioned axis is too short for that estimate, its K_i is pre-bounded in terms of q and is enumerated. Normalized projection coordinates are bounded linear functions of the small lifts, while characters of z, w reduce to characters of β modulo their individual moduli. Thus multiplying by G and $\chi(b)$ gives permitted site twists. There are at most $\sum_{q \leq T} q^{m_{\text{out}}}$ rational

characters and the residue counts have logarithms at most $(\dim t_I) \log T$. Dividing the desired error among these terms and using the product-block coefficient bounds preserves pre cost. No least common multiple of all retained orders is required.

Enlarge the degree-zero seminorm by $\sup_{K_*} |\mathbb{E}_U \phi K_*|$, allowing every finite prime collection and every path satisfying its own bad-product condition. This definition does not assert that a random path satisfies infinitely many tests. The preceding uniform cap and approximation bounds give detection for the enlarged seminorm by site twists alone: if $|v| \leq B$ and $|\mathbb{E}_U K_*| \geq a$, approximate K_* uniformly to $a/(2B)$ by a twist combination of coefficient mass M ; one twist then correlates with v by at least $a/(2M)$. Lemma G.11 still applies: the extra pointwise envelope is just a pre-bounded constant. All these model bounds can therefore be fixed before choosing Q .

H.3 Transferring the score and counting the new slots

Proof of Proposition H.1. We first fix uniform bounds for the approximation and its moduli. We then compare the model terms with comparison weights and recover a positive patch with the required number of slots.

We may assume $0 < \epsilon \leq 1$. First fix the exceptional-product bound D_0 so that the modular failure probability will be less than $\tau/4$, and use it in the enlarged comparison family. Fix $\eta \leq \rho\tau/32$. Lemma G.11, with the uniform detection and comparison bounds above, supplies pre bounds N_0, M_0, b_1 for the number of atoms, coefficient mass, and atom complexity in an approximation at accuracy η . These numerical bounds are fixed before the ensemble scales or the actual atoms.

Choose a pre cutoff Q containing every prime power that can divide an atom modulus of complexity at most b_1 . Prescribe the finite-residue errors and the model-comparison error $\rho/[8(1 + M_0)]$ using these uniform bounds. Choose the later parameters in the stated order; the simultaneous choices are detailed in Section H.4. The modular-rank estimate then gives failure probability less than $\tau/4$ through this cutoff, with the same D_0 .

For this ensemble, apply the enlarged degree-zero approximation to $h_0 = (f - \lambda)T_*(u, \beta)$, extended by zero outside the buffered charts. Definition D.4 gives $|h_0| \leq 1$. We obtain

$$h_0 = \sum_{i=1}^{N_0} c_i J_i + r, \quad \sum_i |c_i| \leq M_0, \quad X_0(r) \leq \eta, \quad \sup_{K_*} |\mathbb{E}_U r K_*| \leq \eta,$$

where X_0 is the degree-zero local-slice seminorm and the comparison family includes every finite prime collection. Markov's inequality gives $\sup_B |\mathbb{E}_{t \in Br}(\psi(t))| \leq \rho/8$ outside a set of paths of probability at most $\tau/4$. For a path satisfying Equation (H.9), use the comparison weight for

$$N_Q = \prod_{p \leq Q} p^{\lfloor \log Q / \log p \rfloor}.$$

The comparisons below use only bounded individual covers, rather than an equidistribution estimate modulo N_Q .

For each model twist one needs

$$\mathbb{E}_U J_i K_* = \mathbb{E}_{t \text{ in the slice}} J_i(\psi(t)) + o(1), \tag{H.10}$$

with error at most $\rho/[8(1 + M_0)]$. Here are the individual comparison steps. Write an atom as

$$J_i(u) = F_i(u/H, y(u)) R_i(u, \beta(u) \bmod q_i),$$

where buffered cutoffs are included in F_i , and F_i and R_i have the permitted bounds. A finite combination of such expressions is handled by including its pre coefficient mass in the accuracy.

The integer maps between β and (z, w) convert the residue factor to these coordinates without increasing its modulus.

Truncate the Fourier series of d_{rat} by individual order at a pre threshold. Lemma H.2 makes its uniform absolute tail as small as needed; the cap for the averaged inactive factor then bounds the resulting comparison error. A retained character of order q and J_i use only the common modulus $\text{lcm}(q, q_i)$. For this term apply the single-site Haar-cover equidistribution in Corollary F.9, jointly with its spatial residues and smooth spatial weight. Its proof from Lemma F.7 applies to these prescribed Fourier approximations. An exact inactive-grid condition is extended in the buffered chart at width comparable to $1/K_j$; because $K_j \leq L^h$, its Fourier cost is an allowed late cost. All other normalized chart maps have height-independent bounds. Thus the error can be chosen uniformly in t_I and in the fixed path's coefficient ranges, and can be averaged over t_I .

For each layer the Haar sheet measure is Lebesgue measure in y_W times counting measure in all z_j , divided by $\text{covol}(\Lambda_W)$. Multiplying by $\kappa \prod_{j \in I} K_j$ therefore leaves Lebesgue measure in y_W , exact counting measure on z_I , and active grid volumes $\prod_{j \notin I} K_j^{-1}$. The spatial residues and the w deck residues are independent uniform variables. Sum an active grid in each residue class required by this term and J_i . Its normalized spacing is at most $\text{lcm}(q, q_i)/K_j$, and $K_j \geq (AL)^h$. The pre Lipschitz bound for GF_i makes these grid sums converge uniformly to Lebesgue measure in z_{act}/K , with independent uniform residues. This also verifies all normalization factors in (H.1).

Restore the full rational density using the same absolute tail. The resulting integral samples real normalized $t_{\parallel}$ for the principal smooth arguments, and independently samples its allowed residue classes for the true integer polynomial output, conditional on the same t_I . Although that output density was defined modulo N_Q , its integral against R_i uses only its marginal modulo q_i , since $q_i \mid N_Q$. By the definition of d_{rat} this marginal is exactly the true polynomial-output law obtained from uniform allowed long-parameter residues modulo q_i .

Finally partition each actual long side into those residue classes. On each class R_i is constant as a function of the parameter residues; normalized Riemann summation for the smooth principal arguments has error bounded by a pre constant times q_i/L . The principal lift arguments differ from the true ones by $O(\sigma)$ uniformly. The constrained spatial law in Section F gives the same bound for the total omitted nonkernel spatial displacement, including the inactive columns. Thus the actual and principal smooth test values differ by at most a pre constant times σ . All residue factors have used the true integer path polynomials, so this comparison makes no claim that small real errors preserve residues. Choosing the truncation accuracy, then σ , then A and the equidistribution scales gives the required error in (H.10).

The productive paths have probability at least τ , and the two discarded sets each have probability at most $\tau/4$. Choose a productive path outside both discarded sets and its productive slice. Equation (H.10) and the two residual bounds give

$$\mathbb{E}_U(f - \lambda)T_*K_* \geq \rho - \rho/8 - \rho/8 - \eta \geq \rho/2 =: S.$$

Remove from d_{rat} the factors at primes that are neither bad nor divisors of D nor below a sufficiently large fixed cutoff. The good-prime factors are pointwise $1 \pm O(p^{-2})$, uniformly in t_I . Their product is within $1 \pm \delta$ after the cutoff is chosen in terms of ϵ . If $\widetilde{K}$ is the resulting weight, then

$$(1 - \delta)\widetilde{K} \leq K_* \leq (1 + \delta)\widetilde{K}.$$

Since fT_* and λT_* are nonnegative,

$$\mathbb{E}(f - \lambda)T_*K_* \leq (1 + \delta)\mathbb{E}fT_*\widetilde{K} - (1 - \delta)\lambda\mathbb{E}T_*\widetilde{K}.$$

For $(1 - \delta)/(1 + \delta) \geq 1 - \epsilon$, it follows that

$$\mathbb{E}(f - (1 - \epsilon)\lambda)T_*\widetilde{K} \geq S/(1 + \delta).$$

Thus the discount preserves an explicit score, without requiring the deleted-prime error to be smaller than S . If P_ϵ is the fixed small-prime cutoff, retained primes either have a bad tested depth, divide D , or are at most P_ϵ . The product of the largest bad powers is at most D_0 , so there are at most $\log D_0 / \log 2$ primes of the first kind. Keeping their factors to the largest tested depths uses one modulus M_* with

$$\log M_* \leq C(\log D_0 + \log D + P_\epsilon) \log Q.$$

This is a permitted polynomial logarithmic cost for fixed ϵ .

Partition the ambient box into allowed residue slices to fix $u \bmod M_*$ and into fine normalized spatial boxes. Freeze only the smooth u/H argument in G , leaving the polynomial arguments of T_* unchanged. Its pre Lipschitz bound and the cap of the remaining discrete factor allow a mesh for which the total score error is at most $S/[2(1 + \delta)]$. Parent sides can be chosen large enough that all nonempty partition pieces have the required relative side bounds. Averaging the normalized slice scores gives one piece of score at least $S' = S/[2(1 + \delta)]$. On that piece the remaining positive weight is a smooth function of the small lifts, supplemented by residue information about β modulo M_* and exact inactive projection coordinates.

Finally put $r_h = \dim W_h$, choose an integral basis matrix B_h of $W_h \cap \mathbb{Z}^{d_h}$ and a representative $c_h \in W_h$, and write $C_h - c_h = B_h p_h(u)$. The coordinates of p_h have ordinary degree at most h ; their coefficients need not be bounded, by Definition D.4. This is the first point at which orientation heights are charged. Write $M = M_*$, $b_h = \max(1, \|B_h\|_{\infty \rightarrow \infty})$, and $q_h = \lceil 16Mb_h \rceil$. Use a smooth partition of the torus of p_h/M into $O(q_h^{r_h})$ cells, with local lifted supports of radius $1/q_h$ and centers a_h . Local lifts also handle cells crossing a torus boundary.

On a cell write $p_h/M = k_h + r_h^{\text{loc}}$. The vector $B_h M r_h^{\text{loc}} - y_h = \beta_h - B_h M k_h$ is integral. Across its contributing points its diameter in maximum norm is at most $2Mb_h/q_h + 1/2 < 1$. It is therefore one fixed integral vector $\beta_{h,0}$, giving

$$\beta_h = B_h M_* k_h + \beta_{h,0} \tag{H.11}$$

Introduce the r_h new weight- h residual coordinates

$$e_h = k_h - (p_h/M - a_h), \quad y_h = B_h M(a_h - e_h) - \beta_{h,0}.$$

Their supports lie in $[-1/4, 1/4]^{r_h}$. Since $B_h M k_h \in W_h \cap M\mathbb{Z}^{d_h}$, all exact projection coordinates and the residues of β_h modulo M are fixed in each cell. The discrete factor of the recovered weight is consequently constant there, and its remaining smooth factor is a function only of the new residuals e_h . Substitute (H.11) into the original d_* -slot patch formula, inserting the new determining slots before old slots of equal weight. This preserves every weighted degree and the triangular order. Multiply the old kernel by the cell cutoffs, the positive recovered smooth weight, and the original buffered chart cutoff evaluated at the recovered y_h . Whenever this last cutoff contributes, the displayed identity constructs a true small integer lift; uniqueness identifies it with the original lift. Thus the new formula has no false lifts. Its kernel depends only on the new and old residuals and has the required support.

Let N_{cell} be the number of product cells. Then

$$\log N_{\text{cell}} \leq C \sum_h r_h (1 + \log M + \log b_h).$$

The normalized chart bounds and the affine formula for y_h show that the kernel's logarithmic Lipschitz bound is polynomial in these same quantities and the previous budgets. Its cap C_* is controlled as well. Summing the nonnegative cell weights gives the weight before this partition, so one cell piece has score at least S'/N_{cell} . Divide its kernel by $\max(1, C_*)$. The resulting positive patch has exactly $d_* + \sum_h \dim W_h$ slots and

$$\log(\text{score}^{-1}) \leq \log(S'^{-1}) + \log N_{\text{cell}} + \log \max(1, C_*).$$

This proves the claimed polynomial logarithmic score, complexity, and slice bounds, with the permitted late height costs. $\square$

H.4 Completion of the constrained sampling theorem

Proof of Proposition F.2. We assemble the preceding estimates for one sampling law. In particular, the prime cutoffs used for transferring a fixed patch will be selected from uniform model bounds before the law, rather than from a model constructed for a different law.

Structural choices. Fix the degree bound s , the positive ambient dimension n , and the dimensions d_h . The cube orders required below are $0 \leq q \leq s+1$; order zero is used for comparison weights. Choose the dedicated block counts large enough for Lemma G.4 at every one of these orders, including the fixed derivative orders used in Proposition G.1. Choose the kernel and modular block counts large enough for the real-minor and integer-cokernel estimates and Lemma H.3. Each requirement is a fixed-degree polynomial in $1+n+\sum_h d_h$, so their maximum is such a polynomial. Choose the enormous-axis exponent larger than

$$\max_{1 \leq h \leq s, 0 \leq q \leq s+1} h(j(h, q) + 1),$$

and enlarge it by a fixed margin. This dominates the power of L in the enormous-axis fiber error. The resulting number m of parameter coordinates is now fixed. None of these choices uses a later accuracy, chart width, or testing budget.

Forecasting all pre bounds. Put the prescribed chart widths, slice costs, local test complexities, input caps, and requested inverse accuracies into one pre budget. The constant-coefficient laws and all principal normalized coefficient widths are then fixed as in Section F. They have positive pre widths; in particular they are not narrowed when σ is chosen. All inverse normalized domain volumes in the product-block estimates are consequently pre-bounded, including on the allowed slices and their bounded-modulus restrictions.

First forecast the detection constants and approximation bounds for each $0 \leq r \leq s$. For transfer also fix a lower score $\rho > 0$, productive fraction $\tau > 0$, and the prescribed fixed density discount. Choose the exceptional-prime product bound D_0 so that the modular exceptional fraction will be less than $\tau/4$. Consider the family of all comparison weights (H.1) with this bound D_0 , allowing every finite prime collection. Its cap and its uniform site-twist approximation bounds were proved above independently of that collection and of the later scales. Enlarge the degree-zero local-slice seminorm by the supremum of its pairings with these weights.

The ordinary and enlarged seminorms satisfy the hypotheses of Lemma G.11 with pre data. Indeed their local parts are dominated by a pre multiple of the one-site density, and their comparison parts have a pre-bounded pointwise envelope. The one-site cap threshold is fixed by the constant-coefficient cap and the full-scale x law. Corollary F.9 makes the mass above that threshold as small as the separator requires by later scale choices. Choose the separator's cap and detection threshold first, its required tail error next, and then its coefficient-mass and number-of-terms bounds. Thus

for every bounded input the required approximation has at most N_0 terms, coefficient mass at most M_0 , and atom complexity at most b_1 , where

$$\log(2 + N_0 + M_0) + b_1 \leq \mathcal{P}(b_{\text{pre}}).$$

These are uniform numerical bounds, not a choice of the actual atoms. The same bounds apply to the enlarged seminorm at accuracy $\eta \leq \rho\tau/32$.

Choose Q from b_1 large enough to contain every prime power that can divide any permitted atom modulus. For example, an upper bound for the modulus itself suffices. Hence $\log Q$ has a pre bound. Choose the modular comparison errors on the single-prime-power and witness-product family after Q , as in the probabilistic modular-rank argument. That family has moduli at most $\max(Q, R_{\text{exc}}^2)$ and at most $Q + R_{\text{exc}}^2$ members; choosing the error per member at most $\tau/[8(Q + R_{\text{exc}}^2)]$ still costs only a pre logarithmic budget. Also prescribe the weight/model comparison error $\rho/[8(1 + M_0)]$, the Fourier truncation errors, and all forward, noninjectivity, cap-tail, and cube errors. Only a fixed number of successive polynomial enlargements has been made. Taking their maximum allows all these requests to be met together.

One choice of later parameters. Choose σ after the preceding pre requirements. The joint total-variation perturbation estimate in Proposition G.1, and the spatial bound (F.3), make the real replacement errors smaller than those requirements. Next choose A large enough for all product-block truncations, long-parameter residue Riemann sums, enormous-axis lattice errors, and the projection-coefficient residue comparisons through the already chosen cutoff Q .

Lemma F.4 and the gap selection in the coefficient construction choose L with

$$A \leq L \leq A^{C_s(1+\sum_h d_h)}$$

for the actual spaces W_h . All Fourier and equidistribution budgets in Lemma F.7 and Corollaries F.9 and F.10 have polynomial logarithms in the pre budget, $\log \sigma^{-1}$, $\log L$, and the logarithmic accuracies, individual moduli, and family counts just prescribed. Replace $\log L$ by $C_s(1 + \sum_h d_h) \log A$ when choosing the rank threshold R . This makes that threshold valid throughout the allowed range, without knowing which gap the spaces select. Finally choose every parent side sufficiently large for the resulting Fourier estimates, integer spatial residue comparisons, fiber counting, boundary errors, and noninjectivity estimate. The narrow nonkernel intervals have widths comparable to $\sigma H_i/(mT_j)$ and therefore also become sufficiently long at this last stage.

For an explicit common majorant, choose $c_s \geq 2$ so that $(2 + x)^{c_s}$ dominates every complete polynomial bound used at one stage below, including its fixed multiplicative constant, and set

$$x_0 = 2 + b_{\text{pre}}, \quad x_{i+1} = (2 + x_i)^{c_s} \quad (0 \leq i < 14). \quad (\text{H.12})$$

The following table records the order of these bounds. Dimensions, structural coefficient-array counts, and complexities are bounded directly. Fourier, model, and family counts, caps, and scales are bounded through their logarithms; positive fractions, correlations, and errors are measured by their inverse logarithms.

Bound	Quantities covered
x_1	Structural dimensions, coefficient counts, chart and slice data.
x_2	D_0 , comparison envelope, domination and one-site caps.
x_3	Separator cap, detecting fraction, threshold, and slice mesh.
x_4	Dual-Gowers output and the required averaged cube power.
x_5	Cube-mixture and comparison-weight Fourier mass, count, and twist complexity.
x_6	Inverse-theorem output complexity and detecting correlation.
x_7	N_0, M_0 , tail error, Q , and witness-family data.
x_8	Model-comparison covers, truncations, and all final pre errors.
x_9	σ^{-1} , using the minor, boundary, and perturbation bounds.
x_{10}	A , using residue, Riemann-sum, and enormous-axis errors.
x_{11}	L , uniformly over its allowed range.
x_{12}	Late ambient Fourier budgets and individual covers.
x_{13}	Rank threshold R .
x_{14}	A sufficient lower bound for every parent side H_i .

Here the product-axis Fourier calculation costs polynomially in the number of axes: if D axes each have coefficient mass at most M , allocate error $\epsilon/[2D(M+1)^{D-1}]$ to each axis. Its inverse logarithm is $\log(2D/\epsilon) + (D-1)\log(M+1)$, and logarithmic term counts add. The ambient Fourier approximation of an a -dimensional Lipschitz function likewise costs $O(a\log(2K+1))$ for cutoff K ; it does not form a net of all Lipschitz functions. The separator uses one detection call: in its notation, coefficient mass M is followed by at most

$$\left\lceil 324C^3(1+C)M^2\eta^{-2} \right\rceil$$

empirical terms. Their number enters later budgets only through its logarithm. Each retained model comparison uses the product of two individual moduli, never the product of all prime powers through Q . Finally, the real perturbation uses fixed-size minors separately on each output axis, so their sublevel exponents depend only on s . These observations, together with the displayed estimates in the preceding lemmas, justify every polynomial enlargement in the table. There are fourteen enlargements, independently of all variable dimensions and term counts. Thus

$$\log \sigma^{-1}, \log A, \log R, \text{ and a sufficient } \log \min_i H_i \leq x_{14} \leq (2 + b_{\text{pre}})^{(2c_s)^{14}}.$$

No bound involves a coefficient of the C_h or a height of W_h . The choices define the single law (F.4) for each center tuple; its normalization is $Z = 1 + o(1) > 0$ uniformly in the centers by Corollary F.9.

Property 1. The spatial supports put every sampled box inside U . On the support of the coefficient density, the coefficientwise small lifts sum to a polynomial staying in the site uniqueness chart. The complementary coefficients are ordinary integers. Thus $y_h \circ \psi$ is fully slow at the actual parameter scales and $\beta_h \circ \psi$ is an integer-coefficient polynomial, both of degree at most h . Corollary F.9 supplies the prescribed noninjectivity bound. Its center-uniform normalization and constant-mode cancellation also give the mixed-center one-site approximation uniformly against unit-bounded inputs. These assertions all concern the law just chosen, with no conditioning on productive paths.

Property 2. Lemma G.10 applies to the law just chosen, with all cube errors and mixture bounds prescribed above. It gives detection uniformly over inputs and path-dependent tests and slices, including the degree-zero case. Its proof uses a common meshed slice and the nonnegative local cube powers, so the local niltests need not be fixed across paths.

The one-site domination and tail error were already prescribed for Lemma G.11. Applying that lemma now gives the actual finite approximations in $X(\phi) = \mathbb{E}_\psi \sup_{B,F} |\mathbb{E}_{t \in B} \phi(\psi(t)) F(t)|$, with the

forecast bounds N_0, M_0, b_1 . The atoms can depend on this law and on ϕ ; their bounds do not. No union bound over inputs or over unrestricted polynomial-orbit coefficients is used.

Property 3. For any fixed patch T_* in the stated class and any f, λ in the stated ranges, the input $h_0 = (f - \lambda)T_*$, extended by zero outside the buffered charts, has cap one. Apply the enlarged degree-zero approximation to this input under the already fixed law. Its actual atoms have modulus bounded by the previously fixed b_1 , so every prime power required by them is already covered by Q . Their residual is small both on all but a $\tau/4$ fraction of local paths and against every admissible comparison weight. The modular argument gives failure probability less than $\tau/4$ for this same law, with the same D_0 . It is not a new sampler or a restriction of the law defining the approximation seminorm.

The proof of Proposition H.1 now applies without another scale choice: its model comparison uses only the individual covers for one retained rational Fourier character and one model atom, whose bounds were forecast above. In particular it does not use equidistribution on the product of all prime powers through Q . The proposition therefore gives the discounted positive score with exactly $\sum_h \dim W_h$ new determining coordinates, and with polynomial logarithmic complexity, inverse score, and slice cost in the allowed augmented budget. This is precisely Property 3, including its linear rank bound and its permitted late height dependence.

The unconstrained version. Set $\mathcal{D} = 1$ and use the unconstrained spatial construction in Section F. For example, fix $\ell_0(s) > (s+1)(s+3)$, enlarging it if needed by a degree-only constant. At order zero the root variable supplies the spatial fiber. For $1 \leq q \leq s+1$, a selected q -column difference minor is a nonzero polynomial in a degree-bounded number of normalized tuple variables. The sublevel estimate controls its small values. The integer-character calculation for the independent free columns gives the tail $C(B'')^\ell \sum_{a>B} a^{q-\ell/q}$, which tends to zero with B because $\ell > q(q+2)$. These bounds use no ambient spatial dimension in the required number of columns. On a good tuple the root variable and these q columns give the controlled spatial pivot in each parent coordinate. Taking the product of the rowwise fiber bounds has only polynomial logarithmic cost in n . The same finite-slice argument therefore proves cube comparison and Property 2, now without lift coordinates or a coefficient tilt. They use only the fixed comparability of the prescribed side lengths to L . The nearly uniform x law, small spatial displacements, and collision bound give Property 1. The smooth integer outer law of (x, V) is dominated by the required uniform product-box law by a pre-bounded factor; the parameter average retains its prescribed uniform law. Thus the same conclusions hold for distinct comparable prime sides chosen in advance. This proves every clause of the proposition. $\square$

I Relative lifting with additive rank

We prove Proposition D.7, with returned rank at most $d_0 + d$. The old patch is removed one weight layer at a time. After a recursive call, we discard the old test and transfer only the returned patch. Removing D old slots requires at most D new lift slots, even though the preparation may introduce more ambient coordinates.

Two statements supply the return from sampled paths to the parent. The weighted major decomposition recovers an ambient slow-plus-rational decomposition from decompositions on many paths. Globalization then replaces the local polynomial orbits by one ambient orbit in the same group, preserving a specified projection. In the patch application, this replacement adds no slots. Positive bump selection and Property 3 of Proposition F.2 perform the remaining transfer. Theorem I.15 assembles these steps and also proves the impossibility alternative. Throughout, we retain the sampler's order of parameters and its uniform complexity bounds.

I.1 The weighted major decomposition

We first recover an ambient decomposition from decompositions on sampled paths. The conclusion separates a polynomial with bounded real coefficients from one with controlled rational denominator, using different variables for the two estimates. The following parameter convention makes this recovery uniform in the later sampling choices.

In the remainder, an *early budget* p bounds dimensions, logarithms of inverse chart widths, logarithmic accuracies, log inverse productive probability, and log inverse score. A bound is *early* if it is at most $\mathcal{P}(p)$ in logarithmic complexity, or $\mathcal{B}(p)$ in size, and is independent of the later sampling parameters and the heights of the spaces W_h . These later parameters must be chosen only after all early input and output budgets have been fixed. The sampler dimension and its structural block counts are fixed from the input dimensions before this numerical budget, as in Proposition F.2; they do not depend on chart widths or requested accuracies. Degree bounds depend only on the fixed degrees. The early numerical bounds may depend on the chart widths, but the rank accounting below uses the actual numbers of patch slots and the inequality $\sum_h \dim W_h \leq D$.

Length thresholds for local symbols. In each local symbol argument, fix a threshold J_{loc} whose logarithm has an early bound. An axis of the current reindexed slice is short if its side length is less than J_{loc} ; every other axis is long. Fixing a short axis costs at most $\log J_{\text{loc}}$, so fixing all short axes has early cost. Homogeneous parts in long variables always use all axes retained after these specializations. In particular, an inactive axis with side at least J_{loc} remains a variable of the symbol. These long variables need not coincide with the parallel variables $t_{\parallel}$ in the forecasting law of Section H. The bounded number of threshold increases used below can be forecast from early data before the later sampling choices.

Proposition I.1 (Weighted major decomposition). *Use the data $U, C_h, W_h, c_h, y_h, \beta_h$ of Proposition F.2, with rank at least R in Definition F.1, and assume Properties 1 and 2 of that proposition on the sampled boxes and the ambient domain U . Assume also the uniform symbol step-drop statement of Lemma B.1. For every fixed degree bound s there is an exponent C_s with the following uniform property. Prescribe an early budget p , sampling dimensions, chart widths, a fraction at least e^{-p} of productive paths, and slow and denominator bounds at most e^p . The output bounds are at most $\exp((2+p)^{C_s})$ before choosing the perturbation size, scale range, rank threshold, or parent side lengths. Choose those later parameters sufficiently in that order, as in Proposition F.2.*

Write $A_h = (C_h)_{[h]}$ for the ordinary degree- h homogeneous part of C_h , and put $A = (A_h)_h$. Give u weight 1 and b_h weight h . Let $D(u, b)$ be any fixed scalar weighted homogeneous polynomial of degree $1 \leq d \leq s$, with no bound on its coefficients. If, after fixing short axes on each productive path, the degree- d homogeneous part of $D(\psi(t), \beta(\psi(t)))$ in all the retained parameter variables is slow plus rational of the prescribed bounds on at least the prescribed fraction of paths, then there exist rational graded spaces $K_h \supset W_h$ of early height and polynomials S, R such that, for $b \in \bigoplus_h K_h$,

$$D(u, b) = S(u/H, b - A(u)) + R(u, b). \quad (\text{I.1})$$

The coefficients of S and a common denominator of R have early bounds; the numerators of R are unrestricted. The assertion is simultaneous for polynomially many scalar polynomials, by intersecting their spaces. Long-variable thresholds and slice costs also have early bounds.

To prove Proposition I.1, we will realize the phase $e(D(u, \beta(u)))$ as an ordinary filtered niltest and apply Lemma B.1. We first construct the group in which this realization takes values.

I.1.1 The translation group and its potential

Fix an integer $d \geq 1$ and finite-dimensional real vector spaces E_h equipped with rational bases, $1 \leq h \leq d$. Write $E = \bigoplus_{h=1}^d E_h$, assigning weight h to a coordinate of E_h . Let $\mathcal{Q}_{<d}(E)$ be the vector space of scalar polynomials in $b \in E$ of weighted degree less than d . A coefficient of a monomial of weight k is assigned group degree $d - k$.

The multiplication below is suggested by polynomial differences. Let $V : E \rightarrow \mathbb{R}$ be weighted homogeneous of degree d , and for fixed $x \in E$ put $F_x(b) = V(b) - V(b - x)$. Its terms of weight d in b cancel, so $F_x \in \mathcal{Q}_{<d}(E)$, and for $x, z \in E$ we have

$$F_x(b - z) + F_z(b) = V(b) - V(b - x - z) = F_{x+z}(b).$$

Thus translating the polynomial coordinate as in the next lemma makes these differences compose with their base translations. Under its top-frequency annihilation hypothesis, Lemma I.3 will show that the relevant subgroup elements have this difference form on their projected space.

Lemma I.2 (Polynomial translation group). *On $\mathcal{G}_d = E \times \mathcal{Q}_{<d}(E)$, the operation*

$$(x, F)(z, Q) = (x + z, F(b - z) + Q(b)) \quad (\text{I.2})$$

defines a connected simply connected graded nilpotent Lie group of degree at most d . Its Lie bracket is

$$[(X, P), (Z, Q)] = (0, \partial_X Q - \partial_Z P). \quad (\text{I.3})$$

In rational coordinate bases, integer x and integer polynomial coefficients form a lattice. For fixed d , all group formulas have polynomially many coordinates and bounded degree in the dimension of E .

Proof. Both associations of a product of three elements have polynomial coordinate $F(b - z - w) + Q(b - w) + R(b)$. The identity is $(0, 0)$ and the inverse is $(-x, -F(b + x))$. Thus the operation is a group operation on a real vector space, with polynomial multiplication and inversion. Expanding a commutator to second order gives (I.3). Differentiation in E_h lowers polynomial weight by h and hence raises group degree by h . Every bracket of total degree greater than d vanishes. The maps which multiply x_h by t^h and the coefficient of b^α by $t^{d-\text{wt}(\alpha)}$ are group dilations, proving the grading. The underlying manifold is a vector space, so it is connected and simply connected.

Integer translations preserve integer polynomial coefficients. This proves closure and discreteness of the stated subgroup. Reduce the x coordinates modulo integers and then the coefficients of F modulo integers to obtain a bounded fundamental region. Consequently the subgroup is cocompact. The number of monomials of ordinary degree at most d in $\dim E$ variables is at most $(1 + \dim E)^d$ up to a constant depending on d ; this also bounds the number and degrees of the formulas just used. $\square$

The next Lemma turns top-frequency annihilation into a polynomial potential. A rational height bound controls both numerators and denominators; a denominator bound alone does not control numerators.

Lemma I.3 (Potential associated with a top-annihilated algebra). *Let $\mathfrak{v}$ be a graded rational Lie subalgebra of the Lie algebra of $\mathcal{G}_d$, and let $K_h \subset E_h$ be its projection in degree h . Put $K = \bigoplus_h K_h$. Suppose a linear functional on the top layer vanishes on $\mathfrak{v}_d$ and takes the value 1 on the constant-polynomial element $(0, 1)$. Then there is a rational weighted homogeneous polynomial $V : K \rightarrow \mathbb{R}$ of weight d such that*

$$F(b) = V(b) - V(b - x) \quad ((x, F) \in \exp \mathfrak{v}, b \in K). \quad (\text{I.4})$$

For fixed d , a polynomial bound on dimensions and on logarithmic rational heights of $\mathfrak{v}$ gives such a bound for V and for a rational polynomial extension of V to E .

Proof. First let $(0, Q) \in \mathfrak{v}_j$ be homogeneous. We claim $Q|_K = 0$. Otherwise a nonzero coefficient of $Q|_K$ can be extracted by successively differentiating in homogeneous coordinate directions of K , with total weight $d - j$. Lift each such direction to an element of the corresponding layer of $\mathfrak{v}$. By (I.3), the iterated bracket with $(0, Q)$ is the same iterated directional derivative, viewed as a pure-polynomial element. It would be a nonzero constant in $\mathfrak{v}_d$, contradicting the hypothesis. This proves the claim.

Choose rational graded linear lifts $(X, P_X) \in \mathfrak{v}$ of $X \in K$. The claim shows that $P_X|_K$ is independent of the lift. Define a polynomial one-form by $\omega_b(X) = P_X(b)$ for $b, X \in K$. The bracket of two chosen lifts has zero base component, so the claim and (I.3) give $\partial_X P_Z = \partial_Z P_X$ on K . Thus ω is closed. Its weighted degree, including the weight of the differential, is d . If $\mathcal{E}_b = \sum_h h b_h$ is the weighted Euler vector field, set

$$V(b) = d^{-1} \omega_b(\mathcal{E}_b).$$

The closedness identity and Euler's identity imply $\partial_X V(b) = \omega_b(X)$. This can also be checked on each coordinate monomial, so it requires no choice of a primitive or integration constant.

The set of pairs with $x \in K$ satisfying (I.4) is a closed subgroup: substitution in (I.2) makes the two differences telescope. Its Lie algebra consists of pairs (X, P) with $X \in K$ and $P|_K = \partial_X V$. We have proved that this Lie algebra contains $\mathfrak{v}$. Since $\exp \mathfrak{v}$ is connected, the subgroup identity follows.

Rational sections, restrictions, and extensions can be found by solving linear systems of polynomial size. The formula for V adds only multiplication by a coordinate and division by the fixed integer d . Clearing denominators and bounding minors therefore gives the asserted rational-height bounds. $\square$

Lemma I.4 (The algebraic major decomposition). *Let $D(u, b)$ be weighted homogeneous of degree d , and let $D_0(b)$ be its part independent of u . Let $A(u)$ be a weighted homogeneous map into K . Suppose the symbol*

$$(A(u), D(u, b) - D_0(b - A(u))) = (e, E)(x, F)(a, Q)$$

has middle factor in the subgroup in Lemma I.3, whose defining algebra has bounded rational height, and e, x, a take values in K . Suppose the left factor has coefficients bounded in normalized variables u_i/H_i , and the right factor has rational coefficients of bounded common denominator. If D_0 has bounded coefficients, then, on $b \in K$,

$$D(u, b) = S(u/H, b - A(u)) + R(u, b),$$

where S has bounded coefficients and R has bounded common denominator. Bounds have polynomial logarithmic dependence on those in the hypotheses, for fixed degree.

Proof. The base identity is $A = e + x + a$. Use Lemma I.3 and the group law twice to obtain

$$\begin{aligned} D(u, b) &= D_0(b - A) + E(u, b - A + e) - V(b - A + e) \\ &\quad + Q(u, b) + V(b - a). \end{aligned}$$

The first three terms are polynomials with bounded coefficients in u/H and $b - A$. The last two have rational coefficients of bounded common denominator. This proves (I.1). Products and substitutions involve a bounded degree and polynomially many coefficients, giving the stated bounds. $\square$

I.1.2 Proof of the weighted major decomposition

Proof of Proposition I.1. We apply Lemma B.1 to ordinary polynomial maps on integer boxes. The polynomial translation group converts the weighted polynomial in the conclusion into this setting.

First subtract the integer parts of all coefficients of D . This leaves its exponential unchanged at integer u, b and changes each leading path evaluation by an integer-coefficient polynomial. It therefore preserves the local major hypothesis. The number of coefficients is polynomial in the dimensions, since d is fixed. We can hence assume they all lie in $[0, 1)$, and restore the removed integer polynomial to R at the end.

Use the buffered site input $\chi(u)e(D(u, \beta(u)))$. On a productive path its top part can be made nearly constant on controlled subboxes and residue classes. Indeed a common residue modulus clears the rational coefficients of the top part, and subdivision in normalized coordinates makes the slow part oscillate by at most $1/100$. Fix only the axes shorter than the early threshold J_{loc} . The other inactive axes remain in the symbol. Since the full substituted polynomial has ordinary degree at most d , the pure degree- d part in the retained variables is independent of the values fixed on short axes. These specializations have early slice cost. Cancel the remaining lower-degree terms by an ordinary polynomial phase of degree at most $d - 1$. Choose χ to equal one on all path supports and to vanish outside the buffered charts. The local correlation is then at least $9/10$. Property 2 of the sampler, applied with degree $d - 1$, gives an early positive ambient correlation with a site twist times a degree- $(d - 1)$ niltest.

Partition normalized spatial coordinates and their bounded-modulus residues so that the smooth spatial factors can be frozen to the accuracy required by this correlation. Choosing one part preserves a correlation of early size. At every tag $h > d$, Fourier expand the smooth lift dependence on the ambient torus cover of the bounded modulus used by the twist. This is the ambient coordinate torus, not a torus basis of W_h ; no height of W_h enters its Fourier cutoff. Convolution with a trigonometric approximate identity approximates a Lipschitz function uniformly. The coefficients remain Lipschitz functions of the unexpanded coordinates, with the same bound up to the fixed truncation cost. Thus the number of terms, coefficient mass, and remaining Lipschitz bounds are early bounded. A residue character of β_h is handled on the same cover by writing $\beta_h = C_h - c_h - y_h$. This gives a bounded rational frequency on $C_h - c_h$ and a smooth factor of y_h .

Set $E_h = \mathbb{R}^{d_h}$ for $1 \leq h \leq s$ and write $D_0(b) = D(0, b)$. Since D uses only tags at most d , we represent these factors on $\mathcal{G}_d$ formed from $\bigoplus_{h \leq d} E_h$. In this realization all base vectors are restricted to those tags. Set

$$x_h = C_h(u) - c_h, \quad F(u, b) = D(u, b) - D_0(b - x(u)).$$

Its integral-part evaluation is $e(F(u, \beta) + D_0(\beta - x)) = e(D(u, \beta))$. Right multiplication by an integral lattice element changes the phase argument by an integer: if (z, Q) is integral, then the new integral part is $\beta + z$ and the argument increases by $Q(\beta + z) \in \mathbb{Z}$. To include residues of β modulo M , use the subgroup lattice with $z \in M\mathbb{Z}^{\dim E}$ and integral Q ; its index and grid bounds are early. On this cover every remaining low-tag twist is a Lipschitz function of $y = x - \beta$ and $\beta \bmod M$. Buffered cutoffs therefore give a well-defined nilmanifold function, including the twist, and remove the jumps where the integral part changes. On bounded group representatives the possible integral parts are bounded. The coefficient reduction therefore makes this function Lipschitz with early bounds. The coefficient of a dummy monomial of weight $k < d$ in F has ordinary u -degree at most $d - k$, while the base x_h has degree at most h . The group law and the logarithmic coordinate change respect these assigned degrees, so the same degree bounds hold in logarithmic coordinates. Hence this is an ordinary filtered polynomial map of degree at most d . The dummy terms of weight d cancel by the definition of D_0 . Taking the homogeneous part in each group grade replaces x_h by A_h

and gives exactly the symbol

$$(A(u), D(u, b) - D_0(b - A(u))).$$

Thus no lower-degree coefficient of C_h enters this symbol.

Consider a resulting term with a highest tag $h > d$ whose frequency does not annihilate W_h . All factors at higher tags are constant on $C_l - c_l \in W_l$. The degree- h factor is a single torus phase; every other factor is an ordinary niltest of degree at most $h - 1$. For the factors of tags at most d , use the nilmanifold realization just constructed. A correlation larger than an arbitrarily prescribed early error would, by Lemma B.1 applied to the product group, force the top of that torus phase to be slow plus bounded-denominator rational: the partner has no top coordinate and the fast algebra is annihilated by the torus frequency. Clearing its early bounded denominator gives an integer row forbidden by high rank. Choose the rank threshold large enough for all these rows and the required error. The total contribution of these terms is negligible, because their total coefficient mass is early bounded. At least one term involving only tags at most d consequently retains early correlation. All errors and cutoffs used in this paragraph were determined before that rank threshold was chosen.

A top-frequency decomposition retains frequency 1 on the constant polynomial direction: every Fourier term of an already equivariant function retains that equivariance. Select a correlated term and apply Lemma B.1 jointly with the degree- $(d - 1)$ partner. The partner has zero grade d , including when $d = 1$ and it is constant. Consequently the selected top functional on the product descends to a functional on the top of $\mathcal{G}_d$, takes value 1 on $(0, 1)$, and annihilates the projection of the fast algebra. Rational projection therefore supplies precisely the algebra required by Lemma I.3. Its base projection contains W_h : otherwise a bounded-height rational row annihilating that projection, but not W_h , would exhibit a forbidden slow-plus-rational top component of C_h . Once this inclusion is known, write the base splitting as $A = e + x + a$ with $x \in K$. Since $A \in W \subseteq K$, each coefficient of $e + a$ lies in K . All these symbol coefficients have positive degree, so Lemma A.5 puts the slow and rational base factors individually in K . Lemma I.4 now applies. Undoing the spatial residue and subbox parametrization multiplies homogeneous coefficient bounds and denominators by early factors only. More explicitly, on $u = u_0 + qv$ the degree- h symbol is $A_h(qv)$; if the new sides are $T_i \asymp \delta_i H_i / q$, undoing this substitution multiplies slow bounds by at most $\prod_i \delta_i^{-\alpha_i}$ and denominators by q^h . Here q and δ_i^{-1} are early bounded. This also justifies each use of the original rank hypothesis after restricting to a spatial residue subbox. For $h > d$ take $K_h = E_h$. For polynomially many simultaneous scalar components, apply the argument separately with a common early budget, intersect the resulting spaces, and restrict their identities. A joint stacked-row computation bounds the rational height of this intersection by an early quantity. $\square$

Remark I.5 (Uniformity of the bounds). The preceding proof uses Property 2 of Proposition F.2 and Lemma B.1 with their stated early bounds. Its Fourier cutoffs use ambient lattice covers of bounded modulus, so this step introduces no additional dependence on the height of W_h . Choosing the local model only after allowing its complexity to depend on that height or the final long scale would, however, invalidate this deduction.

I.2 Globalization in a fixed group

Proposition I.6 (Globalization in a fixed group). *Assume Properties 1 and 2 of Proposition F.2, including its rank hypothesis and order of later parameter choices, the uniform symbol step-drop and separation statements, and their stated polynomial logarithmic bounds. Fix an early budget p and a strong rational projection $\pi : G^0 \rightarrow Y$ of degree $r \leq s$. Include the dimensions, rational structure constants, filtration bases, lattice grids, and matrix of π in the early complexity budget. Let $P_Y(u, b)$*

be a weighted filtered polynomial, with no coefficient bound. Let $\Phi_u : G^0/\Gamma^0 \rightarrow [0, 1]$ have an early uniform Lipschitz bound. Assume that, at each requested early accuracy η , the family $\{\Phi_u\}$ has a supremum-norm η -net of at most $\exp(\mathcal{P}(p + \log(1/\eta)))$ functions. Its bins may be arbitrary site sets. Let $a(u)$ be a fixed real input of early bounded supremum.

There are early output bounds, independent of every later sampling parameter, with the following consequence. Suppose a fraction at least e^{-p} of paths have a permitted slice and a filtered polynomial g_ψ satisfying

$$\pi(g_\psi(t)) = P_Y(\psi(t), \beta(\psi(t))), \quad \mathbb{E}_t a(\psi(t)) \Phi_{\psi(t)}(g_\psi(t) \Gamma^0) \geq e^{-p}.$$

Then there is one weighted filtered polynomial $g^U(u, b)$ in the same group, satisfying the formal identity $\pi g^U = P_Y$, such that an early fraction of the original productive paths have permitted subslices on which replacement by g^U gives early positive score. No coefficient bound on g_ψ or g^U is required.

Proof. We induct on r . For $r = 0$ the group is trivial and its sole map is already fixed. All early budgets below are computed before the later sampling choices. The number of successive polynomial increases of those budgets will be bounded in terms of s .

Top modes and fixed ambient partners. Write $\delta \geq e^{-p}$ for the input score and $B \geq \max(1, \|a\|_\infty)$ for its early bound. Truncate the Fourier expansion in the top torus G_r^0 to uniform error at most $\delta/(100B)$, using Lemma A.4. Call a mode significant on a productive path if its absolute tested average is at least the score divided by one hundred times the number of truncated modes. The sum of nonsignificant terms is then negligible. Let $L \subset G_r^0 \cap \ker \pi$ be the joint connected annihilator of the restrictions of significant modes. Averaging the actual positive test over L retains those modes, preserves the range $[0, 1]$, and retains early positive score. Choose a basis among these significant restricted frequencies. There are at most $\dim(G_r^0 \cap \ker \pi)$ pivots.

Choose these bases in a fixed order. If there are M truncated frequencies and the kernel top layer has dimension D , the number of possible bases, and hence of possible L , is at most $(1 + M)^D$. Pigeonhole L and the pivot basis first. This loses only an early fraction, rather than counting all subsets of the M modes. If there are no pivots, the partner-selection steps below are empty; the product has no partner factors and its fast algebra is the full graded algebra.

For each pivot approximate the external function family in supremum norm and partition sites by a nearest bin. One bin detects the pivot with early correlation. For every possible bin apply the sampler's linear-combination approximation to a times the bin indicator. Such indicators are allowed because its input is an arbitrary bounded function. Use its testing seminorm over all permitted slices and all current local degree- r niltests. The test at this point is only the vertical component of a fixed bin function evaluated at g_ψ ; its ambient partner has not yet been selected. Its degree is r , and its group and function complexity are early bounded. Polynomial coefficients of the orbit do not enter niltest complexity. Conjugation and products, represented on product groups, preserve the degree bound. Choose the error smaller than the pivot threshold times the productive fraction divided by the total early number of bins and modes. Markov's inequality and a union bound make all these approximations valid on all but a negligible fraction of paths, uniformly over their separately chosen slices and tests. Explicitly, when there are pivots, let $J \geq 1$ be the number of masked inputs, let κ be the retained productive probability, and let τ be smaller than half the least correlation threshold after selecting a bin. Normalize each masked input by B before applying Property 2, and choose its approximation so the unnormalized error has X -norm at most $\varepsilon = \kappa\tau/(4J)$. If $e_j(\psi)$ denotes the pathwise supremum error for mask j , then

$$\Pr_{\psi} \{ \max_j e_j(\psi) > \tau \} \leq \sum_j \frac{\mathbb{E}_{\psi} e_j(\psi)}{\tau} \leq \kappa/4.$$

The supremum in Property 2 covers all pivot frequencies as well as the separately chosen slices. Its input test complexity is fixed before any model partner is selected. Net centers can be chosen from the original family, at twice the requested accuracy, so they retain the common Lipschitz and range bounds.

For a retained path and pivot, some model term gives biased pairing of the local vertical function with a site twist J times a fixed ambient partner map $Q_\ell(u)$. Fourier truncate the top of that partner and retain one mode. By subbox and residue subdivision, freeze J to error smaller than the pairing; one cell retains biased pairing without it. Freeze axes shorter than the early threshold required by Lemma B.1. Applying that Lemma to the ordinary local pair gives a splitting of its pure-long symbol into a rational algebra killed by the joint top frequency.

Refer each homogeneous identity back to the original long variables. Translation does not alter it and residue dilation costs only early factors. Thus pieces used for distinct pivots need not intersect: their symbol splittings can be joined by separation. All long sides exceed the fixed degree and interpolation cutoffs. Pigeonhole the partner choices, frequencies and bounded-height rational algebras, which have an early number of possibilities jointly for the polynomially many pivots. An early fraction of productive paths now has fixed choices. Set

$$G = (G^0/L) \times \prod_{\ell} G_{Q_\ell}, \quad F = Y \times \prod_{\ell} G_{Q_\ell}, \quad P_F = (P_Y, (Q_\ell)_\ell).$$

The product has only polynomially many factors, irrespective of the length of the model lists. First project each pair splitting through $G^0 \rightarrow G^0/L$; its top constraint descends because the pivot frequency annihilates L . Pull each resulting fast algebra back to this product, leaving the unused partner coordinates free. Each such inverse-image algebra admits a splitting of the full local symbol: extend its outer factors by identities in those unused coordinates. Lemma A.6 gives a splitting in their intersection; this is the joined fast algebra $\mathfrak{v}$. A top vector with zero projection to F has zero partner coordinates. Its pivot constraints force the original coordinate into $\log L$, so it is zero after quotienting. Consequently $\mathfrak{v}_r \rightarrow (\pi\mathfrak{v})_r$ is injective.

Matching the fixed marked symbol. Write Z for the weighted symbol of P_F and $\mathfrak{v}_F = \pi\mathfrak{v}$. Initially take $K_h = E_h$ and $\mathcal{E} = \mathcal{R} = 1$. Inductively suppose through grade $d-1$ that the logarithm of $\mathcal{E}^{-1}Z\mathcal{R}^{-1}$ belongs to $\mathfrak{v}_F$ on $K = \bigoplus_h K_h$. The left factor has early bounded polynomial coefficients in $u/H, b - A(u)$; the right factor has rational polynomial coefficients of early common denominator.

Raise the long threshold so that the leading evaluations of β_h lie in K_h . Indeed their projections off K_h have rational coefficients of early denominator, since β_h has integer coefficients and K_h has early rational height. They are also negatives of coefficients of the fully slow y_h , since $C_h - c_h \in W_h \subset K_h$. Separation makes each pure-long coefficient zero. The evaluations of the two fixed outer factors are therefore slow and rational ordinary symbols. Here $\beta_h^{\text{top}} - A_h(\psi^{\text{lin}}) = -y_h^{\text{top}}$ is slow, whereas ψ^{lin} and β_h^{top} are integral.

Compare the projected local splitting with these fixed factors. Through grade $d-1$, separation puts the ratios of their corresponding outer factors in $\mathfrak{v}_F$. Lift the ratios within that algebra and absorb them into the local fast factor. Thus the projected local outer factors agree exactly, in those grades, with the evaluations of the fixed ones. The grade- d logarithmic coordinate of the fixed residual, modulo $(\mathfrak{v}_F)_d$, then evaluates to a sum of slow and rational scalar polynomials: all lower discrepancies are zero and the fast coordinate vanishes in the quotient.

Apply Proposition I.1 simultaneously to those scalar coordinates and intersect its spaces with K_h . Rational linear sections lift the two quotient terms to s_d, r_d . Update

$$\mathcal{E} \leftarrow \mathcal{E} \exp(s_d), \quad \mathcal{R} \leftarrow \exp(r_d)\mathcal{R}.$$

At grade d the residual loses exactly $s_d + r_d$ and lower grades are unchanged. This proves the invariant in the next grade. BCH preserves the two allowed types. There are at most s such threshold increases, and restriction to fewer long axes preserves all earlier homogeneous identities.

At the last threshold freeze nonlong axes on favorable fibers of productive slices. The whole projection of β_h off K_h is now constant: each remaining nonconstant coefficient is both rational of early denominator and fully slow. The constant has early bounded norm because it is the projection of $-y_h$, and early denominator because β_h is integral. Pigeonhole these finitely many constants on an early fraction of paths. Let p_h be the rational affine projection onto the resulting fixed translate of K_h and temporarily replace P_F by $P_F^*(u, b) = P_F(u, p(b))$. These maps agree on the retained slices.

Full factors with exact projection. Refilter by Lemma 7.6. Lift $\mathcal{E}, \mathcal{R}$ to full polynomial factors E_F, R_F of P_F^* . Lift their logarithmic layer formulas, replacing $b - A(u)$ on the left by $\text{proj}_K b - (C - c)(u)$, and using the rational formulas in $u, \text{proj}_K b$ on the right. Thus E_F is fully slow and bounded on the sampling charts and R_F has early common denominator. Symbol matching puts every positive-degree coefficient of $P_H = E_F^{-1} P_F^* R_F^{-1}$ in the appropriate $H_{F,i}$: parts in deeper original layers already belong there because $F_{i+1} \subset H_{F,i}$. In the vector quotient $F/H_{F,1}$ only a constant remains. Split it in rational coordinates into a bounded fractional part and an integral part, and insert lifts on the inside of E_F and R_F . The middle map now takes values in H_F .

For each local map construct

$$g = EhR, \quad \pi(E) = E_F, \quad \pi(R) = R_F, \quad \pi(h) = P_H.$$

Here is the exact kernel correction. Write the local symbol as epr , with $p \in \exp \mathfrak{v}$, and write e_F, r_F for the symbols of the prescribed projected outer factors. By Lemma A.6,

$$A = (\pi e)^{-1} e_F \in \exp \mathfrak{v}_F, \quad D = r_F (\pi r)^{-1} \in \exp \mathfrak{v}_F.$$

The first is slow and the second rational. Lift their logarithms through a controlled graded section to $\hat{A}, \hat{D}$ in $\exp \mathfrak{v}$ and replace the splitting by

$$e' = e\hat{A}, \quad p' = \hat{A}^{-1} p \hat{D}^{-1}, \quad r' = \hat{D}r.$$

It now has $\pi e' = e_F$ and $\pi r' = r_F$. Choose a filtered rational logarithmic section ρ of π and put $E_0 = \rho(E_F)$, $R_0 = \rho(R_F)$. The symbols $e'(\text{Sym } E_0)^{-1}$ and $(\text{Sym } R_0)^{-1} r'$ lie in $\ker(\text{gr } \pi)$. Strongness supplies controlled actual-kernel lifts of each graded coefficient: if $X \in \log G_i$ represents one and $\pi X \in \log F_{i+1}$, subtract a section lift of πX in $\log G_{i+1}$. Lift the two symbol discrepancies coefficientwise in this way to kernel polynomial maps K_E, K_R , with no other logarithmic terms. Then $E = K_E E_0$ and $R = R_0 K_R$ have the prescribed projections exactly, retain the slow and rational bounds, and give a middle map whose positive-degree coefficients are adapted to H_i .

For its constant term use the vector quotient $q : G \rightarrow G/H_1$. Since $\pi(H) = H_F$, the condition $\pi(h(0)) \in H_F$ implies $q(h(0)) \in q(\ker \pi)$. Choose a controlled rational linear section of q on this rational subspace, split $q(h(0))$ in a rational basis into a bounded fractional vector and an integral vector, and lift both into $\ker \pi$. Absorb their exponentials on the inside of E and R , respectively. These constant adjustments preserve the symbol and the exact projections; the middle constant now lies in H_1 . Thus $h \in H$ and all three projection identities hold in the groups.

Fix the logarithmic section ρ . The kernel map $E\rho(E_F)^{-1}$ is fully slow and bounded. Early subdivision makes it close to a fixed bounded element k_E from an early net. The rational kernel map $\rho(R_F)^{-1} R$ has early common denominator; residue restriction makes its right lattice coset constant, represented by a fixed bounded rational $k_R \in \ker \pi$. If these common denominators vary

with the path, first pigeonhole their values, of which there are an early number. A cover clears only the finitely many denominators actually selected, rather than every integer below an early bound. Pigeonholing retains early score on an early fraction of paths. Put

$$E^U = k_E \rho(E_F), \quad R^U = \rho(R_F) k_R.$$

Right lattice changes preserve cosets. Left freezing changes the test by at most its Lipschitz constant times the freezing error, uniformly in the unbounded middle map, by right invariance of the upstairs metric.

Lattice reconstruction and external function nets. By Lemma 7.6, the map $i : H \rightarrow H/H_r \times H_F$ is injective onto the group fiber product. We explain why finer lattices make the same assertion usable for nilmanifold functions. Choose a source lattice Γ_H permitting the right translates R^U . This uses only their common denominator: the adjoint formulas for rational elements have bounded denominators independently of the sizes of their numerators, so a common deep integral grid lies in all required conjugate lattices.

In rational logarithmic coordinates let T be a rational left inverse of i_* . If $\mathbb{L}\mathbb{Z}^{\dim H} \subset \log \Gamma_H$, choose a sufficiently divisible early integer D so that $T(D\mathbb{Z}^{\dim(H/H_r) + \dim H_F}) \subset \mathbb{L}\mathbb{Z}^{\dim H}$. Take deep target lattices Δ'', Δ_F whose logarithms lie in that target grid and set $\Delta_H = i^{-1}(\Delta'' \times \Delta_F)$. Rationality implies that Δ_H is a lattice, and the grid choice gives $\Delta_H \subset \Gamma_H$. The original test pulls back to H/Δ_H , which embeds in the product of the two covered nilmanifolds. Rational subgroup reconstruction gives an early Lipschitz bound on the joint image. The quantitative reason is that on bounded representatives any target lattice element outside the rational image has distance at least an early inverse size from it; nearby pairs can therefore be compared using a lattice element in the image and the bounded rational inverse T .

For clarity, fix these lattices for the whole family and write J for the resulting joint image. If ϕ is an original function, e a bounded value of E^U , and r a bounded representative of a right-lattice residue of R^U , the function on J is

$$f_{\phi,e,r}(i(h)) = \phi(ehr\Gamma).$$

All these functions have one early Lipschitz bound L_* , by Lemma A.3; enlarge L_* to be at least one and to dominate the Lipschitz bounds of the original functions ϕ as well. For a fixed r , right invariance also gives

$$\|f_{\phi,e,r} - f_{\phi',e',r}\|_\infty \leq \|\phi - \phi'\|_\infty + \text{Lip}(\phi') d_G(e, e').$$

Use the clipped McShane extension from the same J with the same constant L_* for every function. The infimum formulas show that this operation is nonexpansive in supremum norm. Specialize the extension at the second coordinate $y = P_H(u, \beta(u))\Delta_F$; changing y changes the resulting function on $(H/H_r)/\Delta''$ by at most $L_* d(y, y')$ in supremum norm. Thus, writing $\Psi_{\phi,e,r,y}$ for this specialized function,

$$\|\Psi_{\phi,e,r,y} - \Psi_{\phi',e',r,y'}\|_\infty \leq \|\phi - \phi'\|_\infty + L_* d_G(e, e') + L_* d(y, y').$$

There are an early number of possible right-lattice residues r : Mal'cev reduction of rational elements of one fixed common denominator leaves bounded representatives of controlled denominator. This count is independent of all numerators. The bounded e -region and the compact y -nilmanifold have polynomial dimension and early coordinate bounds. Choose an original function net at accuracy $\eta/6$ and recenter each nonempty bin at a member of the original family, giving accuracy $\eta/3$ with its original range and Lipschitz bounds. Together with grids of mesh $\eta/(3L_*)$ in e and y and the finite choices of r , this gives a net with logarithmic size polynomial in $p + \log(1/\eta)$. The dependence

of e, r, y on the site may be arbitrary: only this parameterized family is netted, not the space of all Lipschitz functions.

Lower-degree induction and restoration. Apply induction to $H/H_r \rightarrow H_F/H_{F,r}$, with marked polynomial $P_H \bmod H_{F,r}$ and the external test just constructed. Its fixed polynomial output lifts jointly with the exact map P_H to one weighted polynomial $h^U \in H$ by the rational filtered fiber isomorphism. Then $E^U h^U R^U$ works for the averaged test and projects to P_F^* . Multiply it on the left by $\rho(P_F(P_F^*)^{-1})$ to restore the formal marked projection P_F ; this correction is the identity on every retained slice. Project to G^0/L and lift polynomially to G^0 by a filtered logarithmic section. The original marked projection is unchanged because $L \subset \ker \pi$.

Finally, if a path has averaged score at least δ and $|a| \leq B$, the set of $l \in L/(L \cap \Gamma^0)$ giving original score at least $\delta/2$ has measure at least $\delta/(2B)$. Otherwise the integral would be smaller than δ , by bounding the score by B on that set and $\delta/2$ off it. Fubini supplies one fixed l working on an early fraction of paths. Its left translation preserves the marked map and gives the desired output.

All dimensions count only polynomially many pivots. Linear systems at each grade are solved jointly. Marked matching has at most s grades and the degree induction at most s steps. Thus only a bounded number of polynomial budget increases are used. These bounds are early and can all be fixed before the later scales. $\square$

Remark I.7 (Uniformity in path-dependent tests). The proof uses the sampler's early-uniform testing seminorm for simultaneous approximation of arbitrary masked inputs against all path-dependent pivot tests. Marked matching, exact reconstruction, and the function-net bound are then deductions from this assertion and Lemma B.1. These are the sampling and symbol inputs to Proposition I.6.

Lemma I.8 (A fixed patch evaluation on many paths). *Suppose an early fraction of paths have positive scores of early size against a real input of early bounded supremum, using patches of degree at most s , rank at most d_* , and early kernel Lipschitz complexity. The bound d_* and all parameter dimensions are included in the early structural budget. After early losses in score and productive fraction, those patches can be replaced by evaluations of one fixed Lipschitz function on one fixed triangular patch group. The local polynomial maps may still vary with the path and have unrestricted coefficients. The fixed function has early complexity and at most d_* patch slots.*

Proof. First pigeonhole the rank and slot weights. Let $D \leq d_*$ be this rank, $L \geq 1$ a common upper bound on kernel Lipschitz constants, and δ a common positive lower bound on score. Normalize all residual polynomial coefficients modulo integers in increasing slot order, using triangular integral changes of slot variables. This preserves the patch value, kernel, and weighted degree. Choose an accuracy $\varepsilon \ll \delta/(1 + \|a\|_\infty)$ and a grid mesh $h \leq \min(1/32, \varepsilon/L)$ depending only on early bounds. Extend each kernel Φ by zero as required in the patch definition. Tensor-product piecewise linear interpolation gives

$$\tilde{\Phi}(x) = \sum_{v \in \mathcal{V}} \Phi(v) \Psi_h(x - v), \quad 0 \leq \Phi(v) \leq 1, \quad \|\tilde{\Phi} - \Phi\|_\infty \leq \varepsilon,$$

where Ψ_h is one nonnegative tensor-product tent of cap one, supported in $[-h, h]^D$. The vertices lie in a fixed slight enlargement of the support and $\log |\mathcal{V}| = O(D \log(2 + L/\varepsilon))$. The joint support lies in $[-1/4 - h, 1/4 + h]^D$, of coordinate width less than one. Triangularity permits at most one integer slot tuple there. Substitution in the patch therefore changes its value by at most ε , with no loss from counting tuples. Its score remains at least $\delta/2$.

All interpolation coefficients are nonnegative. Some translated tent patch consequently has score at least $\delta/(2|\mathcal{V}|)$. Absorb its center into the constant terms of the residual polynomials; every

path now uses the same kernel Ψ_h . These constant shifts leave the homogeneous weight-preserving coefficients unchanged.

Write its triangular change as $S^0 Y$, with S^0 the homogeneous weight-preserving part and Y in the fixed strictly lowering group. All coefficients of S^0 are uniformly bounded and their number is polynomial in D for fixed s . On a bounded fundamental region for Y , triangular inversion gives a uniform early bound for all integral tuples that can contribute for such an S^0 or a nearby coefficient mesh point. There are an early number of such tuples. Polynomial substitution and the Lipschitz bound of Ψ_h give an early uniform continuity bound for

$$S^0 \mapsto \sum_{b \in \mathbb{Z}^D} \Psi_h(S^0 Y(b))$$

in supremum norm on the patch nilmanifold. Mesh these finitely many coefficients sufficiently finely, preserving triangular homogeneous form. The logarithmic mesh size is polynomial in the early budget. A mesh replacement loses less than half the retained score, and pigeonholing its index gives one fixed function on an early fraction of paths. No coefficients of the polynomial maps Y were pigeonholed. $\square$

Corollary I.9 (Globalizing returned patches). *Under the hypotheses of Proposition I.6 and sampling Property 3, let $0 \leq f \leq 1$ and $0 \leq \lambda \leq 1$. A family as in Lemma I.8, with real input $a = f - \lambda$, yields an ambient patch of rank at most $d_* + \sum_h \dim W_h$, with positive score after the prescribed fixed multiplicative discount of the target density. Its logarithmic inverse score, output slice costs, and complexity may include the subspace heights and later parameters, as allowed in sampling Property 3. They have polynomial bounds in that augmented log budget, rather than necessarily early bounds.*

Proof. Apply Lemma I.8 and then Proposition I.6 with trivial initial marked factor. Its fixed weighted polynomial remains in the same triangular group, so its evaluation is a fixed patch formula with d_* slots and formal lift arguments. Sampling Property 3 introduces at most $\sum_h \dim W_h$ slots to determine those arguments, with the asserted score and discount. $\square$

I.3 Removing the lowest patch layer

The removal of an old layer has an elementary part that can be proved without any sampling theorem.

Lemma I.10 (Freezing a lowest layer). *Consider a triangular patch with D lowest-weight slots, all of weight j . Write their residual as*

$$r(t, b) = (I - M)(b - P(t)),$$

where M is strictly lower triangular. Suppose $I(t)$ is an integer-valued polynomial of ordinary degree at most j on the integer parameter box, and

$$\sup_t \|(I - M)(P(t) - I(t) - P(t_*) + I(t_*))\|_\infty \leq \varepsilon < \frac{1}{12}.$$

Suppose the original patch has a contributing tuple at t_ , whose first D entries are b_* . Then the integer polynomial*

$$b^*(t) = b_* + I(t) - I(t_*)$$

is the only possible choice of its first D slots anywhere that the original patch contributes. Substituting it into the higher slots and freezing the first residual at $r(t_, b_*)$ produces a patch with these D slots removed. If the original kernel has Lipschitz constant L_Φ for the maximum norm, the two patch values differ pointwise by at most $L_\Phi \varepsilon$.*

Proof. The displayed oscillation bound shows that the residual of $b^*(t)$ lies within ε of $r(t_*, b_*)$, and hence lies in $[-1/3, 1/3]^D$. If a different integer tuple b first differs in coordinate i , triangularity makes the difference between its i th residual and that of $b^*(t)$ a nonzero integer. The former cannot lie in $[-1/4, 1/4]$, because $1/3 + 1/4 < 1$. Thus every contributing tuple has first block $b^*(t)$.

For this fixed block the higher residual formulas are identical in the original and frozen tests. The union of higher tuples contributing to either kernel has size at most one, by triangularity. For any such tuple the Lipschitz bound gives an error at most $L_\Phi \varepsilon$. The same argument holds when one kernel value is zero, since the kernel is a globally Lipschitz function supported in the prescribed box. Summing over the possible higher tuples therefore gives the pointwise bound. Finally, substitution of a degree-at-most- j polynomial for a weight- j variable preserves the weighted degree bounds of every higher residual. $\square$

Lemma I.11 (Rank-preparation bookkeeping). *Suppose preparation of D coordinates of weight j proceeds by the following operation. At some tag $h \geq 1$, replace a space W_h by a codimension-one subspace and, if $h > 1$, place one one-dimensional contribution in a new tagged copy at $h - 1$. At $h = 1$ absorb that contribution into degree zero. Starting with $W_j = \mathbb{R}^D$ and no other positive tags, at most jD such operations are possible. At every stage*

$$\sum_h \dim W_h \leq D.$$

Only polynomially many ambient copy coordinates are needed when the degree is fixed.

Proof. There are two dimension counts. The unweighted sum $\sum_h \dim W_h$ starts at D . Each operation removes one dimension and adds at most one, so this sum never increases. The weighted sum $\sum_h h \dim W_h$ starts at jD and decreases by at least one at every operation, giving the bound on their number.

These sums count the dimensions of the spaces in which the polynomials take values, not the numbers of their ambient coordinates. A new copy may have many ambient coordinates, but its incoming polynomial is a scalar multiple of one removed vector and has a one-dimensional value space. There are at most jD copies. Each can be placed in a direct sum of the original coordinates and copies from higher tags. The descent has depth at most $j \leq s$, so the total ambient dimension is polynomial in D for fixed s . $\square$

Lemma I.12 (Preparation of an old layer). *Let $D \geq 1$ and let $P : \mathbb{R}^n \rightarrow \mathbb{R}^D$ have degree at most $1 \leq j \leq s$, with arbitrary real coefficients, and prescribe a rank threshold $R \geq 2$ and oscillation tolerance $\delta > 0$. On sufficiently large parent sides, there is a parameter slice $u = \iota(v)$ on which*

$$P \circ \iota = c + I + E + \sum_{h \leq j} S_h C_h.$$

Here c is constant, I has integral nonconstant coefficients, E has oscillation at most δ in maximum norm, C_h has degree at most h , takes values in a rational space $W_h \subseteq V_h$, and is high rank to threshold R in the final indexing. The ambient space at tag h is a direct sum $V_h = \bigoplus_{\nu \in \mathcal{I}_h} \mathbb{R}^D$, with $\mathcal{I}_h$ indexing tagged copies of the original coordinate space, and

$$S_h : V_h \longrightarrow \mathbb{R}^D, \quad S_h((z_\nu)_{\nu \in \mathcal{I}_h}) = \sum_{\nu \in \mathcal{I}_h} z_\nu.$$

Thus $\|S_h\|_{\infty \rightarrow \infty} \leq |\mathcal{I}_h|$, independently of the subspace heights. Moreover $\sum_h \dim W_h \leq D$; all ambient copy dimensions are bounded by a polynomial in $n + D$ for fixed s . Rational log heights of

the spaces W_h and auxiliary sections, and log slice costs, are polynomial in $n+D+\log R+\log(2+\delta^{-1})$. The slice may be selected to preserve the normalized average of any specified real score function on the parent box.

Proof. Start with $C_j = P$ and $W_j = \mathbb{R}^D$. If high rank fails at tag h , there is an integer row λ of norm at most R , nonzero on W_h , for which the top of λC_h is rational plus slow. Choose a rational $e \in W_h$ with $\lambda e = 1$, and replace C_h by $C_h - e\lambda C_h$, whose values lie in $W_h \cap \ker \lambda$. Write the removed term as its rational top, its slow top, and a polynomial of degree at most $h-1$.

Restrict to residue classes clearing the denominators of the rational top, including those of e . Every positive-degree coefficient then becomes integral; the constant is absorbed in c . Subdivide the normalized box so finely that the slow top oscillates by at most a prescribed share of δ , and absorb its value at one point into c . Move the lower-degree term, which takes values in the line spanned by e , into a new copy at tag $h-1$, or absorb it as a constant if $h=1$. Use the actual unmodified score to select a cell of the residue and subbox partition whose normalized average is at least the preceding average. This exists by averaging. Equal or almost equal subdivisions keep every side at a controlled relative size once the parent cutoffs are imposed.

After reindexing, repeat the rank tests at the same R , including tags tested before a subsequent restriction. Lemma I.11 shows that at most jD failures can be removed. At termination every remaining tag is high rank in the final indexing. Previously integral nonconstant coefficients remain integral under integer affine reindexing. Previously allocated oscillation errors can only decrease on restriction. Thus allocating $\delta/(jD)$ per step, with the polynomial norm bound of the copy-summing maps included, gives the asserted total error.

We verify that the rational estimates do not undergo one polynomial exponentiation per cut. Fix a polynomial bound m for all copy coordinates. At tag h let B_h be the block-diagonal basis matrix of its original and incoming spaces, before cuts at that tag, and stack all imposed rows into A_h . The current space is $B_h \ker(A_h B_h)$. Extend earlier rows by zeros whenever a copy arrives. To choose the next section, solve jointly

$$A_h B_h z = 0, \quad \lambda B_h z = 1, \quad e = B_h z.$$

The system is consistent because λ is nonzero on the current space. Clearing the polynomial number of denominators and using one nonzero minor gives

$$\log \text{ht}(e), \log \text{ht}(W_h) \leq m^{C_s} (1 + \log R + \log \text{ht}(B_h)).$$

Crucially, B_h contains sections coming from higher tags, not sections previously chosen at this same tag. Thus the displayed polynomial recurrence is composed at most s times. It bounds both $\|e\|_\infty$ and a common denominator d_e of its coordinates.

For completeness, if $q \leq R$ clears the rational scalar top, the modulus $a = qd_e$ clears the nonconstant coefficients of its product with e on every residue class $u = r + av$: every positive-degree term in the expansion contains a factor a . Its arbitrary numerators do not affect this assertion. In normalized variables the slow top has at most $\binom{n+s}{s}$ monomials with coefficients at most R . If K bounds the copy-summing maps, subdivision into B_0 pieces per coordinate costs an oscillation at most

$$\frac{K s \binom{n+s}{s} R \|e\|_\infty}{B_0}.$$

Choose B_0 large enough to make this at most $\delta/(jD)$. Then $\log(2aB_0)$ is polynomial in the asserted budget. On sufficiently large current sides, nearly equal subdivisions in each residue class give every cell side at least $H_i/(2aB_0)$. Consequently each cut has that polynomial log cost, and summing

over at most jD cuts gives both the claimed cost and a uniform sufficient parent-side cutoff. Copy dimensions are bounded by a fixed polynomial majorant of $D(1 + jD)^j$, by the bounded descent depth, independently of orientations and heights. $\square$

After successful preparation and sampling, the integer polynomial $I(t)$ in Lemma I.10 is the prepared integral-coefficient contribution plus $\sum_h S_h(\beta_h(\psi(t)))$. The small residual oscillation is the sum of the preparation error and the chart widths. Choosing that error small relative to the input absolute score permits the recursive call with the original density threshold, losing only absolute surplus. This choice must be made before the rank threshold and subspace heights become available.

Proposition I.13 (Linear rank accounting). *Assume the following three assertions, with their permitted early or augmented logarithmic cost bounds:*

1. *rank preparation and conditioned sampling remove the lowest D slots of weight j , retaining positive absolute score and producing spaces with $\sum_h \dim W_h \leq D$;*
2. *a family of recursively returned patches of rank at most d_* can be globalized in the same patch group as in Proposition I.6;*
3. *transferring this fixed global patch to the parent costs at most $\sum_h \dim W_h$ new slots, and uses an arbitrarily prescribed fixed multiplicative discount of the target density.*

Assume also the unconditioned base case returns rank at most d_0 and permits an arbitrarily prescribed fixed density discount. Then processing an old patch with d slots returns rank at most $d_0 + d$. For fixed s , its total multiplicative density discount can be made at most any prescribed $\epsilon_0 > 0$.

Proof. Let D_j be the number of original slots of weight j . After the lowest remaining layer j is removed, the recursive call acts on $d - D_j$ old slots. On returning, the old test is discarded: only the returned patch is globalized and charged the new $\sum_h \dim W_h$ slots. Consequently the rank satisfies

$$R(D_1, \dots, D_s) \leq R(0, \dots, 0, D_{j+1}, \dots, D_s) + D_j, \quad R(0, \dots, 0) \leq d_0.$$

Iterating gives $R \leq d_0 + \sum_j D_j = d_0 + d$. There are at most s upward passages and one base passage. Taking each fixed discount $\delta \leq \epsilon_0/(s+1)$ makes their product at least $1 - (s+1)\delta \geq 1 - \epsilon_0$. $\square$

Remark I.14 (Order of parameters in the relative use). The recursive input budget is fixed from the old patch, the dimensions of all possible copies, the desired chart widths, and the retained absolute score. It must not include $\log R$, $\log L$, or the eventual heights of the prepared spaces. The sampler dimension must be fixed from structural dimensions before these recursive accuracy requests; a mere polynomial bound on that dimension allowing dependence on later requested precisions would create a circular requirement. This structural independence is explicitly part of Proposition F.2.

The removed directions e are stored in the unrestricted coefficients of the copied polynomials. The maps S_h only add coordinate copies, and their norms depend on copy counts, not on the rational heights. Therefore the widths needed in Lemma I.10, and the complexity of the remaining kernel after freezing, can be prescribed from the structural budget. The coefficients of the substituted higher residuals remain free. Compute the recursive output, accuracy, and globalization budgets with these data first. Only then choose perturbations, the long-scale range and R , perform Lemma I.12, choose the allowed long scale for its final spaces, and impose parent side cutoffs including its late height and slice costs. At most s old layers are processed, so this forecasting uses bounded depth. This is the order used in the relative induction below, with sampling supplied by Proposition F.2 and step-drop by Lemma B.1.

Theorem I.15 (Relative lifting from the sampling assertions). *Assume Lemmas B.1 and A.5, Proposition F.2, and the scalar transfer of Proposition E.1, including their stated uniform polynomial logarithmic bounds. Then Proposition D.7 holds, with rank at most $d_0 + d$ in its increment alternative. Its exponent E depends only on s , the fixed base dimension n_0 , its fixed side-ratio bound, and the prescribed discount ε_0 .*

Proof. Induct on the number of distinct positive weights in the old patch. Fix a discount $\tau \leq \varepsilon_0/(s+1)$ for each upward passage and the base passage, with $0 < \tau \leq 1/2$. For early bookkeeping replace the absolute rank bound by $\min(d_0, p)$: the rule's complexity bound already bounds every actual returned rank by p . The eventual rank estimate with the original d_0 only becomes weaker. We use the absolute rule solely through its given dimension, cutoff, score, slice, rank, and complexity bounds; no constant below depends on the particular rule.

Base case. If the old patch has no slots, it is a constant in $[0, 1]$. The positive old score gives $\mathbb{E}f \geq a + e^{-p}$: if the constant old patch is c , then $c > 0$ and $\mathbb{E}f - a \geq e^{-p}/c \geq e^{-p}$. Use the unconditioned sampler in dimension exactly n_0 , with distinct comparable prime sides beyond the absolute cutoff. Its spatial law is dominated by the product-box law by an early factor, as specified in the unconditioned variant of Proposition F.2; hence scalar transfer applies to this same ensemble. Put $\sigma = e^{-p}$ and prescribe both the one-site error and the noninjective probability to be at most $\sigma/16$. If Z is the sample mean minus a , then $|Z| \leq 1$ and $\mathbb{E}[Z1_{\text{injective}}] \geq 7\sigma/8$. Since $Z \leq 1$, the fraction of injective samples with $Z \geq \sigma/2$ is at least $\sigma/4$. These samples therefore have mean at least $a + e^{-p}/2$. The structural side-ratio constant in the absolute rule is fixed with $R \geq 2^{n_0+1}$. Choose L divisible by 2^{n_0+1} and apply Bertrand's postulate in n_0 successive disjoint dyadic intervals inside $[L/R, L]$. This gives distinct prime sides comparable to L . Taking $L/R > e^p$ puts every side beyond the rule's cutoff; these are all early requirements on L .

The pullback of f on an injective affine sample is progression-free: a nonzero vector progression in its box has two distinct points, whose images remain distinct, and its image is an arithmetic progression in the parent. If the absolute rule is an impossibility rule, these samples are already a contradiction. Otherwise that rule supplies a patch of rank at most d_0 and early positive score against the pullback minus Λ on a permitted sample slice. Lemma I.8 and the unconditioned case of Proposition I.6 replace these by one fixed ambient patch evaluation, still with at most d_0 slots, on an early fraction of samples.

Apply scalar transfer to this fixed ordinary niltest g , with $\lambda' = (1 - \tau)\Lambda$. It has the required positive lower bound, after an early budget increase. Choose its fixed slack $\epsilon = \tau/(2(1 - \tau))$, so that $(1 + \epsilon)\lambda' = (1 - \tau/2)\Lambda \leq \Lambda$. If every permitted ambient slice had score at most the prescribed small error against $f - \lambda'$, scalar transfer would bound the probability of the observed local positive scores by a smaller early fraction. Here is the finite union bound for the path-dependent slices. Let b bound their total early costs, let δ be their retained score and let ρ be the productive fraction. Each original side is comparable to L , and each restricted side has at least $Le^{-O(b)}$ points. Once this is at least two, its common modulus can be represented by an integer at most $e^{O(b)}$. Enumerate these moduli and residues, and mesh the interval endpoints in units of L at resolution $\delta e^{-O(b)}/C_{n_0}$. The change in the normalized slice measure is at most $\delta/4$ in ℓ^1 distance, after L absorbs the integer-rounding error. The score integrand is bounded by one, so this is also a score bound. There are $N \leq \exp(\mathcal{P}(b + \log \delta^{-1}))$ choices, independently of L . Choose the scalar log budget p_{sc} in advance so that its complexity, modulus, side-length and domination hypotheses hold and

$$e^{-p_{\text{sc}}} < \min\{\delta/4, \rho/(2N)\}.$$

Apply Proposition E.1 to each fixed mesh choice with its own uniform t law. Only the outer law of (x, V) is changed to the actual dominated unconditioned law; no path-dependent reweighting

of t is used. If every permitted ambient slice had score at most the resulting e^{-P} , the union of the local positive-score events would have probability at most $Ne^{-P_{\text{sc}}} < \rho/2$, a contradiction. All endpoint, precision and domination budgets were fixed before choosing L . The contradiction yields an ambient slice with early positive score against $f - (1 - \tau)\Lambda$, with the same d_0 slots. This proves the base increment case.

Removing the lowest weight. Suppose the lowest weight is j and it has D slots. Normalize the patch coefficients by the triangular integral substitutions in Lemma I.8. The kernel is unchanged. At this lowest weight, previous-slot terms are linear with constant coefficients, now of absolute value at most one. Thus its residual block is $(I - M)(b - P(u))$, with $\|I - M\|_{\infty \rightarrow \infty} \leq 1 + D$ and $\deg P \leq j$. The coefficients of P need not be bounded. Forecast the polynomial number of all possible copy coordinates in Lemma I.12, and let K bound $\sum_h \|S_h\|_{\infty \rightarrow \infty}$ for every such configuration. Put $\sigma = e^{-P}$ and prescribe the preparation error δ and common chart radius w so that

$$(1 + D)(\delta + 2Kw) \leq \min\{1/24, \sigma/(16 \max(1, L_\Phi))\}.$$

Only the early kernel, score and structural data enter these choices, not the heights of the eventually prepared spaces.

The constrained sampler dimension is fixed structurally. Consequently the inductive input dimensions, kernel complexity, desired inverse score, and the original absolute rule's bounds can all be put in an early polynomial budget before choosing its testing accuracies or long scales. Compute the induction output and globalization budgets first. Then choose the later parameters in the sampler's required order, choose R for the entire allowed long-scale range, and perform Lemma I.12. The initial side cutoff includes its polynomial late height and slice costs. The final prepared spaces admit one of the allowed long scales, and their high rank holds at the threshold already chosen for the whole range.

Preparation preserves the actual old score, which is at least σ . Prescribe the mixed-center one-site error and the bad-path probability to be at most $\sigma/16$ each. For the sampled old score Z , $|Z| \leq 1$ and $\mathbb{E}[Z1_{\text{good}}] \geq 7\sigma/8$. Therefore at least a $\sigma/4$ fraction of paths are good and have $Z \geq \sigma/2$. Averaging in the centers gives one fixed center tuple with at least that productive fraction. On each such path choose a point where the old patch contributes. The prepared integral part plus $\sum_h S_h(\beta_h(\psi(t)))$ is the integer polynomial $I(t)$ of Lemma I.10: the prepared integral part stays integral under the affine integer map, and each $\beta_h \circ \psi$ has integer coefficients and degree at most $h \leq j$. Constant centers cancel in differences. The residual oscillation is at most $(1 + D)(\delta + 2Kw)$, so freezing loses at most $\sigma/16$ in score. The remaining patch has $d - D$ slots and score at least $7\sigma/16$ against $f - a$. Its kernel is $\Phi(r_*, \cdot)$, with the same support and no larger Lipschitz constant. Its residual coefficients may be large, as permitted in the patch definition.

Let p_{child} include the structural dimension, frozen kernel complexity, p , and $\log(16/(7\sigma))$, and let H_{child} be the already computed inductive side cutoff for that budget. Choose $L > H_{\text{child}}$. Freeze axes with $T_i < H_{\text{child}}$ on a favorable fiber. This preserves normalized score, and each fixed axis costs at most $\log H_{\text{child}}$; their total cost is early. Omit these singleton coordinates for induction, then reinsert their fixed values when recording its output as a slice of $\mathcal{T}$. At least one free axis remains because the sampler has a side of length L , and every free side is at least H_{child} . The same absolute rule is valid with budget $p_{\text{child}} \geq p$: its cutoff, slice, complexity and inverse-score bounds only become weaker, and $a \geq e^{-p_{\text{child}}}$. Apply induction to this positive-dimensional box and patch. In the impossibility case induction contradicts the retained score, so that alternative is proved at every depth. In the increment case it returns patches and slices of early complexity and positive score at the inductively discounted target, with rank at most $d_0 + d - D$.

Discard the old test and globalize precisely these returned patches by Corollary I.9. This costs

at most $\sum_h \dim W_h \leq D$ slots and one additional fixed discount τ . Therefore the rank is at most

$$d_0 + (d - D) + D = d_0 + d.$$

There are at most s old-weight passages plus the base passage, so the product of the target factors is at least $(1 - \tau)^{s+1} \geq 1 - \varepsilon_0$. Lowering the target to the one requested only increases its positive score.

Here is an explicit form of the final budget recurrence. Let $F_q(p)$ majorize all required logarithmic bounds when at most q distinct old weights remain, including the sufficient parent log side. The base construction gives a polynomial F_0 . Structural dimensions, the frozen kernel and its retained score admit $p_{\text{child}} \leq (2 + p)^{c_s}$, independently of R, L and heights. After computing the child output, all globalization and testing requirements fit in

$$b_* \leq (2 + p + F_{q-1}((2 + p)^{c_s}))^{c_s}.$$

The quantitative sampler hypotheses choose perturbation, A and R with logarithms bounded by a fixed polynomial in b_* , valid throughout $A \leq L \leq A^{C_s(1+\sum_h d_h)}$. Preparation's joint height estimate then bounds its heights and slice costs by a polynomial in $p + \log R$. The allowed augmented output costs, and the parent cutoff absorbing every restriction, are consequently bounded by one more fixed polynomial in b_* . Increasing a fixed exponent if necessary gives

$$F_q(p) \leq (2 + p + F_{q-1}((2 + p)^{c_s}))^{C_s}.$$

The polynomial number of preparation cuts adds costs; it does not add compositions to this recurrence. Since $q \leq s$, there is one exponent E , depending only on the fixed parameters, bounding all sufficient initial side cutoffs in logarithm, final complexities, log inverse scores and slice costs by $(2 + p)^E$. This is the full uniform assertion of Proposition D.7. $\square$

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